

Doctoral Thesis

Investigating the Relationship Between Multiword Sequences and L2 Oral
Fluency: The Impact of Lexical Bundle Length on Fluency and Proficiency

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Summary

Lexical bundles (LBs) are crucial in second language (L2) oral fluency and proficiency, yet their impact in terms of length is under-researched. Previous studies have predominantly focused on shorter LBs (typically two- and three-word sequences), limiting our understanding of learners' phraseological knowledge and its impact on oral fluency. Responding to calls from recent literature (e.g., Hougham et al., 2024a; Tavakoli & Uchihara, 2020), this thesis addresses this gap by investigating how LB usage varies across proficiency levels, the relationship between different LB lengths and aspects of oral fluency, the predictive power of LB usage on oral presentation scores, and the influence of presentation topics on LB use. Four experimental studies explore the following questions: How does LB usage vary across different proficiency levels? ([Chapter 3](#)); To what extent do longer vs. shorter LBs influence fluency aspects (speed, breakdown, and repair)? ([Chapter 4](#)); To what extent do longer LBs predict oral presentation scores compared to shorter LBs? ([Chapter 5](#)); Does presentation topic influence the diversity and frequency of high MI-score bigrams, and does this influence vary with proficiency? ([Chapter 6](#)). This thesis aims to shed light on how LB length influences L2 fluency and proficiency, focusing on the predictive role of shorter vs. longer LBs in shaping fluency across proficiency levels.

Participants were 150 L2 English learners enrolled in a pre-session course at a UK university, completing a 7-minute oral presentation assessed using grading descriptors. There were 50 participants each at IELTS levels 6.5, 7, and 7.5. I used a two-pronged approach to LB measurement: 1) Text-externally, I calculated scores for two- and three-word units (proportion, frequency, and association strength) using TAALES (Kyle & Crossley, 2015) with the Corpus of Contemporary American English (Davies, 2009) as a reference; and 2) Text-internally, I identified frequent and strongly associated LBs using AntConc (Anthony, 2020) and Collocate

(Barlow, 2015). I measured three- to five-word sequence usage within the learner corpus using an innovative LB refinement procedure and scoring system. Robust regression, dominance analysis, and random forests techniques isolated the unique contribution of LB length to three aspects of fluency (speed, breakdown, repair) and oral proficiency scores.

The findings show that both shorter and longer LBs are integral to L2 fluency and proficiency, with usage increasing alongside proficiency levels. Higher proficiency learners used both shorter and longer LBs more frequently, with a shift toward higher mutual information (MI) scores, indicating richer language use. Longer LBs contributed to smoother speech by increasing articulation rates and reducing mid-clause pauses, though their complexity led to a modest trade-off in speed fluency. Shorter LBs had mixed effects: bigram MI reduced mid-clause pauses, while bigram frequency and trigram MI were linked to increased end-ASU pauses, reflecting planning demands. Presentation topics significantly influenced LB usage, with specialized topics eliciting higher MI scores, highlighting the context-dependent nature of L2 fluency. These results underline the complementary roles of shorter and longer LBs in shaping fluency, supporting theoretical models of L2 speech production that emphasize pre-constructed language units for cognitive resource management (e.g., Levelt, 1989; Skehan, 2014). The trade-offs between LB complexity and fluency suggest that longer sequences may introduce processing demands, warranting further investigation. The thesis confirms that LBs (particularly high-MI bigrams) are predictive of oral proficiency scores and play a key role in fluent speech across proficiency levels. The results suggest incorporating high-MI bigrams and longer sequences into curriculum design, targeted practice, and assessment criteria. Future research should expand participant diversity and sample size, employ longitudinal designs, and explore additional LB measures to better capture L2 fluency and LB-use dynamics.

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Glossary

Analysis of Variance (ANOVA)	A statistical measure used to analyze the differences among means.
Chunking	The process by which learners store and retrieve language as multiword units rather than individual words. This process reduces cognitive load and facilitates fluent speech.
Cognitive Fluency	The smooth operation of the cognitive processes (i.e., speed and efficiency of word-meaning retrieval) underlying second language (L2) speech acts
Confidence Interval (CI)	The mean of an estimate +/- the variation in the estimate. A CI is considered statistically significant when it does not include zero, indicating that the effect or difference observed is unlikely to be due to random chance.
Lexical Bundles (LBs)	A contiguous type of multiword sequence identified primarily by their frequency (i.e., a minimum number of occurrences of the unit in a corpus) and dispersion/range (i.e., a minimum number of texts in which the units occur or a minimum number of learners using them).
Multiword Sequences (MWSs)	A broad category that encompasses different types of commonly occurring combinations of words in language, including idioms (e.g., <i>hit the sack</i>), phrasal verbs (e.g., <i>look into</i>), and lexical bundles (e.g., <i>at the end of</i>). The term MWS is often used interchangeably with MWEs (multiword expressions), MWUs (multiword units), and FSs (formulaic sequences), among other acronyms.
Mutual Information (MI)	A statistical measure that quantifies the association strength between word pairs.
Perceived Fluency	Listener-based subjective judgements of fluency
Phraseological Knowledge	The knowledge of multiword sequences, including how to use them appropriately in speech and writing.
Utterance Fluency	The temporal aspects of speech (i.e., speech rate, hesitations, and pausing phenomena) that can be measured objectively.

Chapter 1: Introduction

1.1 Understanding L2 Fluency: The Importance of Multiword Sequences

How can second language (L2) learners achieve the kind of fluency that allows them to convey their thoughts effortlessly and smoothly? What roles do multiword sequences (MWSs) play in helping learners sound more natural? How can mastering these phrases enhance communicative competence? MWSs such as lexical bundles (LBs) are arguably key to understanding how language learners develop fluency and achieve more natural speech. MWSs help speakers maintain the flow of speech, reduce cognitive load, and sound more fluent by allowing them to retrieve preconstructed chunks rather than constructing each sentence word by word from scratch. For learners, mastering these sequences can lead to more confident communication in both academic and real-world settings, allowing them to reach their language learning goals more effectively.

A major challenge in language learning is helping learners communicate more effectively and smoothly to achieve their language learning goals. This research area is particularly important given the increasing emphasis on spoken fluency skills in today's interconnected world, where effective communication is key. Moreover, with rapid advancements in generative AI, the need for effective communication skills has grown significantly. Language learners must now be able to speak fluently and naturally, not only to engage in real-world interactions but also to take full advantage of emerging learning opportunities and to excel in both professional and personal areas of interest. By focusing on the role of MWSs, the current thesis aims to provide empirical insights into how learners can improve their spoken fluency, ultimately contributing to their overall communicative competence and success in real-world situations such as results on high-stakes language assessments.

1.2 Background and Context

In recent years, there has been a growing interest in studying MWSs in L2 oral proficiency. Foundational research by Pawley and Syder (1983) emphasized the critical role of lexical phrases, or “chunks,” in achieving L1-like fluency. They argued that a large part of first language (L1) speakers’ fluency stems from their use of these prefabricated chunks, which reduce the cognitive load during speech production by allowing speakers to retrieve whole phrases from memory rather than constructing sentences word-by-word. This process, known as “chunking” (Ellis, 1996), enables more fluent and efficient language use and is crucial for effective communication. Building on these insights, subsequent research has highlighted the significance of MWSs in L2 learning and speaking proficiency (Schmidt, 1992; Nation, 2013). LBs, defined as frequent, contiguous MWSs, have gained particular attention for their role in speech fluency (McGuire & Larson-Hall, 2021; Tavakoli & Uchihara, 2020) and oral proficiency (Garner & Crossley, 2018; Kyle & Crossley, 2015; Zhang et al., 2021). The findings from these studies suggest that incorporating MWS-focused activities in language teaching can significantly enhance students’ speaking skills. Despite the growing evidence of the importance of MWSs, there are still significant gaps in our understanding of how LBs of different types contribute to fluency.

1.3 Addressing a Major Gap in Existing Research

Although research findings in this area have been insightful, previous studies have predominantly focused on shorter LBs (typically two- and three-word sequences), thus potentially limiting our understanding of learners’ phraseological knowledge and its impact on oral proficiency. Exploring this gap by examining both longer (e.g., *as you can see*) and shorter (e.g., *there are*) LBs is critical, as longer LBs could offer enhanced proficiency and show

processing efficiencies that are essential for fluent speech production. Responding to calls from recent literature (e.g., Hougham et al., 2024a; Tavakoli & Uchihara, 2020), the current thesis addresses this gap by investigating how the usage of LBs varies across proficiency levels, the extent to which LBs of different lengths are associated with aspects of oral fluency, the predictive power of both shorter and longer LB usage on oral presentation scores among learners with varying proficiency levels, and the extent to which presentation topics influence the use of LBs. I aim to isolate the unique contributions of longer LBs in comparison to shorter LBs using an innovative two-pronged (text-internal and text-external) approach to measure LBs, thus providing a deeper understanding of how LB length influences oral fluency and proficiency. Addressing this gap is crucial for advancing our theoretical understanding of phraseological knowledge in L2 learners and for developing practical teaching strategies that leverage the benefits of various types of LBs to enhance oral fluency.

1.4 Research Questions and Objectives

The research questions guiding the current thesis are designed to explore how LBs of different lengths contribute to L2 oral fluency and proficiency across various proficiency levels. Specifically, the empirical research seeks to answer the following four questions: (1) How does LB usage, differentiated by length, vary across different proficiency levels? ([Chapter 3](#)); (2) To what extent is the use of longer versus shorter LBs associated with the three aspects of oral fluency: speed, breakdown, and repair? ([Chapter 4](#)); (3) To what extent do longer LBs predict raw oral presentation scores compared to shorter LBs? ([Chapter 5](#)); (4) does the choice of presentation topic influence the diversity and frequency of high-MI (mutual information) bigrams used by L2 English speakers, and does this influence vary with proficiency scores?

(Chapter 6). These questions are intended to provide a comprehensive view of the role LBs play in oral fluency and proficiency.

The objectives of this research are to understand the relationships between LB usage and various aspects of L2 oral fluency, identify the specific effects of different LB lengths on fluency measures, validate and refine existing theoretical frameworks related to phraseological knowledge and fluency development, and provide practical suggestions for pedagogical and research purposes. I aim to contribute to a deeper understanding of how shorter and longer LBs influence oral fluency, particularly in relation to proficiency levels. This knowledge can be used to enhance teaching practices by highlighting which types of LBs are most beneficial for learners at different stages, thus providing empirical support for instructional approaches that emphasize MWS use as a pathway to improve L2 fluency for real-world purposes.

1.5 Structure of the Thesis

The current thesis is organized as follows: Chapter 2 reviews existing literature on the LB-fluency link, highlighting theoretical frameworks, key empirical findings, and limitations of previous studies. Chapter 3 explores how LB usage varies across proficiency levels. Chapter 4 examines the influence of LB length on aspects of L2 fluency, analysing shorter and longer bundles' contributions. Chapter 5 investigates the extent to which LBs of different lengths can predict oral presentation scores. Chapter 6 explores whether presentation topics affect the use of sophisticated LBs. Chapter 7 presents a detailed discussion and synthesis of the findings of the four experimental chapters in relation to existing theories and previous findings, linking them back to the theoretical models and empirical studies reviewed in Chapter 2. Chapter 8 concludes the thesis by summarizing the main contributions and implications for L2 fluency development in terms of synergizing theory and practical, pedagogical applications.

Chapter 2: Literature Review and Theoretical Framework

To investigate the relationship between the use of MWSs and oral fluency, it is essential to define both concepts. My aim in this chapter is to examine how oral fluency and MWSs are defined and measured, explore theoretical frameworks explaining the MWS-fluency relationship, review key findings from previous studies on this relationship, and provide a foundation for the experimental chapters (3, 4, 5, and 6) that follow this literature review.

The literature review is divided into six sections. First, I discuss how oral fluency is conceptualized in the literature and how it is operationalized in the current thesis. This first section also explores how MWSs and LBs are defined and measured according to various approaches and techniques. Next, I review several studies that explore MWS processing advantages and identification methods, addressing the recurring issue of the structural status of LBs. Additionally, a meta-analysis on the processing advantage of MWSs is reviewed. Following this, the review presents theoretical frameworks that explain why and how MWS use relates to oral fluency. Subsequently, key findings from previous studies examining the MWS-fluency link are reviewed, focusing on studies that have examined the use of MWSs in relation to L2 oral fluency. The review then synthesizes the findings and discussions from the previous sections, highlighting limitations and gaps. Finally, the literature review concludes with a summary of the key points from all sections and suggests the need for a more detailed experimental investigation of the claims discussed throughout this chapter.

2.1 L2 Oral Fluency

L2 fluency can be defined broadly as general L2 proficiency, with many using the terms “fluency” and “speaking ability” interchangeably (Tavakoli & Hunter, 2018). Alternatively, it

can be considered in a narrower sense, as a component of oral proficiency (Lennon, 1990). Segalowitz (2016) outlines three types of L2 fluency: cognitive, utterance, and perceived fluency. Cognitive fluency means the smooth operation of the cognitive processes (i.e., speed and efficiency of word-meaning retrieval) underlying L2 speech acts. Utterance fluency refers to the temporal aspects of speech (i.e., speech rate, hesitations, and pausing phenomena). Perceived fluency refers to listener-based subjective judgements of fluency. Given these distinctions, the current thesis focuses on fluency in the narrow sense, specifically on utterance fluency.

Utterance fluency can be conceptualized in terms of three dimensions, where fine-grained analysis of audible speech is objectively measured according to: (a) speed (i.e., the flow and continuity of speech), (b) breakdown (i.e., pauses that disrupt the speech flow), and (c) repair (i.e., self-monitoring mechanisms such as self-corrections and reformulations) (e.g., Skehan, 2003; Suzuki et al., 2021; Tavakoli & Skehan, 2005). This “objective” approach contrasts with “subjective” methods (e.g., Boers et al., 2006; Derwing et al., 2004; Saito et al., 2018) that rely on human judgements to measure perceived fluency based on various assessment criteria such as fluency, range of expression, and accuracy. For instance, Boers et al. (2006) asked an experienced teacher to rate speech samples using an assessment sheet that considered fluency, range of expression, and accuracy. Similarly, Derwing et al. (2004) enlisted untrained listeners to judge speech samples according to the flow and smoothness of speech, focusing on temporal variables like silent pauses rather than overall proficiency. These listeners rated the speech samples on a 9-point scale from 1 (extremely fluent) to 9 (extremely dysfluent). As the field of language testing shifts toward more objective measurements of L2 ability (Tavakoli et al., 2017), the use of more objective measures of fluency becomes increasingly important (Thomson et al., 2019).

Studies using the objective approach have found that fluent and dysfluent speech differ significantly in terms of pause location. Fluent speech is characterized by less frequent pausing in mid-clause positions, whereas dysfluent speech often contains more mid-clause pauses (Tavakoli, 2011; Kahng, 2014). Furthermore, mid-clause pauses have been found to be related to proficiency, with more proficient learners producing fewer silent pauses within clauses (de Jong, 2016a), and to perceived fluency (Kahng, 2017; Suzuki et al., 2021). These studies show that pause location is an important indicator of successful fluent speech production. Bringing these threads together, the current thesis examines fluency in the narrow sense, emphasizing the importance of investigating objective L2 fluency measurements—specifically speed, repair, and breakdown, including pause location in terms of mid- versus end-clause position—in the fields of language teaching and testing (Tavakoli et al., 2017; Thomson et al., 2019).

2.2 Multiword Sequences and Lexical Bundles

MWS is an umbrella term encompassing combinations of words that occur together in language such as idioms (e.g., *break a leg*), collocations (e.g., *fast food*), phrasal verbs (e.g., *cheer up*), proverbs (e.g., *the squeaky wheel gets the grease*) and LBs (e.g., *one of the*) (Garner & Crossley, 2018). We can broadly identify MWSs according to one of two approaches: a phraseological or a frequency-based approach (Granger & Paquot, 2008; Nesselhauf, 2004). The phraseological approach defines MWSs using linguistic criteria on a continuum ranging from free combinations to pure idioms (Cowie, 1981). The phraseological approach originated with the pioneering work of Russian phraseologists' (e.g., Vinogradov, 1947, cited in Cowie, 1981) in developing a framework of descriptive categories in the 1940s. This approach often relies on L1-speaker intuition in defining and measuring MWSs. Previous studies taking a phraseological approach (e.g., Boers et al., 2006) often involve the use of subjective (human) raters who

measure MWSs by counting the number of multiword chunks they consider to be formulaic sequences. Such subjectivity might cause a relatively low agreement between experienced judges because the level of inter-rater reliability ($r < .60$), for instance, reported by Boers et al. (2006) is less than the median inter-rater reliability ($r = .92$) in L2 acquisition research reported by Plonsky and Derrick (2016).

In contrast, the frequency-based approach uses corpus-based automated extraction techniques to identify a wider range of recurrent word combinations according to quantitative criteria such as frequency threshold (i.e., a minimum number of occurrences of the unit in a corpus) and range (i.e., a minimum number of texts in which the units occur or a minimum number of learners using them). Researchers often refer to these recurrent word combinations as lexical bundles or n-grams where n can be two, three, four, or more consecutive words, including structurally complete word combinations (e.g., *you know what I mean*) and incomplete ones (e.g., *so I think the*), “regardless of their idiomaticity, and regardless of their structural status” (Biber et al., 1999, p. 990). LBs are a type of MWS that have become the focus of an increasing number of learner corpus studies because of objective, frequency-based measurement techniques (Paquot & Granger, 2012; Granger, 2019). By using the lexical bundle approach, we can acquire a large quantity of data for quantitative investigation, which serves as an ideal starting point for phraseological explorations (Ebeling & Hasselgård, 2015). In the current thesis, the term LB is used interchangeably with n-gram, referring to sequences identified using automatic extraction software with specified criteria such as minimum frequency, minimum range, and mutual information (MI) scores. MI scores evaluate the strength of association between pairs of words. In the review of research and the discussion that follows, I strive to distinguish between LBs and other types of MWSs wherever possible. However, it is important to recognize that the

conceptual boundaries between LBs and other MWSs are sometimes ambiguous. For instance, the automatic extraction of bigrams with high MI scores could be seen as imbuing LBs with a distinct collocational quality.

When considering LB production in learner corpus research, two approaches of analysis can be taken: text-internal or text-external. Studies taking a text-*internal* approach (e.g., Biber & Gray, 2013) analyze LB production using only data within learner corpora (i.e., electronic collections of learner-produced text), tallying the number of adjacent sequences of a specified number of words. An example of a study that takes a text-internal approach is Biber and Gray (2013), which looked at the use of both lexical bundles and collocations in written and spoken responses to the TOEFL Internet-based Test (iBT). Biber and Gray (2013) only looked at word sequences that are repeated within the learner corpus itself, focusing on sequences that are common in student writing and speaking, rather than those which are target-like (i.e., commonly used by L1 speakers). One of their findings was that many lexical bundles were borrowed from task prompts.

In the text-*external* approach, on the other hand, the formulaic status of word combinations is determined according to the frequency of words co-occurring in an external reference corpus (Gablasova et al., 2017). In other words, an external reference corpus is used to identify and extract MWSs that co-occur in the learner corpus. The text-external approach is thus concerned with the extent to which L2 learners use word combinations that appear in the language used by L1 speakers in a target language community. An example of a longitudinal study that took a text-external approach is Garner and Crossley (2018), which adopted corpus-based measures to identify recurrent two- and three-word strings. Garner and Crossley examined the extent to which L2 speakers use target-like MWSs. Their study confirmed that L2 users

produce a gradually increasing proportion of two- and three-word units frequently found in an L1-speaker reference corpus. Another example of a study taking a text-*external* approach is Tavakoli and Uchihara (2020), which is reviewed in Section 2.5.2.

Both above-mentioned approaches (text-internal and text-external), however, have limitations. Myles and Cordier (2017) argued that the text-internal approach is limited because the frequency cut-offs adopted are arbitrary and likely to be low in learner-produced texts in a single data set, and raw frequency alone is an inadequate measure of formulaic language. The text-external approach is limited because certain widely used software programs (e.g., the Tool for the Automatic Analysis of Lexical Sophistication: TAALES; Kyle & Crossley, 2015; Kyle et al., 2018) used to determine the frequency of co-occurrence can only analyze shorter (two- and three-word) bundles. A further limitation of text-external techniques is that frequency of co-occurrence in external corpora is no guarantee that word bundles have psycholinguistic reality for the specific learners investigated (Ellis et al., 2009; Myles & Cordier, 2017). The current thesis aims to overcome these limitations by adopting a frequency-based approach using text-internal techniques to measure the contribution of longer LBs and using text-external techniques to measure the contribution of shorter LBs.

One major recurring issue in the LB literature is the structural status of LBs. The frequency-based approach, in which LBs are automatically extracted from a corpus, means that the vast majority of extracted LBs are semantically and/or structurally *incomplete* (Biber, 2006). Several studies have examined LBs in various attempts to characterize academic discourse (e.g., Biber, 2006; Biber et al., 2004; Cortes, 2004; Hyland, 2008). These studies have produced LB lists with many semantically and structurally incomplete bundles such as *to do with the* (Simpson-Vlach & Ellis, 2010; Liu, 2012). Such incomplete bundles are “neither terribly

functional nor pedagogically compelling” (Simpson-Vlach & Ellis, 2010, p. 493). Subsequent research has reported mixed findings: Such incomplete bundles could provide a processing advantage in terms of speed fluency, as their composition might leave memory traces in the brain (Tremblay et al., 2011). However, they are unlikely to be stored or represented as wholes in the mental lexicon, nor would they provide a processing advantage compared to more complete formulaic sequences (Jeong & Jiang, 2019). To tackle the problem of structural incompleteness, Liu (2012) proposed extracting LBs and identifying core constructions from those which appear incomplete (e.g., *to do with the* can be reduced to *to do with*). In response to Liu’s (2012, p. 27) call for a “more effective approach for identifying useful LBs,” Wood and Appel (2014) introduced a new method that goes beyond the traditional LB extraction method, aiming to produce a more educationally valuable list of recurrent sequences (3-5 words in length), which they call multiword constructions (MWCs).

2.3 Studies Investigating MWS Processing Advantages and Identification Methods

As Tremblay et al.’s (2011) study offers foundational insights into the processing advantages of longer LBs, it is essential to review their findings first to establish a basis for understanding subsequent research. Following this, Wood and Appel’s (2014) innovative approach, which proved effective in identifying more structurally and semantically complete units, and Jeong and Jiang’s (2019) findings, which further justify the methodological choice of adopting Appel and Wood’s LB refinement method, will be summarized. Additionally, Yi and Zhong’s (2024) meta-analysis will be reviewed, as it offers a thorough examination of the processing advantages of MWSs across various learner populations. Therefore, the following sections (2.3.1 , 2.3.2 , 2.3.3 , and 2.3.4) provide a summary of Tremblay et al.’s study, Wood and Appel’s method, Jeong and Jiang’s findings, and Yi and Zhong’s meta-analysis,

respectively. As will be explained in Section 3.3.2 in experimental Chapter 3, I have adopted Wood and Appel's LB identification and refinement method in my experimental Chapters 3-5.

2.3.1 Tremblay et al. (2011): Processing Advantages of LBs

Understanding the processing advantages of longer LBs is crucial for the current thesis. Tremblay et al. (2011) found that longer LBs offer significant benefits in terms of processing efficiency in receptive tasks, supporting the hypothesis that these bundles could enhance oral fluency. Tremblay et al. defined LBs as frequently recurring strings of words that often span traditional syntactic boundaries. Their study aimed to determine if LBs are processed faster than matched control sentence fragments and whether this processing advantage implies holistic storage similar to single words. Rooted in the frequency effect—a phenomenon where frequently encountered words are recognized more quickly and easily than less frequent words (Rastle, 2007; Bybee, 2007; Baayen, 2007)—their study explored whether these frequency effects apply to multiword sequences. Challenging the traditional Chomskyan view that sentences are typically one-off constructions, Tremblay et al. investigated if LBs show a processing advantage due to their frequent occurrence, building on Bybee's (2006) argument that frequency effects significantly influence the processing of repeated multiword sequences.

To frame their study, the authors referenced several studies showing the processing advantages for frequent words and idioms. For instance, previous research highlighted how frequency affects the processing of single words (e.g., Kutas & Federmeier, 2007) and idioms (e.g., Cronk & Schweigert, 1992). Despite these findings, little research had been conducted on nonidiomatic LBs. Previous studies provided mixed results on whether LBs are stored and processed holistically. Bod (2001) showed that high-frequency three-word sentences such as *I like it* were processed faster than low-frequency sentences such as *I keep it*. Jiang and Nekrasova

(2007) found that L1 and non-L1 speakers responded to three-word LBs (e.g., *the fact is*) faster and more accurately than to non-LBs (NLBs) (e.g., *in your case*). In contrast, Schmitt et al. (2004) examined the production of MWSs such as *in the middle of the* and found no correlation between frequency of occurrence and performance in a dictation test. Tremblay et al. highlighted the need for more rigorous experimental designs to distinguish LBs from non-LBs in terms of whether they are stored and/or processed holistically.

In designing their self-paced reading experiments, Tremblay et al. (2011) were inspired by Schmitt and Underwood (2004), who examined the processing of formulaic sequences (e.g., idioms, lexical phrases) using a word-by-word self-paced reading task. Schmitt and Underwood hypothesized that terminal words (i.e., the final words) in formulaic sequences would be read faster than in nonformulaic contexts. However, they found no significant differences in reading times and speculated that the word-by-word nature of the task might disrupt holistic processing. In contrast, other studies (e.g., Gibbs et al., 1997) found evidence for holistic processing using similar word-by-word tasks. Tremblay et al. suspected that the lack of direct comparisons between entire formulaic and nonformulaic sequences might have affected Schmitt and Underwood's results. Therefore, Tremblay et al. designed their experiments to directly compare reading times for LBs and nearly equivalent non-LBs using word-by-word, portion-by-portion, and sentence-by-sentence self-paced reading tasks.

Participants in Tremblay et al.'s study were undergraduate students from the University of Alberta. Each of the three self-paced reading experiments involved 20 different participants. For the recall experiments, 50 participants each were recruited. The participants were L1 speakers of North American English. For materials, the researchers selected 20 LBs and their NLB counterparts from Biber et al. (1999), ensuring the NLB control sentences had higher

frequency and shorter pivot words than the LBs. For example, the sentence *He sat in the middle of the bullet train* contains the LB *in the middle of the*, whereas its NLB counterpart is *He sat in the front of the bullet train*, with the pivot word *front* being more frequent and shorter than *middle*. Semantic ratings were used to filter the most suitable sentence pairs, resulting in nine pairs for the final analysis. The stimuli were carefully controlled to ensure comparability between the LBs and NLBs.

Tremblay et al.'s study comprised five distinct experiments. The first three were self-paced reading experiments, whereas the latter two focused on recall. The self-paced reading experiments were designed to measure the reaction times (RTs) of participants as they read sentences containing either LBs or NLBs. The self-paced reading experiments were conducted as follows:

- **Experiment 1:** Word-by-word presentation.
- **Experiment 2:** Portion-by-portion presentation.
- **Experiment 3:** Sentence-by-sentence presentation.

In each experiment, participants read sentences and subsequently answered comprehension questions. RTs were recorded and analyzed using linear mixed-effects regression models to account for subject, item, group, and set variability.

The recall experiments aimed to assess the cognitive load associated with processing LBs versus NLBs. These experiments were structured as follows:

- **Experiment 4:** Auditory word and sentence recall.
- **Experiment 5:** Visual word and sentence recall.

Participants listened to or read sentences containing either LBs or NLBs, followed by a list of additional words. After hearing or reading each sentence and its associated words, participants were asked to recall and type the sentence as well as many of the additional words as possible. The accuracy of sentence recall and the number of additional words recalled were analyzed (using linear mixed-effects regression) to determine the cognitive load associated with processing LBs compared to NLBs.

Results of the self-paced reading experiments were:

- **Experiment 1** (word-by-word presentation): LBs were read 13 milliseconds faster than NLBs (M LB = 401 ms, M NLB = 414 ms; $t[1,421] = 2.8$, $p_{\text{MCMC}} < .01$). This suggests a modest processing advantage for LBs at the word level.
- **Experiment 2** (portion-by-portion presentation): LBs were read 124 milliseconds faster than NLBs (M LB = 933 ms, M NLB = 1,057 ms; $t[305] = 3.1$, $p_{\text{MCMC}} < .01$). This increase in processing speed suggests that when chunks of text are presented together, the hypothesized holistic storage of LBs becomes more evident.
- **Experiment 3** (sentence-by-sentence presentation): LBs were read 389 milliseconds faster than NLBs (M LB = 3,303 ms, M NLB = 3,692 ms; $t[416] = 3.7$, $p < .0001$). The substantial processing advantage observed here suggests the efficiency of holistic processing for entire sentences containing LBs.

The results across all three self-paced reading experiments showed a consistent processing advantage for LBs over NLBs. Notably, the magnitude of this advantage increased from the word-by-word to the sentence-by-sentence presentations, suggesting that LBs benefit from holistic processing. As well as supporting the holistic storage hypothesis for LBs, these

results could also reflect high-frequency processing or task-specific effects in self-paced reading. To further explore these findings, Tremblay et al. conducted two word and sentence recall experiments to assess whether the observed processing advantages for LBs extend beyond the self-paced reading task and to determine their impact on cognitive load during recall.

Results of the recall experiments were:

- **Experiment 4** (auditory recall): Sentences containing LBs were more accurately recalled than those with NLBs ($M_{LB} = 30\%$, $M_{NLB} = 8\%$; $z[769] = -6.8$, $p < .001$). This significant difference suggests that LBs facilitate better recall, likely due to their holistic representation. There was no significant difference in the number of additional words recalled ($M_{LB} = 3.44$, $M_{NLB} = 3.41$; $t[252] = -0.2$, $p_{MCMC} > .05$). This suggests that although LBs enhance sentence recall, they do not necessarily reduce the cognitive load for recalling additional information.
- **Experiment 5** (visual recall): Sentences containing LBs were more accurately recalled than those with NLBs ($M_{LB} = 25\%$, $M_{NLB} = 10\%$; $z[747] = -5.4$, $p < .001$). Similar to the auditory recall, this finding supports the robust recall advantage of LBs. Additionally, participants recalled significantly more additional words after sentences containing LBs than those with NLBs ($M_{LB} = 2.67$, $M_{NLB} = 2.33$; $t[341] = -2.5$, $p_{MCMC} < .05$), showing that LBs not only aid in sentence recall but also enhance the recall of subsequent words.

The recall experiments in both auditory and visual modalities showed that sentences containing LBs were remembered more readily than those without LBs, supporting the hypothesis that LBs are stored and retrieved as whole units. However, it is also possible that

LBs, being processed more quickly than NLBs, allowed participants more time for mnemonic processes, such as using mental strategies to remember the information. The authors acknowledged that using frequency counts from the British National Corpus for American English speakers could introduce bias due to potential differences in language usage between British and American English. However, their reanalysis using the Corpus of Contemporary American English yielded similar results, mitigating this concern. They also noted potential issues with frequency thresholds and the frequency of smaller chunks within LBs, suggesting that future research should use a regression-type design to address these concerns.

Despite the robust findings, several potential weaknesses should be considered. First, the study's participants were all L1 English-speaking undergraduates, limiting the generalizability of the findings to other populations, such as older adults or individuals with different educational backgrounds. Additionally, Tremblay et al. made no distinction between structurally complete and incomplete units in their test materials. A significant portion of the test items involved repeated use of similar expressions (e.g., *do know/want* for non-LBs and *don't want/think/know/worry* for LBs). This repetition reduces the representativeness of a diverse array of LB structures and complicates the determination of whether the observed effects arose from the LBs or embedded formulas, as noted by Jeong and Jiang (2019). This affects the clarity and specificity of the study's conclusions. Furthermore, the experiments were limited to self-paced reading and recall tasks, which may not capture the full complexity of LB processing in naturalistic settings. Expanding the range of tasks and methods could provide a more comprehensive understanding of LB processing.

Tremblay et al. concluded that LBs provide online processing advantages over NLBs, suggesting holistic storage and retrieval. This aligns with usage-based models of language, which

posit that frequently encountered multiword sequences are stored as single units in the mental lexicon. Despite the strong evidence for holistic processing, the possibility of sped-up computation (i.e., faster assembly of sequences during processing) remains. Tremblay et al.'s findings highlight the need for further research to distinguish between holistic retrieval and efficient computation and to explore how sentences are constructed from these building blocks. Their study contributes to our understanding of LBs as fundamental elements in language processing, supporting the view that both holistic and compositional processes play a role in linguistic cognition.

2.3.2 Wood and Appel (2014): Identifying Pedagogically Useful LBs

Wood and Appel examined multiword constructions (MWCs) in first-year university textbooks and popular English for Academic Purposes (EAP) textbooks. Borrowing the term MWC from Liu (2012) but using it in a more restricted way, Wood and Appel distinguish between LBs and MWCs, defining MWCs as refined versions of LBs based on frequency, range, and pedagogical usefulness. Their study aimed to (a) identify MWCs in first-year engineering and business textbooks, (b) check if these MWCs also exist in popular EAP textbooks, and (c) evaluate the treatment of MWCs in EAP textbooks. For their methodology, Wood and Appel compiled a 1.6-million-word corpus from first-year business and engineering textbooks used at a large Canadian university. They selected five textbooks from each subject. Additionally, they compiled smaller collections from five intermediate/advanced EAP textbooks, focusing on reading passages and instructional language (151,839 words and 108,123 words, respectively). They examined these to see if they contained any of the identified MWCs from the business and engineering textbooks and whether they were treated pedagogically.

To identify LBs, Wood and Appel extracted a list of four-word sequences using Wordsmith software. To be included, LBs had to meet minimum frequency and range criteria (25 occurrences per million words, and occurring in at least two textbooks, one from each major). They then used two additional methods to address significant overlap among sequences and better identify sequences of different lengths with pedagogical value:

1. They split each four-word structure into its two constituent four-word sequences (i.e., the four-word sequence *as long as the* was split into *as long as* and *long as the*) and compared their frequency. If one four-word sequence was at least twice as frequent as the other, it was classified as the “root” structure, and the remaining word was classified as a commonly occurring word. This way, they identified core MWCs with variable slots at the periphery [e.g., *as long as (the)*].
2. They compared each four-word structure for overlaps. When two four-word sequences contained identical three-word structures (e.g., *at the end of* and *the end of the*), they classified the common three-word string as the “root” structure and put variable slots at each end [e.g., *(at) the end of (the)*].

Wood and Appel found that the first method above produced fewer structurally or semantically incomplete units, suggesting it is effective in identifying MWCs. For this reason, I adopted the first method of LB identification and refinement in the current thesis.

2.3.3 Jeong and Jiang (2019): Representation and Processing of LBs

Jeong and Jiang examined the holistic representation and processing of corpus-derived MWSs (i.e., LBs or n-grams). The goal of their study was to determine whether different types of

MWSs are stored and processed as whole units or as individual words combined analytically in the mental lexicon.

The authors distinguished between two types of MWSs: lexical bundles (LBs) and formulaic sequences (formulas). They described LBs as structurally incomplete since they span across phrasal boundaries (e.g., *as can be seen*). In contrast, they defined formulaic sequences as structurally complete, forming a complete phrase (e.g., *to begin with*) with a unified semantic representation (e.g., *to begin with* = *firstly*). Jeong and Jiang emphasized the significance of structural completeness in the representation and processing of LBs versus formulas in the mental lexicon. They explained that LBs and formulas differ because LBs are less likely to possess a unified semantic representation, whereas formulas tend to have a single semantic association that serves as a holistic unit in the mental lexicon. For instance, the formula *at the same time* has the single meaning of *meantime* attached to it, whereas the LB *the presence of a* does not have a single meaning attached to it, as it is a noun phrase with an unfinished prepositional phrase. For the form-meaning mapping process to occur and for holistic storage and processing to be feasible, Jeong and Jiang contended that an MWS must have a single meaning attached to it. Without a single meaning, a multiword unit is arguably less likely to be stored as a whole unit in the mental lexicon. If a single meaning is attached to a multiword phrase, it is more likely that the phrase can be stored and processed as a whole unit in one's mental lexicon, which would be beneficial in terms of efficient retrieval and use in real-time speech production.

Jeong and Jiang reviewed various studies that explored whether MWSs are stored as chunks or as individual words in the mental lexicon. They argued that the findings of much

previous research are limited and unclear because researchers have not differentiated between distinct types of multiword units in their materials (i.e., structurally complete formulas and structurally incomplete LBs were included in the same materials). For instance, two studies (Nekrasova, 2009; Tremblay et al., 2011, reviewed in Section 2.3.1) focusing on representation and processing of LBs provided evidence supporting the holistic processing of LBs. However, Nekrasova's (2009) test materials included formulas like *at the same time*. Tremblay et al.'s (2011) test materials also had limitations in the type of LB structures they used (e.g., they repeatedly used non-LBs such as *do think/worry*). Furthermore, Tremblay et al.'s study only tested L1 English speakers, thus limiting its generalizability.

Participants in Jeong and Jiang's study were university students in the United States. They were recruited in two groups: (a) 30 L1 English speakers, and (b) 28 L2 English speakers. The L2 speakers were highly proficient in English and spoke various L1s such as Arabic and Chinese.

Jeong and Jiang used a word-monitoring task to examine the holistic representation of LBs. In this task, a target word is first selected and displayed. The target word is then preceded by a sequence of words displayed one at a time, embedded within a list of random words. The participant's task is to monitor the visually displayed word sequence and press a button when they see the target word. For the monitoring task items, Jeong and Jiang selected 30 LBs from the *Written Academic Formulas List* by Simpson-Vlach & Ellis (2010), all consisting of three- or four-word units (e.g., *an increase in the*). Jeong and Jiang created thirty non-LBs by replacing a word in a selected LB with another word of similar length and frequency (e.g., *an industry in the*). Additionally, the authors selected 30 formulaic expressions (referred to as "formulas") from

three previous studies (Biber et al., 1999; Jiang & Nekrasova, 2007; Martinez & Schmitt, 2012) that had already identified target formulaic sequences using corpus-based methods. These formulas were also three or four words in length and structurally complete in the sense that they form a complete phrase with a single meaning (e.g., *in the first place* has the single meaning of *firstly*). They created thirty control expressions (i.e., non-formulas) using the same procedure as for non-LBs. In summary, they devised four conditions for the study: (i) the LB condition, (ii) the control condition for LBs, (iii) the formula condition, and (iv) the control condition for formulas. The authors embedded each LB or formula and its respective control expression in carefully selected lists of random words (seven or eight words in length) that functioned as carrier contexts for their experiment.

For the task procedure, participants were tested individually on 100 test items. For each test item, they were first shown a target word and then a sequence of words. Some sequences contained a target word to be monitored, whereas others did not. Participants were asked to monitor for the target word and press a “yes” key as soon as they saw it. If they did not see the target word, they were instructed to press “no.”

For their results, the researchers found that the participants did not respond significantly faster to LBs than their control sequences for either group. However, for formula items, participants responded to the target words in formulas significantly faster than their control sequences for both groups. In other words, Jeong and Jiang found that LBs did not show a processing advantage whereas formulaic sequences did.

In their discussion, Jeong and Jiang noted that the observed processing advantage for formulas echoed the findings from earlier studies focusing on structurally complete MWSs (e.g.,

Conklin & Schmitt, 2008; Jiang & Nekrasova, 2007; Underwood et al., 2004). For instance, Jiang and Nekrasova (2007) found that both L1 and L2 speakers responded to formulas faster than to controlled phrases in a phrase acceptability task.

The lack of a processing advantage for LBs aligned with the results of some previous studies (e.g., Schmitt et al., 2004; Tremblay & Tucker, 2011) but contrasted with the findings of some other studies (e.g., Nekrasova, 2009; Tremblay et al., 2011). Jeong and Jiang provided a potential explanation for the contrast in findings: previous studies did not distinguish between structurally complete and incomplete units in their materials, so it is difficult to see if the observed processing advantage was caused by the LBs or formulas in those studies.

Jeong and Jiang explained that the observed processing advantage in their study was a result of the preactivation of the target word in the word monitoring task. If the phrase is represented as a whole unit in the mental lexicon, then the processing of the initial words in a phrase (e.g., *at the same*) can activate the whole phrase (*at the same time*). Thus, their study provides evidence for the holistic representation of formulas and argues against the holistic representation of LBs. The authors interpreted this evidence as suggesting that high frequency of occurrence is not sufficient for a multiword unit to be processed as a single unit. Structural completeness and semantic unity are also important factors for holistic representation. Formulas are more likely to have a semantic unit associated with them, making it easier for the form-meaning mapping process to occur and for these formulas to be represented as whole units in the mental lexicon. Although LBs occur frequently, they are less likely to have a unified semantic unit associated with them, as they are structurally incomplete. Therefore, LBs are not likely to be represented as whole units in the mental lexicon.

While Jeong and Jiang's study offers valuable insights, it is important to consider several limitations that may affect the generalizability and comprehensiveness of their findings. First, the study involved university students in the US, which may not represent the full range of English speakers, including those with different educational backgrounds and age groups. Additionally, the sample size was relatively small, which may limit the generalizability of the findings. Second, the study only used one type of task (word monitoring) to assess the holistic representation and processing of lexical bundles. A word monitoring task may not fully capture the complexity of how MWSs are stored and processed in the mental lexicon. Other tasks could provide a more comprehensive understanding of these processes. Third, the selection of LBs was limited. The study did not differentiate between different types of LBs, such as those with high mutual information scores. This leaves room for potential discrepancies in the processing of different types of LBs that were not included in the study.

Despite their study's limitations, Jeong and Jiang concluded that LBs may not be viable linguistic units in language learning and teaching. Their study emphasized the importance of distinguishing between structurally complete and incomplete multiword units in the study of their holistic representation. This distinction is crucial for avoiding conflicting or unclear findings.

2.3.4 Yi and Zhong (2024): The Processing Advantage of MWSs: A Meta-Analysis

Yi and Zhong (2024) conducted a meta-analysis to examine the processing advantage of MWSs over novel word combinations. Although research supports the efficient processing of MWSs, inconsistencies remain due to varying methodologies, leaving questions about the extent of this advantage and its influencing factors. Addressing these gaps, Yi and Zhong (2024) analyzed 35 studies to clarify the conditions under which MWSs show a processing advantage.

They focused on three variables: statistical regularities, MWS type, and task explicitness. First, they examined statistical regularities, because higher phrasal frequency and stronger association strength have been linked to enhanced processing efficiency (Gries & Ellis, 2015). Second, they considered MWS type because subtypes like idioms and LBs differ in structural and semantic properties, which can influence processing (Columbus & Wood, 2010; Jeong & Jiang, 2019). Third, Yi and Zhong included task explicitness as a variable since different tasks vary in the degree to which they tap into implicit or explicit knowledge, and this can significantly affect processing outcomes, particularly for L2 speakers (Suzuki & DeKeyser, 2017; Yi, 2018). By analyzing these variables, the study aimed to determine whether the processing advantage is restricted to specific statistical profiles, how it varies across MWS types, and whether task explicitness matters.

While both L1 and L2 speakers generally process MWSs faster than novel word combinations, the degree of this advantage varies by MWS subtype and the speaker's language background. For example, idioms often show a clear processing advantage for L1 speakers (Siyanova-Chanturia et al., 2011) but not always for L2 speakers (Carrol & Conklin, 2014), and the processing of LBs may benefit L1 speakers more consistently than L2 speakers (Jeong & Jiang, 2019). Research also suggests L1 speakers may be more influenced by association strength, while L2 speakers may be more sensitive to phrasal frequency (Ellis et al., 2008). These differences highlight the need to examine how these variables impact processing in L1 versus L2 speakers.

To ensure the inclusion of relevant studies, the authors searched multiple databases and applied a range of criteria. For example, studies had to be published in English, employ online tasks performed under significant time pressure, and focus on perception tasks (e.g., lexical

decision). Thirty-five studies met the inclusion requirements, which collectively provided 130 effect sizes and data from 1,981 participants. They systematically coded the included studies, focusing on moderator variables (statistical regularities, MWS type, and task explicitness), language background, and descriptive statistics necessary for calculating effect sizes. Statistical regularities were divided into two main components: phrasal frequency and association strength. To measure association strength, the authors used metrics such as Mutual Information (MI), forward delta P, and backward delta P, which have been validated as reliable indicators for both L1 and L2 speakers (Gries, 2013; Öksüz et al., 2021). MI captures non-directional associations, while delta P measures the predictability of words in a sequence. To maintain consistency and accuracy, the authors reanalyzed corpus data using the British National Corpus (BNC), the most commonly used corpus in the included studies. They calculated mean values for phrasal frequency and association strength, categorizing these into low, medium, high, and very high categories based on quartile rankings.

The studies were further categorized based on MWS types, which included collocations, idioms, lexical bundles, binominals, and phrasal verbs. To ensure consistency, the authors verified and, in some cases, relabeled the MWSs according to their definitions. The experimental tasks employed across the included studies varied and included eye-tracking reading (ET), grammaticality judgment (GJT), lexical decision (LDT), phrasal acceptability judgment (PJT), self-paced reading (SPR), and word monitoring (WM). These tasks were categorized based on three factors that measure task explicitness: time pressure, focus of attention, and metalinguistic knowledge, following established practices (Ellis, 2005; Suzuki & DeKeyser, 2015, 2017). Tasks completed without time constraints (i.e., at participants' own pace) were categorized as most explicit, while those with time pressure were categorized as less explicit. Similarly, tasks that

required minimal contextual support or involved analytic linguistic rules were also considered more explicit. The tasks were scored across these categories, with totals of 4-5 points indicating low explicitness and 6-8 points indicating high explicitness.

The meta-analysis utilized advanced statistical techniques, including three-level meta-regression models and cluster robust variance estimation, to calculate effect sizes and examine the moderating effects of the identified variables. The overall results confirmed a significant processing advantage for MWSs over novel word combinations, with a weighted average effect size of Hedges's $g = 0.417$ (95% CI = [0.276, 0.558], $p < .001$). This processing advantage was observed in both L1 and L2 speakers, with L1 speakers showing a larger effect size ($g = 0.475$) compared to L2 speakers ($g = 0.348$). The difference in processing advantage between the two groups was statistically significant ($F = 11.9$, $p = .003$), indicating that although both groups benefited from MWSs, L1 speakers processed them more efficiently. Further analysis revealed that both phrasal frequency and association strength significantly moderated the processing advantage of MWSs for both L1 and L2 speakers. Higher frequency MWSs consistently demonstrated greater processing advantages. For example, the processing advantage peaked in the very-high-frequency category for both L1 ($b = 0.455$, $p = .029$) and L2 speakers ($b = 0.483$, $p = .006$). Association strength (measured by MI) also played a key role. MWSs with medium MI scores provided a significant processing advantage, with L1 speakers benefiting more ($b = 0.260$, $p = .025$) and L2 speakers also showing a significant advantage ($b = 0.335$, $p = .033$) in this MI category. Interestingly, the processing advantage followed an inverted U-shaped pattern for association strength, particularly for L2 speakers, with the greatest processing efficiency observed at moderate levels of association strength. This suggests that overly strong or weak

associations might reduce processing efficiency, possibly because of the lower familiarity of such sequences.

The processing advantage also varied across different types of MWSs. Phrasal verbs showed the largest effect sizes (L1: $g = 3.309$; L2: $g = 0.605$), indicating a strong processing advantage, while idioms and LBs showed smaller advantages. L2 speakers did not benefit as much as L1 speakers when it came to understanding idioms, showing that L2 speakers may have difficulty with figurative language. However, collocations provided a consistent processing advantage for both L1 and L2 speakers. This consistent advantage may be attributed to the compositional nature of collocations, which makes them easier to process than more figurative or structurally incomplete MWSs, such as idioms or LBs.

The study found that the explicitness of tasks did not significantly moderate the processing advantage of MWSs for L1 speakers. Their processing advantage remained robust regardless of task explicitness. In contrast, for L2 speakers, the processing advantage significantly increased when tasks were more explicit ($b = 0.302$, $p = .020$). This finding suggests that while L1 speakers might process MWSs implicitly, L2 speakers benefit more from tasks that draw explicit attention to MWSs. This result aligns with previous studies that have suggested L2 speakers might require more explicit instruction to fully benefit from MWSs, given their reliance on more analytical processing strategies (Cieślicka, 2006; Wray, 2012).

Yi and Zhong (2024) emphasized that MWSs are processed significantly faster than novel word combinations, showing a moderate overall effect size (Hedges's $g = 0.417$, $p < .001$), particularly for L1 speakers. This finding supports usage-based theories, which propose that MWSs reduce cognitive effort and are represented as units in the mental lexicon, thereby facilitating faster processing (Christiansen & Arnon, 2017; Goldberg, 1995). The study also

found that the processing advantage of MWSs is influenced by statistical regularities and MWS type, but remains consistent across varying levels of task explicitness for L1 speakers. For L2 speakers, however, the advantage is enhanced by more explicit tasks, suggesting that L2 learners benefit from instructional approaches that highlight MWSs.

Regarding differences between L1 and L2 speakers, the study found that L1 speakers consistently showed a greater overall processing advantage across various MWS types and levels of statistical regularities. While both groups benefited from increased phrasal frequency and association strength, L1 speakers were more sensitive to the direction of association, particularly with backward delta P, indicating a greater sensitivity to the co-occurring probability of word sequences. L1 speakers also demonstrated a processing advantage for idioms, a benefit not observed in L2 speakers. This difference likely stems from the distinct cognitive strategies employed by each group, with L1 speakers relying on direct retrieval of idiomatic expressions (van Lancker Sidtis, 2012), while L2 speakers tend to analyze idioms literally to compute the meaning of such units (Cieślicka, 2006), which slows their processing. Conversely, L2 speakers showed a greater processing advantage for collocations and lexical bundles, suggesting that while they struggle with the figurative meanings of idioms, they find structurally and semantically simpler MWSs easier to process. Lastly, the authors noted that while task explicitness did not moderate the processing advantage for L1 speakers, it significantly enhanced the advantage for L2 speakers. This finding highlights L2 learners' cognitive approach of analyzing MWSs into individual words (Cieślicka, 2006; Wray, 2012) and suggests that L2 speakers may benefit more from explicit instructional methods that emphasize certain MWSs.

While Yi and Zhong's (2024) study represents a significant step forward in resolving inconsistencies in the literature regarding the processing of MWSs, several limitations should be

considered. First, they only included studies that used perception tasks (e.g., lexical decision) and excluded those using production tasks (e.g., word naming). Second, they included only one study (i.e., Kim & Kim, 2012) on L1 speakers and L2 speakers' processing of phrasal verbs, which limits the robustness of the conclusions regarding this specific MWS type. Third, they only examined the moderating effects of statistical regularities, MWS type, task explicitness, and language background on the processing advantage of MWSs.

Yi and Zhong (2024) presented strong evidence for the processing advantage of MWSs. However, it is important for future studies to address these limitations to improve our understanding of language processing and apply these findings in real-world situations. Specifically, future studies should consider including production tasks (e.g., word naming) to determine whether the processing advantage of MWSs extends beyond perception tasks. Additionally, more research is needed to explore the processing of phrasal verbs across a broader range of studies to confirm the findings related to this specific MWS type. Additionally, expanding the scope of moderators beyond statistical regularities, MWS type, task explicitness, and language background would provide a more comprehensive understanding of the factors that influence the processing of MWSs. As the authors suggested, another factor that might influence the processing of MWSs is language aptitude (Yi, 2018).

2.4 Theoretical Frameworks Explaining the LB-Fluency Relationship

Understanding the relationship between LBs and fluency in L2 speech production requires examining various theoretical perspectives. This section explores foundational theories, cognitive models, and usage-based approaches that collectively provide a comprehensive view of how LBs contribute to oral fluency.

2.4.1 Foundations of LBs and Fluency

Pawley and Syder's (1983) lexical phrase theory provides foundational insights into the role and importance of multiword sequences in achieving language proficiency and fluency. According to Pawley and Syder, the ability of a second language speaker to effectively use fixed and semi-fixed phrases, commonly referred to as "chunks" of language, is essential in order to attain fluency comparable to that of a first-language speaker. Such lexical phrases ease the cognitive load during speech production and serve as essential building blocks for fluent and idiomatic language use. This early perspective is significant as it suggests that the repertoire of lexical phrases is a key factor distinguishing more proficient speakers from less proficient ones, enabling more efficient processing and production of language.

In line with Pawley and Syder's view, Sinclair's (1991) Idiom Principle also emphasizes the prevalence and importance of fixed or semi-fixed expressions in language. According to Sinclair, much of what we say is made up of ready-made phrases rather than novel constructions, which supports the idea that reliance on these pre-fabricated chunks reduces cognitive load and enhances fluency. Along similar lines, Renouf and Sinclair (1991) introduce the notion of collocational frameworks—continuous sequences with one or more free slots—suggesting that words in a given frame are not random but belong to certain semantic categories related to that structure. For instance, "as + ? + as," where the slot is typically filled with adjectives like *big*, *important*, *strong*, and *good*. These collocational frameworks further support the argument that structured patterns in language facilitate fluent and coherent speech. Ellis (1996) also discusses the concept of "chunking," where sequences of words or phrases are processed as single units, reducing the cognitive load during speech production and enhancing fluency. These foundational

theories highlight the critical role of pre-fabricated language units in fluent speech production, setting the stage for more detailed cognitive models of speech production.

2.4.2 Cognitive Models of Speech Production

Levelt's (1989) model, Kormos' (2006) revision of Levelt's model, and Skehan's (1998) trade-off hypothesis provide detailed accounts of the cognitive processes involved in speech production, explaining how LBs can facilitate these processes.

Levelt's (1989) three-stage model of speech production provides further theoretical justification for the link between LBs and fluency. Levelt proposes three main stages of speech production: conceptualization, formulation, and articulation. First, speakers plan speech content at the conceptualization stage. Second, speakers encode lexical and grammatical items in the mental lexicon at the formulation stage, at which point appropriate lemmas are activated and put into syntactic surface structures, followed by morphological and phonetic encoding. The third and final articulation stage requires the phonetic plan to be implemented and speech is produced (Levelt, 1989).

Kormos (2006) later refined Levelt's model to include L2 speakers, addressing the unique challenges they face. The L2 mental lexicon is considered smaller, less structured, less easily accessed, and with fewer formulaic expressions. Theoretically, speakers with a larger MWS repertoire in their mental lexicon can retrieve longer MWSs at a similar processing rate to those required for single-word lexical item retrieval at the formulation stage. Retrieval of longer MWSs may free up cognitive resources and processing time at the formulation stage. This could allow speakers to prepare for other processing needs such as phonological encoding, further message generation and articulation, thus enabling them to enjoy a processing advantage. A

larger MWS repertoire may enable L2 learners to decrease the demands on cognitive resources that can be employed in simultaneous processing during other speech production aspects (e.g., lexical and grammatical accuracy or complexity; Kormos, 2006; Skehan, 1998, 2009). Understanding and utilizing MWSs can enhance oral fluency, especially during lexical selection at the formulation stage (Kormos, 2006; Levelt, 1992).

Conversely, speakers with a smaller MWS repertoire might not have this processing advantage, as they might use up cognitive resources trying to retrieve single lexical items one by one during the formulation stage. Speakers with a limited range of MWSs are more likely to show hesitations or pauses within MWSs. Skehan's (1998) trade-off hypothesis helps explain this by positing that language learners must balance their attentional resources between complexity, accuracy, and fluency during language production. According to Skehan's hypothesis, achieving fluency often involves a trade-off where resources allocated to fluency may reduce those available for complexity or accuracy. The use of LBs or other MWSs can mitigate this trade-off by providing ready-made chunks of language that reduce the cognitive load during speech production. This allows learners to allocate more resources to maintaining fluency without significantly compromising accuracy or complexity. In this way, LBs can serve as a strategic tool for managing the cognitive demands of speech production, supporting more fluent and coherent language use (Skehan, 1998). These cognitive models emphasize the significance of efficient retrieval and processing of LBs in fluent speech production, particularly for L2 learners.

2.4.3 Usage-Based Theories

Usage-based theory, as articulated by Ellis (2002) and supported by Bybee (2006), further complements our understanding of how language proficiency develops from the experience and frequency of usage. Ellis (2002) introduced the notion that linguistic competence is significantly influenced by the accumulation of exposure to MWSs, which reinforces their retrieval and production. This perspective aligns with the proceduralization of language use, where proficiency emerges from the frequent and familiar use of language structures. Bybee (2006) suggests that repeated exposure to MWSs not only facilitates their entrenchment in the learner's mental lexicon but also highlights the importance of frequency and familiarity for L2 learning.

Together, these theoretical perspectives—Pawley and Syder's lexical phrases theory, Levelt's speech production model (and Kormos' revision for L2 speakers), Skehan's trade-off hypothesis, and the usage-based theories proposed by Ellis and Bybee—form a multifaceted view of the mechanisms underlying the relationship between LB use and L2 oral fluency. The current thesis aims to bridge and synergize these theoretical insights with empirical evidence, exploring how the length, frequency, and association strength of LBs contribute to oral fluency. I seek to illuminate the relationship between LB use and oral fluency, grounded in a broad spectrum of linguistic theory.

2.5 Studies Examining the LB-Fluency Relationship

To date, most learner-corpus-based studies have focused on LBs used in writing rather than speaking (e.g., Appel & Wood, 2016; Siyanova-Chanturia & Spina, 2020; Staples et al. 2013). Such studies highlight the complexity of the MWS-L2-proficiency link, but general trends

have emerged, including (a) the quantity of LBs produced by learners decreases as proficiency increases (Appel & Wood, 2016; Staples et al., 2013) or with time spent in an English-speaking country (Groom, 2009). Less proficient learners overuse bundle tokens and under-use bundle types, a trend resembling the use of “phraseological teddy bears” (i.e., overusing high-frequency phrases with which one feels comfortable; Hasselgård, 2019). Research has demonstrated that proficient learners have a firmer and more creative command of lower-frequency bundles whose constituent words are non-associated (Siyanova-Chanturia & Spina, 2020); and (b) less proficient learners rely more on LBs copied from writing prompts or source texts due to their relatively limited lexical repertoires (e.g., Appel & Wood, 2016; Staples et al., 2013). Overall, these studies suggest that much remains to be understood about the relationship between LB use and general writing proficiency. While most existing learner-corpus-based studies have focused on the use of LBs in written contexts, the insights they provide into the relationship between MWS-L2-proficiency potentially apply to the spoken domain as well. The understanding that learners’ reliance on LBs decreases with proficiency or extended exposure to an English-speaking environment, and that proficiency dictates a learner’s command over lower-frequency bundles, offers a compelling framework to consider for spoken data. The current thesis seeks to bridge this gap by exploring how these patterns manifest in speech fluency, which has been less frequently examined. There is potential that the dynamics of LB use in writing, as detailed in these studies, could find parallels in oral production, providing richer insights into speech fluency and its intricacies.

Several recent studies have examined the link between LB use and general speaking proficiency from a *text-external* perspective, focusing on the extent to which L2 learners use L1 target-like two- and three-word (bi- and trigram) measures in terms of quantitative indices

(frequency, proportion, and association) (e.g., Garner & Crossley, 2018; Kyle & Crossley, 2015; Zhang et al., 2021). The findings from these studies differ from the typical trend from writing-focused studies (which reported that less proficient L2 writers (over)use a greater number of high-frequency bundles). Kyle and Crossley (2015) found significant correlations between (human-rated) oral proficiency scores and several n-gram scores, of which the strongest predictor of speaking proficiency came from high-frequency trigrams, suggesting that more skillful L2 speakers use a larger number of highly frequent trigrams. Garner and Crossley (2018) found that beginning-level L2 learners indicated the greatest increase in oral production of high-frequency bigrams over the course of their four-month longitudinal study. Zhang et al. (2021) reported that several n-gram measures (e.g., bigram proportion and association: MI and *t* scores) significantly correlated with (human-rated) oral proficiency scores on story retelling and monologic tasks. Such studies highlight the important role of proficiency in the development of LB use but also suggest that further research is required to bring clarity to this research area.

Relatively few studies (e.g., Biber & Gray, 2013; De Cock, 2004) have examined LB use in spoken corpora from a text-*internal* perspective. De Cock (2004) examined two- to six-word bundle use among advanced EFL learners compared to L1 speakers. She found that learners' preferred bundles were less interactional and included relatively few vagueness markers (e.g., *or something, kind of*) compared to L1 speakers. Biber and Gray's (2013) study of spoken and written responses to the TOEFL iBT revealed a slightly more complex pattern than other n-gram studies. They reported that intermediate-level test-takers produced a greater number of bundles (four-word units) than the lower and higher proficiency groups, suggesting a general developmental progression in which lower-level test-takers use a smaller number of bundles, middle-level test-takers overuse a larger number of bundles, and high-scoring test-takers show

greater control and creativity in using the bundles they have acquired (p. 37). In summary, these studies indicate a multifaceted view and highlight the necessity for further research into how language utilization differs among individuals from divergent linguistic backgrounds.

While the aforementioned studies have offered insights into the patterns of LB use in spoken corpora, a broader context emerges when we consider the significant relationship between MWS use and speech fluency. This is evident across various teaching contexts. Studies that show significant and positive relationships between MWS use and speech fluency, including L2 proficiency in the broader sense, come from a range of different teaching contexts (e.g., Boers et al., 2006; McGuire & Larson-Hall, 2018, 2021; Stengers et al., 2011; Suzuki et al., 2022; Tavakoli, 2011; Tavakoli & Uchihara, 2020; Uchihara et al., 2022; Wood, 2009, 2010). To provide a clear overview of some of the key findings regarding the relationship between MWS use and oral fluency, Table 2.1 summarizes several key studies in chronological order. This summary highlights the diverse teaching contexts, sample sizes, key results, and noteworthy details offering a brief overview of the positive relationships identified between MWS use and speech fluency.

To thoroughly understand this relationship, it is essential to provide comprehensive reviews of key studies that have explored this phenomenon in diverse educational settings. The following Sections, [2.5.1](#) to [2.5.5](#), will present in-depth analyses of five selected studies that exemplify the significant impact of MWS use on oral fluency. These studies were chosen for their methodological rigor, the diversity of contexts they represent, and their contributions to the field.

Table 2.1

Summary of 10 Key Studies on the MWS-Fluency Relationship (in Chronological Order)

Year	Study	Context	Sample Size	Key Findings	Noteworthy Details
2006	Boers et al.	Belgian EFL	32	Strong links between the number of MWSs used in story retelling tasks and perceived oral ability scores.	Highlighted the importance of MWS use in improving perceived oral proficiency.
2009	Wood	Canadian ESL	1	MWS-focused instruction led to increased MWS use and spoken fluency over six weeks.	Case study demonstrating rapid improvement in fluency through targeted teaching.
2010	Wood	Canadian ESL	11	Similar results to Wood (2009) with a larger sample size over six months.	Reinforced findings from the 2009 study with a more robust sample size and extended period.
2011	Stengers et al.	Belgian EFL	60	Strong links between the number of MWSs used in story retelling tasks and perceived oral ability scores.	Confirmed Boers et al.'s (2006) findings in a similar context.
2011	Tavakoli	UK ESL	80	L2 learners rarely paused in the middle of multiword units, supporting the notion that lexical chunks facilitate fluency.	Provided evidence of the role of lexical chunks in reducing pauses in speech.
2018	McGuire & Larson-Hall	American ESL	19	Moderately strong relationship between MWS use and fluency measures replicating Wood's (2009) study.	Replicated findings in a different context, reinforcing the validity of Wood's results.
2020	Tavakoli & Uchihara	UK ESL	56	Greater LB use linked to higher speaking ability scores and improved fluency aspects such as faster articulation rate.	First systematic study linking specific LBs to utterance fluency across proficiency levels.
2022	Uchihara et al.	Japanese EFL	40	Speakers providing more low-frequency MWS responses spoke more rapidly with fewer silent pauses.	Highlighted the importance of low-frequency MWSs in improving speech rate and reducing pauses.
2022	Suzuki et al.	Japanese EFL	50	Task repetitions facilitated MWS proceduralization, influencing mid-clause pauses (fewer but longer pauses within clauses).	Indicated both positive and negative effects of MWS reuse on mid-clause pauses, suggesting ongoing encoding restructuring.
2024	Hougham et al.	Japanese EFL	50	Longer LBs enhanced speech fluency by reducing pauses and repairs; frequent two-word combinations increased speech rate.	Provided comprehensive analysis of the impact of LBs of various lengths and types on aspects of utterance fluency.

2.5.1 Wood (2009): Effects of Focused Instruction of FSs on Fluent Expression in L2

Narratives: A Case Study

Wood's (2009) pedagogical intervention study was conducted in a Canadian university ESL context. His study investigated the effect of MWS-focused instruction on a single learner participant's MWS use in relation to her oral fluency. Wood illuminated the extent to which language teachers can teach MWSs and facilitate fluent speech. For Wood, teaching MWSs incorporated three key concepts: (1) noticing, (2) automatization, and (3) memorization.

Wood (2009) suggested that MWS teaching needs more explicit and rigorous approaches to support MWS uptake. This specific motivation stems from "noticing." Noticing refers to when learners consciously recognize and attend to language structures and their meanings. Although the concept of noticing seems promising, research related to its benefits remains mixed. Lewis's (1993, 2000) Lexical Approach, for example, emphasized the importance of encouraging learners to "notice" MWSs. Boers et al. (2006) investigated whether learners in an experimental group noticed MWSs. Boers et al. found that their experimental group was judged as more proficient and more fluent than a comparison group. The comparison group's attention was not explicitly directed to the input text's phraseological aspects. However, as they acknowledged, the experimental group outperformed the comparison group in MWS usage. This outperformance was observed during the first half of a final interview conversation based on a short text they had earlier been asked to read. The outperformance was largely attributed to recycling/usage of phrases encountered in the text. Boers et al.'s intervention may therefore have equipped the treatment group with a strategic advantage. The prompts may have stemmed from the text they used in their responses to the final interview task. Whether students acquired the noticed MWSs

from the main course materials, however, remains unclear. Stengers et al.'s (2010) replication of Boers et al. (2006) suggests a

relationship between MWS use and perceived fluency. Stengers et al., however, did not report their experimental/noticing group using more MWSs or being more fluent than their comparison group. Counter to such findings, other research investigating MWS noticing comes from L2 reading research (Bishop, 2004). Bishop reported a lack of MWS noticing in the read texts.

Wood's approach also incorporated the idea of automatization or automaticity. Automaticity "refers to the speed and ease of handling utterances; the greater the automaticity, the faster the recognition and production of grammatically correct and communicatively appropriate utterances" (Gatbonton & Segalowitz, 1988, p. 474). Gatbonton and Segalowitz (1988, 2005) sought to promote automaticity in the classroom. To promote automaticity, they developed a task-based teaching methodology involving three stages: creative automatization, language consolidation, and free communication. To be successful in developing the automatization of utterances, Gatbonton and Segalowitz assert that the main tasks used must be genuinely communicative and repetitive, and focused on practicing functionally formulaic utterances. The use of computers to assist in the automatization of MWS knowledge and skills also holds great potential (Segalowitz & Gatbonton, 1995), though this too is an underexplored area where future research is much needed.

In addition to noticing and automatization, Wood included the concept of memorization in his pedagogical approach. Studies indicate that memorization plays a key role in the learning and production of MWSs. O'Brien et al. (2007) found a significant relationship between phonological memory and L2 fluency gains in adult L2 learners, suggesting that the ability to store phonological utterances in short-term memory can facilitate MWS learning. Wray (2004)

found that memorizing a lengthy set of specific MWSs in a short time period (four days) not only helped a learner to perform a cooking demonstration competently in L2 Welsh but also enabled the learner to remember much of the material over a long-term period (nine months). Wray and Fitzpatrick (2008) found that memorizing and practicing L2 conversations considerably helped intermediate/advanced learners in real-life interaction in their L2. Ding (2007) found that his Chinese participants perceived text memorization/imitation practice as the most effective English learning method in China. Ding discovered that such practice enabled learners to acquire MWSs for productive use, to improve their pronunciation, and to achieve their goals, including winning prizes in nationwide speaking competitions and debate tournaments in English. Wray and Pegg (2009) examined the proportion of potentially memorized material in the performance of IELTS (International English Language Testing System) candidates. Wray and Pegg found that, although memorization might seem a rather off-putting and impractical approach, it can facilitate noticing and new language transfer from working memory to long-term memory: “Such transfer can lie at the heart of truly successful learning” (p. 208), as it makes MWSs retrievable for use when needed. Thus, memorization can be seen as a key strategy for various purposes including preparation for high-stakes assessments.

As for methodology, the participant in Wood’s (2009) study was a Japanese female doing a 1-year study abroad and homestay in Canada. She had recently already completed her undergraduate degree in Japan. She participated in a 6-week fluency instruction course in an intermediate-level ESL program at a Canadian university. As a pretest-posttest, she was asked to talk about a topic of personal interest, but she was not given any preparation time, nor was she allowed to use notes. She produced the first monologue on the first day of the fluency instruction course. For her first monologue, she chose to talk about her experience attending a Celine Dion

concert in Osaka. For her second monologue, which she produced a week after the end of the workshop, she chose to talk about her summer vacation memories as a child in Kumamoto. Both monologues were audio-recorded.

As for the pedagogical intervention, Wood conducted a 6-week fluency workshop consisting of 90-minute sessions per week (nine hours in total). He provided MWS-focused materials, and a sequence of tasks focused on the production of monologic speech. The tasks were implemented in the following sequence:

1. *Input stage*: The learners listened to recordings of L1 speakers telling personal stories. Two L1 model narratives were used: (i) a reaction to an item the speaker had read in a newspaper, and (ii) a childhood memory/reminiscence of summer cottage vacations. First, the learners listened to the recording. Next, they discussed the content of the recording. Then, they listened again while reading a transcript and marking fluency (e.g., pauses) and MWS features on the transcript. Wood drew their attention to key MWSs that served fluency functions.
2. *Automatization stage*: The learners engaged in various speaking activities to automatize (i.e., to make automatic or habitual) the encountered MWSs. The main speaking activities were shadowing, dictogloss (Wajnryb, 1990), mingle jigsaw (Wood, 1998), and chat circle.
3. *Practice and production stage*: Wood guided learners to prepare their own brief talk on a topic of personal interest for use in Nation's (1989) 4/3/2 speaking activity. The learners practiced telling their story to a partner: the first round for *four* minutes, the second round with a different partner for *three* minutes, and the third round with another new partner

for just *two* minutes. This activity pushed them to produce the same talk under increasingly shorter time limits, thus encouraging them to speak more fluently each time.

4. *Free talk stage*: In groups, learners brainstormed topics related to the themes of the model narratives. The topics were then randomly circulated among the members who took turns listening to each other talking spontaneously about each selected topic. Learners then commented and reflected on the speed, hesitations and smoothness (or roughness) of their productions.

As for analysis, Wood analyzed the participant's pre- and posttest productions in terms of fluency by calculating (i) the speech rate (i.e., the number of syllables uttered per minute) and (ii) the mean length of runs (i.e., the total number of syllables uttered divided by the number of runs). Wood defined a fluent run as a hesitation-free stretch of speech without pauses longer than 0.3 seconds. Wood also analyzed the participant's output to determine how many of the MWSs used by the learner were taken from the L1 narrative models studied in the workshop. Wood took a phraseological approach in deciding whether a sequence used by the participant was identified as formulaic. First, Wood used his intuition/judgment to identify MWSs used by the participant that also appeared to serve fluency or discourse functions in the L1 narrative models studied in the workshop. He also applied the following checklist based largely on Wray and Namba's (2003) extensive examination of judgment criteria for identifying MWSs in speech corpora:

1. Phonological coherence and reduction
2. The taxonomy developed by Nattinger and DeCarrico (1992), a classification consisting of four broad lexical phrase categories distinguished by their pragmatic function: (i) polywords, which function as single words, without variability or lexical insertions (e.g.,

so far so good); (ii) institutionalized expressions, which are sentence-length, invariable, and usually continuous (e.g., *nice meeting you*); (iii) phrasal constraints, which have variations of lexical and phrase categories, and are generally continuous (e.g., *a ____ ago*, *the ____er the ____er*); and (iv) sentence builders, which contribute to the construction of full sentences with fillable slots (e.g., *I think that X, not only X but Y*) (pp. 38–45).

3. Greater length/complexity than other output
4. Semantic irregularity, as in idioms and metaphors
5. Syntactic irregularity

It is worth noting that Wood also used the above criteria checklist to identify formulaicity in his earlier (2006, 2007) and later (2010, 2020) research. Wray (2002) has advocated the use of such checklists together with L1 speaker common sense as an effective way to identify MWSs, especially when analyzing small samples of spoken data.

Wood also accepted idiosyncratic and non-standard (i.e., non-target-like) MWSs in his analyses on the grounds that the cognitive stress of the narrative production task may have caused the participant to slightly deviate from standard MWS forms. Previous studies suggest that such deviations from L1 speaker norms have little link to fluency measures but instead are linked with difficulty in realizing pragmatic aims (Prodromou, 2007), or gaps in linguistic competence (Wray & Fitzpatrick, 2008).

As for results, Wood found strong gains in fluency measures from the pre- to the posttest: mean length of run increased by 26.3%, and speech rate increased by 13.8%. Strong gains were also found in the quantity of speech produced. More formulaic sequences were produced in the second speech sample, many of which came from the L1 speaker models studied in the

workshop. In the first speech sample, the participant produced 18 MWSs (only 2 of which also appeared in the L1 workshop models), whereas in sample two she produced 52 MWSs (18 of which also occurred in the L1 speaker models). The percentage of syllables from MWSs also increased from 11.3% in sample one to 12.5% in sample two. The second speech sample showed a wider range of functions and types of MWSs used as well as greater length, detail and complexity than the first sample, especially those taken from one of the workshop models: the mean length of MWSs in sample one was 3.17 words and in sample two was 4.46 words, thus showing an increase of 40.7%. On the whole, the participant showed that she was able to incorporate many of the MWSs from the L1 models—especially the second model—into her own narrative production.

It is important to note several limitations of Wood's study. First, the evidence is limited due to the small sample size ($N = 1$) and the absence of a comparison group, restricting the generalizability of the findings. Second, Wood only measured speed fluency, neglecting other crucial aspects of fluency such as breakdown and repair fluency. Third, the unspecified length of the speech samples further complicates the evaluation of fluency improvements.

Fourth, regarding the phraseological approach taken by Wood to identify MWSs in the speech samples, the checklist of criteria he used relied on his intuition and subjective judgment. Wray (2002, p. 23) points out some challenges with this type of approach. For example, inconsistent judgment may occur due to rater fatigue or changes in judgment thresholds over time, and different judges may apply the criteria differently, thus perhaps making it hard to replicate or validate the findings. Fifth, one of the main criteria Wood used to identify sequences as formulaic in his study was “phonological coherence,” thus possibly introducing some

circularity in the conclusion that MWSs contribute to fluency (Boers & Lindstromberg, 2012; Thomson et al., 2019).

Sixth, some of the word strings identified by Wood appeared idiosyncratic and not entirely target-like (e.g., *almost the all*), thus suggesting a need for more or better guidance at the MWS-focused instruction stage and a need for future studies to distinguish between non-target-like MWS use and the use of standard multiword expressions that are produced in a hesitation-free manner (Thomson et al., 2019). Last but not least, as Wood himself acknowledges, a large majority (15 out of 18) of the MWSs borrowed from the L1 speaker models in speech sample two were taken from the second L1 model that was studied and practiced intensively in the workshop. The second L1 model (a childhood summer vacation memory) was very similar in genre to the topic the participant chose (her childhood Kumamoto summer vacation reminiscence) for her second monologue. It may thus be that the second speech sample was likely not “spontaneous” speech but rather was quite shaped (semi-planned at the level of topic and lexis) by the workshop practice materials/activities. The strong gains may thus have resulted in large part due to a practice effect: the participant chose to speak on a topic of personal relevance based on the same model topic she had studied and practiced in the workshop. It is unclear at what exact point the participant was asked to choose the topic of her second speech. If she had been given advance notice (e.g., one week?) that she could choose either of the L1 model topics studied in the workshop (or any other topic of her choice), then she would have had considerable time to plan her second speech, which may well have positively influenced the results of Wood’s study.

Despite its limitations, overall, Wood’s study demonstrates that MWS-focused instruction using L1 speaker models can result in increased MWS use and increased speaking

fluency. Wood observed clear gains in speed fluency and more extensive and complex MWS use over a six-week period.

2.5.2 Tavakoli and Uchihara (2020): To What Extent Are MWSs Associated With Oral Fluency?

Tavakoli and Uchihara (2020) investigated the extent to which L2 speakers' use of MWSs relates to oral fluency across different proficiency levels. Their study examined this relationship systematically over a range of assessed proficiency levels, aiming to enhance our understanding of speech production processes in L2 learners. Their work is important because no MWS-fluency study so far has methodically investigated this link across a range of assessed proficiency levels. The primary goal of their study was to determine whether a link exists between the use of MWSs and various aspects of oral fluency. By illuminating this connection, they aimed to inform language testing practices and contribute to more valid evaluations of learner MWS knowledge and its impact on fluency. To achieve this, following previous studies (e.g., Garner & Crossley, 2018; Kyle & Crossley, 2015), Tavakoli and Uchihara adopted a frequency-based, text-external approach (as outlined in Section 2.2) using an external reference corpus to identify target-like two- and three-word units used by participants in their study. This method allowed them to assess the extent to which L2 learners employ target-like word combinations in their speech.

Tavakoli and Uchihara's study reported on data from 56 samples of test-takers performances on the speaking part of a standardized test of proficiency (the Test of English for Educational Purposes: TEEP). Their participants were pre-degree university applicants who took the TEEP test as one of the requirements to enter a UK University. Three-minute recordings of

their speaking performances were assessed across four levels of L2 proficiency according to the Common European Framework of Reference (CEFR, low B1 to C1 level).

To precisely measure the acoustic aspects of fluency (e.g., pauses), Tavakoli and Uchihara used *PRAAT* software (Boersma & Weenink, 2013). While previous studies (e.g., Wood, 2009) that have investigated the MWS-fluency link have measured fluency in a somewhat limited way focusing on only one aspect (e.g., speed fluency), Tavakoli and Uchihara's study is notable in that they investigated one objective measure from each aspect of utterance fluency (speed, breakdown, and repair). For each fluency aspect, there are several possible measures. For example, speech rate, articulation rate and phonation time ratio can be used to measure the speed of speech. However, researchers (e.g., Bosker et al., 2013) have cautioned against using different measures from each aspect, as multiple measures might correlate and overlap with each other. To avoid the problem of overlapping measures, Tavakoli and Uchihara chose to use just one measure of each aspect of fluency:

1. *Articulation rate* (i.e., the total number of syllables divided by the total amount of speaking time per minute (excluding pauses)) was selected as a measure of speed fluency.
2. *Frequency of silent pauses per minute* was used to investigate breakdown fluency. They also investigated the location of pauses to see whether pauses occurred in the middle or at the end of clauses. While some previous studies have defined a pause as an occurrence of silence longer than 1 second (Foster & Skehan, 1996) or longer than 0.4 seconds (Freed, 2000), many recent fluency studies (including Tavakoli & Uchihara, 2020) set a much lower threshold of .25 seconds. This shift toward a lower threshold emerged from the

findings of studies (e.g., de Jong & Bosker, 2013) that reported a pause of longer than .25 seconds can affect L1 speakers' perceptions of fluency.

3. *The total number of repairs, namely hesitations, repetitions, reformulations, and self-corrections per minute*, was used to examine repair fluency (Skehan, 2009).

For their quantitative analysis, Tavakoli and Uchihara examined the amount and type of MWS use at the four different proficiency levels and the strength of the associations (correlations) between MWS use and fluency aspects. Following earlier studies (Garner & Crossley, 2018; Kyle & Crossley, 2015), Tavakoli and Uchihara adopted a frequency-based text-external approach (as described above) and used three n-gram indices (proportion, frequency, and association strength, described below) focusing on two- and three-word contiguous sequences (i.e., bi- and trigrams). They used the *Tool for the Automatic Analysis of Lexical Sophistication 2.0* (TAALES, Kyle et al., 2018) to calculate n-gram scores, using the spoken subsection of the *Corpus of Contemporary American English* (COCA: Davies, 2009) as a reference corpus.

As for the meaning of each n-gram score, the proportion score is the proportion of bi- and trigrams in the learner's spoken data that are also found in the external reference corpus. Unlike n-gram proportion scores, which base their scoring on the absence or presence of learner-produced n-grams in the reference corpus, frequency scores base their scoring on a continuous scale, ensuring that all MWSs are among the 30,000 most frequent MWSs in COCA. The two association measures used by Tavakoli and Uchihara were mutual information (MI) and *t* score, each offering insights into different types of word combinations. MI tends to highlight combinations that may be less frequent but whose component words are rarely found apart (e.g.,

ultimate arbiter, tectonic plates). In contrast, *t* score emphasizes very frequent combinations, with rankings that closely resemble those based on raw frequency (e.g., *good example, hard work*; Durrant & Schmitt, 2009).

As for quantitative results, overall, Tavakoli and Uchiyama's study showed significant positive relationships between the use of MWSs and aspects of oral fluency over the range of different proficiency levels. They found that greater use of MWSs was associated with higher oral proficiency scores and with some aspects of fluency (articulation rate and fewer mid-clause pauses). They reported a linear relationship between some of their n-gram measures across proficiency levels—an important finding because it suggests that learners of higher proficiency produce a larger proportion of frequent n-grams and more frequent n-grams, confirming the important job that MWSs perform in fluent speech. They found that high-frequency n-grams correlated positively with articulation rate (i.e., the speed of speech), suggesting that fluent speakers tend to produce more high-frequency n-grams ($r = .396$, $p = .003$ for pruned articulation rate and $r = .325$, $p = .014$ for unpruned articulation rate). They also found that n-gram proportion correlated negatively with the frequency of mid-clause pauses, but the correlation ($r = -.271$, $p = .041$) did not reach statistical significance with their corrected alpha level.¹ This result implies that speakers who produced a higher proportion of target-like n-grams paused less frequently in the middle of clauses, although the effect size was small.

In addition, they found a non-significant positive correlation between end-clause pauses and strongly associated MWSs ($r = .302$, $p = .024$), suggesting that speakers who produced n-grams of high mutual information quality also paused more frequently at the end of clauses. Finally, they found a negative correlation between n-gram association strength and repair

frequency ($r = -.308$, $p = .021$), which also missed the corrected alpha level. This implies that speakers who produced strongly associated MWSs tended to make fewer repairs, though the effect size was again small.

For their qualitative results, they reported three main findings:

1. All participants tended to use MWSs that were specifically applicable to task completion, many of which seemed to have been taken from the directions given to guide the learners on what to say to complete the speaking task. They observed that more advanced learners borrowed a large number of task-specific MWSs from the task instructions, demonstrating the well-known learner tendency of text-mining (i.e., making use of MWSs from source texts to L2 output; e.g., Hoang & Boers, 2016).
2. They found that lower proficiency learners tended to borrow the n-grams and use them repeatedly in their current form, whereas higher proficiency learners tended to avoid exact repetition from the instructions and use MWSs more creatively. This finding is in line with the well-documented key strategy of avoidance of redundancy and repetition of MWS use among more advanced learners (e.g., see Granger, 2019), implying that the ability to manipulate MWSs and avoid word-for-word repetition is a key trait of higher-level L2 speakers.
3. They found that higher-level learners used discourse markers more frequently. This finding is aligned with the well-documented tendency of more proficient learners to use more discourse markers (Tavakoli, 2018).

In their discussion, Tavakoli and Uchihara pointed out that, in response to their first research question, the quantitative evidence they found suggests a potentially linear relationship between many of the n-gram measures and level of proficiency, providing support for the notion that MWSs play a key role in proficiency. In response to their second research question, Tavakoli and Uchihara reported that their findings shed light on the formulation stage and the articulation stage of the speech production process. First, at the formulation stage, their finding that high-frequency n-grams correlated positively with articulation rate (i.e., speed fluency) suggests that fluent speakers tend to produce more high-frequency n-grams (particularly more trigrams), which is aligned with psycholinguistic research evidence (Ellis, 2012; Siyanova-Chanturia & Van Lancker Sidtis, 2018) suggesting that MWSs are stored and retrieved as individual items that allow for more efficient language processing. Tavakoli and Uchihara suggest that this processing efficiency might enhance speed fluency not only at the formulation stage but also at the articulation stage where phraseological knowledge might enhance the articulation of individual words (e.g., through phonetic reduction), speeding up their overall articulation rate. Furthermore, they note that their finding of a positive correlation between end-clause pausing and the MI measure is important because it suggests that speakers who used a larger number of MWSs paused less frequently at mid-clause position. Because mid-clause pausing is thought to be linked to the formulation stage of speech production (Skehan, 2014), this finding supports the view that more phraseologically-proficient learners are able to retrieve prefabricated chunks as whole units without mid-clause pausing.

Tavakoli and Uchihara noted several limitations of their study that are worth addressing in future research. First, a potential limitation relates to their small sample size ($N = 56$), with only four proficiency levels (ranging from low B1 to C1 on the CEFR scale). Looking at a wider

range of proficiency levels may better illuminate the relationship between MWS use and fluency aspects, as patterns of MWS use might be different at lower or higher levels (e.g., lower proficiency learners might tend to use fewer MWSs, possibly hindering their speed of speech). Larger sample sizes in different contexts (e.g., EFL) will help ensure the generalizability of the results in future research. Second, measurement of MWSs was limited to corpus-derived frequency-based measures, without examining other pre-defined characteristics of formulaic sequences such as syntactic relations (e.g., adjective + noun) or phonological features (e.g., intonation units).

Another potential weakness of their text-external approach (i.e., using an external reference corpus) is that many of the identified sequences likely do not have any psycholinguistic reality/validity for the individual learners in the study. As Myles and Cordier (2017) point out, the fact that a sequence is frequent in an external corpus is not a guarantee of its being processed holistically within individual learners. The automatic extraction of the most frequent n-grams in a given corpus does not take account of *phonological coherence*, nor semantic or functional coherence: For example, a bigram sequence such as *and the* might be very frequent, but is it a “processing unit” (PU, to borrow the term coined by Myles & Cordier), given its lack of semantic or functional unity and unidentified phonological status? Future research will need to look at text-internal measures to identify which MWSs are learner-internal PUs, that is, which have psycholinguistic reality for individual learners. Third, an additional potential limitation of Tavakoli and Uchihara’s study relates to their measurement of only relatively short (two- and three-word) sequences. More research is needed to explore the relationship between longer (e.g., four- and five-word) sequences, proficiency levels, and aspects of utterance fluency. Fourth, their MWS measures were based on n-gram tokens rather than types. Fifth, their measurement of

MWSs did not distinguish between structurally complete n-grams (e.g., *on the other hand*) and structurally incomplete ones (e.g., *one of the*).

2.5.3 Uchihara et al. (2022): To What Extent is Collocation Knowledge Associated With Oral Proficiency? A Corpus-Based Approach to Word Association

Uchihara et al. (2022) explored the relationship between L2 learners' collocational knowledge and their oral proficiency. While previous studies have shown that both L1 and L2 speakers process collocations more rapidly and accurately than novel word strings in various language tasks (Ellis et al., 2008; Tremblay et al., 2011; Siyanova-Chanturia & Van Lancker Sidtis, 2018), the focus has largely been on collocation "use" rather than "knowledge". Uchihara et al. (2022) address this gap by utilizing a word association task and corpus-based measures to analyze learners' collocational knowledge as a distinct aspect of language proficiency.

Corpus-based research has shown that L1 writers tend to use more sophisticated and low-frequency collocations (indicated by higher MI scores) compared to L2 writers (Durrant & Schmitt, 2009). Similar findings were observed comparing L2 learners of different proficiency levels (Granger & Bestgen, 2014). Moreover, proficient L2 speakers are more likely to use strongly associated, sophisticated collocations (measured by higher MI scores) in spontaneous oral production (Eguchi & Kyle, 2020; Saito, 2020). Despite these insights, existing research has not adequately measured collocational knowledge. Uchihara et al. highlight the need for separate elicitation tasks, such as lexical decision tasks, to measure vocabulary knowledge more accurately (Gyllstad & Schmitt, 2018).

Word association tasks have been used to assess L2 collocation knowledge (e.g., Clenton, 2015; Fitzpatrick, 2006). In these tasks, participants respond to stimulus words with the first words that come to mind. Researchers then analyze and categorize the relationships between the stimuli and responses (e.g., collocationally or phonologically) using human judgement (Fitzpatrick, 2013). Studies using this approach have shown that L1 speakers produce more collocational responses than non-L1 speakers (Fitzpatrick, 2006), and these responses correlate with higher L2 proficiency as measured by receptive vocabulary size (Clenton, 2015; Fitzpatrick, 2006).

An alternative to the human judgement approach is the corpus-based approach, which uses co-occurrence frequency data as well as association strength data from language corpora to identify collocational associations, allowing measurement of collocational strength in a more objective way, though further research is needed to confirm the corpus-based approach's validity. Only one study, by Zareva and Wolter (2012), has used corpus-based measures to examine the collocation-proficiency relationship using a word association task and corpus-based collocation measures. Zareva and Wolter (2012) found that intermediate learners produced more collocational responses than advanced learners or L1 speakers, challenging the assumption that higher proficiency correlates with more collocational associations. However, their reliance on binary identification and *t*-scores limits their findings, highlighting the need for more nuanced measures. Uchihara et al.'s study thus advances our understanding by focusing on collocational knowledge rather than mere usage, providing empirical evidence for the role of collocations in producing fluent and lexically rich speech.

Forty Japanese university students participated in Uchihara et al.'s study. These participants were chosen to represent L2 learners at an intermediate-to-advanced proficiency level. Their study employed two main tasks: (1) A word association task (Lex30; Meara & Fitzpatrick, 2000), in which participants were presented with stimulus words and asked to provide the first four words that came to mind within 30 seconds. This task aimed to elicit spontaneous collocational knowledge. (2) A spontaneous speaking task (Suitcase Story picture narrative; Derwing et al., 2009): Participants described a series of pictures, which was used to measure their oral proficiency.

To evaluate oral proficiency and collocational knowledge, Uchihara et al. used both subjective and objective measures. For oral proficiency measures, they used human judgments to assess perceived fluency (speech rate) and lexical richness (variety and sophistication). They also used three objective fluency measures: articulation rate, silent pause ratio, and filled pause ratio. Lexical richness was also measured using three objective lexical measures: diversity, frequency, and range. Diversity scores were automatically computed using Coh-Metrix (McNamara et al., 2014), and frequency and range scores were automatically computed using TAALES 2.2. (Kyle et al., 2018). For corpus-based collocation measures, they used absolute frequency count, t-score, and MI score to analyze the frequency and strength of collocations.

Uchihara et al. examined the validity of the word association task (Lex30) for eliciting collocations, showing that over 80% of participant responses were accurately scored using corpus-based metrics. They confirmed high interrater reliability for fluency (optimal speed) and lexical richness ratings (Cronbach's $\alpha = .90$ and $.85$, respectively). They employed a Bayesian approach for statistical modeling, allowing for robust estimation of correlations and regression

analyses. This analytical method provided a detailed understanding of the relationships between collocation knowledge and oral proficiency, factoring in vocabulary size as a potential covariate.

Uchihara et al.'s results showed significant relationships between collocational knowledge and oral proficiency. In terms of fluency measures, participants who used more low-frequency collocations in the word association task (i.e., lower collocation frequency scores) spoke faster ($\rho = -.312$) and with fewer silent pauses ($\rho = .329$) and were perceived as optimally fluent ($\rho = -.371$). In terms of lexical richness measures, participants who provided more strongly associated collocations (as measured by MI scores) used more sophisticated lexical items (i.e., words that appear rarely and in a narrow range; $\rho = -.347$) and more low-frequency vocabulary ($\rho = -.339$) and were perceived (rated) as more lexically proficient ($\rho = .344$). The study also found that collocation knowledge remained a distinct predictor of oral proficiency even after accounting for vocabulary size. This highlights the importance of collocational knowledge as an independent component of language proficiency.

Uchihara et al. (2022) discussed their findings in the context of existing research, supporting the notion that not only knowledge of low-frequency words (Hilton, 2008; Uchihara & Saito, 2019) but also knowledge of collocations (Saito, 2020; Tavakoli & Uchihara, 2020) play a key role in fluent and lexically rich speech (i.e., single-word item knowledge and multiword item knowledge are linked with higher oral proficiency). They argued that learners who have a larger repertoire of collocations in their mind might (1) experience processing advantages (Siyanova-Chanturia & Van Lancker Sidtis, 2018), enabling them to access and retrieve appropriate lemmas more efficiently during the formulation stage (Kormos, 2006); (2) allocate more attentional resources to other aspects of language processing, such as lexical or

grammatical accuracy or complexity (Boers et al., 2006; Skehan, 2009); (3) speak the L2 more quickly with fewer pauses (de Jong, 2016a); and (4) give listeners the impression of being fluent (Uchihara & Saito, 2019). Uchihara et al.'s findings also aligned with previous research suggesting that advanced language users have a better command of collocations with higher MI scores (Durrant & Schmitt, 2009; Eguchi & Kyle, 2020; Ellis et al., 2008; Granger & Bestgen, 2014; Saito, 2020; Sonbul, 2015).

One emerging question Uchihara et al. discussed is why MI score was exclusively related to lexical richness measures and not fluency measures, despite collocation frequency being related to fluency but not lexical richness in their study. One possible reason they pondered is that the two types of collocation measures tap into different aspects of phrasal competence and reflect different stages of L2 oral development. The MI score, indicating a higher level of phraseological sophistication, is more relevant to later stages of L2 development, which includes lexical richness (Saito et al., 2016; Tavakoli, 2018). In contrast, raw frequency score may be more pertinent to the initial stages of L2 development, particularly temporal features like fluency. The authors also discussed that from a theoretical standpoint, the link they found between raw frequency score and fluency can be explained by the power law of L2 practice (DeKeyser, 2001; Ellis, 2002). The power law of L2 practice says that learners with knowledge of more low-frequency collocations likely have had extensive exposure to high-frequency collocations, making these collocations more entrenched and easily retrievable for L2 use. This entrenched and retrievable knowledge enables the development of oral fluency. Previous research has indicated that oral fluency can improve rapidly with intensive repetition practice, such as short-term, frequent practice sessions (Suzuki, 2020). Such rapid improvement is often attributed to the proceduralization of frequent phraseological expressions, which then form the

basis for fluent oral production (Tavakoli & Uchiyara, 2020). In contrast, improving lexical richness involves more than just proceduralization of existing lexical resources through short-term intensive fluency practice (Suzuki, 2020) or immersion (Tavakoli, 2018). It requires expanding vocabulary by learning lower-frequency words and deepening various aspects of lexical knowledge, such as contextual constraints and grammatical functions. Thus, lexical richness necessitates both breadth and depth of word knowledge, beyond mere form and meaning.

Uchiyara et al. concluded that collocational knowledge is a significant predictor of oral proficiency, highlighting the importance of phraseological competence in language learning and teaching. Their findings suggest that L2 learners should be encouraged to develop their knowledge of collocations to enhance their fluency and lexical richness. The study provides a foundation for future research to further investigate the role of collocations in language proficiency and to develop pedagogical strategies that emphasize the acquisition of collocational knowledge.

Uchiyara et al.'s study makes a significant contribution to the field by exploring the relationship between L2 learners' collocational knowledge and oral proficiency using innovative methods, such as word association tasks and corpus-based measures. These strengths provide a more objective and nuanced understanding of how collocational knowledge influences fluency and lexical richness in spontaneous speech. However, several limitations should be considered. First, the study's sample, consisting solely of 40 Japanese university students, limits the generalizability and robustness of the findings. The small sample size may create some uncertainty in the results, while the lack of diversity may restrict the applicability of the results to

other L2 learner populations. Future research should aim to include larger and more diverse samples, which would not only increase the reliability of the findings but also ensure that the results are more broadly applicable across different cultural and linguistic contexts. Second, the use of a picture narrative task, although effective for eliciting spontaneous speech, may not fully capture the complexity and range of oral proficiency. Third, the study relied on specific corpus-based measures, such as frequency and MI scores, which may not fully encompass the multifaceted nature of collocational knowledge. Additionally, the study's cross-sectional design limits the ability to draw causal inferences about the relationship between collocational knowledge and oral proficiency. Longitudinal studies would be beneficial to explore how this relationship evolves over time. In summary, while Uchihara et al.'s study offers important insights, future research should address these limitations by increasing sample size and diversity, expanding measurement tools, and employing longitudinal designs to better understand the role of collocational knowledge in L2 proficiency.

2.5.4 Suzuki et al. (2022): Does the Reuse of Constructions Promote Fluency Development in Task Repetition? A Usage-Based Perspective

Suzuki et al. (2022) examined the relationship between the reuse of linguistic constructions during task repetition and fluency development in L2 learners. Rooted in the usage-based constructionist approach, which emphasizes the importance of frequency and repetition in language acquisition (Goldberg, 2006; Tomasello, 2003), their study addressed a significant gap in the literature concerning the proceduralization of linguistic knowledge through repeated task performance. According to the usage-based constructionist approach, repeated use of specific constructions reinforces them in a learner's mental lexicon (Langacker, 2008),

facilitating faster and more efficient retrieval during language production. This strengthening and facilitation process is known as proceduralization, and it allows declarative knowledge to become more automated with practice (DeKeyser, 2018).

The authors noted that while previous research has highlighted the benefits of task-repetition practice for fluency improvement (Bygate, 2018; de Jong & Perfetti, 2011; de Jong & Tillman, 2018), there has been insufficient exploration into how specific types of linguistic constructions—ranging from concrete lexical items to abstract syntactic patterns—are entrenched through repetition and how this process relates to different aspects of utterance fluency (e.g., speed, breakdown, and repair). Suzuki et al. specifically aimed to extend existing research in this area in three key respects. First, they aimed to validate the Natural Language Processing (NLP) approach employed by de Jong and Tillman (2018), which had primarily been used with ESL learners, by applying it to EFL learners. The NLP technique involves the analysis of language data to measure the degree of linguistic construction reuse across task repetitions. De Jong and Tillman used this approach to quantify the similarity of linguistic constructions across repeated tasks and found that higher reuse of specific constructions was correlated with improvements in fluency, particularly in terms of a greater proportion of time filled with speech (i.e., fewer and/or shorter pauses) and shorter mean syllable duration (i.e., a faster speech rate). By applying this methodology (described in detail below) to a different learner population, Suzuki et al. aimed to assess its broader applicability across varied L2 contexts.

Second, Suzuki et al.'s study introduced a novel comparison between blocked and interleaved task repetition. While prior studies had mainly focused on conditions such as constant versus decreasing time or content-repetition versus no-content-repetition, Suzuki et al. explored the potential pedagogical benefits of blocked vs. interleaved repetition schedules.

Blocked practice involves repeating the same task consecutively, which is hypothesized to enhance the reuse and proceduralization of specific constructions, leading to faster and more efficient speech production. On the other hand, interleaved practice alternates between different tasks within the same session, potentially promoting more flexible language use. By comparing these two approaches, Suzuki et al. aimed to provide insights into how varying the structure of task repetition might differentially influence the reuse of linguistic constructions and subsequent fluency development.

Finally, Suzuki et al. employed a pretest-posttest design to investigate how the reuse of constructions during training might contribute to fluency improvements in novel (untrained) tasks—an area that has been relatively under-explored. By examining how constructional entrenchment during practice could transfer to new contexts, their study aimed to enhance our understanding of the proceduralization process and its impact on fluency development, building on the limited existing research in this area (e.g., de Jong & Perfetti, 2011).

Suzuki et al.'s task-repetition intervention study involved 40 Japanese EFL learners who were randomly assigned to one of two practice schedules: blocked or interleaved. The blocked group performed the same oral narrative task three times per day as part of a single training session (e.g., Day 1: AAA, Day 2: BBB, Day 3: CCC), while the interleaved group performed different tasks in a rotating sequence (e.g., Day 1: ABC, Day 2: ABC, Day 3: ABC). Before and after the three training sessions, the participants were given a pretest and posttest to assess their fluency development. This design allowed the researchers to compare the effects of the two repetition schedules on the reuse of linguistic constructions and the subsequent impact on fluency development. The tasks involved narrating stories based on cartoon prompts designed to elicit spontaneous speech. The participants' performances were audio-recorded and transcribed

for detailed linguistic analysis. Following de Jong and Tillman's (2018) two-step analysis procedure, the researchers (a) extracted three types of constructions from the transcripts: unigrams (i.e., single words), trigrams (i.e., sequences of three words) at a concrete level, and parts of speech trigrams (e.g., "preposition determiner noun") at an abstract level. They then (b) calculated similarity scores between repeated tasks performances (cosine similarity). Fluency was assessed using several measures: mean syllable duration (speed fluency), frequency and duration of mid- and end-clause pauses (breakdown fluency), and frequency of repetitions and repairs (repair fluency). The researchers used independent samples *t*-tests to compare the similarity scores between the two practice schedules (blocked vs. interleaved). Robust correlational analyses were also conducted to examine the relationship between construction reuse (similarity scores) and fluency development from the first to the third task performance (during-training fluency change) as well as from the pretest to the posttest.

Suzuki et al.'s study found that the blocked practice group showed significantly higher frequencies of construction reuse compared to the interleaved practice group. Specifically, the blocked group reused lexical trigrams and parts of speech trigrams more frequently, with large effect sizes for both (lexical trigrams: $t(35.27) = 4.19, p < .001, d = 1.20, 95\% \text{ CI } [0.57, 1.81]$; POS trigrams: $t(37.52) = 2.96, p = .01, d = 0.85, 95\% \text{ CI } [0.25, 1.43]$). This increased reuse was associated with varied fluency outcomes during the training phase. In the blocked practice condition, no significant correlations were found, though small negative correlations suggested that greater reuse of all three construction types might contribute to more continuous speech, as indicated by shorter mid-clause pause durations ($r = -0.27$ to -0.34). In contrast, the interleaved practice condition revealed a significant positive correlation between mean syllable duration and the reuse of lexical and POS trigrams ($r = 0.37, p = 0.06; r = 0.49, p = 0.01$), indicating that

greater reuse in this group was unexpectedly linked to *lower* speed fluency. The increased reuse of constructions was also associated with varied fluency outcomes from the pretest to the posttest. In the blocked practice condition, although correlations were small and not statistically significant, there were potentially meaningful associations between construction reuse and breakdown fluency measures (mid-clause pause duration and frequency, as well as end-clause pause duration). In contrast, the interleaved practice condition showed stronger and more significant correlations, including a negative correlation between lexical trigram reuse and mean syllable duration ($r = -0.40$, $p = .045$), suggesting that greater reuse led to higher speed fluency on the posttest for the interleaved group.

For discussion, Suzuki et al. noted that blocked practice led to greater constructional recycling compared to interleaved practice, likely due to the repetitive nature of the tasks, which facilitated the reuse of both concrete (lexical unigrams and trigrams) and abstract (parts of speech trigrams) constructions. While this contributed to improved breakdown fluency, it also risked reinforcing ungrammatical constructions due to the absence of corrective feedback (Boers, 2014; Thai & Boers, 2016). The study suggests that while blocked practice may enhance breakdown fluency, integrating accuracy-focused interventions is necessary to prevent the reinforcement of errors (Lynch, 2018; Tran & Saito, 2021). Regardless of practice condition (blocked vs. interleaved), learners who repeatedly used the same constructions during training produced fewer but longer mid-clause pauses. This pattern suggests that constructional recycling facilitated linguistic encoding, leading to more efficient retrieval processes. However, the trade-off between fewer pauses and longer duration of pauses may indicate ongoing developmental changes in fluency, as learners balanced encoding efficiency with processing challenges. An additional point for discussion is that higher constructional recycling was associated with longer

end-clause pauses, especially in the interleaved practice condition. The authors attributed this to the cognitive demands of re-engaging in conceptual planning between task repetitions. While longer end-clause pauses might seem counterintuitive to fluency development, they may reflect deeper engagement in encoding complex grammatical constructions, particularly in the interleaved condition where more cognitive resources were required.

The strength of Suzuki et al.'s study lies in its innovative approach to examining the effects of task repetition on construction reuse and fluency development. Using a controlled experimental design with clearly defined practice schedules (blocked vs. interleaved) and pretest-posttest allowed for a clear comparison of the two approaches, providing valuable insights into the mechanisms underlying fluency development in L2 learners. However, there are some limitations to consider. First, the study's sample size was relatively small and limited to Japanese EFL learners, which may affect the generalizability of the findings. The homogeneity of the participant group means that the results cannot be generalized to other L2 learner populations with different linguistic backgrounds. Additionally, the focus on oral narrative tasks raises questions about the transferability of the findings to other types of speaking activities, such as dialogues or discussions, which may involve different cognitive demands and linguistic constructions. The use of a more varied and open-ended task design, such as interactive speaking activities, could provide a broader range of linguistic constructions for analysis, particularly at the abstract level. Another limitation was the study's exclusive focus on specific linguistic units (lexical unigrams and trigrams, parts of speech trigrams), which may not capture the full range of constructions that contribute to fluency. Future research could benefit from exploring a broader range of linguistic constructions and their role in fluency development. In addition, the short duration of the intervention, with only three task performances per prompt, suggests a need for

more longitudinal studies to better understand the effects of extended construction reuse on fluency development and underlying proceduralization. Furthermore, while the study focused on fluency, the lack of accuracy measures means that the potential reinforcement of incorrect constructions remains a concern. Future research should include accuracy assessments and consider integrating accuracy enhancement techniques, such as corrective feedback, to ensure that repeated constructions are both fluent and correct.

While their study provided valuable contributions to our understanding of the relationship between construction reuse and fluency development, addressing these limitations in future research could enhance the robustness and generalizability of the findings. Suzuki et al.'s study shows that language teachers should think about practice schedules in L2 instruction. Different schedules affect how students learn language and become fluent. Additionally, implementing methods to improve accuracy when repeating tasks could reduce the chances of making the same mistakes.

2.5.5 Hougham et al. (2024a): Disentangling the Contributions of Shorter Versus Longer LBs to L2 Oral Fluency

Hougham et al. (2024a) explored the role of LBs in enhancing L2 oral fluency. In recent decades, there has been a lot of focus on MWSs in language learning, particularly LBs, which are important for speaking fluently (Biber et al., 1999; Garner & Crossley, 2018). However, most previous studies have primarily focused on shorter LBs (two- and three-word sequences), due to their frequent occurrence and ease of identification in both spoken and written discourse (Tavakoli & Uchihara, 2020), potentially overlooking the contributions that longer LBs might make to

fluency. Fluent speakers often use longer phrases like *in light of recent events*, which can help them speak more smoothly and quickly.

Addressing this gap in the literature, Hougham et al. (2024a) aimed to investigate the relative contributions of both shorter and longer LBs to different aspects of L2 oral fluency, specifically focusing on speed, breakdown, and repair fluency. Their study was motivated by the recognition that previous methodological approaches often failed to capture the full spectrum of LBs' impact on fluency, particularly neglecting longer sequences. Their study responded to calls for more comprehensive research on the role of longer LBs in L2 speech production (Tavakoli & Uchiyama, 2020). By including both shorter and longer LBs in their analysis, the researchers aimed to provide a deeper understanding of how these language units contribute to fluency, with potential implications for pedagogical strategies, language assessment, and instructional materials development. Through the integration of longer language bursts into these specific areas, their findings have the potential to enhance learners' ability to produce more fluent and coherent speech.

Participants in Hougham et al.'s study were 50 undergraduates, whose proficiency levels ranged from lower-intermediate to intermediate, at two universities in Japan. The study employed three distinct speaking tasks: a narrative task, a descriptive task, and an argumentative task. These tasks were chosen to cover a broad spectrum of speaking contexts, allowing the researchers to examine the impact of LB usage across different types of speech. The tasks were presented with bilingual instructions and images to ensure clarity and contextual understanding for the participants. Hougham et al. measured three aspects of utterance fluency: speed fluency, breakdown fluency, and repair fluency. Speed fluency was measured by calculating the mean

syllable duration (i.e., total phonation time divided by the number of syllables produced during each task). Breakdown fluency was assessed by analyzing the frequency of pauses, particularly focusing on end-clause and mid-clause pauses, which are known to disrupt the flow of speech (Kahng, 2014). Repair fluency was evaluated based on the frequency of total repairs such as self-corrections and reformulations, which are indicators of a speaker's ability to monitor and adjust their speech in real-time. Fluency measures were analyzed using PRAAT software for precise pause analysis.

The researchers measured the usage of LBs using a dual approach of analyzing the text internally and externally. The *text-internal* approach involved identifying longer LBs within the learner corpus (i.e., transcriptions of the participants' speech) using AntConc software (Anthony, 2022), whereas the *text-external* approach compared the participant speech with speech from the spoken subsection of COCA as an external reference corpus to measure the use of shorter LBs (bi- and trigrams), using TAALES software (Kyle et al., 2018) for n-gram indices (proportion, frequency, and association strength). The researchers then employed robust multiple regression analysis, dominance analysis, and random forests techniques to analyze the relationship between LB usage and the different aspects of fluency, isolating the unique contribution of longer vs. shorter LBs to aspects of fluency.

Hougham et al.'s findings provided several key insights into the relationship between LB usage and various aspects of L2 oral fluency. In terms of speed fluency, their analysis showed that participants who frequently used bigram combinations tended to speak at a faster rate. Specifically, bigram frequency was positively associated with mean syllable duration, with a regression coefficient of $b = 313.41$ (95% CI [40.22, 586.60], $t = 2.25$), indicating that an

increase in the use of high-frequency bigrams corresponded to a faster speech rate. In contrast, participants who used bigrams of higher collocational quality, as indicated by higher scores, tended to speak more slowly. This was evidenced by a negative association between bigram MI and mean syllable duration ($b = -120.23$ (95% CI $[-229.36, -11.10]$, $t = -2.16$). This finding was reinforced by the fact that bigram MI was the only MWS variable that was confirmed important by random forests (Boruta algorithm). Although the use of longer word sequences (four-to-five-word MI measure) showed a slight positive effect on speech rate, this effect was not statistically significant ($b = 13.07$, 95% CI $[-46.53, 72.67]$, $t = 0.43$), as indicated by the confidence interval crossing zero.

In terms of breakdown fluency, the study found that four-to-five-word MI was inversely related to the frequency of mid-clause pauses. Although the regression analysis did not reach statistical significance ($b = -9.99$, 95% CI $[-21.34, 1.37]$, $t = -1.72$), the dominance analysis highlighted four-to-five-word MI as the most influential predictor, contributing 39.59% of the variance in mid-clause pauses. Furthermore, the random forests (Boruta) analysis confirmed four-to-five-word MI as the most important variable. This suggests that increased use of longer word sequences could reduce the frequency of pauses within clauses, thereby promoting smoother speech flow. Although the effects on end-clause pauses were not statistically significant at conventional thresholds, the analysis showed that four-to-five-word MI was again the most influential factor, with a dominance weight of 34.17%. This indicates a potential, albeit tentative, effect of longer LBs on reducing pauses at the ends of clauses, hinting at their role in enhancing overall speech fluency.

Regarding repair fluency, the study found a significant negative association between the frequency of total repairs and the use of longer sequences (four-to-five-word MI). The regression analysis revealed that as participants used more four-to-five-word sequences, their need for speech repairs decreased ($b = -2.41$ (95% CI $[-4.24, -0.59]$, $t = -2.60$). Four-to-five-word MI emerged as the strongest predictor, accounting for 42.75% of the variance in total repair frequency according to the dominance analysis. Four-to-five-word MI was also confirmed as an important variable by Boruta. This finding highlights the role of longer LBs in reducing repairs or disruptions in speech, allowing for more fluid and uninterrupted oral production.

Overall, the research findings indicate that using shorter and more frequent language sequences can improve the speed of speaking, while longer language bursts are important for reducing pauses within sentences and the need for corrections. This ultimately leads to smoother and more coherent speech production. Their study challenges the traditional emphasis on shorter LBs by demonstrating that longer LBs (four- and five-word sequences with high MI scores) significantly impact fluency. Specifically, these longer bundles were found to slightly increase speech speed, marginally reduce mid-clause pauses, and significantly decrease the frequency of speech disruptions and repairs. This indicates a complex relationship between bundle length and fluency among intermediate EFL learners, suggesting that incorporating longer LBs into language teaching could be instrumental in fostering more fluent speech.

Additionally, the study exposed unforeseen obstacles when employing shorter, target-like trigrams (three-word sequences). Contrary to expectations, the frequent use of these trigrams was associated with an increase in pausing during speech, which may indicate additional cognitive demands on learners. Although this effect did not reach statistical significance, it suggests the

importance of tailoring fluency instruction to different proficiency levels, ensuring that learners are well prepared to manage the complexities associated with various types of LBs.

In their discussion, Hougham et al. relate these findings to existing L2 speech production models, suggesting that longer LBs may facilitate more efficient processing, leading to improved fluency. Theoretically, Hougham et al.'s study contributes to the understanding of how different lengths and types of LBs impact fluency. It supports the view that both shorter and longer LBs play key roles in fluent speech production, promoting an expanded focus on both shorter and longer LBs. The findings align with usage-based models of language, which posit that frequently encountered multiword sequences are stored as single units in the mental lexicon (Bybee, 2006).

Moreover, Hougham et al. stress the importance of their findings for teaching, advocating for a more comprehensive integration of longer language units in language instruction. The significant role of longer LBs in reducing speech repairs and improving overall fluency suggests that they could be a valuable addition to language teaching curricula, L2 fluency training in particular. The authors recommended that language educators incorporate both shorter and longer LBs into their teaching strategies, suggesting the potential for more comprehensive, data-driven instructional strategies to effectively enhance L2 oral fluency.

Hougham et al.'s study is particularly noteworthy for its rigorous methodology and its novel exploration of the impact that both shorter and longer LBs have on L2 oral fluency. By employing advanced statistical techniques, including robust regression alongside dominance analysis and random forests analysis, the researchers were able to provide nuanced insights into how different lengths of LBs contribute to various aspects of fluency. Additionally, the use of both text-internal and text-external methods to analyze LB usage further strengthens the study's

contributions, offering a more comprehensive understanding of the role these language units play in fluent speech production.

Despite its contributions, Hougham et al.'s study has several limitations. First, the participants were drawn from just one L1 background, which may limit the extent to which the results can be applied to broader populations of EFL learners. Second, some of the study's regression analyses show fairly wide confidence intervals, indicating a degree of uncertainty in the effect sizes observed. Third, the frequency-based automatic extraction techniques used to identify LBs in the study might not have captured all MWSs, particularly those that are used infrequently or idiosyncratically. Additionally, some LBs identified in the study (e.g., those borrowed from task prompts) may not represent genuine learner-produced expressions. This limitation suggests a need for more refined approaches to identifying and analyzing LBs, particularly methods that can accurately capture both common and less frequent bundles within learner-produced texts. Moreover, the use of MI scores as a measure for longer LBs raises concerns. MI scores, originally developed for two-word collocations, do not account for the order of words and may not reliably measure the collocational strength of longer, more complex sequences.

In light of these limitations, several directions for future research should be considered. Studies should focus on the LB-fluency relationship among learners from diverse linguistic backgrounds and employ larger sample sizes to improve the reliability of the findings. Increasing the sample size in future studies could help to narrow the confidence intervals and improve the precision and reliability of the statistical estimates. Additionally, the development and use of refined LB rating techniques could help identify more structurally and semantically complete

units. Finally, the potential of mixed-effects models as a more sophisticated statistical technique for analyzing learner corpus data could offer deeper insights into the nuances of LB usage and fluency across different proficiency levels and contexts. In addition to their suggestion for further research to explore more sophisticated measures, the authors also emphasized the importance of finding alternative methods, apart from MI, that can effectively capture the intricate structure and function of longer LBs.

2.6 Discussion

2.6.1 *Synthesis of Findings From the Current Literature Review*

According to the reviewed literature, there is a growing consensus that MWSs, especially LBs, play a crucial role in fostering fluency in speaking a second language. Across numerous empirical studies, MWSs have been associated with enhanced speech production, including faster articulation rates and reduced pause frequency (e.g., Tavakoli & Uchihara, 2020; Uchihara et al., 2022). This suggests that MWSs contribute to smoother, more fluid speech, likely because their holistic retrieval reduces the cognitive load required during the speech production process (Levelt, 1989; Kormos, 2006).

In terms of patterns, one recurring theme is that more proficient speakers tend to use shorter MWSs (e.g., bi- and trigrams; Tavakoli & Uchihara, 2020) and longer MWSs (e.g., four- to five-word sequences; Wood, 2009; Hougham et al., 2024a) more frequently in speech. Tavakoli and Uchihara (2020) reported that greater use of shorter MWSs was associated with higher oral proficiency scores and specific aspects of fluency, such as faster articulation rates and fewer mid-clause pauses. They showed a linear relationship between some n-gram measures and proficiency levels, showing that more proficient learners produce a larger proportion of frequent

n-grams, highlighting the important role that shorter MWSs play in fluent speech. In addition to shorter sequences, longer MWSs also contribute to fluency improvements. Wood (2009) demonstrated that as the learner in his study improved her fluency through targeted instruction, she shifted from producing shorter, more familiar MWSs to longer, more complex ones (the mean length of MWSs in her second speech sample was 4.46 words, which was 40.7% longer than her first sample). Similarly, Hougham et al. (2024a) found that L2 speakers who were more fluent, as measured by fewer mid-clause pauses and speech repairs, tended to incorporate longer lexical bundles into their speech. The findings from these studies suggest that more proficient speakers are able to retrieve and deploy these longer sequences with greater ease, freeing up cognitive resources that allow for more fluid speech production. These findings align with usage-based theories, which suggest that the frequency and familiarity of these longer sequences help streamline cognitive processing and facilitate smoother, more automatic speech production (Bybee, 2006).

The reviewed literature also indicates that the processing advantages of LBs are not limited to advanced learners. Wood (2009) provided evidence of this by demonstrating how an intermediate-level learner improved her fluency through targeted instruction on MWSs. Although not a beginner, the learner in Wood's study experienced measurable gains in fluency, shifting from shorter to longer MWSs after the targeted intervention. Similarly, Hougham et al. (2024a) investigated lower-intermediate to intermediate level speakers and found that the use of longer lexical bundles was linked to reductions in speech repairs and mid-clause pauses, contributing to improved fluency. These findings suggest that MWS use can benefit learners at various proficiency levels.

2.6.2 Limitations of Previous Studies on the MWS-Fluency Relationship

While the reviewed studies support the hypothesis of a positive relationship between MWS use and speech fluency, they might be limited in at least six important ways. First and foremost, studies focusing on relatively short (two- and/or three-word) sequences may not fully capture learners' actual phraseological knowledge and how it relates to aspects of oral fluency and proficiency scores. This is exemplified by multiple studies (e.g., Garner & Crossley, 2018; Kyle & Crossley, 2015; Kyle et al., 2018; McGuire & Larson-Hall, 2021; Suzuki et al., 2022; Tavakoli & Uchihara, 2020; Zhang et al., 2021). For example, using longer LBs might be more beneficial for improving aspects of oral fluency and increasing assessment scores. Emphasizing the potential importance of longer LBs, Tremblay et al. (2011) demonstrated that longer (four- and five-word) LBs offer online processing advantages over non-LBs in receptive tasks. Hougham et al. (2024a) also showed that longer LBs of high collocational quality enhanced various aspects of fluency. Despite this understanding, the effects of longer LBs on achieving fluency in speech production have not been fully explored. Given what we know from Pawley and Syder's insights into the significance of lexical phrases, the comprehensive framework provided by Levelt's speech production model, as well as the principles of usage-based theory by researchers such as Bybee and Ellis, it is reasonable to hypothesize that employing longer LBs can lead to enhanced processing efficiency. The lack of studies examining longer LBs in comparison to shorter LBs is a gaping hole in the research that needs to be addressed.

Second, many previous studies have had methodological or contextual limitations, such as relying on subjective human judgments of speaking (e.g., Boers et al., 2006; Stengers et al., 2011; Zhang et al., 2021), focusing on only one aspect of fluency (e.g., speed fluency in Thomson, 2017; McGuire & Larson-Hall, 2018, 2021; Wood, 2009), or measuring MWSs

subjectively using a criteria checklist and L1 speaker intuition (e.g., McGuire & Larson-Hall, 2018; Wood, 2009). With some exceptions (e.g., Suzuki et al., 2022; Tavakoli & Uchihara, 2020), there is a paucity of studies designed to examine the link between LB use and objectively measurable aspects of utterance fluency (speed, breakdown, and repair). Suzuki et al. (2022) and Tavakoli and Uchihara (2020) aside, no previous study has examined the link between LB use and various aspects of utterance fluency, including, importantly, pause location (mid- versus end-clause position). The paucity of studies exploring the connection between LB use and objectively measurable aspects of utterance fluency is a significant gap in the research that requires attention.

Third, most previous research is restricted to investigating the MWS-fluency link with a learner-*external* approach (i.e., examining learners' use of selected sequences that are thought to be formulaic in L1-speaker English and identified in advance as formulaic, or quantifying a text's formulaicity by checking the frequency of all of its constituent word sequences against an external reference corpus) (e.g., Garner & Crossley, 2018; Tavakoli & Uchihara, 2020; Zhang et al., 2021). Only one study by Hougham et al. (2024a) has systematically attempted to employ both text-external and text-internal methods to analyze learner-produced LBs in relation to aspects of oral fluency, but their study did not examine oral proficiency scores. Fourth, some previous studies have suffered from small sample sizes (e.g., $N = 19$ in McGuire & Larson-Hall, 2018; $N = 1$ in Wood, 2009; $N = 11$ in Wood, 2010). Fifth, the LBs examined in studies based on learner corpora frequently have varying lengths, typically ranging from two to six words. Additionally, the criteria for extracting these LBs, such as frequency and dispersion, differ significantly across different studies. Therefore, it is important to view the aforementioned

findings and general patterns as hypotheses that require further testing using alternative corpus data within diverse contexts across various proficiency levels.

2.6.3 Outline of the Experimental Chapters That Follow

Considering these gaps and motivated by the lack of studies investigating the impact of longer LBs on oral fluency, the current thesis seeks to evaluate the influence of longer LBs on fluency, building upon previous research in various significant ways. This thesis is among the few studies focusing on the use of longer LBs in relation to shorter LBs and objectively measured aspects of utterance fluency (speed, breakdown, and repair), including, importantly, pause location (mid-ASU versus end-ASU position). To explore the relationships between various LB and fluency measures and their importance in predicting L2 oral fluency and proficiency, robust statistical techniques (e.g., robust regression, dominance analysis, and random forests analysis) are used. Additionally, I employ a larger sample size than most previous studies and systematically compare different lengths of LBs using varied extraction methods (text-internal and text-external) across proficiency levels, aiming for robust and generalizable findings. My research questions and experiments in the following chapters are designed to build on and extend the work of Tavakoli and Uchihara (2020) and Hougham et al. (2024a). The following is a summary of the research questions for the four experimental chapters that follow.

Chapter 3 addresses the need for a more comprehensive analysis of LB usage in relation to language proficiency, as highlighted by Hougham et al. (2024a) and Tavakoli and Uchihara (2020). It also conceptually replicates the study conducted by Tavakoli and Uchihara (2020). The research question for Chapter 3 is: How does LB usage, differentiated by length, vary across different proficiency levels? Chapter 4 continues the conceptual replication of Tavakoli and

Uchihara (2020) by disentangling the relationship between LBs of varying lengths and various aspects of oral fluency, expanding on Tavakoli and Uchihara's foundational work. The research question for Chapter 4 is: To what extent is the use of longer vs. shorter LBs associated with the three aspects of oral fluency (speed, breakdown, and repair)? Chapter 5 shifts focus from analyzing the relationship between LB usage and fluency to investigating how lexical sequences of various lengths predict performance in high-stakes oral assessments. The research question for Chapter 5 is: To what extent do longer LBs predict raw oral presentation scores compared to shorter LBs? Chapter 6 examines whether certain presentation topics and sub-topics affect the use of sophisticated lexical bundles. The research questions for Chapter 6 are: (1) Does the choice of presentation topic influence the diversity and frequency of high MI-score bigrams used by L2 English speakers? And (2) does this influence vary with proficiency scores? In my seventh chapter, I aim to discuss the findings from the experimental chapters, how they compare with related studies and theories from my review of the literature, and practical implications. Finally, my concluding chapter brings together and reflects on the main findings and implications of the thesis and discusses possible areas for future research.

2.7 Conclusion

The literature review has analyzed how different lengths and types of MWSs can improve processing abilities and determine their relationship to L2 oral fluency and proficiency measures. The review suggests that MWSs can provide a processing advantage, depending on various factors such as length and associational strength. The hypothesis that longer LBs have distinct processing advantages in terms of oral fluency and proficiency is therefore closely and empirically examined in the following experimental chapters.

Chapter 3: Exploring How LB Usage Differs Across Proficiency Levels

The study of how language proficiency is reflected in multiword sequence usage is crucial for advancing language education and assessment (Hougham et al., 2024a reviewed in Section 2.5.5; Hougham et al., 2024b; Tavakoli & Uchihara, 2020 reviewed in Section 2.5.2; Wood, 2009, reviewed in Section 2.5.1). However, research to date (e.g., Tavakoli & Uchihara, 2020) has not fully explored LB usage across different proficiency levels, leaving a critical gap in the field. This chapter seeks to bridge this gap by expanding the scope of investigation beyond the typical bi- and trigram units to include refined three-to-five-word LBs, thus offering a nuanced perspective on language proficiency.

Prior research (Tavakoli & Uchihara, 2020) provides valuable insights into the relationship between LB usage and proficiency levels, highlighting a general trend of increasing LB use with higher proficiency. However, their study focused on shorter (two- and three-word) LBs, potentially overlooking the subtleties and complexities that longer (e.g., four- and five-word) LBs might reveal about a learner's language proficiency. This chapter therefore addresses the literature gap concerning the comprehensive assessment of LBs across proficiency levels, including a detailed comparison of longer versus shorter LBs. The current chapter responds to calls (e.g., Hougham et al., 2024a; Tavakoli & Uchihara, 2020) for a more differentiated analysis of LB usage as it relates to language proficiency. By extending the range of LBs examined and introducing robust statistical techniques, the current chapter aims to provide a richer and deeper understanding of the relationship between multiword sequence use and language proficiency assessment.

Including longer LBs is an important step in capturing the sophisticated linguistic structures that are hallmarks of language use. These longer structures (e.g., *as you can see*) often embody complex grammatical patterns and a nuanced application of language in context, which shorter LBs (e.g., *there are*) may inadequately represent. Prior research (Hougham et al., 2024a) has reported that longer LBs marginally enhance speed fluency as well as play a significant role in reducing speech disruptions and repairs. However, their study suffered from several shortcomings that could limit the generalizability of their findings: they did not examine differences in LB use across proficiency levels; their study's participants came from a narrow range of linguistic backgrounds (lower-intermediate Japanese EFL learners); and their LB measurement procedure identified many LBs that do not comprise structurally complete grammatical or syntactical units. By examining the use of longer LBs in comparison with shorter LBs across proficiency levels in a different teaching context, the experiment detailed in this chapter therefore provides insights into the intricate linguistic competencies that develop as learners progress, which are essential for a more complete understanding of language proficiency.

The current chapter also introduces a methodological innovation by utilizing text-internal measurement and refinement techniques to identify and examine longer LBs, in tandem with text-external measures for shorter units. This dual approach (outlined in Section 3.3 below) not only enhances the depth of analysis but also represents a methodological advancement in the field of language assessment because it identifies more of the structurally complete sequences used by participants in my dataset.

Adhering to the framework of conceptual replication as outlined by LeBel et al. (2018), the current chapter and the next chapter represent a “far” type of conceptual replication since they test similar hypotheses and use similar independent and dependent variables as Tavakoli and Uchiyara (2020), but with modifications to their experimental conditions. The current chapter employs a broader range of methods and a larger sample size to explore how LB usage, differentiated by length, varies across different proficiency levels (IELTS 6.5–7.5). Sections 3.2 to 3.5 below detail how the current chapter's methods replicate, extend, and deviate from Tavakoli and Uchiyara (2020). For a comprehensive side-by-side comparison of the methods used in Tavakoli and Uchiyara (2020) versus my Chapters 3 and 4, readers are directed to Appendix A. By employing rigorous statistical techniques such as robust ANOVA with bootstrapping and means trimming, the current chapter seeks to offer fresh insights into the generalizability and reliability of the original findings, advancing our understanding of the relationship between LBs and oral fluency.

The research question guiding the current chapter is: How does LB usage, differentiated by length, vary across different proficiency levels? Based on the results observed in Tavakoli and Uchiyara (2020), I hypothesize that usage of both longer and shorter LBs increases at higher proficiency levels.

3.1 Participants

The participants were 150 second language learners taking a pre-session course at a UK university (Queen Mary University of London). They provided access to their oral presentations via the written consent form (see Appendix B), and the Research Ethics Committee of Queen Mary University approved this research (approval number: QMREC2414a). The participants

were from 20 different L1 backgrounds (see Table 3.1), with most of the participants being L1 Chinese ($n = 101$), L1 Arabic ($n = 9$), or L1 Turkish ($n = 9$). Of the participants, 60 (40%) were male, and 90 (60%) were female. Participants' raw presentation scores (outlined in the next section) were first converted into IELTS bands ranging from 6.5 to 7.5. Next, they were categorized into three groups according to these bands. From a larger pool, 50 individuals were randomly selected for each level, ensuring balanced representation across the three bands. There were 50 participants at IELTS level 6.5, 50 at IELTS level 7, and 50 at IELTS level 7.5 (see Table 2).

Table 3.1

Participants by L1 Backgrounds (N = 150)

Nationality	N	Nationality	N	Nationality	N
Chinese	101	Indonesian	2	Taiwanese	1
Saudi Arabian	9	Bangladeshi	1	Chilean	1
Turkish	9	German	1	Egyptian	1
Japanese	7	Bahraini	1	Georgian	1
Thai	6	Kuwaiti	1	Pakistani	1
Indian	2	Ghanian	1	Russian	1
Colombian	2	Italian	1		

Table 3.2

Participants' IELTS-Rated Speaking Proficiency Levels

IELTS speaking level	6.5	7	7.5	Total
<i>N</i>	50	50	50	150

3.2 Oral Presentation Tasks and Proficiency Scores

All participants completed oral presentations comprising a 7-min presentation (monologue) in small groups with the aid of PowerPoint slides as part of the pre-session course. It is important to note that these presentations were not conducted specifically for the purposes of the experiments in the current thesis but were completed as a requirement of the pre-session course at Queen Mary University of London. Their outputs were video recorded online and assessed by teachers using grading descriptors (in equal measure): presentation content, presentation structure, seminar leadership, language fluency, language accuracy (see the full descriptors in Appendix C). The oral presentation tasks covered a wide range of sub-topics under the overarching theme of globalization, which had been the focal point of the participants' 5-week academic English course. Although globalization served as the central theme, participants were encouraged to explore this broad topic through various lenses, ranging from economic and legal perspectives (e.g., how globalization affects development and international trade law) to more specialized areas (e.g., acoustic telemetry in fisheries). This diverse range of sub-topics allowed participants to delve into areas aligned with their academic interests and expertise. The students represented a diverse range of disciplines, including humanities, law, science, technology, engineering, and mathematics. From the video recordings, 3-min speech samples were taken starting at the 30-sec timestamp. As the 3-min sections of speech analyzed came from the "presentation" part of their seminar, I am specifically analyzing monologic oral presentations rather than freer dialogic speech. These samples were transcribed using Sonix.ai software, and the transcriptions were checked for accuracy by a research assistant and double-checked by the author. I used the transcripts in the lexical analyses (described in detail below), and I used the IELTS-banded scores as the measure of oral proficiency in the current chapter's analysis.

3.3 Measuring LBs: A Two-Pronged Approach

The methods described in Sections 3.3 and 3.4 are applied consistently in Chapters 3-5, using both text-internal and text-external techniques to analyze the contributions of shorter and longer LBs. These methods are not restated in Chapters 4 and 5 to avoid repetition.

3.3.1 Text-External Lexical Bundle Analysis and Measures

Following previous studies (Garner & Crossley, 2018; Tavakoli & Uchihara, 2020), I used three n-gram indices (proportion, frequency, and association) to objectively measure the use of shorter LBs, specifically two- and three-word contiguous sequences (i.e., bi- and trigram tokens) in my learner corpus. TAALES version 2.0 was used to calculate three kinds of n-gram scores, producing six score indices (two proportion, two frequency, and two association indices). The metrics I chose differed from Tavakoli and Uchihara's in only one respect: In contrast to Tavakoli and Uchihara (2020), I opted not to use *t*-scores for measuring associations, as recent findings suggest they do not provide a reliable measure of association (Gries, 2022). For my external reference corpus, I chose the spoken subsection of the Corpus of Contemporary American English (COCA, Davies, 2009), which comprises 79 million words from transcriptions of a wide range of TV and radio programs. My choice of this spoken corpus was in alignment with research findings showing a gap in L2 learners' spoken and written vocabulary sizes (Uchihara & Harada, 2018) and differences in lexical profiles between spoken and written modes (Dang et al., 2017). I maintained the consistency between the modality in which L2 words were elicited and the modality of the reference corpus based on practice used in previous studies (e.g., Uchihara & Clenton, 2020; Uchihara et al., 2022).

In the current chapter, proportion score indices measure the proportion of bi- and trigrams in my learner speech sample data, which are also found among the 30,000 most frequent bi- and trigrams in the external reference corpus (COCA). Higher proportion scores show that participants in the sample produced a higher percentage of high-frequency, target-like bi- and trigrams. Frequency score indices measure the number of high-frequency, target-like n-grams produced by my participants. Logarithmic bi- and trigram scores, instead of raw frequency scores, were used to control for Zipfian effects common in word frequency lists (Kyle & Crossley, 2015; Tavakoli & Uchihara, 2020). Higher frequency scores show that participants in my sample produced a larger number of high-frequency target-like bi- and trigrams. Association score indices measure the association strength between individual words within bigrams and trigrams. Of the five association measures available in TAALES, I used the mutual information (MI) score. MI score measures the strength of association between two words. While higher MI scores indicate words are more strongly associated, MI highlights word pairs which are relatively infrequent (Schmitt, 2010, p. 130). Before n-gram analysis using TAALES, all the transcripts were cleaned by correcting any misspellings and mispronunciations and removing any markings of filled pausing (i.e., *ums*, *uhs*, etc.). The resulting transcripts ranged between 216 and 436 words ($M = 310.78$, $SD = 47.32$).

3.3.2 Text-Internal Lexical Bundle Identification and Refinement Procedures

I adopted a text-internal approach to isolate and measure the unique contribution of longer LBs to fluency aspects. As a first step, I conducted frequency analyses using AntConc (Anthony, 2022) to generate lists of the most frequently used four-word LBs in the learner corpus. The frequency and dispersion thresholds used to identify lexical bundles vary from study to study, but “the actual frequency cut offs are somewhat arbitrary” (Hyland, 2008, p. 8)

depending on both the size and specificity of the corpus. For relatively small spoken corpora like the one in this thesis, a raw cut-off frequency is often used, ranging from two to ten occurrences (e.g., Altenberg, 1998; Biber & Barbieri, 2007; De Cock, 1998). Given the small size of the spoken corpus in the current thesis (46617 words), for four-word combinations to qualify as lexical bundles, I used a cut-off point of three or more occurrences in at least three texts, following Biber & Barbieri (2007). These minimum figures help to ensure that the identified bundles are not idiosyncrasies confined to occurrences produced by an individual speaker. This resulted in 447 instances of four-word LBs that met these criteria. To deal with the issues of overlap and structural incompleteness among these instances, I used a refinement procedure developed by previous researchers (Wood & Appel, 2014) aiming to identify more structurally complete and useful LBs. I split each four-word sequence (e.g., *I will give you*) into two constituent three-word clusters (e.g., *I will give*, *will give you*). The frequency of the two three-word clusters in the corpus were identified and compared. If the frequency of one three-word cluster was at least double the frequency of the other, the more frequent cluster was classified as the root structure and the fourth word was considered as a word that commonly occurred with that structure and was put in parentheses. For example, *I will give* occurred over two times (freq = 41) more than *will give you* (freq = 20) in the current data set. Therefore, the final resultant structure is in this case: *I will give (you)*. Another example from the current data set is the sequence *the first part is*. Since *the first part* (freq = 41) occurred over two times more frequently than *first part is* (freq = 17), the final resultant structure is *the first part (is)*. This refinement procedure resulted in a list of 119 multiword structures, comprised mainly of core three-word bundles and four-word structures with a minority comprising longer five-word structures. Table 3.3 shows the number of three-word (55), four-word (58) and five-word structures (6) identified.

As can be seen in the full list of 119 multiword structures in [Appendix D](#), many of the resultant three-word structures are semantically or structurally complete units (e.g., *(with) the development of*), which suggests the refinement procedure was effective in identifying more core structures.

Table 3.3

Number and Examples of Structures Identified at Different Lengths

Lexical Bundle	Three-Word Units	Four-Word Units	Five-Word Units	Total
Number of Units	55	58	6	119
Examples	<i>I'm going to the first part I will give</i>	<i>I would like to at the end of at the same time</i>	<i>if you have any questions the presentation will be brief in a socially responsible way</i>	

Note. Contracted forms (e.g., *I'm*) were counted as a single word; for example, *I'm going to* was considered to be a trigram rather than a quadgram.

Note that because this refinement procedure entails going beyond the traditional LB approach, it is important to describe the newly identified structures in a way that distinguishes them from traditional LBs (and other multiword structures). Recall (from [Section 2.2](#)) that the traditional LB approach typically identifies multiword sequences within two parameters: frequency and range. For this reason, when making modifications or refinements to traditional LB methods, previous researchers (e.g., Simpson-Vlach & Ellis, 2010; Wood & Appel, 2014) have used different terminology to refer to such refined or modified LBs. Simpson-Vlach and Ellis (2010), for instance, started with an LB approach and then applied additional criteria (e.g., human ratings of formula teaching worth combined with MI as a measure of collocation strength) aiming to identify more useful multiword units for teaching purposes. Simpson-Vlach and Ellis adopted the general term “formulaic language” to describe the word combinations identified in their study. Another example is Wood and Appel (2014), who developed the LB

refinement procedure used in the experiments in the current thesis. Wood and Appel adopted the general term “multiword constructions” to refer to the refined LBs identified in their study. “Formulaic language” and “multiword constructions” are also used as umbrella terms in other studies, such as Liu (2012). Because of the fuzzy nature of boundaries between various types of semi-fixed multiword combinations, and because each study’s identification and refinement methods are different, there is no consensus in the literature as to which terms apply in all cases. Although it is challenging to pin down a consistent definition of MWSs across all studies, most researchers agree it is important to distinguish between different MWSs where possible and to report clearly how MWSs are identified in each study to facilitate comparisons of research findings across studies. In the current thesis, to avoid terminological confusion, I use the term “refined LBs” to refer to the refined LB list (see [Appendix D](#)) produced by the above-described refinement procedure.

3.4 Scoring the Use of Text-Internal LBs

To perform quantitative analyses and compare the relationships between different types of LBs (unrefined text-external vs. refined text-internal), it is essential to quantify the usage of the list of refined LBs. Following the methodology established by Hougham et al. (2024a), I assigned one point for each identified refined LB used by each participant. I tallied up the total number of points for each participant, arriving at a three-five-word usage score for each participant. As for text-internal MI scores, I extracted MI scores for sequences of various lengths using the Collocate 2.0 software program (Barlow, 2015). MI scores for sequences ranging from three to five words have been used in several studies (e.g., Ellis et al., 2008; Hougham et al., 2024a; Simpson-Vlach & Ellis, 2010) because they provide a reliable measure of phrasal

coherence. For each refined bundle used by each participant, I awarded the corresponding MI score. I then tallied all MI scores and assigned each participant a total MI score.

3.5 Statistical Analyses

I analyzed four fluency measures, six text-*external* n-gram (two- and three-word) measures, and two text-*internal* refined n-gram (three-to-five-word) measures. To address the research question regarding how LB usage varies across different proficiency levels, a one-way ANOVA was selected as the most suitable statistical test because the ANOVA is used when we want to test whether the scores or levels of three or more groups differ statistically (Larson-Hall, 2015). Prior to running the ANOVA, I checked the assumptions for a parametric one-way ANOVA and found that two of the fluency measures in this thesis (see Section 4.2 in Chapter 4 for fluency measures) violated the assumption of homogeneity of variances (Levene's test) across the three proficiency groups, and the assumption of normality was violated as outliers were present among all measures (for detailed results, see Appendix E). I therefore conducted a series of one-way ANOVAs with 20% trimmed means and bootstrapping in line with Larson-Hall (2015). Using robust methods, such as trimmed means and bootstrapping, is an objective and principled way to deal with outliers and will perform better than if data had been simply cleaned up (Larson-Hall, 2015). As its name suggests, means-trimming involves removing a specific percentage (e.g., 5%, 20%) from the tail ends of the distribution. By doing so, the mean scores being compared in a statistical test such as ANOVA are less sensitive to outliers (Loewen & Plonsky, 2016).² Bootstrapping is a resampling procedure that randomly resamples from a data set to create a result that is more stable, aiming to emulate the results that could be obtained from a much larger sample (Efron, 1979; Larson-Hall & Herrington, 2010).

In R, I used the “t1waybt” and “mcppb20” functions in the “WRS2” package, which is developed specifically for conducting robust statistical analyses (Mair & Wilcox, 2020). The “t1waybt” function uses a two-step method based on a combination of 20% trimmed means and bootstrapping developed by Wilcox (2017a) to control the Type I error rate when dealing with skewed distributions or outliers. I implemented 10,000 bootstrap samples in the “t1waybt” function. The “mcppb20” function performs corresponding pairwise bootstrap post-hoc comparisons (Dudek, 2023; Mair & Wilcox, 2020). In what follows, I present the descriptive statistics first, then I report the robust ANOVA and post-hoc results to address the research question.

3.6 Results

Table 3.4, which is modelled after Table 2 in Tavakoli and Uchiyama (2020, pp. 16-17), shows the means and standard deviations for the different LB measures across the proficiency levels (IELTS 6.5-7.5). As for bigrams and trigrams, the numbers suggest that frequency, proportion, and the association of MI increase as proficiency increases (except for trigram MI, which decreased a little). As for longer (text-internal three-five-word usage and MI) LBs, the figures show an overall increase in their usage as proficiency develops. I will provide a more in-depth analysis of these observations in the Discussion (Section 3.7).

Table 3.5 shows the one-way ANOVA and post hoc test results. The ANOVAs showed that there were significant differences for four MWS measures across the proficiency levels: bigram MI, $F = 7.466$, $p = .001$; trigram proportion, $F = 3.428$, $p = .042$; three-to-five-word usage, $F = 5.906$, $p = .004$; three-to-five-word MI, $F = 5.842$, $p = .005$. There were no significant differences between the other four n-gram measures. Post hoc tests using a two-step method

Table 3.4*Descriptive Statistics for All N-Grams at Different Levels of Proficiency (N = 150)*

Measure	Level ^a	<i>M</i>	<i>SD</i>	Median	Minimum	Maximum
Bigram log frequency	6.5	1.31	0.12	1.30	1.04	1.73
	7	1.31	0.10	1.31	1.07	1.53
	7.5	1.34	0.12	1.33	1.04	1.57
Bigram proportion	6.5	0.44	0.06	0.45	0.28	0.60
	7	0.42	0.07	0.41	0.26	0.56
	7.5	0.45	0.07	0.44	0.29	0.63
Bigram association MI	6.5	1.49	0.14	1.48	1.13	1.84
	7	1.55	0.17	1.56	1.23	1.97
	7.5	1.60	0.18	1.59	1.26	2.08
Trigram log frequency	6.5	0.68	0.12	0.67	0.41	0.95
	7	0.66	0.12	0.67	0.44	0.94
	7.5	0.69	0.14	0.67	0.46	1.07
Trigram proportion	6.5	0.12	0.03	0.12	0.06	0.26
	7	0.11	0.04	0.11	0.04	0.22
	7.5	0.13	0.04	0.13	0.07	0.26
Trigram association MI	6.5	2.54	0.27	2.56	1.91	3.04
	7	2.52	0.23	2.50	1.94	2.95
	7.5	2.53	0.27	2.51	2.03	3.20
Three-to-five-word usage	6.5	9.20	4.16	8.00	3.00	20.00
	7	9.50	5.05	9.00	1.00	22.00
	7.5	12.18	5.56	12.00	2.00	27.00
Three-to-five-word MI	6.5	120.50	67.38	107.61	37.30	311.52
	7	119.71	66.32	98.92	19.87	293.22
	7.5	156.96	81.22	147.29	24.80	399.56

^aProficiency level of oral presentations assessed according to IELTS descriptors. *SD* = standard deviation. MI = mutual information.

based on 20% trimmed means and bootstrapping with Bonferroni correction (.05/8 = 0.006 for eight comparisons per measure) indicated that bigram MI scores were higher among 7.5-level speakers compared with 6.5-level speakers at the $p < .001$ level. The results also showed that longer three-five-word usage and MI scores were higher at level 7.5 compared with level 6.5 ($p < .001$). Several other differences between levels were found (bigram MI scores were higher at level 7 than level 6.5; three-five-word usage and MI scores were higher at level 7.5 than level 7),

Table 3.5*One-Way ANOVA and Post-Hoc Results for N-Gram Measures Across Different Speaking Levels*

Measure	<i>F</i>	<i>P</i>	6.5-7	6.5-7.5	7-7.5
Bigram frequency	1.109	0.337	NS	NS	NS
Bigram proportion	2.007	0.150	NS	NS	NS
Bigram MI	7.466	0.001	*	**	NS
Trigram frequency	0.224	0.800	NS	NS	NS
Trigram proportion	3.428	0.042	NS	NS	*
Trigram MI	0.566	0.569	NS	NS	NS
Three-five-word usage	5.906	0.004	NS	**	*
Three-five-word MI	5.842	0.005	NS	**	*

Note. MI = mutual information. NS = not significant. * $p < .05$, ** $p < .01$

but these differences did not reach the corrected alpha level (see [Appendix F](#) for post hoc effect sizes, confidence intervals, and p -values). Figures 3.1-3.8 show the observed relationships between n-gram measures and speaking proficiency levels.

To further explore potential interaction effects between proficiency level and LB length, I conducted a series of complementary two-way ANOVAs ([Appendix G](#)). Results of the two-way ANOVAs confirmed a significant interaction, indicating that longer bundles (three-to-five-word) show particularly strong increases at higher proficiency levels, consistent with the one-way ANOVA findings. Full tables and post hoc comparisons are provided in [Appendix G](#).

3.7 Discussion

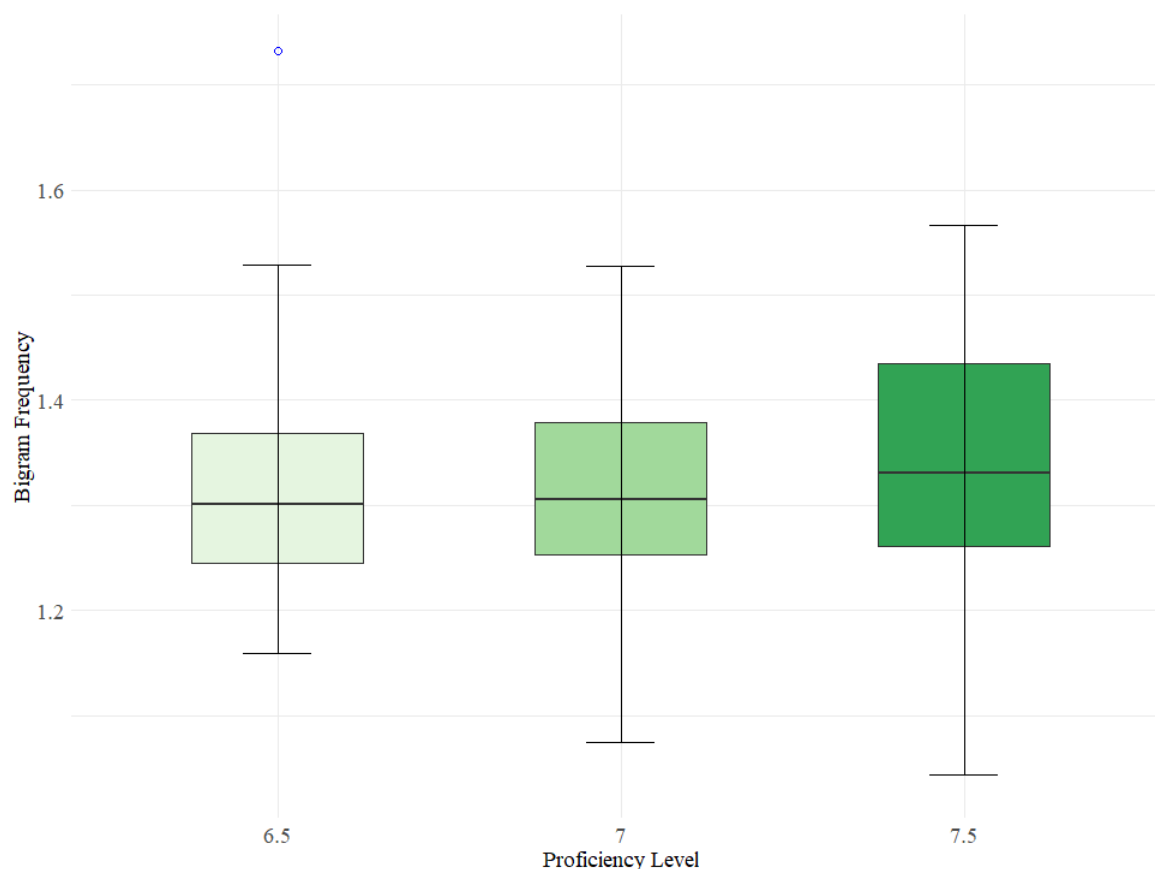
This chapter examined how LB usage, differentiated by length, varies across different proficiency levels. To address this question, it analyzed descriptive statistics and conducted a series of robust ANOVAs. The findings reveal both consistencies and inconsistencies with Tavakoli and Uchihara (2020). Consistent with their study, I observed an increase in LB usage, particularly in bigram and trigram frequency, as proficiency levels increased. This trend highlights the connection between proficiency development and the use of shorter LBs. However, my analysis diverged in terms of MI statistics.

Contrary to Tavakoli and Uchihara, who reported a decrease in MI with higher proficiency levels, my findings show an increase in bigram MI as proficiency improves, suggesting a more sophisticated use of LBs by advanced learners. This inconsistency could be attributed to differences in datasets, methodologies, or proficiency categorizations, particularly as my participants were at higher proficiency levels (IELTS 6.5 to 7.5) compared to Tavakoli and Uchihara's participants (CEFR low B1 to C1). To provide a clearer understanding of the proficiency levels of my participants in comparison to previous studies, Figure 3.9 illustrates the relationship between IELTS bands and CEFR levels, showing how my participants (IELTS 6.5-7.5) fall into the higher end of the CEFR framework. This visualization helps contextualize the differences in LB usage patterns observed in the current chapter compared to Tavakoli and Uchihara (2020). A more detailed discussion on the above-mentioned divergence is provided in Chapter 5 ([Section 5.4](#)), where methodological and sample size differences between studies are further explored.

Regarding the post hoc analyses, Tavakoli and Uchihara's study revealed patterns at the $p < .008$ level that suggest a linear relationship between proficiency level and most n-gram

Figure 3.1

Bigram Frequency Plotted Against IELTS-Rated Speaking Proficiency Level

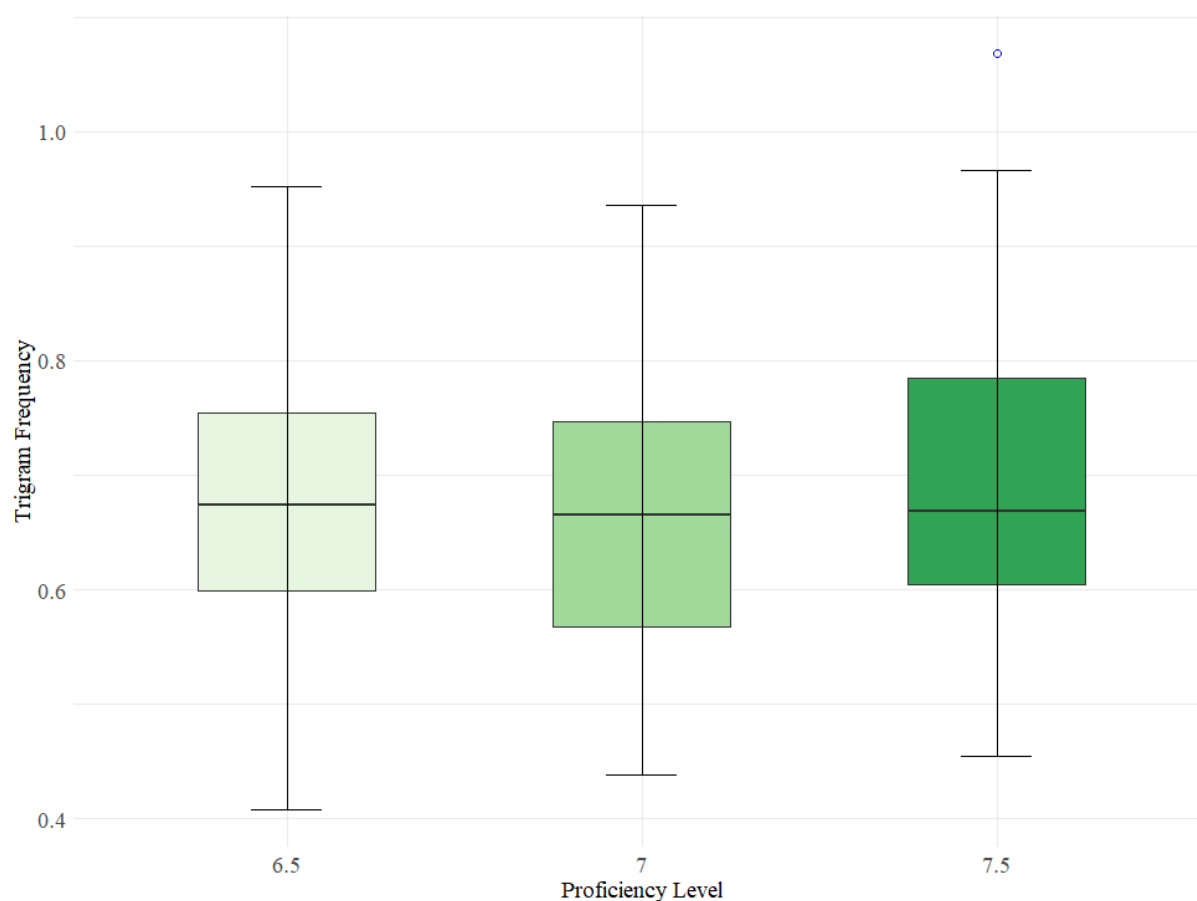


measures. Similarly, my study's post hoc tests, with a more stringent Bonferroni correction ($p < .006$ level), indicated higher bigram MI, higher three-to-five-word usage, and higher three-to-five-word MI at higher proficiency levels (IELTS 7.5 compared to 6.5), confirming a proficiency-related progression.

My study's findings also substantially align with my hypothesis that the usage of both shorter and longer LBs increases at higher proficiency levels. The pattern of increased LB usage is clearly evident in the data presented in Table 3.2 (as well as in Figures 3.4, 3.5, 3.7, and 3.8),

Figure 3.2

Trigram Frequency Plotted Against IELTS-Rated Speaking Proficiency Level

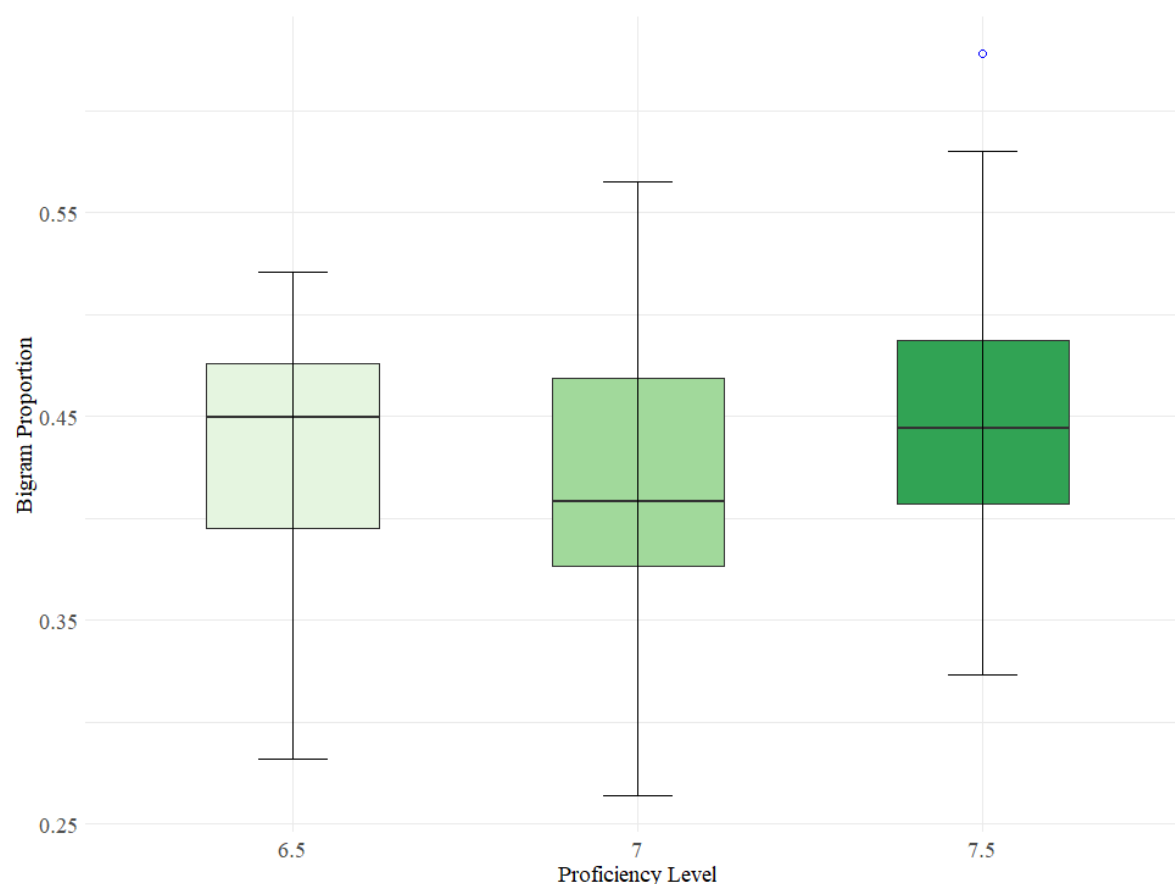


where an increase in the use of bigrams, trigrams, and especially longer three-to-five-word LBs is observed as proficiency advances from IELTS level 6.5 to 7.5. Specifically, the increased mean scores for bigram MI (from 1.49 at level 6.5 to 1.60 at level 7.5), trigram proportion (from 0.12 to 0.13), and notably, the more significant jump in three-to-five-word usage (from 9.20 to 12.18) and three-to-five-word MI (from 120.50 to 156.96), strongly support my hypothesis.

These patterns of increased LB usage are further substantiated by the significant differences identified in the ANOVA findings (Table 3.5), particularly for bigram MI,

Figure 3.3

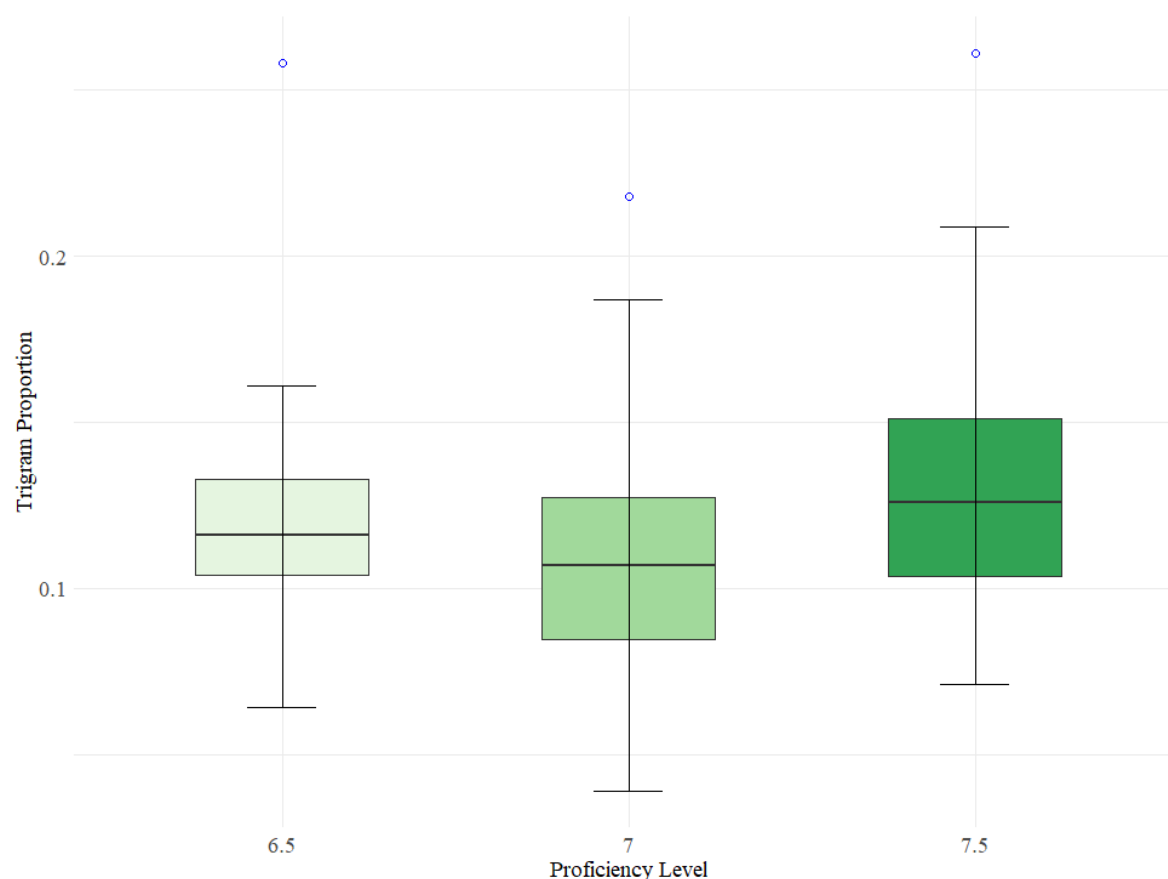
Bigram Proportion Plotted Against IELTS-Rated Speaking Proficiency Level



trigram proportion, and both usage and MI of longer LBs. The increased usage of both shorter and longer LBs not only exemplifies Kormos's (2006) extension of Levelt's model, which posits a more efficient lexical selection process at higher proficiency levels, but also shows an emerging mastery of complex linguistic patterns, in alignment with usage-based theory (Bybee, 2006; Ellis, 2002). The significant increases in longer LBs offer unique advantages in enhancing fluency by boosting articulation rates and reducing mid-clause pauses (as shown in Chapter Four). This dual impact strongly supports my hypothesis of a positive association between

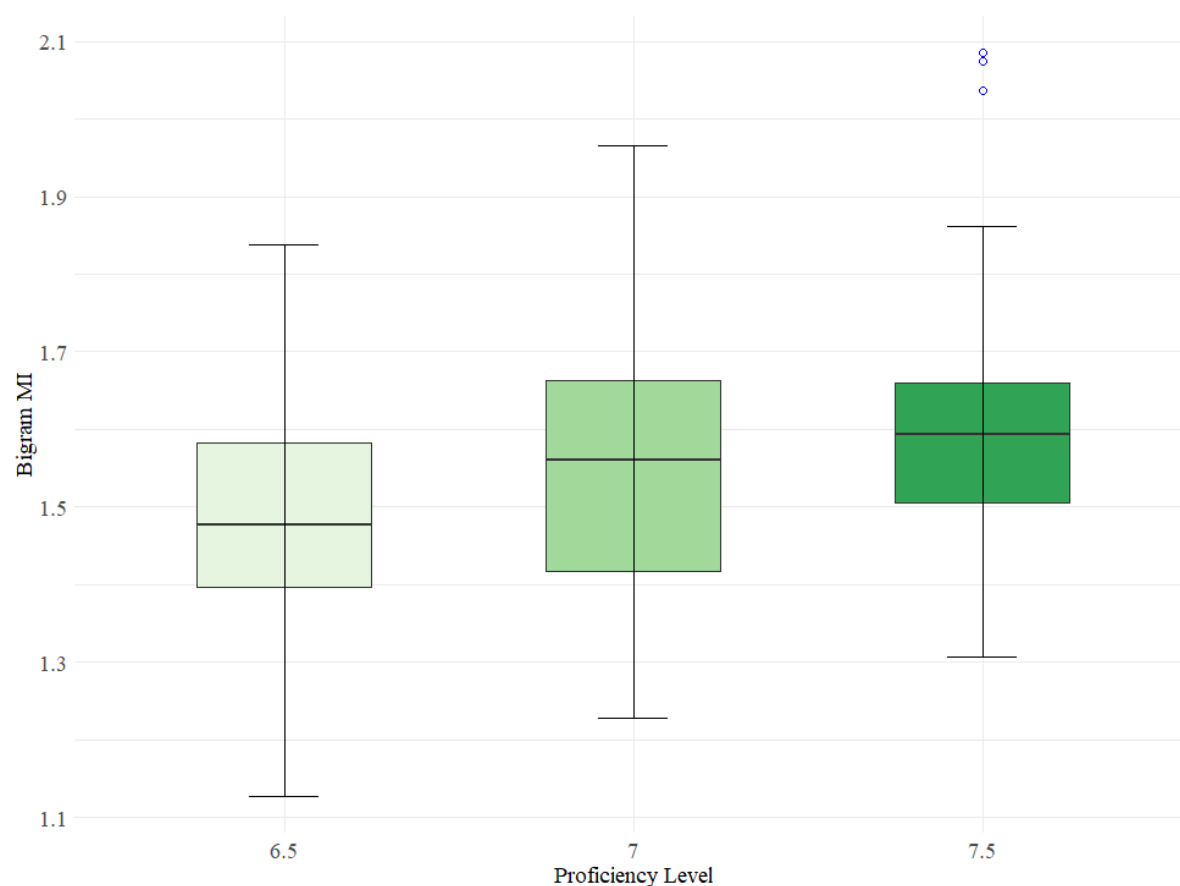
Figure 3.4

Trigram Proportion Plotted Against IELTS-Rated Speaking Proficiency Level



proficiency levels and the use of LBs of various lengths. By integrating these empirical findings with theoretical perspectives, it becomes clear that advanced learners use LBs of various types and lengths with greater ease and frequency, facilitating more fluent language use.

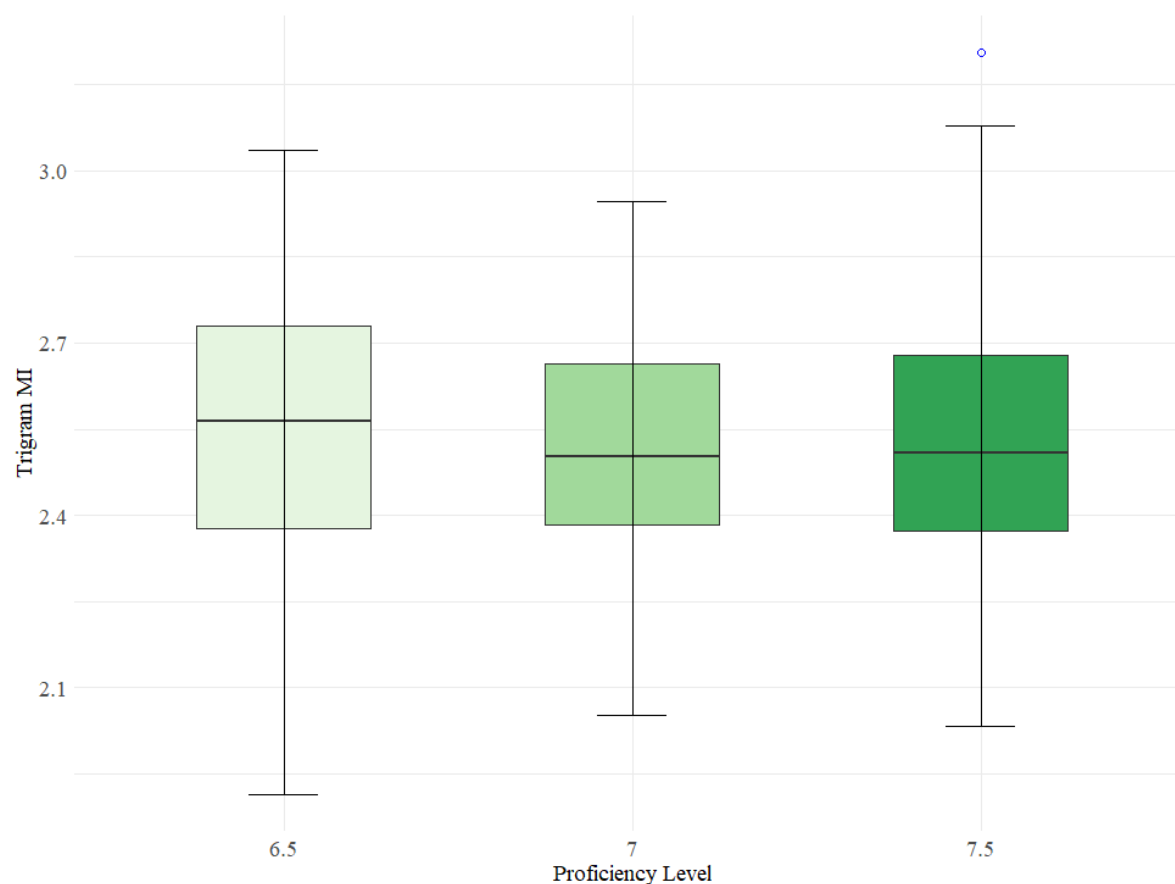
Earlier studies (Durrant & Schmitt, 2009; Ellis et al., 2015; Schmitt, 2012) have consistently shown that L1 users produce collocations with higher MI scores compared with L2 users. My research builds on these findings by showing how proficiency impacts the use of sophisticated collocational patterns. Specifically, I found that more proficient L2 users employ

Figure 3.5*Bigram MI Plotted Against IELTS-Rated Speaking Proficiency Level*

higher MI-score collocations, therefore narrowing the gap with L1 speakers' usage. This suggests that cognitive efficiency and strategic linguistic deployment play crucial roles in advancing language proficiency. By exploring the use of high MI-score collocations among more proficient L2 users, the current chapter contributes to a deeper understanding of LB usage across proficiency levels, suggesting these nuanced patterns of language use warrant further investigation. Importantly, the patterns of increased LB usage with proficiency not only support my hypothesis but also contribute to discussions on the linguistic sophistication of higher-level

Figure 3.6

Trigram MI Plotted Against IELTS-Rated Speaking Proficiency Level

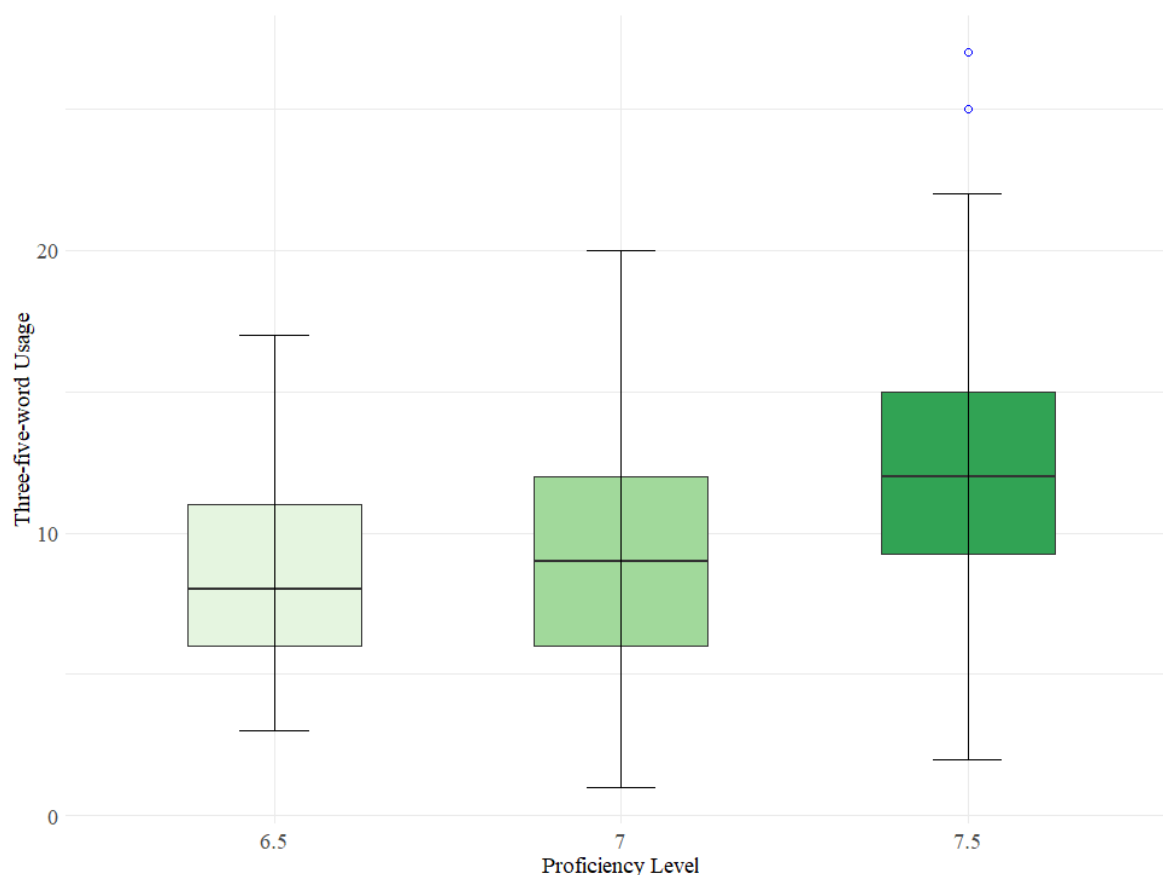


language users. The increase in the MI of bigrams and the expanded use of longer LBs suggest a more complex and nuanced use of language as learners progress. These results therefore not only align with my hypothesis but also enhance our understanding of language development, extending beyond the typical focus on bi- and trigram units.

In summary, the findings in this chapter substantially align with my hypothesis that higher proficiency levels are associated with greater and more sophisticated LB usage. The analysis presented here enhances the existing literature by providing a richer and more

Figure 3.7

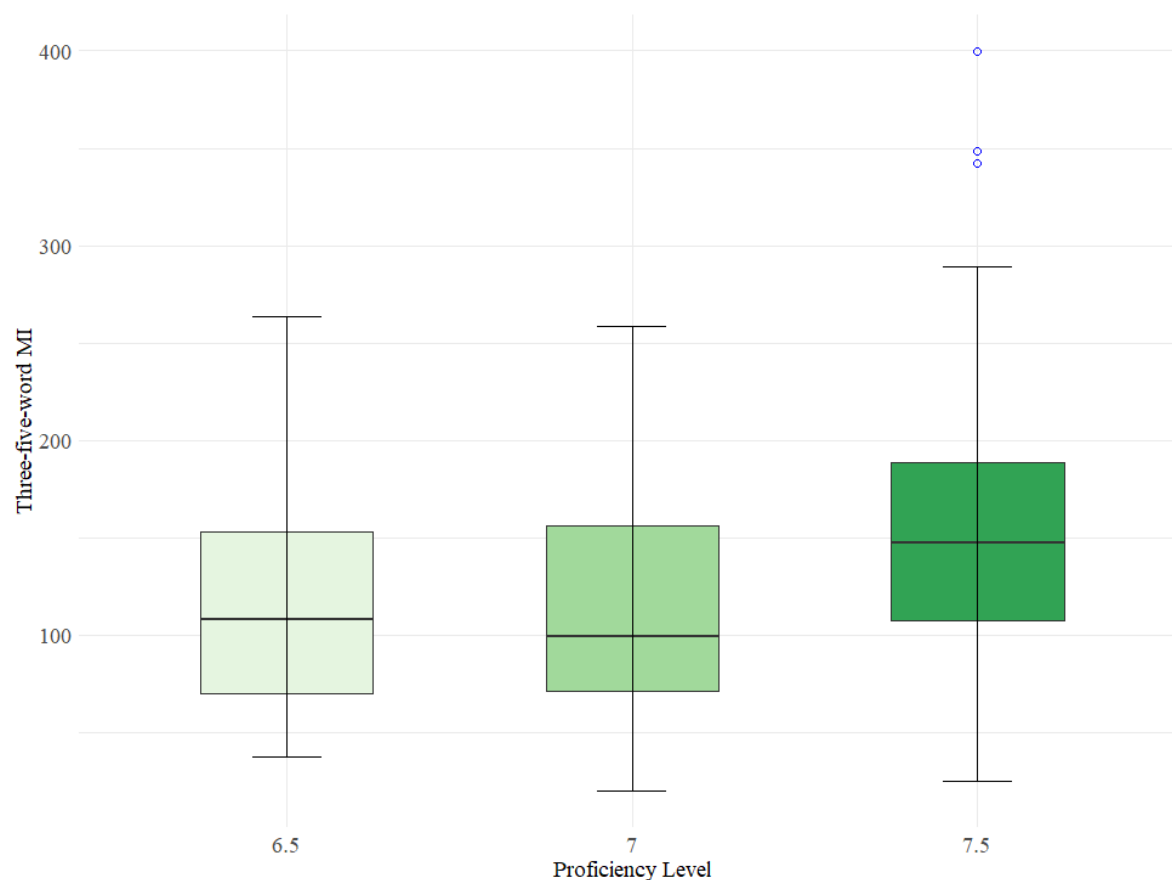
Three-to-Five-Word Usage Plotted Against IELTS-Rated Speaking Proficiency Level



differentiated picture of how LB usage evolves at higher levels of language learning. It reaffirms the importance of examining a wide range of LBs and highlights the need for innovative methodological approaches to capture the complexities of language proficiency. By investigating both shorter and longer LBs across higher proficiency levels, this chapter's analysis enhances our understanding of lexical sophistication in advanced language use and provides potentially valuable insights for language assessment and teaching.

Figure 3.8

Three-to-Five-Word MI Plotted Against IELTS-Rated Speaking Proficiency Level

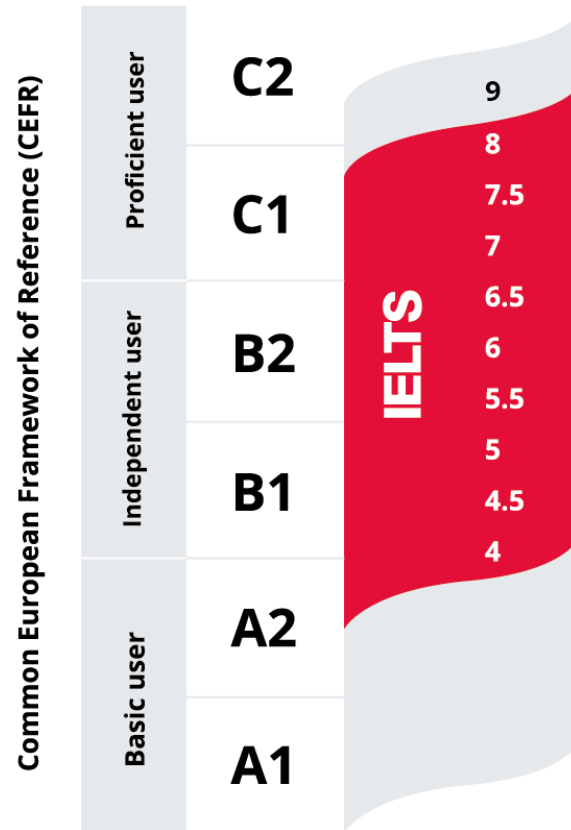


3.8 Conclusion

In conclusion, the study presented in this chapter has provided detailed insights into the patterns of refined LB usage that correspond to various levels of language proficiency. By extending beyond the scope of Tavakoli and Uchiyara (2020), it has illuminated the intricate relationship between proficiency and both the frequency and complexity of LBs. Through rigorous statistical analysis, it has substantiated the progression of linguistic sophistication with higher proficiency levels, highlighting the pivotal role of both shorter and longer LBs in language acquisition and competency. The findings set the stage for the subsequent chapters,

Figure 3.9

Alignment of IELTS Bands With the CEFR Scale (Source: IELTS Official Website)



<https://ielts.org/organisations/ielts-for-organisations/compare-ielts/ielts-and-the-cefr>

which transition from descriptive to predictive analysis. In Chapter 4, the focus will be on LB usage as a predictor of aspects of oral fluency, and in Chapter 5, the focus will shift to the potential of LB usage as a predictor of assessment scores. These inquiries aim to yield insights of practical significance for language assessment and pedagogical practices. Building on the foundation laid in this chapter, the forthcoming analyses aim to bridge the gap between empirical observation and the forecasting of language fluency and proficiency, enriching our understanding of the factors that contribute to successful language performance in high-stakes academic settings.

Chapter 4: Assessing the Impact of Varying Lengths of LBs on Oral Fluency

In Chapter 3, I explored how LB usage varies across different proficiency levels, specifically within the IELTS bands of 6.5 to 7.5. Chapter 3 addressed one research question: How does usage of LBs of different lengths vary across different proficiency levels? Using robust one-way ANOVAs, I observed clear patterns in the use of LBs across proficiency levels. Key findings from Chapter 3 highlighted that as proficiency increased, so did the frequency and range of LB usage, particularly for longer, three-to-five-word bundles. The results reported in Chapter 3 contribute to a nuanced understanding of the linguistic characteristics that distinguish between different proficiency levels.

Chapter 4 builds on the intricate exploration of LBs across proficiency levels in high-stakes assessments. It aims to dissect the relationship between LBs of varying lengths and various aspects of oral fluency, expanding on Tavakoli and Uchiyama's (2020) foundational work. Examining both longer (e.g., *as you can see*) and shorter (e.g., *there are*) LBs is crucial for a comprehensive understanding of fluency, as longer LBs potentially offer processing efficiencies vital for fluent speech production. Responding to calls (e.g., Hougham et al., 2024a; Tavakoli & Uchiyama, 2020) for a detailed examination of the influence of LB length on aspects of oral fluency, the current chapter continues the conceptual replication of Tavakoli and Uchiyama (2020) and extends their inquiry by comparing longer versus shorter LBs in relation to L2 oral fluency across proficiency levels. Specifically, it assesses the relative contributions of LBs of different lengths to various aspects of oral fluency. Maintaining the innovative methodological approach introduced in Chapter 3 (drawing on the work of Hougham et al., 2024a), this chapter uses both text-internal and text-external techniques to measure LBs. Building on Chapter 3's methods, I employ robust regression, dominance analysis and random

forests analysis to gauge the unique contributions of shorter versus longer LBs. This mix of statistical methods provides a detailed and triangulated analysis of how LBs influence various aspects of oral fluency. This approach explores the nuanced effects of LB length on fluency, broadening the scope beyond the typical focus on shorter LBs and contributing to discussions on the complex dynamics of phraseological knowledge and L2 fluency.

Adhering to the framework of conceptual replication as outlined by LeBel et al. (2018), this chapter continues the “far” type of conceptual replication initiated in the previous chapter. While it tests similar hypotheses and uses similar independent and dependent variables to those of Tavakoli and Uchihara (2020), it incorporates modifications to their experimental conditions. The current chapter employs a broader range of methods and a larger sample size to assess the impact of varying lengths of LBs on different aspects of oral fluency (speed, breakdown, repair) across proficiency levels (IELTS 6.5-7.5). Sections 4.1 to 4.3 below detail how this chapter's methods replicate, extend, and deviate from Tavakoli and Uchihara (2020). For a comprehensive side-by-side comparison of the methods used in Tavakoli and Uchihara (2020) versus my Chapters 3 and 4, readers are directed to [Appendix A](#). By employing robust statistical techniques such as regression, dominance analysis, and random forests analysis, the current chapter seeks to offer fresh insights into the generalizability and reliability of the original findings, advancing our understanding of the relationship between LBs and various aspects of oral fluency.

The research question guiding the current chapter focuses on the association between LBs of varying lengths and the three dimensions of utterance fluency: Speed, breakdown, and repair (for explanation of these dimensions, see Chapter 2, Section 2.1). The research question is: To what extent is the use of longer vs. shorter LBs associated with the three aspects of oral fluency (speed, breakdown, repair)? Based on theoretical frameworks of speech production (Kormos,

2006; Levelt, 1989), I hypothesize that both longer and shorter LBs are positively associated with speed fluency, with longer LBs potentially reducing hesitations and the need for repairs.

4.1 Participants, Presentation Tasks, LB Measurements and Scoring

The current chapter used the same participants as in Chapter 3 (i.e., 150 L2 learners taking a pre-session course at Queen Mary University of London; see Section 3.1 for a detailed description of their L1 backgrounds and speaking proficiency level groupings; research ethics approval number: QMREC2414a). The same oral presentation tasks, LB measurements and scoring are also used (see Sections 3.1 to 3.4 for detailed descriptions). Additionally, the banded scores are used as the measure of proficiency levels, in order to facilitate comparisons with previous research (Tavakoli & Uchihara, 2020) that used banded scores.

4.2 Measuring Fluency

Following de Jong and Bosker (2013), and Tavakoli and Uchihara (2020), I define a silent pause as an occurrence of silence lasting 250 milliseconds or more. I used the same fluency measures as Tavakoli and Uchihara: (a) articulation rate to assess speed fluency, (b) the frequency of total repairs (reformulations, self-corrections, repetitions and hesitations) per minute of speaking time (excluding silent pauses) to measure repair fluency, and (c) frequency of pauses within clauses and at the end of clauses per minute of speaking time (silent pauses excluded) to measure breakdown fluency. Using frequency measures corrected for speaking time, instead of total phonation time, proves advantageous because it removes the effect of silent pausing time and improves accuracy (de Jong, 2016b). As for the pause analysis procedure, I followed the methodology precisely as reported in Hougham et al. (2024a). I employed an expert who has experience of using PRAAT scripts (Boersma & Weenink, 2016). First, the

“mark_pauses.praat” script (from the Speech Corpus Toolkit for Praat, Lennes, 2021) was used to compute the silent pause measures automatically. The automatic pause measurements were then checked and edited manually while listening to the audio recordings to ensure a high degree of precision using spectrograms created in PRAAT. The script “calculate_segment_durations.praat” (Lennes, 2021) was then used to calculate the pause segment durations automatically. To investigate pause location in terms of occurrences in mid- and end-clause location, I transcribed the audio recordings using AI-powered transcription software (www.sonix.ai).³ A research assistant then checked the transcriptions, and I double-checked them for accuracy. I then divided the unpruned transcriptions into analysis of speech units (ASUs, Foster et al., 2000).⁴ The pause location analysis then entailed several iterative steps, including listening to the audio recordings, examining the spectrograms created in PRAAT, inspecting the transcriptions, and marking pauses in either mid- or end-ASU position on the transcripts, closely mirroring the approach by Hougham et al. (2024a).

4.3 Statistical Analyses

Initially, I examined the assumptions underlying regression models. Among the four outcome variables (fluency measures), three yielded a Shapiro-Wilk p -value of less than 0.001, indicating a significant deviation from normal distributions. To identify influential outliers within the data, I used Cook’s distance, facilitating the assessment of the impact of individual data points on the regression model. Several data points from each model emerged as influential outliers, surpassing the Cook’s distance threshold. Their presence implies a potential influence on the estimated regression coefficients (see [Appendix H](#) and [Appendix I](#) for the detailed results of these checks). Subsequently, to examine the relationship between fluency variables and n-gram measures, I chose a robust regression employing MM estimation, using the “rlm” function

in the MASS package in R (R Development Core Team, 2019).⁵ Guided by Larson-Hall (2015), I selected this approach as it accommodates datasets with non-normal distributions and outliers. I executed four distinct regression models, each harnessing the same eight predictor variables (LB measures) and one of the four criterion variables (fluency measures).

To make the results more robust, I decided it was relevant to conduct additional analyses on the relative importance of each predictor variable in explaining the variance in the model. I pursued this inquiry through a dominance analysis (DA), using the “calc.relimp” function from the “relaimpo” package in R (Grömping, 2006). DA can effectively address correlations among predictor variables (PVs) and can help in better understanding the unique contribution of each PV to the criterion variable in multiple regression analysis, as opposed to relying solely on possibly misleading standardized beta coefficients (Mizumoto, 2022). DA facilitates comprehension by computing dominance weights for each predictor, which show the mean impact of a variable on the predictability of all potential subsets of predictors, consequently presenting a thorough picture of the influence of each predictor on the outcome.

To achieve a more precise estimation of variable importance in the multiple regression model, it is important to conduct dominance analysis in combination with random forests analysis (Mizumoto, 2022). The random forests approach is a nonparametric machine learning model, meaning it can offer more precise outcomes when multiple regression assumptions are violated (Liakhovitski et al., 2010). Using random forests analysis allows researchers to acquire a nuanced perspective on variable importance. Hence, following the guidelines by Mizumoto (2022), I integrated the random forests analysis using the Boruta package in R (Kursa & Rudnicki, 2010). The Boruta algorithm, specifically designed for feature ranking based on

random forests analysis, runs the random forest multiple times. It labels features (or predictors) as “confirmed”, “rejected”, or “tentative” based on their significance compared to randomized shadow features. “Confirmed” predictors are deemed significant, “rejected” ones are considered unimportant, and “tentative” labels are reserved for predictors whose importance remains uncertain. I show these using boxplots in the figures presented in the following section.

Integrating Boruta with robust regression and DA allows for a comprehensive analysis where each method compensates for the limitations of the others. Robust regression ensures that the model is not unduly influenced by outliers, providing reliable coefficient estimates even when the data distribution is non-normal. DA helps clarify the relative importance of each predictor by showing their unique contributions to the model while addressing correlations among predictors. Meanwhile, Boruta offers a rigorous feature selection process that ranks predictors based on their importance, independent of the distributional assumptions. By using Boruta, I can validate which predictors are truly significant, providing a layer of verification to the results obtained from robust regression and DA. This combination of methods allows for triangulation, where the results from one method can support and validate the findings of the others, possibly enhancing the overall reliability and robustness of my conclusions. In what follows, I present the descriptive statistics for the fluency measures first, then I report the four distinct regression models in combination with DAs and random forests analysis to address my research question. Readers are directed to Chapter 3 for the descriptive statistics for the LB measures (see Table 3.2).

4.4 Results

Table 4.1, which is modelled after Table 2 in Tavakoli and Uchiyama (2020, pp. 16-17), shows the means and standard deviations for the different fluency measures across the three proficiency levels (IELTS 6.5 to 7.5). The figures in this table suggest that speakers at higher proficiency levels speak at a higher articulation rate. The figures also show that as proficiency increases, the frequency of pauses at mid-clause position decreases and the frequency of total repairs decreases. Regarding the frequency of end-clause pauses, there is no clear trend across proficiency levels.

Table 4.1

Descriptive Statistics for All Fluency Measures at Different Levels of Proficiency (N = 150)

Measure	Level ^a	<i>M</i>	<i>SD</i>	Median	Minimum	Maximum
Articulation rate	6.5	226.19	21.83	225.98	179.01	289.05
	7	234.82	23.13	234.12	196.96	290.79
	7.5	233.70	37.87	228.68	181.59	401.12
Frequency of mid-ASU pauses	6.5	21.82	6.76	20.14	11.71	36.25
	7	18.94	6.02	18.91	6.07	30.11
	7.5	17.47	6.69	16.62	5.79	32.94
Frequency of end-ASU pauses	6.5	8.32	2.05	8.22	2.92	15.13
	7	8.48	2.21	8.09	2.61	15.37
	7.5	7.95	2.14	7.86	3.56	12.82
Frequency of total repair	6.5	3.33	2.78	2.85	0.00	13.40
	7	2.41	1.79	1.90	0.00	8.10
	7.5	2.04	1.31	2.00	0.00	4.90

^aProficiency level of oral presentations assessed according to IELTS descriptors. *M* = mean. *SD* = standard deviation.

Table 4.2 shows the results from the robust regression analysis with the criterion variable being articulation rate, and the corresponding dominance weights. Robust regression identified three-to-five-word usage as a potential predictor that notably increased articulation rate (*b* =

2.03; dominance weight: 32.75%), with the confidence interval nearing significance, as it narrowly spans zero (95% CI [−0.17, 4.23]. Bigram frequency also showed a positive effect on articulation rate ($b = 18.76$; dominance weight = 17.77%), though its wide confidence interval crossing zero (95% CI [−39.45, 76.98]) warrants cautious interpretation. Trigram MI also showed a positive association ($b = 16.56$) with a considerable dominance weight (14.98%), but it too was statistically non-significant given its wide CI crossing zero (95% CI [−3.30, 36.43]). In contrast, three-to-five-word MI slightly reduced articulation rate ($b = -0.08$) with a considerable dominance weight of 22.30%, yet its CI also crossed zero (95% CI [−0.24, 0.06]) indicating statistical non-significance. Other predictors such as bigram and trigram proportion showed minimal effects, all marked by wide CIs that include zero and lower dominance weights, showing uncertainties in their impacts.

Table 4.2

Robust Regression and Dominance Analyses (Criterion: Articulation Rate)

Predictor	b	95% CI	SE	t	Dominance weight (%)
Intercept	169.30	62.24, 276.38	54.63	3.10	
Bigram frequency	18.76	−39.45, 76.98	29.70	0.63	0.010 (17.77%)
Bigram proportion	−8.15	−138.56, 122.26	66.54	−0.12	0.002 (3.14%)
Bigram MI	−7.14	−37.39, 23.10	15.43	−0.46	0.001 (1.05%)
Trigram frequency	11.41	−36.16, 58.99	24.27	0.47	0.002 (4.01%)
Trigram proportion	−70.96	−369.73, 227.79	152.43	−0.47	0.002 (4.01%)
Trigram MI	16.56	−3.30, 36.43	10.14	1.63	0.009 (14.98%)
Three-to-five-word usage	2.03	−0.17, 4.23	1.12	1.81	0.019 (32.75%)
Three-to-five-word MI	−0.08	−0.24, 0.06	0.08	−1.19	0.013 (22.30%)
Total					0.057 (100%)

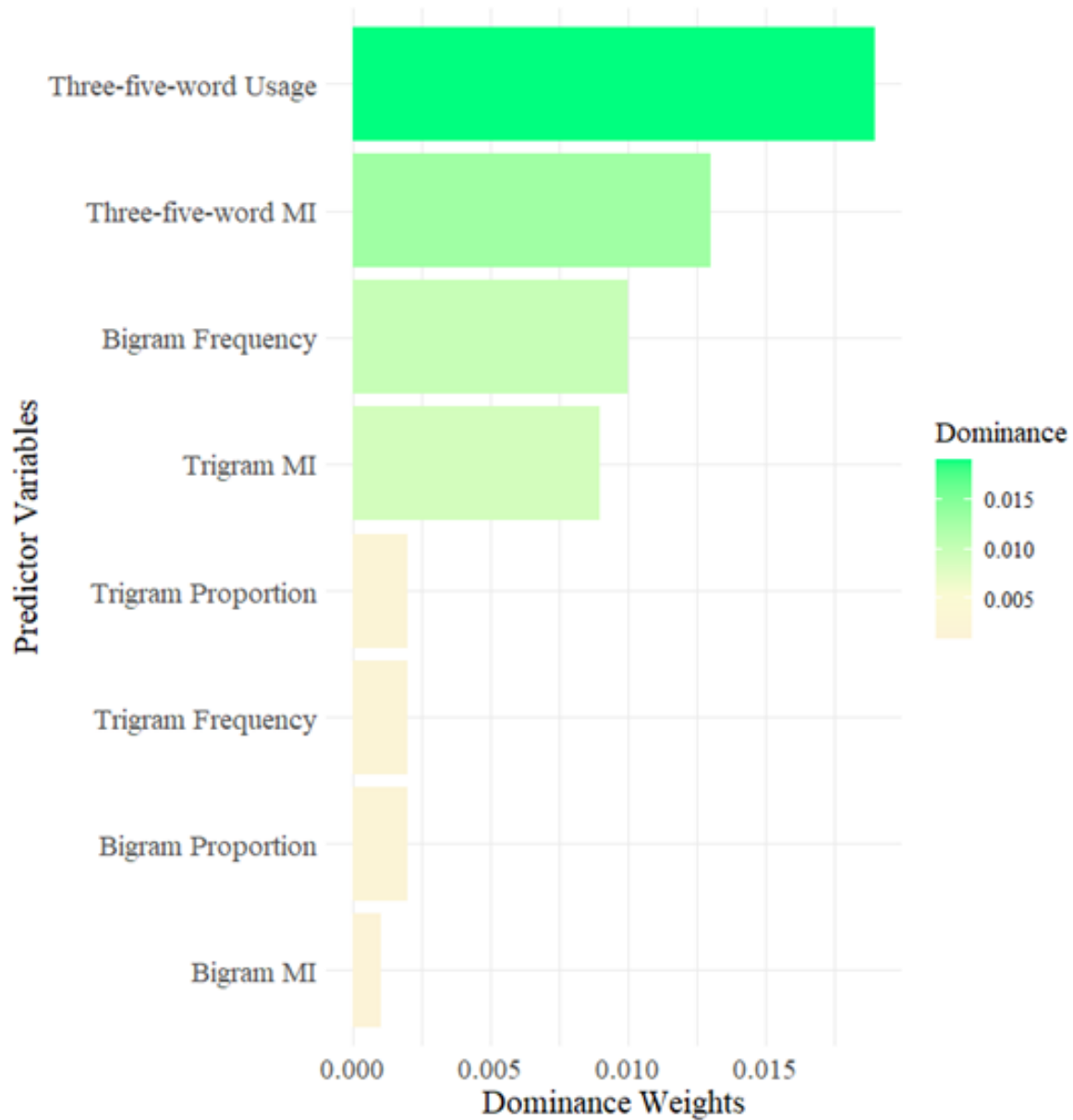
Note. $N = 150$. MI = mutual information. CI = confidence interval. SE = standard error.

The dominance weights from Table 4.2 are shown in Figure 4.1 (below) arranged from highest (three-to-five-word usage) to lowest (bigram MI), facilitating interpretation through a color gradient from spring green (highest weight) to misty rose (lowest weight). Figure 4.2 shows a different perspective via a variable importance plot from random forests, corroborating some of the findings from the DA. The Boruta analysis confirms the significance of three-to-five-word MI and three-to-five-word usage as critical variables, aligning with their dominance weights and highlighted on the right side of the “shadow max” variable in Figure 4.2. Conversely, variables like bigram frequency, bigram MI, and trigram metrics were rejected by Boruta, with trigram proportion noted for its tentative importance. These visualizations highlight the potentially prominent role of three-to-five-word phrases in articulation rate, as supported by both dominance and Boruta analyses. Detailed results and the corresponding R code for the Boruta and random forests analyses are available in [Appendix J](#).

Table 4.3 shows significant insights into mid-ASU pause frequencies. Three-to-five-word usage emerges as a key factor in reducing pauses, with a notable negative impact ($b = -0.58$; $t = -2.01$). Its significant influence is evidenced by the 95% CI of -1.15 to -0.01 , which does not span zero, and it holds the highest dominance weight in the model (40.36%). This marks it as a significant factor in decreasing mid-ASU pauses. Bigram MI also shows significant pause reduction ($b = -8.75$; $t = -2.19$), supported by a 95% CI that does not span zero $[-16.59, -0.91]$. However, it has a lower dominance weight (16.01%) compared to three-to-five-word usage, and the wider CI suggests that the estimate is less precise. Conversely, three-to-five-word MI ($b = 0.01$; $t = 0.76$; dominance weight: 22.55%) and trigram MI ($b = 4.68$; $t = 1.78$; dominance weight: 11.87%) present contrasting effects, showing a slight potential to increase mid-ASU

Figure 4.1

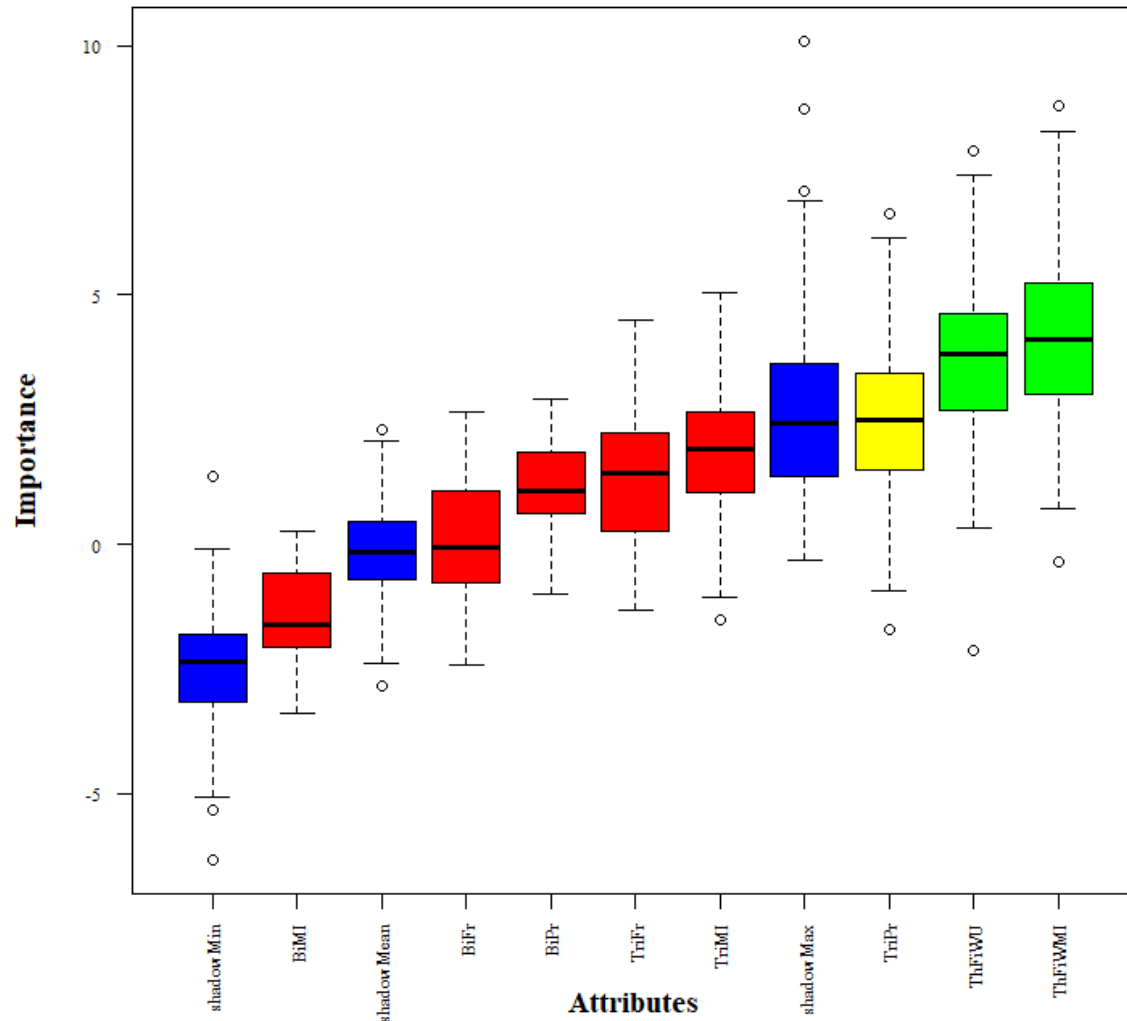
Plot of Dominance Weights in Descending Order (Criterion: Articulation Rate)



pauses. The CIs for three-to-five-word MI $[-0.02, 0.05]$ and trigram MI $[-0.47, 9.83]$ suggest nearing significance, indicating a potential but not statistically confirmed effect. Other predictors, such as bigram and trigram proportion, show negligible effects, as shown by their

Figure 4.2

Variable Importance Plot From Random Forests Analysis (Criterion: Articulation Rate)



Note. In the boxplots generated by the Boruta algorithm, green indicates “confirmed” variables, red signifies “rejected” variables, yellow represents “tentative” variables, and blue is for randomized shadow variables, which serve as a reference to assess the importance of the 8 original variables against random chance.

wide CIs spanning zero and their lower dominance weights.

Table 4.3*Robust Regression and Dominance Analyses (Criterion: Freq. of Mid-ASU Pauses)*

Predictor	<i>b</i>	95% CI	SE	<i>t</i>	Dominance weight (%)
Intercept	23.19	−4.58, 50.96	14.17	1.64	
Bigram frequency	4.81	−10.29, 19.91	7.70	0.62	0.004 (2.87%)
Bigram proportion	−12.97	−46.80, 20.85	17.26	−0.75	0.005 (3.20%)
Bigram MI	−8.75*	−16.59, −0.91	4.00	−2.19	0.024 (16.01%)
Trigram frequency	−1.55	−13.89, 10.79	6.30	−0.25	0.001 (0.40%)
Trigram proportion	15.71	−61.77, 93.19	39.53	0.40	0.004 (2.74%)
Trigram MI	4.68	−0.47, 9.83	2.63	1.78	0.018 (11.87%)
Three-to-five-word usage	−0.58*	−1.15, −0.01	0.29	−2.01	0.061 (40.36%)
Three-to-five-word MI	0.01	−0.02, 0.05	0.02	0.76	0.034 (22.55%)
Total					0.150 (100%)

Note. *N* = 150. MI = mutual information. CI = confidence interval. *SE* = standard error. *Results are marked significant if the 95% CI excludes zero.

Figure 4.3 displays the dominance weights from Table 4.3, starting with the most influential predictor, three-to-five-word usage, showing its leading role. Similarly, Figure 4.4's importance plot of variables from random forests (Boruta) corroborates these findings, confirming the key impact of three-to-five-word MI, three-to-five-word usage, and trigram MI as important variables. These findings, consistent across both dominance and random forests analyses, highlight the key role of three-to-five-word sequences and trigram MI in regulating mid-ASU pauses. Bigram MI, marked as “tentative” by Boruta, reflects ambiguity in its influence, while others like bigram frequency and proportion are deemed less impactful. This coherent validation across analyses highlights three-to-five-word usage's significance in influencing mid-ASU pause frequency, reinforcing the reliability of the outcomes and the nuanced understanding of speech fluency dynamics.

Figure 4.3

Dominance Weights in Descending Order (Criterion: Freq. of Mid-ASU Pauses)

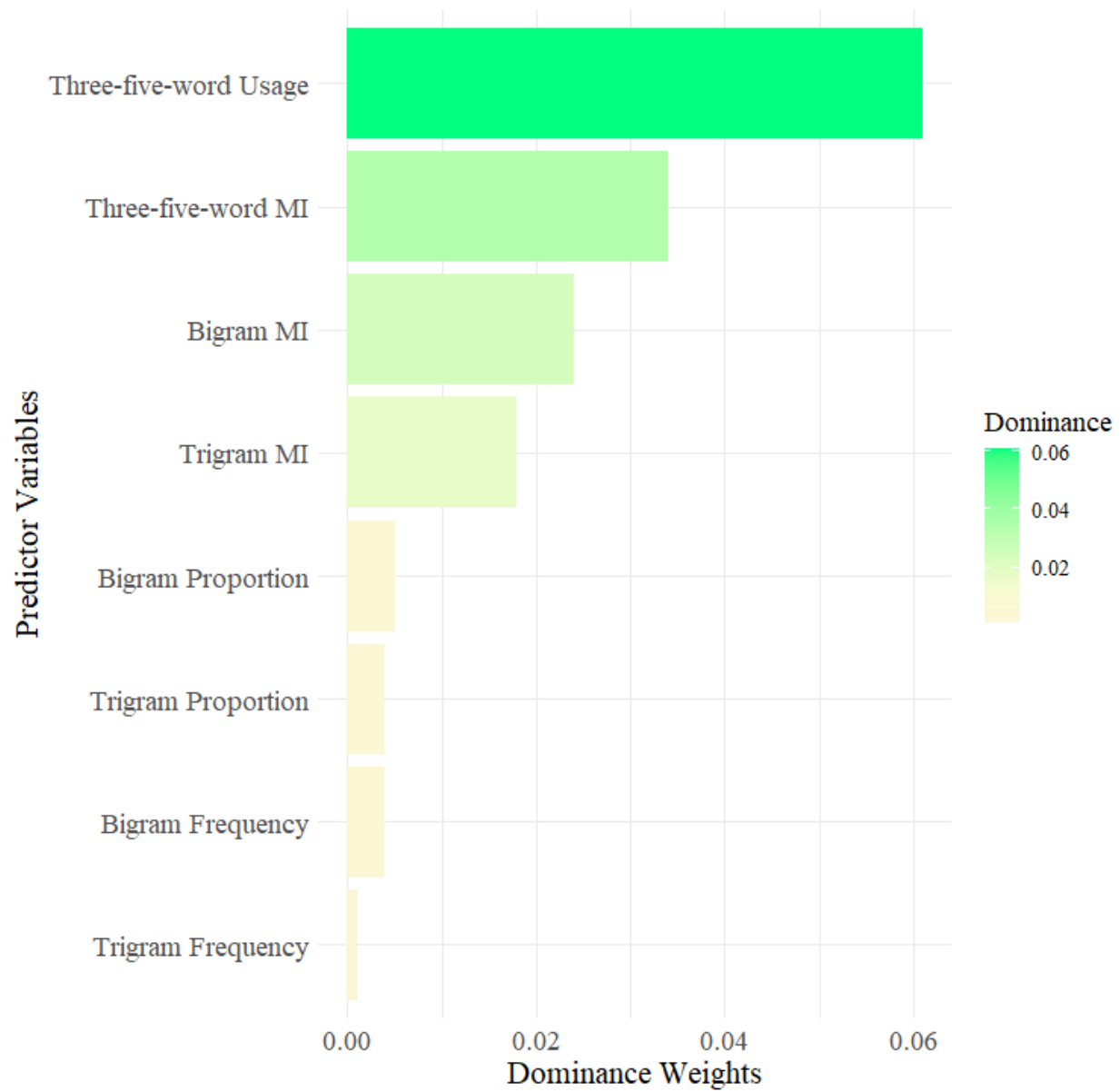


Figure 4.4

Variable Importance Plot From Random Forests Analysis (Criterion: Freq. of Mid-ASU Pauses)

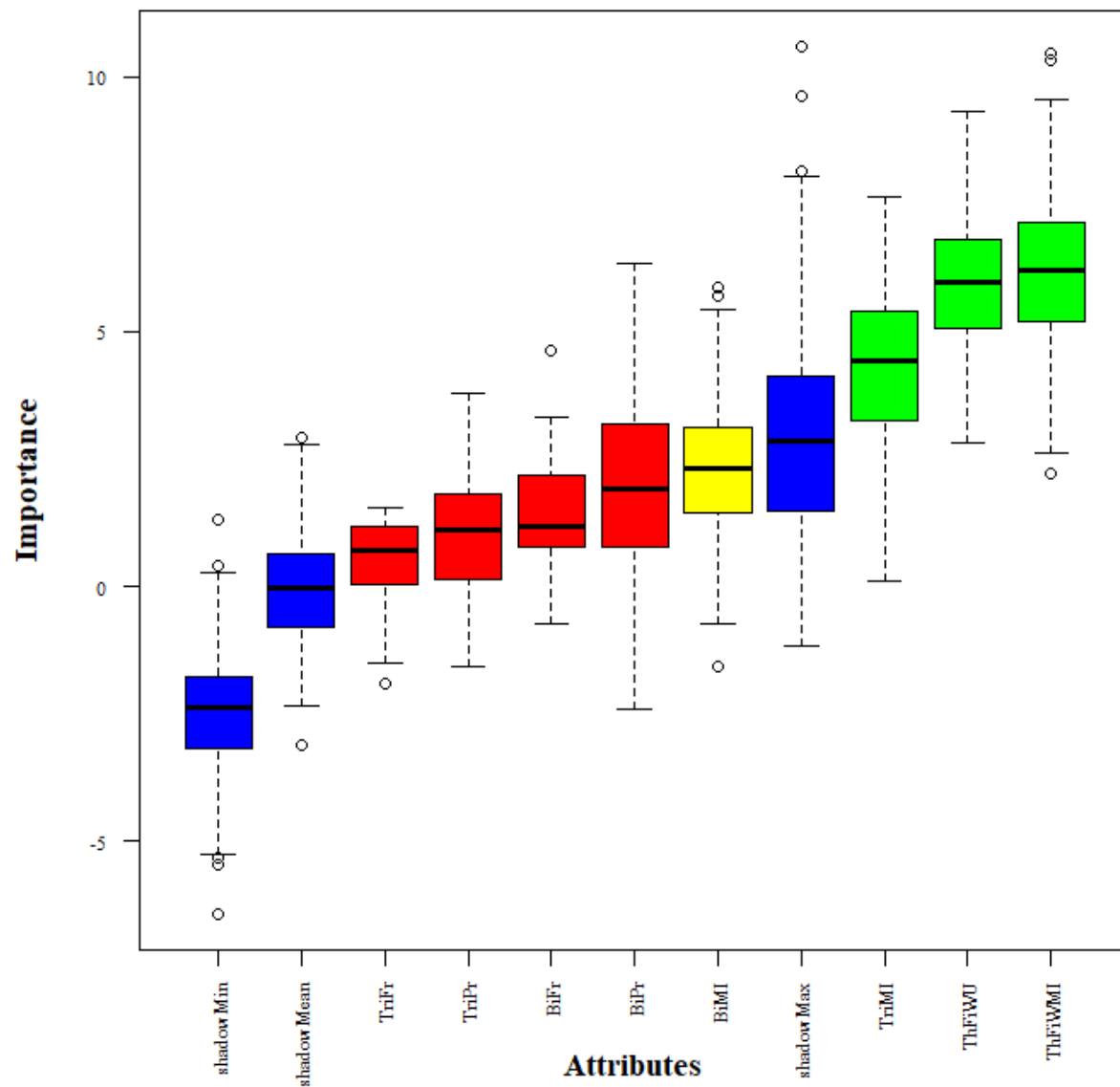


Table 4.4's analysis on end-ASU pause frequencies identifies bigram frequency as a significant contributor, with a positive b -value ($b = 6.76$; $t = 2.91$) suggesting it increases pauses, a finding robustly reinforced by its 95% CI of 2.20 to 11.33 and substantial dominance weight of 40.12%. This positions it as the most impactful predictor in increasing end-ASU pause frequencies. Next, in terms of impact, trigram MI also increases pauses ($b = 1.87$; $t = 2.36$) with a 95% CI of 0.31 to 3.43 and a dominance weight of 23.02%, marking it as another key factor in pause occurrence. Conversely, predictors such as bigram proportion, bigram MI, trigram frequency, trigram proportion, alongside three-five-word usage and MI, display lesser effects. This is evidenced by their relatively lower t -values and dominance weights, and notably, their CIs span zero, suggesting these variables contribute minimally to explaining variations in end-ASU pause frequencies.

Table 4.4

Robust Regression and Dominance Analyses (Criterion: Freq. of End-ASU Pauses)

Predictor	b	95% CI	SE	t	Dominance weight (%)
Intercept	-2.54	-10.93, 5.85	4.28	-0.59	
Bigram frequency	6.76*	2.20, 11.33	2.33	2.91	0.053 (40.12%)
Bigram proportion	-4.34	-14.56, 5.88	5.21	-0.83	0.020 (15.05%)
Bigram MI	-0.18	-2.55, 2.19	1.21	-0.15	0.002 (1.60%)
Trigram frequency	-0.84	-4.57, 2.89	1.90	-0.44	0.004 (3.27%)
Trigram proportion	-7.21	-30.63, 16.20	11.94	-0.60	0.008 (6.23%)
Trigram MI	1.87*	0.31, 3.43	0.79	2.36	0.030 (23.02%)
Three-to-five-word usage	0.01	-0.16, 0.18	0.09	0.09	0.006 (4.33%)
Three-to-five-word MI	0.00	-0.01, 0.02	0.01	0.74	0.008, (6.38%)
Total					0.132 (100%)

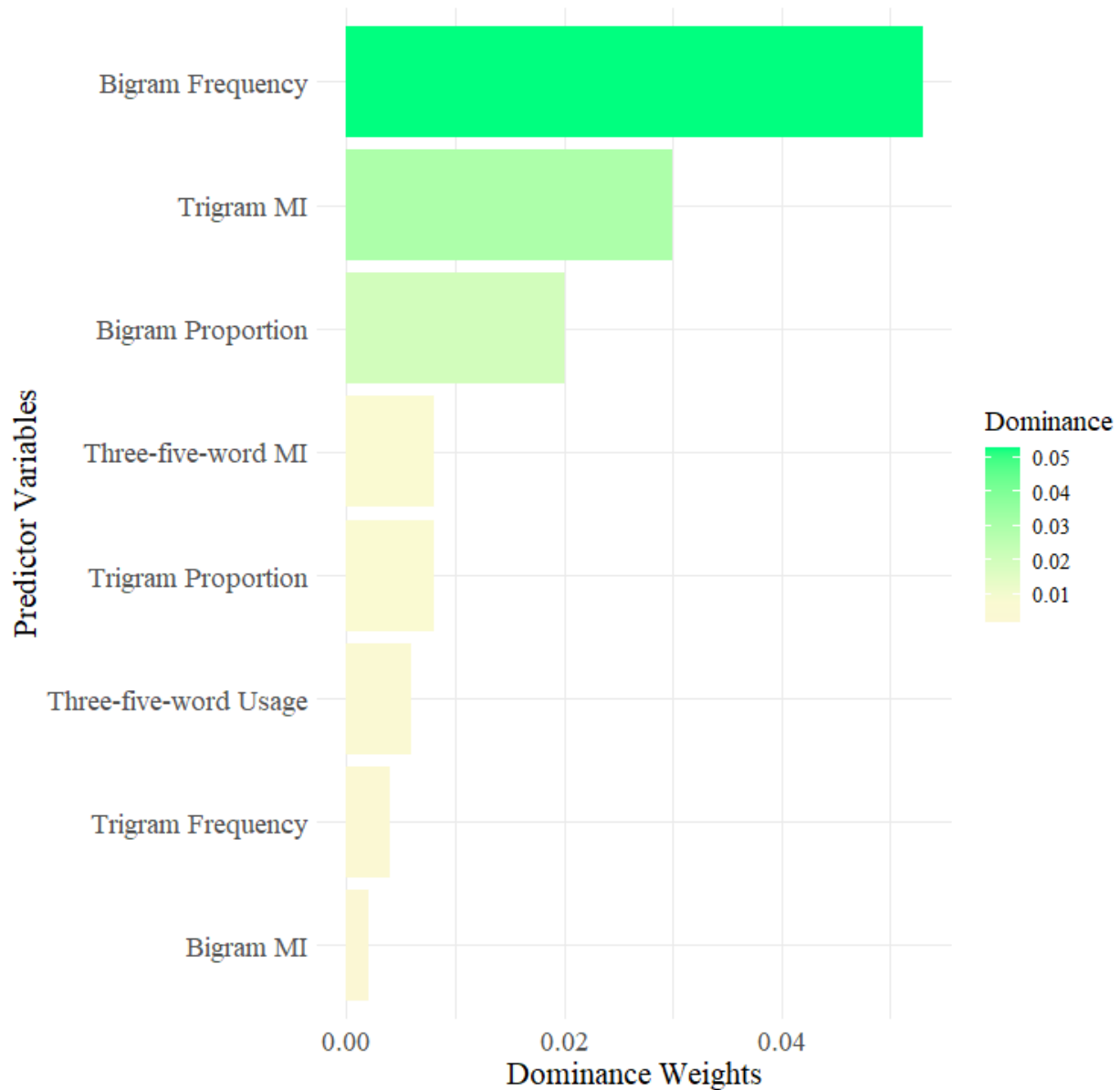
Note. $N = 150$. MI = mutual information. CI = confidence interval. SE = standard error. *Results are marked significant if the 95% CI excludes zero.

The variable importance plot in Figure 4.6, derived from the Boruta analysis, presents a stringent evaluation of variable importance that contrasts with the findings from Table 4.4 and Figure 4.5. It categorically deems all predictors, including bigram frequency and trigram MI, as insignificant, showing a stringent evaluation of variable selection. This shows a divergence from the regression analysis, which highlighted the impact of these predictors on end-ASU pause frequencies. Essentially, the random forests (Boruta) analysis calls for a more prudent interpretation of these effects, emphasizing the absence of confirmed importance for any predictors. Such a rigorous approach enriches our understanding of variable impacts on end-ASU pauses, urging caution in their assessment.

Table 4.5's robust regression and dominance analyses focus on the frequency of total repairs. Notably, bigram frequency stands out as leading to significantly more speech repairs, with a b -value of 3.71 and a 95% CI that does not span zero [0.04, 7.38]. Despite its relatively low dominance weight (3.48%), its statistical significance underscores its role as a predictor of increased speech repairs. Bigram MI has a notable dominance weight of 55.26%, despite its modest effect size ($b = -0.48$) and a CI that spans zero ($-2.38, 1.43$). This is visually reinforced in Figure 4.7, where bigram MI's prominence is clear. The Boruta analysis, illustrated in Figure 4.8, further confirms its leading role, marking bigram MI as a primary attribute for predicting repair frequency, while rejecting bigram frequency's contribution. Trigram MI follows in significance with a b -value of -0.34 , a CI that spans zero ($-1.60, 0.91$), and a dominance weight of 13.06%, with its importance also confirmed by Boruta as an important predictor. Bigram proportion, with a CI spanning zero ($-5.51, 10.94$), still shows relevance with a dominance weight of 4.98% and a positive effect size ($b = 2.72$), suggesting its variable but notable impact

Figure 4.5

Dominance Weights in Descending Order (Criterion: Frequency of End-ASU Pauses)



on repair frequency, a sentiment echoed by the Boruta analysis. Other predictors, including trigram proportion, three-to-five-word usage, and three-to-five-word MI, contribute to the model with dominance weights ranging from 5.55% to 7.99%, all having CIs that span zero, reflecting statistically insignificant influences.

Figure 4.6

Variable Importance Plot From Random Forests Analysis (Criterion: Freq of End-ASU Pauses)

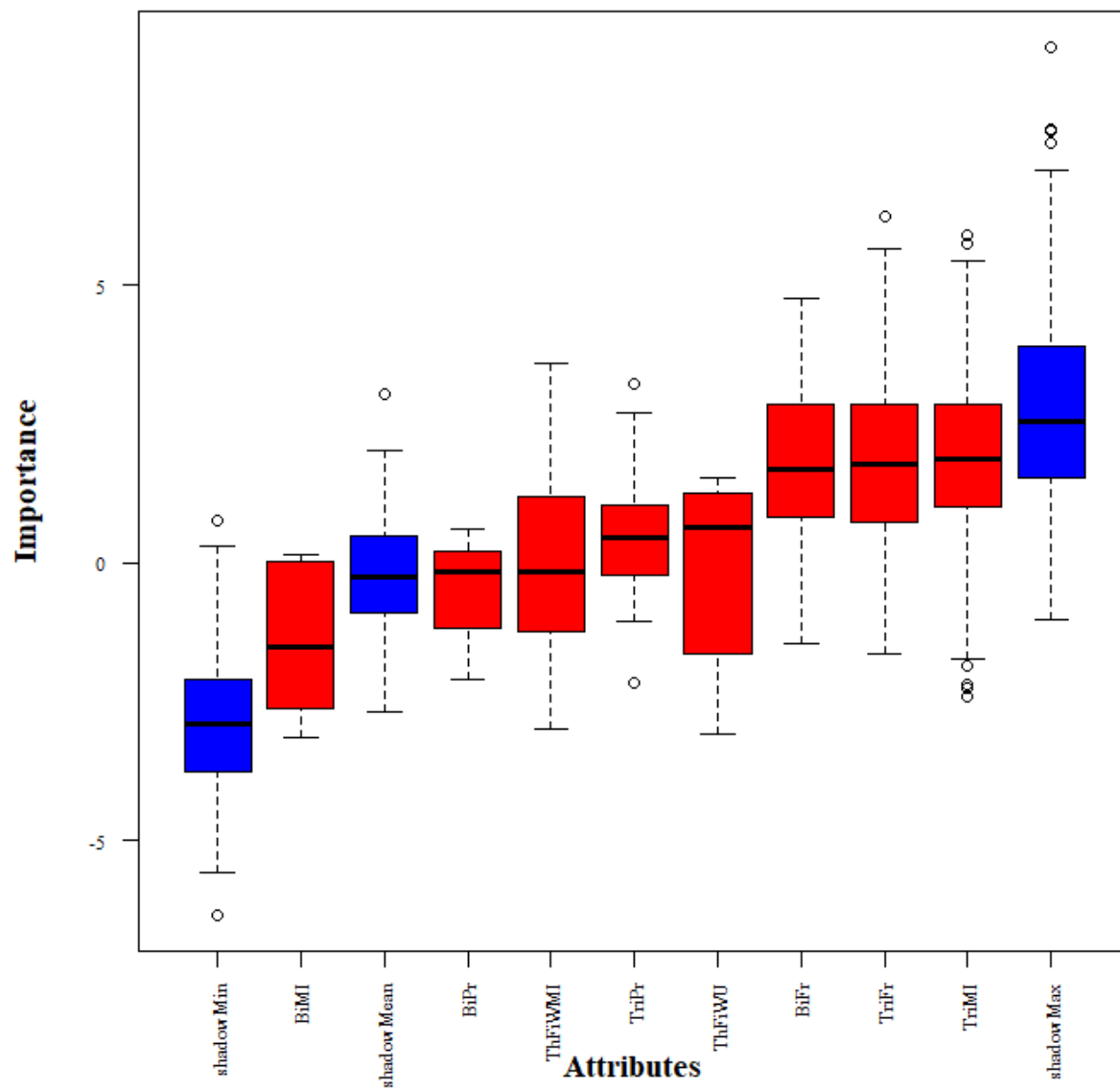


Table 4.5*Robust Regression and Dominance Analyses (Criterion: Freq. of Total Repairs)*

Predictor	<i>b</i>	95% CI	SE	<i>t</i>	Dominance weight (%)
Intercept	0.72	−6.03, 7.47	3.44	0.21	
Bigram frequency	3.71*	0.04, 7.38	1.87	1.98	0.004 (3.48%)
Bigram proportion	2.72	−5.51, 10.94	4.20	0.65	0.005 (4.98%)
Bigram MI	−0.48	−2.38, 1.43	0.97	−0.49	0.059 (55.26%)
Trigram frequency	−0.81	−3.81, 2.19	1.53	−0.53	0.004 (3.38%)
Trigram proportion	−15.07	−33.91, 3.76	9.61	−1.57	0.009 (7.99%)
Trigram MI	−0.34	−1.60, 0.91	0.64	−0.54	0.014 (13.06%)
Three-to-five-word usage	−0.05	−0.19, 0.09	0.07	−0.73	0.006 (5.55%)
Three-to-five-word MI	0.00	−0.01, 0.01	0.00	0.00	0.007 (6.30%)
Total					0.106 (100%)

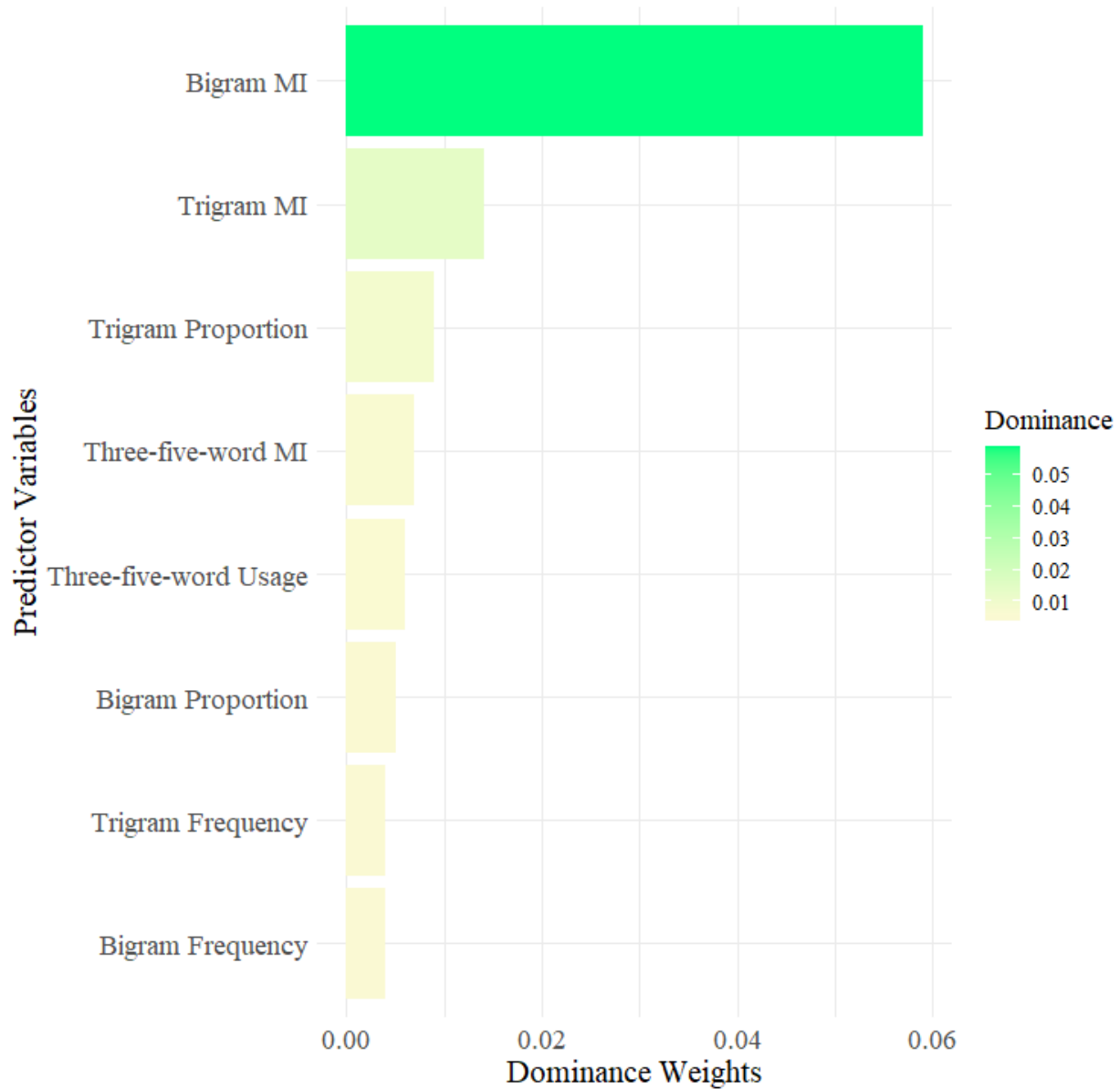
Note. *N* = 150. MI = mutual information. CI = confidence interval. SE = standard error. *Results are marked significant if the 95% CI excludes zero.

4.5 Discussion

The overarching aim of the analysis presented in the current chapter has been to explore the impact of LB usage on oral fluency across a broad dataset. My investigation was driven by one research question exploring the extent to which LB usage is associated with three dimensions of utterance fluency: speed, breakdown, and repair. The subsequent discussion is organized to tackle this research question. Section 4.5.1 covers the relationship between articulation rate and lexical bundles, analyzing both longer and shorter LB measures. Section 4.5.2 examines mid- and end-ASU pauses, starting with the influence of longer LBs on mid-ASU pauses, followed by their impact on end-ASU pauses, then discussing shorter LB measures in a similar order. Section 4.5.3 addresses repair fluency, exploring the influence of both longer

Figure 4.7

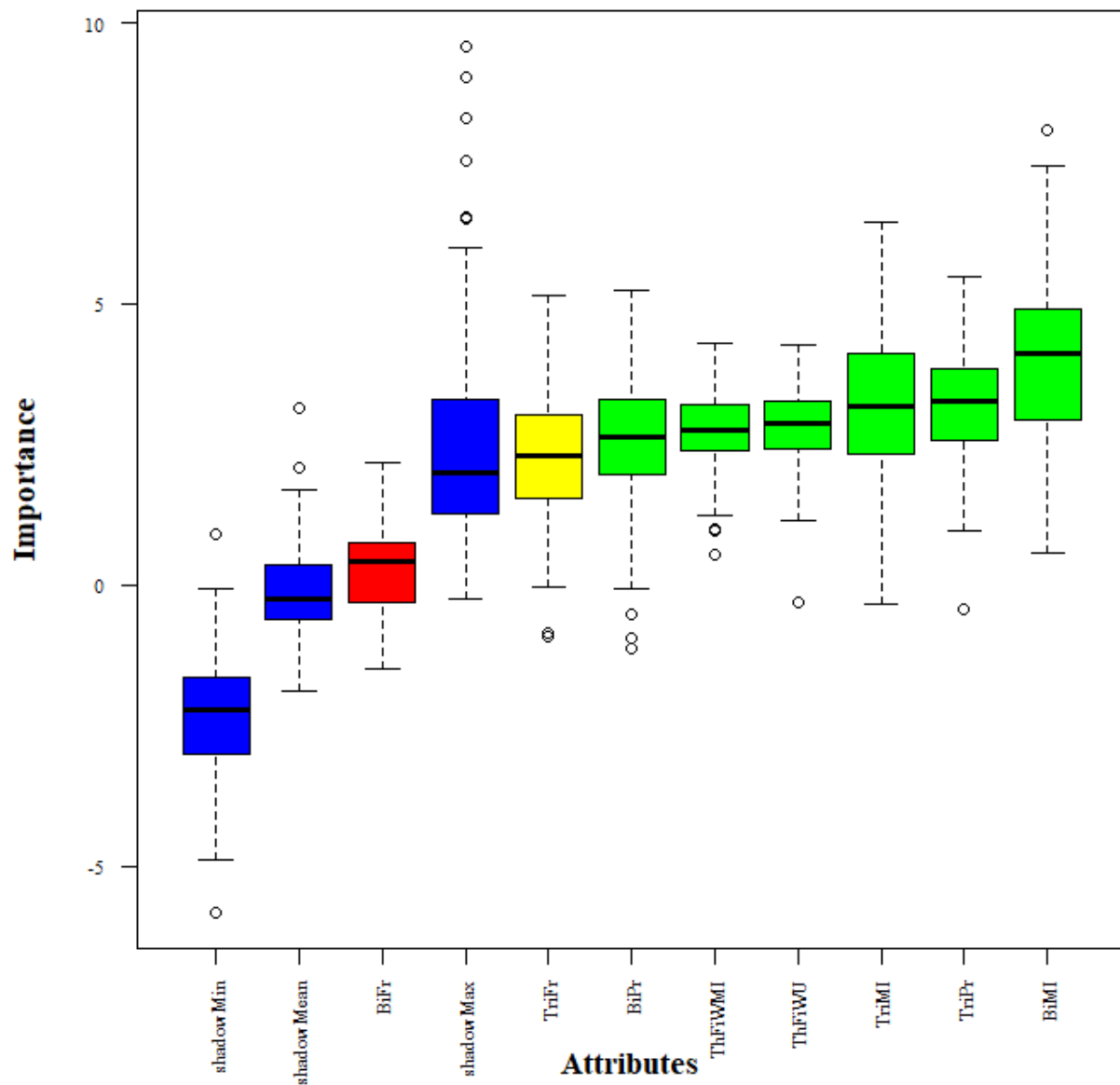
Dominance Weights in Descending Order (Criterion: Frequency of Total Repairs)



and shorter LBs on the frequency of speech repairs. The final paragraphs in each section discuss whether the findings align with existing research.

Figure 4.8

Variable Importance Plot From Random Forests Analysis (Criterion: Freq. of Total Repairs)



4.5.1 Articulation Rate and Lexical Bundles

The robust regression and dominance analyses revealed relationships between the use of three-to-five-word LBs and articulation rate. Specifically, as shown in the results, three-to-five-word usage was associated with an increase in articulation rate, with a dominance weight of 32.75%. This finding aligns with my hypothesis that frequent use of longer sequences correlates with speed fluency by facilitating more efficient information conveyance. However, given that the confidence interval includes zero, these results should be interpreted cautiously, as the effect might be negligible. Conversely, the complexity of three-to-five-word MI showed a slight negative effect on articulation rate, potentially hindering fluency by introducing additional cognitive processing demands. The Boruta analysis confirmed three-to-five-word MI and three-to-five-word usage as the most important variables influencing articulation rate, corroborating the findings from dominance analysis. The impact of shorter LBs, such as bigram frequency and trigram MI, was less clear due to statistical uncertainty, as indicated by their wide confidence intervals, and the lack of confirmation of their importance in the Boruta analysis.

These findings align to some extent with Tavakoli and Uchihara (2020), who found significant correlations between high-frequency trigram and articulation rate, both pruned ($r = .396$, $p = .003$) and unpruned ($r = .325$, $p = .014$). Both Tavakoli and Uchihara's study and the analysis in this chapter confirm that fluent speakers frequently use high-frequency n-grams, linking lexical frequency to increased speech speed. In the current chapter, bigram frequency positively impacted articulation rate, consistent with Tavakoli and Uchihara's findings, though the wide CI in my analysis suggest a lack of statistical certainty.

In addition to the alignment with Tavakoli and Uchihara (2020), the results of the current chapter are also aligned with Hougham et al. (2024a), particularly regarding bigram frequency

and bigram MI. Hougham et al. found that bigram frequency was a significant predictor of faster speech rates, with a positive association with mean syllable duration ($b = 313.41$, 95% CI [40.22, 586.60], $t = 2.25$), accounting for 18.12% of the variance in their model. This finding aligns with the current chapter's results, where bigram frequency positively influenced articulation rate, although wide CIs suggest cautious interpretation. Similarly, Hougham et al. reported a significant negative effect of bigram MI on speed fluency ($b = -120.23$, 95% CI [-229.36, -11.10], $t = -2.16$), accounting for 22.65% of the variance. This aligns with my findings, where higher MI in bigrams slightly reduced articulation rate (albeit not statistically significantly), indicating that collocational strength may introduce cognitive processing demands, thereby slowing speech.

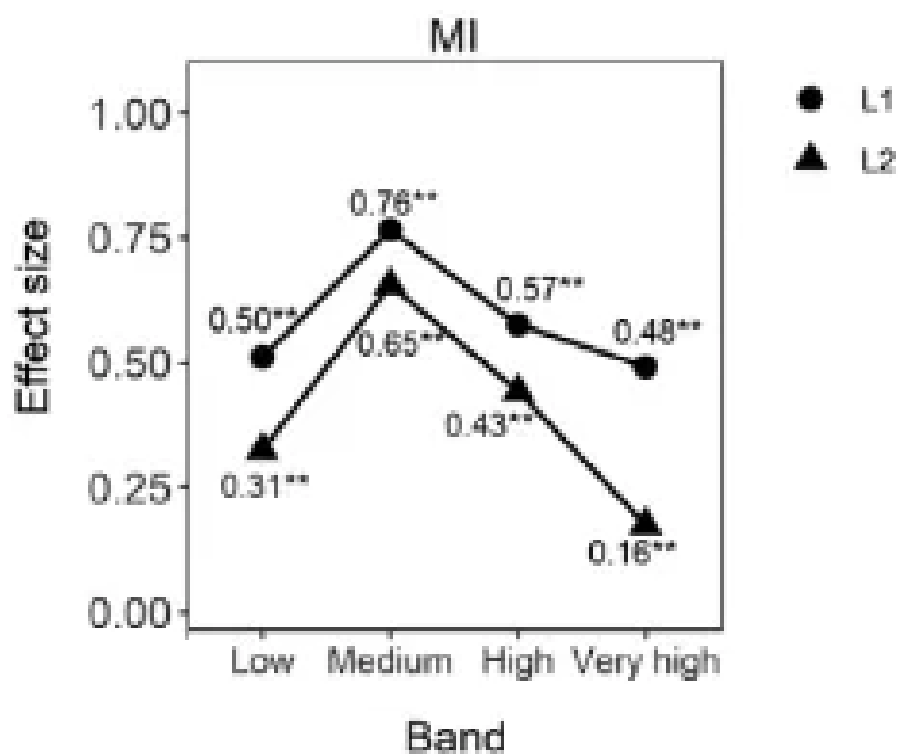
Regarding longer sequences, Hougham et al. (2024a) observed a positive but statistically non-significant effect of four-to-five-word MI ($b = 13.07$, 95% CI [-46.53, 72.67]) on speed fluency. Similarly, the current analysis found that three-to-five-word usage increased articulation rate ($b = 2.03$; dominance weight: 32.75%), although the confidence interval narrowly spanned zero (95% CI [-0.17, 4.23]), indicating marginal statistical significance. This suggests that while longer LBs appear to enhance speed fluency by facilitating more efficient speech production, their complexity may introduce cognitive processing demands that offset these benefits. This observation is further supported by the slight negative effect of three-to-five-word MI on articulation rate found in the current analysis ($b = -0.08$, dominance weight: 22.30%).

In line with Yi and Zhong's (2024) findings, the inverted u-shaped relationship between LB complexity and processing advantage may explain why high-MI LBs slightly reduce speech speed (see Figure 4.9, adapted from Yi & Zhong, 2024, p. 14.). According to Yi and Zhong's (2024) meta-analysis, moderate LB complexity (as measured by MI) is associated with optimal

processing efficiency, whereas very high MI scores reduce efficiency. This inverted u-shaped pattern suggests that while moderate LB complexity facilitates fluency, very high complexity increases processing demands, slowing articulation speed. This non-linear relationship helps explain the analysis in the current chapter, where longer LBs improved articulation rate, but those with higher MI seemed to impose greater cognitive demands.

Figure 4.9

Inverted U-Shaped Relationship Between LB Complexity (MI) and Processing Advantage (Effect Size) in L1 and L2 Speakers (Adapted From Yi & Zhong, 2024, p. 14)



Levelt's (1989) theory on the advantages of using MWSs supports the idea that longer linguistic units have a noticeable effect on how quickly we articulate. This is relevant for L2 speakers, as Kormos (2006) and Skehan (2014) have highlighted the advantage of a larger MWS

repertoire in freeing up cognitive resources for other processing needs. Skehan's trade-off hypothesis helps explain the balance between complexity and fluency, particularly why the use of high-MI sequences may slightly reduce articulation rate. While using longer, prefabricated chunks can enhance fluency by managing cognitive load, their complexity (MI) might also offset these fluency benefits. Furthermore, these theories, along with usage-based perspectives (Ellis, 2002; Bybee, 2006), highlight the efficiency of using prefabricated language chunks and the cognitive trade-offs involved in managing complex word combinations. My results, particularly regarding the use of longer LBs, provide some support for these theories, as indicated by the positive relationship between three-to-five-word usage and articulation rate, and by the trade-off between three-to-five-word MI and articulation rate. Nevertheless, the inclusion of zero in the confidence intervals means that we should interpret the results with caution, suggesting the need to conduct additional research to confirm these patterns more conclusively.

In summary, the findings highlight the role of longer LBs in enhancing speed fluency, particularly with three-to-five-word usage increasing articulation rate. This positive effect was supported by dominance and random forests (Boruta) analyses, and showed trends toward significance in robust regression, suggesting potential benefits for faster speech production. Conversely, three-to-five-word MI showed a subtle negative effect on articulation rate, indicating a potential trade-off where complexity may impose cognitive demands that reduce fluency. These two longer LB measures were consistently highlighted as important across multiple analyses, whereas shorter LB measures showed less consistent support. These findings highlight the importance of longer LBs for speed fluency and suggest further research to confirm these patterns in different contexts or larger samples, as well as implications for language proficiency and teaching practices.

4.5.2 Breakdown Fluency and Lexical Bundles

My investigation into the role of longer (three-to-five-word) LBs in breakdown fluency showed nuanced influences on mid-ASU pauses. Regression and dominance analyses robustly identified three-to-five-word usage as a dominant predictor, significantly reducing mid-ASU pause frequency ($b = -0.58$, 95% CI $[-1.15, -0.01]$, $t = -2.01$, dominance weight = 40.36%), thus enhancing speech smoothness. The Boruta analysis further confirmed the significant role of three-to-five-word usage in reducing mid-clause pause phenomena. Interestingly, while three-to-five-word MI showed a notable dominance weight of 22.55% and a confirmed Boruta classification, it did not significantly impact mid-ASU pause frequency as indicated by its regression coefficient and CI ($b = 0.01$, 95% CI $[-0.02, 0.05]$, $t = 0.76$). This implies that although the presence of complex LBs is important, it may not directly correspond to a higher or lower frequency of pauses in the middle of utterances, suggesting a more nuanced influence on the smoothness of speech. Regarding end-ASU pauses, neither of these longer LB measures showed a significant effect in influencing them in any of the analyses.

When compared with Hougham et al. (2024a), the current findings on mid-clause pauses are aligned, with only a slight difference in the level of statistical significance. Hougham et al. found that four-to-five-word MI was nearly statistically significant in predicting mid-ASU pauses, and it carried the highest dominance weight (39.59%), indicating its potential to reduce breakdown fluency disruptions. This converges with my finding that longer LBs play an important role in facilitating smoother speech. My findings in the current chapter support this notion, indicating a significant impact of three-to-five-word usage on mid-ASU pauses (as noted above), reflecting a robust relationship between frequent use of longer LBs and reduced mid-clause pause frequency. While the cognitive demands associated with high-MI longer bundles

might explain the non-significance in Hougham et al.'s findings, the current chapter shows a clearer reduction in mid-ASU pauses associated with the use of longer LBs.

Regarding shorter LBs, my findings indicate that they have a varying effect on breakdown fluency. Specifically, bigram MI showed a significant negative association ($b = -8.75$, 95% CI $[-16.59, -0.91]$, $t = -2.19$, dominance weight = 16.01%), suggesting that increased usage leads to fewer mid-ASU pauses, likely due to the cognitive ease in accessing strongly associated lexical pairs. However, bigram MI was classified as tentative in the Boruta analysis, thus requiring cautious interpretation. On the other hand, trigram MI demonstrated a positive impact ($b = 4.68$, 95% CI $[-0.47, 9.83]$, $t = 1.78$, dominance weight = 11.87%), albeit nearing statistical significance. This suggests that high MI-score trigrams could potentially result in increased mid-ASU pauses possibly due to the cognitive complexities involved in processing strongly associated three-word units. These findings highlight a complex interplay between LB length, structural complexity, and pauses within clauses.

The dynamics of mid-ASU pauses observed in my study partially align with those of Tavakoli and Uchihara (2020), who reported varying impacts of n-grams on pause frequencies. Tavakoli and Uchihara identified a non-significant negative correlation between n-gram proportion and mid-clause pauses ($r = -.271$). This trend, suggesting reduced mid-clause pauses with more frequent n-gram use, aligns with my findings where bigram MI and three-to-five-word usage similarly decrease mid-ASU pauses. This reduction in mid-clause pause frequency can be attributed to the formulation stage of the speech production process (Skehan, 2014), reinforcing the long-held view that mid-clause pauses in L2 speech are indicative of speakers grappling with the lexical and morphosyntactic demands inherent in speech processing (Kahng, 2014). This effect is likely because these speakers can retrieve prefabricated chunks as whole units, creating

“zones of safety” (Boers et al., 2006, p. 247), and are thus more likely to pause at the junctures between MWSs, which facilitates smoother speech flow. This finding suggests that more phraseologically proficient speakers tend to pause less frequently at mid-clause positions. In contrast, Tavakoli and Uchihara (2020) and Hougham et al. (2024a) found that bigram MI contributed to increasing mid-clause pauses, though neither of their findings were statistically significant. My results diverge from these earlier findings, indicating that bigram MI instead reduces mid-ASU pauses. This divergence is theoretically significant because it challenges the assumption that more tightly associated bigrams necessarily hinder fluency. Instead, the findings from Section 4.4 suggest that high-MI bigrams may facilitate fluid speech production more effectively than previously recognized.

It is worth highlighting that, while bigram MI shows a slightly stronger statistical significance (as indicated by *t*-values and CIs), the use of longer (three-to-five-word) bundles plays a more substantial role in my regression model, holding a higher dominance weight (40.36% for three-to-five-word usage vs. 16.01% for bigram MI). This suggests that longer LBs have a more important impact on reducing mid-ASU pause frequency compared to shorter LBs, supporting the notion that the use of longer MWSs can significantly enhance the smoothness of speech flow.

Regarding end-ASU pauses, regression analysis highlighted bigram frequency ($b = 6.76$, 95% CI [2.20, 11.33], $t = 2.91$, dominance weight = 40.12%) as the most significant predictor, suggesting that the frequent use of bigrams leads to increased end-ASU pauses. Regression and dominance analyses also highlighted trigram MI ($b = 1.87$, 95% CI [0.31, 3.43], $t = 2.36$, dominance weight = 23.02%) as another significant predictor, suggesting that the use of strongly associated trigrams leads to increased end-ASU pauses. Despite these findings, the Boruta

analysis did not validate any variables as significant predictors of end-ASU pauses. This divergence between the regression and Boruta analyses indicates a nuanced complexity in understanding the exact influence of these predictors. It suggests that although certain LBs show statistical significance in the regression model, their practical implication and influence may require a more conservative interpretation when it comes to end-ASU pauses.

My findings regarding end-ASU pauses align to some extent with those reported by Tavakoli and Uchihara (2020), who observed a trend indicating that higher MI scores in LBs might be associated with increased end-clause pauses, although this was not statistically significant at their corrected alpha level. Similarly, my regression analysis identified bigram frequency and trigram MI as significant predictors of increased end-ASU pauses. This contrast with the negative associations observed for mid-ASU pauses suggests a difference in how LBs function depending on their position within a sentence. Although shorter LBs like bigrams reduce pauses mid-sentence, possibly due to their frequent and fluent usage, their increased occurrence at sentence ends might reflect a preparatory pause for upcoming speech segments, aligning with the cognitive effort required to process or plan subsequent material. Hougham et al. (2024a), on the other hand, found nearly statistically significant results regarding end-ASU pauses ($b = -4.10$, 95% CI $[-8.53, 0.33]$), with their dominance analysis also highlighting the potential importance of four-to-five-word MI (dominance weight: 34.17%), suggesting that longer bundles may still help reduce pauses at the ends of clauses. This nuanced impact of shorter LBs and the potential role of longer LBs reflect the complex interplay between LB length, cognitive demands, and fluency development. Consistent with usage-based theory, my findings suggest that fluency develops from frequent and experienced use of LBs, yet the employment of various LB types may temporarily disrupt speech flow, especially at end-clause

positions. This points to the multi-faceted challenges in language proficiency development, particularly as learners manage the cognitive load associated with integrating these structural elements into their speech.

4.5.3 Repair Fluency and Lexical Bundles

My analysis examined the influence of both longer and shorter LBs on repair fluency. Although the results of regression and dominance analyses showed that longer LBs are not significant influencers, the Boruta analysis confirmed their subtle influence on the frequency of total repairs, supporting my hypothesis that usage of three-to-five-word LBs reduces speech repairs. Regarding shorter LBs, bigram frequency stands out as having a significant positive influence with a CI that does not span zero (i.e., increased use of frequent bigrams leads to increased frequency of repairs), despite its relatively low dominance weight (3.48%). However, bigram frequency also stands out as the only variable confirmed as unimportant by the Boruta analysis, thus casting doubt on its actual impact on repair fluency. Bigram and trigram MI's influence on enhancing repair fluency, although notable with dominance weights of 55.26% and 13.06, respectively, requires cautious interpretation due to the CIs spanning zero. The Boruta analysis confirmed the importance of both bigram MI and trigram MI in reducing speech repairs, with bigram MI appearing to be the most important measure. These findings suggest a potential relationship between higher MI-score bigrams and trigrams and fewer repairs, consistent with trends observed by Tavakoli and Uchihara, albeit not conclusively. These observations highlight the potential for linguistic sophistication and strategic lexical pattern use to influence repair fluency, as supported by usage-based theory (Ellis, 2002; Bybee, 2006). Nevertheless, the fact that there are confidence intervals that include zero for multiple predictors requires careful

interpretation of these results, highlighting the need for additional research to establish these patterns with more certainty.

The findings of the current chapter diverge from Hougham et al. (2024a) in two important ways. First, while Hougham et al. identified four-to-five-word MI as a strong predictor for reducing repairs ($b = -2.41$, 95% CI $[-4.24, -0.59]$, $t = -2.18$, dominance weight = 42.75%), the current chapter did not find a significant impact of three-to-five-word usage or three-to-five-word MI on repair fluency, as the results were minimal and non-significant. This discrepancy suggests that the role of longer sequences in reducing repairs may depend on specific task conditions or speaker characteristics. Second, as noted above, the current chapter identified bigram frequency as a significant predictor of increased repairs, a finding not reported in Hougham et al. (2024a). This suggests that more frequent use of bigrams may lead to a higher number of repairs, possibly reflecting an over-reliance on common lexical bundles, which may disrupt speech flow. This divergence highlights the complexity of how different lengths and types of LBs influence repair fluency and suggests that further exploration is needed to fully understand these relationships.

4.6 Conclusion

This chapter examined the impact of LB length on aspects of oral fluency across proficiency levels, using robust regression, dominance analysis, and random forests analysis. The findings suggest that both shorter and longer LBs are used more frequently as learners' proficiency increases. Notably, longer LBs (three-to-five-word units) tend to enhance speed fluency and reduce mid-clause pauses, indicating unique processing efficiencies. In contrast, shorter LBs show more mixed effects: (1) Frequent bigrams and high MI-score trigrams subtly yet positively influence articulation rates; (2) high MI-score bigrams significantly reduce mid-

ASU pauses; and (3) frequent bigrams lead to significantly more speech repairs. These findings emphasize the distinct and key influences of various LB lengths in oral fluency, underscoring their importance in language proficiency development. My use of an innovative LB refinement technique has enabled a more precise analysis of how LBs of varying lengths influence oral fluency. By building on previous findings and showing the varying impacts of LB lengths, my research contributes to a richer understanding of their purposeful use in language learning, supporting the integration of varied LB types into EFL pedagogy to enhance advanced language proficiency.

As the current chapter has shown, the differential impacts of shorter and longer LBs on aspects of fluency highlight the need for further investigation into their broader implications, particularly in how these linguistic features might predict outcomes in high-stakes language assessments. These issues, raised by the intricate relationships explored in Chapters 3 and 4, set the stage for the ensuing Chapter 5, where I will examine the predictive validity of LBs in forecasting oral proficiency scores, continuing the exploration of their role in language assessment across proficiency levels.

Chapter 5: Exploring the Extent to Which LB Usage Predicts Presentation Scores

In Chapter 4, I examined the intricate relationship between the use of LBs of varying lengths and different aspects of oral fluency (speed, breakdown, and repair) across different proficiency levels (IELTS 6.5 to 7.5). The findings emphasized the key role that both shorter and longer LBs play in enhancing various dimensions of fluency, with longer LBs showing particular promise in improving articulation rate and reducing mid-clause pauses. These results provided valuable insights into the nuanced ways LBs contribute to L2 speech production, highlighting their importance in fluency development.

Building upon the foundations laid in Chapters 3 and 4, Chapter 5 shifts focus from analyzing the relationship between LB usage and fluency to further investigating how these linguistic features predict performance in high-stakes language assessments. The current chapter represents a logical progression from the prior chapters, moving from understanding the characteristics and fluency impacts of LBs of various lengths to assessing their practical utility in predicting proficiency outcomes. Chapter 5 is driven by several key objectives that build on the work done in Chapters 3 and 4. First, it seeks to establish the predictive validity of LB usage, showing that these linguistic constructs are not merely reflective of proficiency but can actively forecast it. This is important for the integration of LB usage into language education, curriculum development, and evaluative descriptors. Additionally, the chapter ensures methodological continuity by using predictive models directly informed by the empirical findings of the previous chapters, allowing for a seamless transition in the analysis. By using the LB measures already examined, the current chapter aims to project assessment outcomes while maintaining a strong connection to the established data trends. Finally, the chapter seeks to advance practical implications by identifying the LB characteristics most closely associated with higher

proficiency scores. This could provide educators and evaluators with valuable insights, potentially guiding the development of effective teaching strategies and the refinement of assessment tools.

To address these objectives, Chapter 5 employs advanced statistical procedures, consistent with the rigorous methods utilized in Chapter 4. I apply robust regression to handle the variability within the continuous raw score data, supplemented by dominance analysis and random forests analysis to parse out the relative importance of each LB variable. This comprehensive approach allows for a detailed and triangulated examination of the structures that most significantly impact proficiency as measured by high-stakes assessments.

The research question guiding the current chapter is: To what extent do longer LBs predict raw oral presentation scores compared to shorter LBs? Based on the findings of Tavakoli and Uchihara (2020) and theoretical frameworks of speech production (e.g., Kormos, 2006), I hypothesize that longer LBs will significantly contribute to higher oral proficiency scores, potentially more so than shorter LBs, due to their ability to enhance processing efficiency and fluency.

5.1 Participants, Presentation Tasks, Proficiency Scores, LB Measurements and Scoring

The current chapter used the same participants as in Chapter 3 and 4 (i.e., 150 L2 learners taking a pre-session course at Queen Mary University of London, see Section 3.1 for a detailed description of their L1 backgrounds and speaking proficiency level groupings; research ethics approval number: QMREC2414a). The same oral presentation tasks, proficiency scores, LB measurements and scoring are also used in the current chapter (readers are referred to Sections 3.2 to 3.4 for detailed descriptions). The only difference is that the current chapter used the raw

scores instead of banded scores as the measure of oral proficiency. Using raw scores allows for a finer-grained analysis through robust multiple regression and other advanced statistical techniques, like dominance analysis and random forests analysis, enhancing statistical power and predictive accuracy while avoiding information loss that can occur with banded scores.

5.2 Statistical Analyses

I analyzed six text-external n-gram (two- and three-word) measures and two text-internal refined n-gram (three-to-five-word) measures. To explore my research question, that the use of LBs of varying length can predict oral presentation raw scores, I selected regression analysis. Initially, I examined the assumptions underlying regression models. The presentation raw scores variable yielded a Shapiro-Wilk p -value of .04, showing a significant deviation from a normal distribution. To identify influential outliers within the data, Cook's distance was employed with a conservative threshold of $4/n$, facilitating the assessment of the impact of individual data points on the regression model. Several data points emerged as influential outliers, surpassing the Cook's distance threshold. Their presence implies a potential influence on the estimated regression coefficients (see [Appendix H](#) and Appendix I for detailed results of these checks). Subsequently, guided by Larson-Hall (2016), I conducted a robust regression using MM estimation with the “rlm” function in the MASS package in R (R Development Core Team 2019). I chose the “rlm” function because it aptly accommodates datasets with non-normal distributions and outliers. The regression model harnessed the eight predictor variables (PVs) and the one criterion variable (presentation raw scores). I then examined the relative importance of each PV through dominance analysis in combination with random forest analysis using the Boruta algorithm (see Section 4.3 in Chapter 4 for a detailed explanation and justification of these combined methods). In what follows, I present the descriptive statistics first, then I report

the robust regression model in combination with DA and random forests analysis to address my research question.

5.3 Results

Table 5.1 shows the descriptive statistics for the presentation raw scores and different LB measures used in this chapter's analysis. The table includes the mean (M), standard deviation (SD), median, minimum, and maximum values for each measure. These statistics summarize the distribution and central tendency of the presentation scores and n-gram measures, highlighting the variability and range of the data.

I created a boxplot to visualize the distribution of presentation scores across different L1 groups (Figure 5.1). The boxplot provides a clear display of the central tendency and variability within each group. Although a Kruskal-Wallis test did not find statistically significant differences in presentation scores across L1 groups (Kruskal-Wallis chi-squared = 29.825, $df = 19$, p -value = 0.054), the boxplot offers valuable insights into the data distribution. It shows that most L1 groups have similar median scores, but there is considerable variability within some groups. The blue dots represent individual data points, highlighting the spread of scores within each group.

Table 5.2 shows the findings from the robust regression and dominance analyses with the criterion variable being presentation raw scores (see Appendix J for R code and detailed results of the dominance analysis). The robust regression analysis reveals multiple predictors impacting the presentation raw scores. Bigram MI displayed a strong positive association ($b = 14.38$, 95% CI [8.01, 20.76], $t = 4.42$), accounting for a dominant 58.63% of the variance in presentation raw

scores. This implies that learners using more bigrams with higher MI scores tend to score higher on their presentations, holding other variables constant. Examples of bigrams with high MI scores produced by high-scoring participants include *carbon dioxide*, *Kyoto Protocol*, and *global warming* (for more examples, see Appendix L). Three-to-five-word usage also showed a significant positive influence on the model ($b = 0.53$, 95% CI [0.06, 0.99], $t = 2.22$), contributing to 18.80% of the variance in presentation raw scores. This suggests that individuals employing specific three-to-five-word phrases achieve higher presentation raw scores. Other predictors like trigram frequency, trigram proportion, and trigram MI indicated lesser influence, each contributing less than 4% to the total dominance weight.

Table 5.1

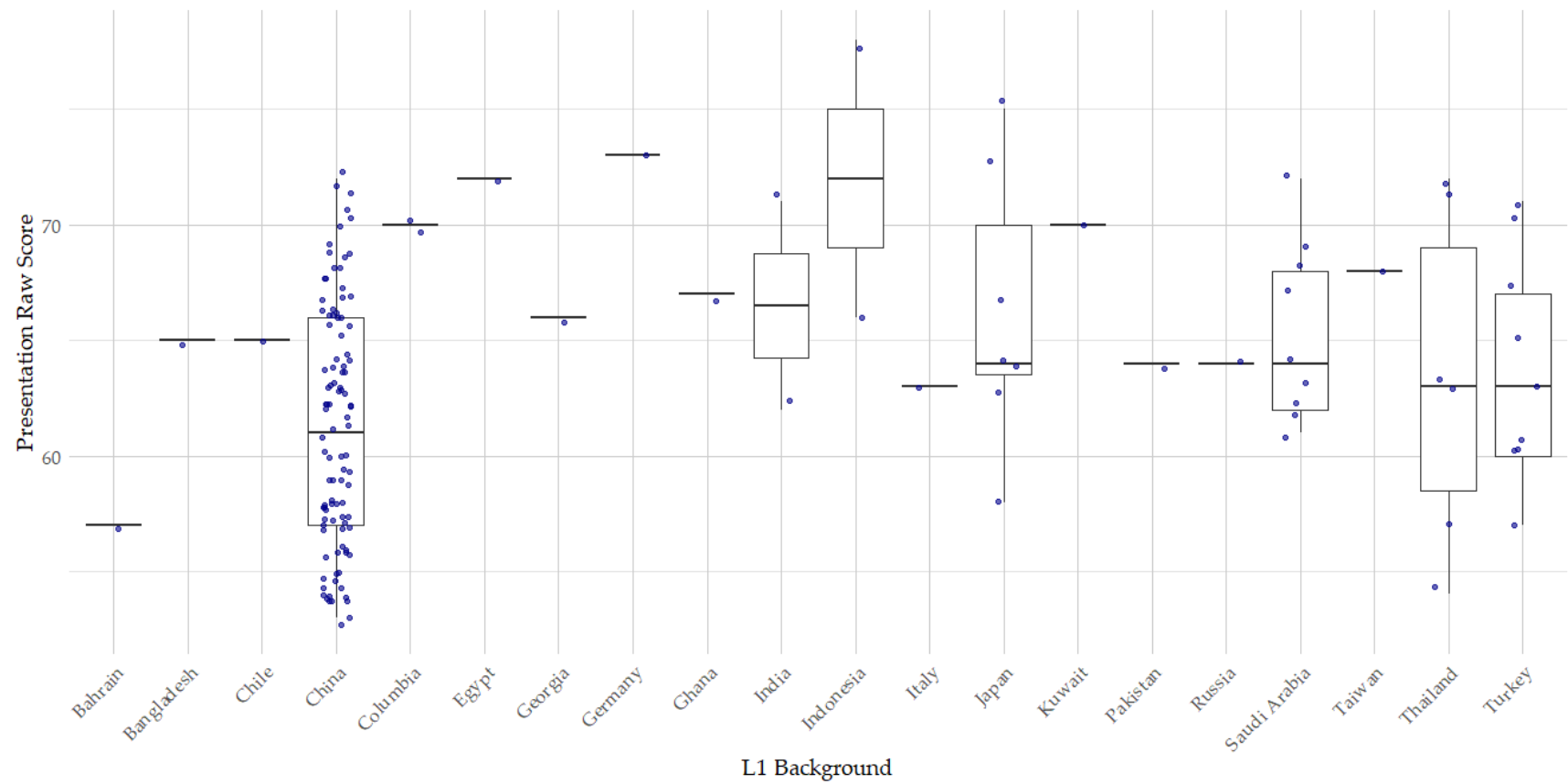
Descriptive Statistics for Presentation Scores and N-Gram Measures

Measure	<i>M</i>	<i>SD</i>	Median	Minimum	Maximum
Presentation raw scores	62.71	5.53	63.00	53.00	78.00
Bigram log frequency	1.32	0.12	1.31	1.04	1.73
Bigram proportion	0.44	0.07	0.44	0.26	0.63
Bigram association MI	1.54	0.17	1.54	1.13	2.09
Trigram log frequency	0.68	0.13	0.67	0.41	1.07
Trigram proportion	0.12	0.04	0.12	0.04	0.26
Trigram association MI	2.53	0.25	2.54	1.91	3.21
Three-to-five-word usage	10.29	5.10	10.00	1.00	27.00
Three-to-five-word MI	132.39	73.57	117.97	19.87	399.56

Note. *M* = mean. *SD* = standard deviation. MI = mutual information.

Figure 5.1

Presentation Scores by L1 Background



In Figure 5.2, the dominance weights are presented in descending order, starting from the predictor variable with the highest weight (bigram MI) and ending with the one with the lowest weight (trigram frequency). Figure 5.2 helps us visualize each predictor's relative importance, highlighting bigram MI as the most important predictor among all variables in the current chapter's analysis. Figure 5.3 shows a variable importance plot derived from random forests analysis using the Boruta algorithm. The Boruta results confirm the importance of five variables: Bigram MI, three-to-five-word usage, trigram proportion, three-to-five-word MI, and bigram proportion. Two attributes, trigram frequency and trigram MI, were confirmed as unimportant, while bigram frequency remained tentative. Detailed results and R code are available in Appendix J.

Table 5.2

Robust Regression and Dominance Analyses (Criterion: Presentation Raw Scores)

Predictor	<i>b</i>	95% CI	<i>SE</i>	<i>t</i>	Dominance weight
Intercept	39.02	16.45, 61.58	11.51	3.39	
Bigram frequency	8.10	−4.17, 20.37	6.26	1.29	0.008 (3.41%)
Bigram proportion	−12.09	−39.57, 15.39	14.02	−0.86	0.006 (2.64%)
Bigram MI	14.38*	8.01, 20.76	3.25	4.42	0.129 (58.63%)
Trigram frequency	−3.07	−13.10, 6.95	5.11	−0.60	0.003 (1.23%)
Trigram proportion	4.15	−58.81, 67.11	32.12	0.13	0.007 (3.23%)
Trigram MI	−2.05	−6.24, 2.14	2.14	−0.96	0.007 (3.28%)
Three-to-five-word usage	0.53*	0.06, 0.99	0.24	2.22	0.041 (18.80%)
Three-to-five-word MI	−0.02	−0.05, 0.01	0.02	−1.25	0.019 (8.78%)
Total					0.220 (100%)

Note. *N* = 150. MI = mutual information. CI = confidence interval. *SE* = standard error. *Results are marked significant if the 95% CI excludes zero.

Figure 5.2

Dominance Weights in Descending Order (Criterion: Presentation Raw Scores)

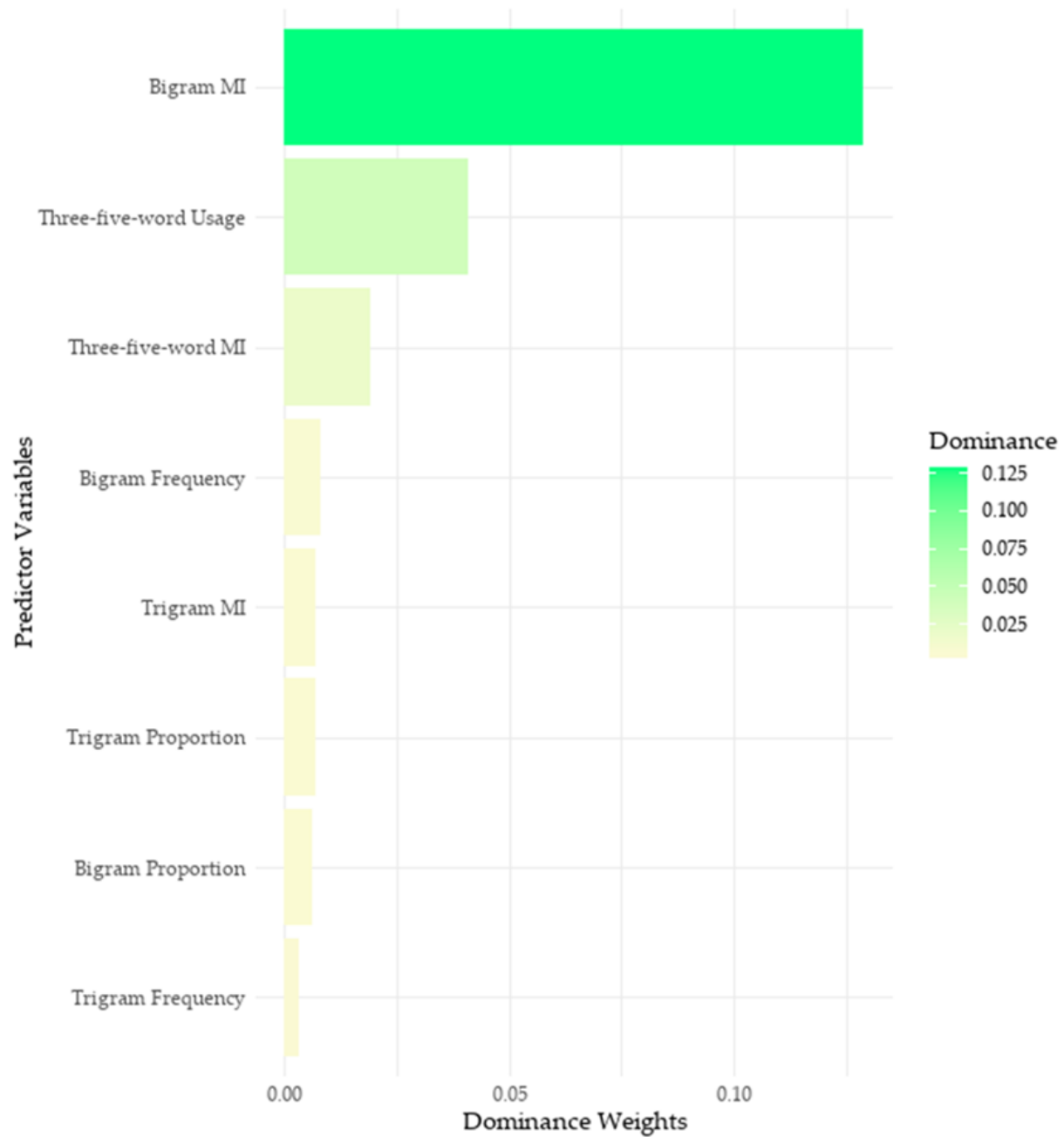
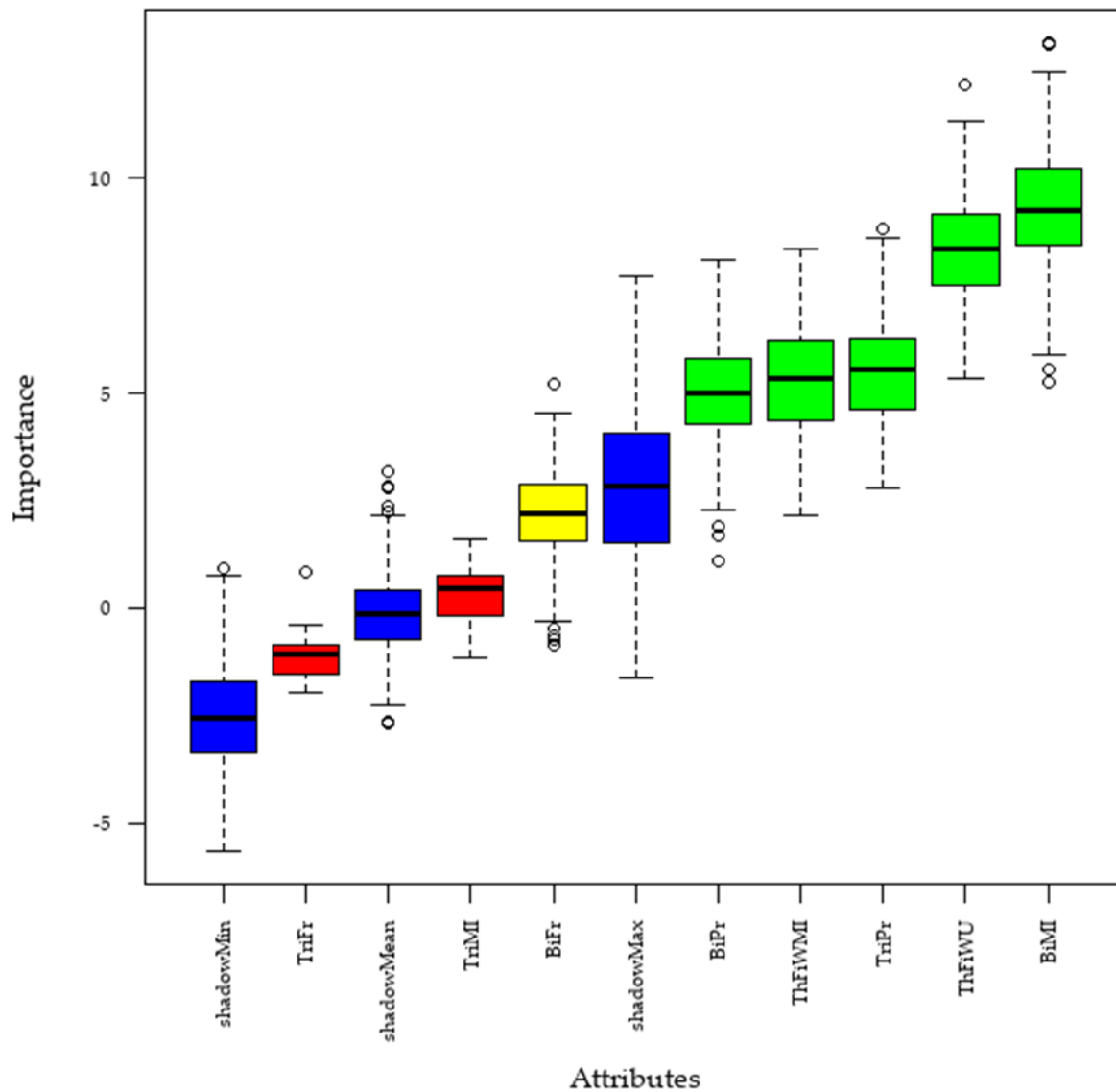


Figure 5.3

Variable Importance Plot From Random Forests Analysis (Criterion: Presentation Raw Scores)



Note. In the Boruta algorithm box plots, green indicates “confirmed” variables, red “rejected” variables, yellow “tentative” variables, and blue “randomized shadow” variables, which serve as a reference to assess the importance of the original variables against random chance.

In addition to these analyses, a performance comparison was conducted to evaluate the importance of bigram MI in predicting presentation scores. This comparison examined two models: Model A, which includes bigram MI, and Model B, which excludes it. The findings highlight the substantial impact of Bigram MI on model performance, confirming that bigram MI is an essential parameter for accurately predicting presentation scores. Full details of this comparison, including performance metrics such as adjusted R^2 and AIC (Akaike information criterion) are provided in [Appendix K](#).

To summarize the results presented in this section, the Boruta analysis, along with dominance and robust regression analyses, together highlight the substantial influence of bigram MI and three-to-five-word usage on presentation scores. These consistent findings across different analytical techniques underscore the importance of specific lexical choices, especially bigram MI, in determining speaking performance, thus offering a triangulated insight into how LBs impact speaking proficiency.

5.4 Discussion

The overarching aim of this chapter has been to explore the impact of LB usage on oral proficiency using a broad dataset and a novel LB refinement technique. My research question sought to determine the extent to which LB usage of varying lengths predicts raw presentation scores. The robust regression and dominance analyses identified bigram MI and three-to-five-word LB usage as significant predictors of presentation success. The consistency of these findings across different approaches, including random forests (Boruta algorithm), reinforces their importance. The following discussion therefore focuses on these two LB measures and addresses the research question in relation to theoretical frameworks, previous findings, and my hypothesis.

My finding that learners who effectively use text-external high-MI bigrams achieve higher proficiency levels are consistent with Pawley and Syder's (1983) lexical phrases theory. They argued in their seminal work that L1-like selection of expressions, which includes collocations and idiomatic phrases captured by measures like bigram MI, is an integral aspect of speaking a language fluently. Their lexical phrases theory posits that a speaker's proficiency is marked by the ability to produce sequences of words that L1 speakers recognize as familiar, suggesting that L1-like proficiency is to some extent a function of memory for lexically stored sequences rather than just rules for combining words. The findings reported in the current chapter align with and extend this notion by suggesting that not just the presence of LBs, but also their "rare exclusivity", which is the main practical effect of the MI score (Gablasova et al., 2017, p. 10), differentiates more proficient speakers from their less proficient counterparts. Rare exclusivity means that two words often appear together in certain contexts but are not commonly found together in everyday language. In other words, when these two words do show up together, it is in a specific or specialized way that makes their combination stand out. This suggests that the speaker has a more advanced and precise understanding of the language because they are using these less common word pairs effectively. It is like knowing a secret handshake: not everyone uses it, but those who do are part of a particular group that knows something special. Similarly, using these rare word combinations signals higher proficiency in language. More examples of high MI-score bigrams from the current dataset include *21st century*, *intellectual property*, and *human beings* (for more examples, see [Appendix L](#)). The empirical evidence outlined in Chapters 3 and 5 supports the view that language proficiency, especially in productive skills such as speaking, involves the use of linguistically sophisticated and exclusive language patterns that are characteristic of L1 speaker usage. The findings reported

here suggest that, as L2 proficiency increases, so does the use of high MI-score bigrams, indicating a narrowing gap in collocational usage between L2 and L1 speakers. These findings highlight the importance of integrating specific types of LBs, particularly those with high MI, into language assessment and instruction, as they reflect not just lexical range but the use of more sophisticated and strongly associated lexical patterns that are characteristic of higher proficiency.

The results from the study reported in the current chapter also align with Kormos's (2006) extension of Levelt's (1989) model to L2 speakers by illustrating the formulation stage's critical role in speech production. Specifically, the increase in text-external bigram MI and longer text-internal LB usage with proficiency suggests a more efficient lexical selection process among higher proficiency learners, echoing the notion that advanced speakers can activate and employ appropriate lemmas with greater ease. This efficiency likely contributes to the freeing up of cognitive resources for other processing needs, which is essential for achieving fluency. It provides empirical support for the theory that a larger MWS repertoire enables L2 learners to enjoy a processing advantage by reducing the demands on cognitive resources, facilitating more fluent and sophisticated language production. In addition, my finding that bigram MI and longer (three-to-five-word) LB usage predict higher scores echoes the tenets of usage-based theory, specifically as it relates to language proficiency. Usage-based theory suggests that language learning is exemplified through the increased use and understanding of recurrent patterns or constructions in language, which are acquired through exposure and use (Bybee, 2006; Ellis, 2002). Bigram MI within the usage-based framework can be seen as a proxy for the type of patterned use that is a hallmark of language proficiency. Bigram MI measures the exclusivity and specificity of the co-occurrence between two words, reflecting how their joint appearance

significantly exceeds what would be anticipated based on their independent distribution across texts. This measure not only highlights word pairs that share a unique connection but also underscores the meaningful combinations that are preferentially utilized by more proficient speakers. High bigram MI scores in the current chapter's analysis suggest that more proficient language users are more likely to employ exclusive and information-rich lexical sequences. In a sense, bigrams with high MI scores are those that are entrenched in the linguistic repertoire of proficient speakers, reflecting common usage patterns.

My findings, highlighting the predictive power of bigram MI for proficiency levels, are also consistent with key insights from corpus-based SLA studies that have explored collocational usage differences between L1 and L2 writers. These findings, while encouraging, are for writing while the current thesis considers speaking. Notably, research in this area has consistently found that L1 users tend to produce collocations with higher MI-score values compared to L2 users (Durrant and Schmitt, 2009; Ellis et al., 2015; Schmitt, 2012). Schmitt (2012, p. 6) has emphasized this disparity, concluding that the absence of high MI collocations is a distinctive marker of non-L1 versus L1 production. My findings also echo those of previous research comparing bigram usage among L2 writers across different proficiency levels. Granger and Bestgen (2014) compared the use of collocations by intermediate and advanced L2 writers, finding that essays scoring in the advanced range of the CEFR had higher proportions of bigrams with high MI scores than essays scoring in the intermediate range. These convergent findings emphasize the significance of the MI-score as a measure for distinguishing between the collocational choices of L1 and L2 users, as well as between intermediate and advanced L2 users, highlighting its utility and growing importance in language research. The differential use of high MI-score collocations between L1 and L2 users and between intermediate- and higher-

proficiency L2 users highlights a key aspect of language proficiency: the ability to employ exclusive, less frequent word combinations beyond common collocations.

However, a notable divergence exists between my finding on bigram MI and the existing literature on spoken production. Specifically, Tavakoli and Uchiyara (2020) reported a general decrease in MI with rising proficiency, suggesting greater variability and diversity in LB combinations as learners become more proficient. Conversely, my results showed an increase in bigram MI with proficiency levels (except for a slight decrease in trigram MI). Although both studies used TAALES software to measure bigram MI text-externally, the difference in findings could be due to the distinct datasets, methodologies, or proficiency level groupings employed in the two studies (see Figure 3.9 in Chapter 3 for an infographic showing how IELTS and CEFR proficiency levels align). The divergence can be explained by differences in tasks, as Tavakoli and Uchiyara used spontaneous spoken tasks, while my study focused on structured oral presentations, potentially encouraging more frequent use of fixed expressions and associated bigrams. Another factor that may contribute to the divergent findings is the difference in sample sizes. Tavakoli and Uchiyara had relatively fewer participants at each proficiency level (Low B1 = 11, High B1 = 14, B2 = 17, C1 = 14), while my study involved a much larger and more evenly distributed sample, with 50 participants at each proficiency level (IELTS 6.5-7.5). The larger sample size in my study could have provided more statistical power, allowing for a clearer picture of the relationship between bigram MI and proficiency levels. This divergence highlights the importance of task types, sample sizes, and proficiency categorizations when analyzing LB usage. Further research is needed to explore how these variables interact to influence MI scores and L2 oral proficiency. The current analysis contributes a novel perspective by documenting the pattern of bigram MI increase as learners progress. Such contrasting results underscore the need

for further research to explore the intricate dynamics of bigram MI use and proficiency levels, hopefully enriching our understanding of effective language use by L2 speakers.

Regarding my hypothesis that the use of LBs of all lengths would positively correlate with raw oral presentation scores, the results show a mixed but insightful picture. The robust regression and dominance analysis, corroborated by the Boruta algorithm, strongly support the hypothesis in the case of bigram MI and three-to-five-word usage. Bigram MI, which emerged as the most powerful and dominant factor, accounts for a significant portion of the variance in presentation scores. This likely reflects the importance of conciseness in effective oral communication, where the use of meaningful and compact word pairs (bigrams with high mutual information) enables speakers to convey their points succinctly, engaging the audience effectively. The positive influence of three-to-five-word usage also highlights the value of specificity. These slightly longer LBs, which I measured text-internally, are critical for articulating complex or specific ideas in a clear and focused manner, enhancing the speech's informativeness without becoming overly wordy. However, my hypothesis finds less support with other types of LBs. Bigram proportion, for instance, showed a negative non-significant association with presentation scores, and trigram-related measures (frequency and MI) had less influence. This suggests that, while certain LBs positively correlate with presentation performance, not all LB types show the same level of predictive power. These findings partially validate my hypothesis, underscoring the significant predictive power of specific types of LBs, particularly bigram MI, in relation to raw presentation scores.

In summary, although my hypothesis that all lengths of LBs would positively predict oral proficiency scores was only partially supported, the analysis presented in the current chapter contributes insights into the specific lexical features (bigram MI in particular) that are most

predictive of higher proficiency scores. The current chapter's findings enhance the existing literature by providing a nuanced picture of how LB usage evolves at higher levels of language learning. My findings also contribute to research methodology by showing the effectiveness of combining robust statistical techniques with machine learning algorithms like Boruta to strengthen the robustness of educational research. The convergence of evidence from different analytic techniques strengthens the reliability of the findings in this chapter. Using robust regression helped to mitigate the influence of outliers, dominance analysis provided insights into the relative importance of predictors, and the Boruta algorithm offered an additional layer of confirmation about which LB attributes are truly influential. Such methodological triangulation strengthens the study's credibility, allowing for more confident conclusions regarding the predictive power of LB usage for speaking proficiency.

5.5 Limitations and Suggestions for Future Research

Although the results are encouraging, there is still room for future research to overcome the limitations of this chapter. One notable area that was not explored is the potential influence of different presentation topics on LB usage and presentation scores. This unexplored area might shed further light on findings such as the increased use of high MI-score bigrams among higher-scoring learners. For example, Gablasova et al. (2017) emphasized that the MI-score favors less frequent and more specialized combinations, such as technical terms, which can vary significantly depending on presentation topic. They noted that "the technical nature of a specific topic can influence the strength of collocations as measured by the MI score," often revealing hidden patterns when only generalized MI-score rankings are considered across whole corpora (Gablasova et al., 2017, p.20). This observation could be highly pertinent to my research, indicating that the frequency and variety of high MI-score word pairs I have observed might not

only reflect the speakers' language proficiency but also the specific vocabulary requirements of their chosen topics. Technical presentations, for instance, might necessitate the use of specialized vocabulary, increasing the MI scores of the collocations used. Future studies could benefit from incorporating an analysis of presentation topics, examining how the choice of topic influences the use of high MI-score bigrams across proficiency levels. This line of inquiry is included in the ensuing Chapter 6, where I analyze how presentation topics influence the use of high MI-score bigrams across proficiency levels. Further potential limitations of the current chapter's analysis are discussed in Chapter 7 (Section 7.2).

5.6 Pedagogical Implications and Suggestions

The findings support including bigram MI and longer LBs in assessing and teaching English proficiency, especially in speaking. They highlight the importance of focusing on the quality of LBs, particularly high MI bigrams, rather than just their quantity. This insight can inform instructional design to enhance learners' speaking proficiency. Here are a few practical steps for implementing these findings in the classroom:

1. *Select relevant texts:* Choose texts that align with students' interests and proficiency levels.
2. *Identify high MI bigrams:* Use Tom Cobb's "Phrase Extractor" tool (<https://www.lex tutor.ca/multiwords/phrase/>) to extract high MI bigrams from the selected texts. These bigrams often serve as the foundation for longer LBs and expressions, making them a useful starting point.
3. *Practice and raise awareness:*
 - a. *Develop exercises:* Create exercises that target high MI bigrams and longer LBs.

- b. Highlight during activities: Have students notice and highlight high-MI bigrams and longer LBs during reading and listening activities.
- c. Assessment criteria: Include criteria related to use of high-quality LBs in speaking (and writing) grading rubrics.

By integrating and building on these steps, educators can effectively enhance students' proficiency and awareness of high-quality LBs in English.

5.7 Conclusion

This study has provided insights into the patterns of LB usage that correspond to higher oral presentation scores. By extending beyond the scope of Tavakoli and Uchihara (2020), it has illuminated the complex relationship between proficiency and both the frequency and complexity of LBs. Through detailed statistical analysis, it has established the link between linguistic sophistication and presentation scores, emphasizing the importance of both shorter and longer LBs in speaking proficiency. The current chapter's findings make a persuasive case for the critical role of LB usage, particularly bigram MI, in predicting English language proficiency and presentation performance. These findings enhance our understanding of the relationship between lexical choice and speaking performance, offering practical insights for language assessment and teaching. This chapter has highlighted bigram MI (and to some extent three-to-five-word usage) as a reliable indicator of English language scores on oral academic presentations. This is consistent with linguistic theories first put forward in the 1980s (Pawley & Syder, 1983; Levelt, 1989) that emphasize the efficient use of pre-formed chunks of language as a requirement of fluent speech and therefore a hallmark of proficiency. The implications for language teaching are

considerable, suggesting that educators should include a focus on teaching strategies that improve learners' awareness and command of high-MI bigrams.

In the upcoming chapter, I will explore how the choice of presentation topics influences the diversity of high MI-score bigrams and whether these factors correlate with oral presentation scores. The chapter will examine the notable area discussed in Section 5.5 by exploring whether specific topics elicit more sophisticated language use, providing a deeper understanding of the contextual factors that impact lexical sophistication and presentation performance. By employing a mixed-methods approach, I aim to uncover the nuanced interactions between topic choice, language complexity, and learner proficiency, further enriching our understanding of effective language learning strategies.

Chapter 6: Examining the Extent to Which Topic Choice Influences Lexical Sophistication

Chapter 5 demonstrated that bigram MI and three-to-five-word usage are significant predictors of presentation scores, with bigram MI being the strongest predictor. This finding suggests that the use of high MI-score bigrams significantly enhances speaking performance. However, the potential influence of different presentation topics on LB usage was not explored. Building on this, the current chapter aims to examine whether the choice of presentation topic influences the diversity and frequency of high MI-score bigrams used by L2 English speakers, and whether this influence varies with proficiency scores. Specifically, it investigates if certain topics, represented by predefined pathways (1. Humanities and Social Science [HSS], 2. Law, and 3. Science, Technology, Engineering, and Math [STEM]) and sub-topics with these pathways, affect the use of sophisticated lexical bundles.

The research question guiding the current chapter is: Does the choice of presentation topic influence the diversity and frequency of high MI-score bigrams used by L2 English speakers, and does this influence vary with proficiency scores? This question can be broken down into two specific sub-questions:

1. Does the choice of presentation topic influence the diversity and frequency of high MI-score bigrams used by L2 English speakers?
2. Does this influence vary with proficiency scores?

My hypothesis is that presentations on more specialized and technical topics, such as those found in the STEM and Law pathways, will show higher diversity and frequency of high MI-score bigrams compared to broader topics, and this effect will be more pronounced at higher

proficiency levels. This hypothesis is based on the idea that specialized topics require more exclusive and sophisticated language, which is reflected in higher MI-score bigrams.

6.1 Participants and Data

The current chapter uses the same participants as in Chapters 3-5 (i.e., 150 L2 learners taking a pre-session course at Queen Mary University of London, see Section 3.1 for a detailed description of their L1 backgrounds and speaking proficiency level groupings; research ethics approval number: QMREC2414a). The same oral presentation tasks, proficiency scores, and scoring are also used (refer to Sections 3.2 to 3.5 for detailed descriptions). In the same way as Chapter 5, the current chapter used the raw scores instead of banded scores as the measure of oral proficiency for the quantitative analysis. However, for the qualitative analysis, I refer to the banded scores for clarity. The LB measurement used in the current chapter is bigram MI scores, since bigram MI scores proved to be the strongest predictor of presentation scores (see Table 5.2 in Chapter 5). It could thus prove interesting to examine bigram MI in relation to presentation topics.

6.2 Oral Presentation Tasks and Topics

As described in Chapter 3 (Section 3.2), the oral presentation tasks covered a wide range of sub-topics under the overarching theme of globalization, which had been the focal point of the participants' 5-week academic English course. Although globalization served as the central theme, participants were encouraged to explore this broad topic through various lenses, ranging from economic and legal perspectives (e.g., how globalization affects international trade law) to more specialized areas (e.g., acoustic telemetry in fisheries). This range of sub-topics allowed participants to explore areas related to their academic interests and expertise.

The students came from a wide range of disciplines, including humanities, law, science, technology, engineering, and mathematics. They were grouped into three pre-defined pathways at Queen Mary University of London: (1) HSS, (2) Law, and (3) STEM. The Law pathway represents a coherent and specialized subject area, while the HSS and STEM pathways encompass a broader mix of related topics. These pathways provide a structured way to categorize a wide range of topics. Therefore, I used pathways as a proxy for presentation topics as a starting point of analysis. This approach allowed me to establish a foundational understanding of how thematic categories influence linguistic complexity.

6.3 Selecting Representative Cases for Qualitative Analysis

To explore the context in which high MI-score bigrams are used within each pathway and across different proficiency levels, I created a list of sub-categories within each pathway based on keywords from the presentation titles and transcripts (the full list of subcategories is shown in Table 6.2). Since some presentations discussed more than one topic, I tagged them with each topic discussed. I then selected representative cases for a qualitative analysis based on the following criteria:

1. Representation across IELTS levels: To capture a diverse range of language proficiency, at least one participant from each IELTS level (6.5, 7, 7.5) was selected across the pathways.
2. Diversity of mean MI scores: Mean MI scores were categorized into three levels based on quartiles: Low MI scores (below the 25th percentile [5.83]), moderate MI scores (between the 25th percentile [5.83] and the 75th percentile [7.09]), and high MI scores (above the 75th percentile [7.09]). The qualitative analysis focused on low mean MI

scores from the HSS pathway, moderate mean MI scores from the Law pathway, and high mean MI scores from the STEM pathway. These specific choices complement the quantitative results which showed significantly higher MI scores for STEM compared to HSS, for Law compared to HSS, and for STEM compared to Law (see ANOVA results in Table 6.3).

3. Variety of topics: Participants were chosen from different sub-categories within each pathway to ensure a diverse analysis.

Based on these criteria, the following participants were selected:

- HSS Pathway: Three participants from “globalization and economic development” at IELTS level 6.5 with low mean MI scores (below the 25th percentile, 5.83)
- Law Pathway: One participant from “trade and economic regulations” at IELTS level 7 with a moderate mean MI score (between the 25th percentile, 5.83, and the 75th percentile, 7.09)
- STEM Pathway: Two participants from “healthcare” at IELTS level 7.5 with high mean MI scores (above the 75th percentile, 7.09)

This selection ensures a balanced and illustrative qualitative analysis, providing insights into the use of high MI-score bigrams across different pathways and proficiency levels, and helping to address my second research sub-question.

6.4 Selecting an MI Score Threshold

To compare bigram use across pathways and subtopics, I initially selected an MI-score cutoff of 3, aligning with previous studies (e.g., Ballance, 2022; Webb & Chang, 2022) that have

justified the use of MI scores of 3 or higher to distinguish meaningful collocations from less significant word pairs. Researchers (e.g., Church & Hanks, 1990) have observed that word pairs with $MI > 3$ often represent meaningful or contextually notable combinations, making them particularly relevant for linguistic analysis. However, my initial analysis showed no significant differences across pathways for an MI-score threshold of 3 (see ANOVA results in Table 6.3). I therefore examined MI-score thresholds of 4 and 5 aiming to capture significant differences among collocations at higher levels of association strength and specificity across pathways. My follow-up analyses showed significant differences only at the MI score cutoff of 5 (see Table 6.3), so I focused on MI scores of 5 and higher throughout the current chapter. In the experiment detailed in this chapter, an MI-score threshold of 5 is justified as higher MI score word pairs are more exclusive and specialized and may highlight the use of domain-specific terminology in technical presentations, such as some of those in the STEM pathway. The cutoff of 5 may also better differentiate proficiency levels, as more proficient speakers tend to use more sophisticated language. This approach keeps the analysis focused on the most important and relevant results for the research sub-questions addressed in the current chapter.

6.5 Statistical Analyses

The assumptions for a parametric one-way ANOVA were tested for normality using the Shapiro-Wilk test and for homogeneity of variances using Levene's test. The Shapiro-Wilk tests showed significant deviations from normality for all MI score thresholds (3-5) across all pathways ($p < 0.001$). Levene's test showed that the assumption of homogeneity of variances was not violated for the MI-score threshold of 3 ($p = 0.2343$) but was violated for thresholds of 4 ($p = 0.01655$) and 5 ($p < 0.001$) (see [Appendix M](#) for detailed results of these assumption checks). Given these violations, robust ANOVA methods were considered appropriate for the

analysis. To address research sub-question one, I conducted a series of one-way ANOVAs with 20% trimmed means and bootstrapping, mirroring the methodology outlined in Chapter 3 (Section 3.5). Robust methods, such as trimmed means and bootstrapping, are effective in handling outliers and skewed distributions (Larson-Hall, 2015; Loewen & Plonsky, 2016). Specifically, the “`t1waybt`” and “`mcppb20`” functions from the “WRS2” package in R were used for the analysis (Mair & Wilcox, 2019).

To address research sub-question two, I conducted a robust regression analysis to examine the interaction between presentation topic and proficiency scores. This analysis used the same robust regression method employed in Chapter 4, utilizing MM estimation with the “`rlm`” function from the MASS package in R (R Development Core Team, 2019). As previously discussed (in Chapter 4, Section 4.3), this approach is guided by Larson-Hall (2015) and is well-suited for datasets with non-normal distributions and outliers. The regression model focused on the broad pathways of presentation topics. Interaction terms were included to assess whether the influence of these broad pathways on MI scores varied with proficiency scores. This model allowed me to determine if the effect of one variable (pathway) on the outcome (MI score) changed depending on the level of another variable (presentation raw score).⁶

In the following, I present the results in three sections. First, I explore the influence of presentation topic on the diversity and frequency of high MI-score bigrams, including descriptive statistics, robust ANOVA, and post-hoc test results. Second, I examine the interaction between presentation topics and proficiency scores using robust regression analysis to determine how proficiency impacts MI scores across different pathways. Third, I provide a qualitative analysis of selected participants’ transcripts, focusing on the context in which high MI-score bigrams (MI scores above 5) are used across different pathways and proficiency levels.

6.6 Results

6.6.1 Influence of Presentation Topic on the Diversity and Frequency of High MI-Score

Bigrams

To understand the distribution of MI scores across different pathways, I computed descriptive statistics for MI scores at the threshold of 5, as shown in Table 6.1.⁷ Table 6.1 shows that the mean MI scores increase from HSS to Law to STEM, with STEM showing the highest mean MI score (6.86). This trend suggests that presentations in the STEM pathway tend to use higher MI-score bigrams compared to those in HSS and Law. Figure 6.1 illustrates the distribution of MI scores across pathways for the bigram MI score threshold of 5.

Table 6.1

Descriptive Statistics for Bigram MI Scores Across Pathways (Threshold = 5)

Pathway	<i>M</i>	<i>SD</i>	Median	Minimum	Maximum
HSS	6.24	1.06	5.93	5	11.5
Law	6.47	1.26	6.44	5.01	11.2
STEM	6.86	1.59	6.52	5.01	11.4

To understand the distribution of MI scores in the subcategories within each pathway, I computed descriptive statistics, as shown in Table 6.2. This table illustrates the variation in mean MI scores across different subcategories within each pathway. For instance, the mean MI scores for subcategories within the HSS pathway range from 5.83 (globalization and armed conflict) to 8.46 (environment and sustainability). Similarly, the Law and STEM pathways also exhibit a broad range of mean MI scores, showing diverse usage of high MI-score bigrams across different topics.

Figure 6.1

Box Plot of MI Scores Across Pathways (Threshold = 5)

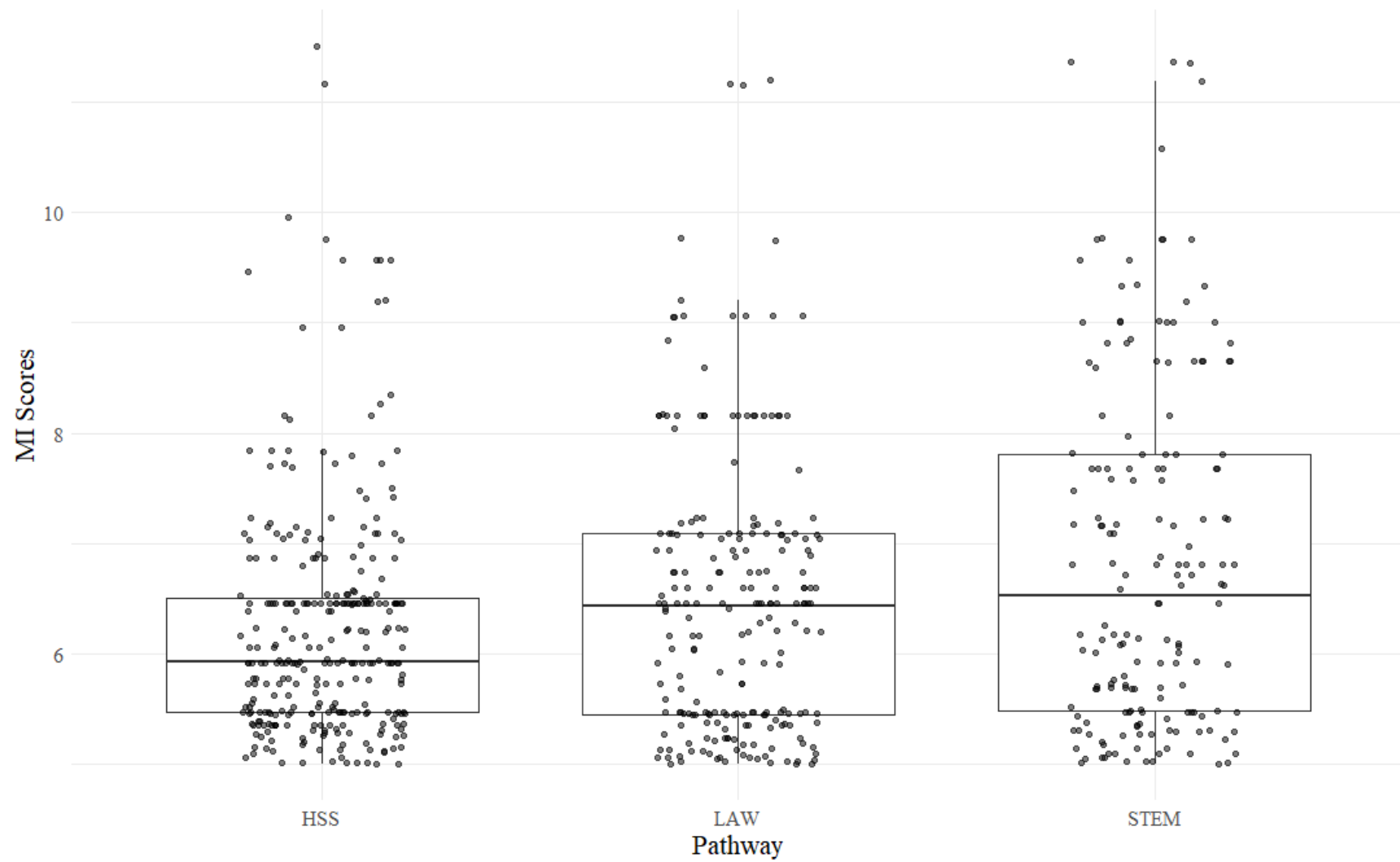


Table 6.2 also shows the number of participants at each IELTS proficiency level within each subcategory (see the three columns on the right side of the table). Notably, the “globalization and economic development” subcategory in the HSS pathway has a relatively high number of participants at IELTS 6.5 (13 participants), showing a concentration of lower proficiency levels in this topic. In contrast, the STEM pathway shows a higher number of participants at IELTS 7 and 7.5, especially in the “healthcare” subcategory, which has 10 participants at IELTS 7 and 4 participants at IELTS 7.5. This suggests that higher proficiency students are more engaged in certain STEM topics. These trends highlight the distribution of linguistic proficiency among the participants, which may influence the complexity of their language use.

Table 6.3 presents the robust ANOVA and post-hoc test results for MI scores across pathways at MI thresholds of 3, 4, and 5. The ANOVAs showed significant differences at the MI score threshold of 5, $F = 6.4647$, $p = .002$, but not for thresholds of 3 or 4. Post-hoc tests using a two-step method based on 20% trimmed means and bootstrapping with 10,000 samples revealed that MI scores were significantly higher for STEM compared to HSS ($p = .027$) and for Law compared to HSS ($p < .001$). The difference between STEM and Law was not significant ($p = .054$) (see [Appendix N](#) for detailed post-hoc results, including effect sizes and confidence intervals). According to the descriptive statistics reported in Table 6.1, the STEM pathway has the highest overall MI scores, followed by the Law and HSS pathways.

6.6.2 Influence of Proficiency Scores on the Relationship Between Topic Choice and Lexical Sophistication

A robust regression analysis was conducted to examine the interaction between presentation topic (pathway) and proficiency scores (presentation score) on MI scores. The

model included interaction terms to assess whether the influence of presentation topics on MI scores varied with proficiency scores. The results, as summarized in Table 6.4, show that at lower proficiency levels, the MI scores for the Law ($t = -3.46$) and STEM ($t = -6.01$) pathways are lower compared to the reference pathway, HSS. However, the positive interaction terms for pathway (Law) and presentation score ($t = 3.58$) and pathway (STEM) and presentation score ($t = 6.38$) show that as proficiency scores increase, the MI scores for Law and STEM pathways increase at a higher rate compared to the HSS pathway. These findings suggest a significant interaction effect where proficiency scores are more strongly associated with increases in MI scores for the Law and STEM pathways than for the HSS pathway. Specifically, while presentations in Law and STEM pathways start with lower MI scores at lower proficiency levels, they benefit more from higher proficiency scores in terms of the diversity and frequency of high MI score bigrams used. This implies that proficiency levels have a differential impact on lexical sophistication across pathways, with a more pronounced effect observed in the Law and STEM pathways.

Figure 6.2 illustrates these interactions by plotting the predicted MI scores against the presentation scores for each pathway. The plot clearly shows that, although the Law and STEM pathways start with lower MI scores, their scores increase more rapidly with higher proficiency scores compared to the HSS pathway. This visual representation highlights the significant interaction effects identified in the robust regression analysis.

Table 6.2*Descriptive Statistics of MI Scores Across Pathways and Subcategories (Threshold = 5)*

Pathway	Subcategory	M	SD	Median	Min	Max	No. at IELTS 6.5	No. at IELTS 7	No. at IELTS 7.5
HSS	Environment and sustainability	8.46	1.13	8.16	6.58	9.57	1	0	0
HSS	Globalization and corporate social responsibility	6.47	1.44	5.92	5.02	11.51	7	5	7
HSS	Globalization and cultural development	6.36	1.07	6.23	5.24	9.20	0	1	2
HSS	Globalization and global poverty	6.33	1.14	6.32	5.06	9.95	9	4	0
HSS	Film studies	6.15	0.96	6.04	5.24	7.42	0	1	1
HSS	Globalization and economic development	6.15	0.72	6.06	5.00	8.96	13	5	6
HSS	Globalization and armed conflict	5.83	0.71	5.73	5.11	8.26	2	1	2
Law	International law and arbitration	6.43	1.22	6.44	5.01	11.19	1	8	6
Law	Human rights and social justice	6.75	1.27	6.74	5.01	9.75	0	4	6
Law	Intellectual property and corporate law	6.55	1.19	6.46	5.01	9.20	1	1	8
Law	Trade and economic regulations	6.28	1.11	6.28	5.03	9.77	0	3	6
Law	Labor and employment law	6.04	0.84	6.39	5.08	8.04	0	4	1
Law	Healthcare	6.93	1.32	7.09	5.01	9.75	0	1	3
Law	Globalization and economic development	5.63	0.45	5.37	5.13	6.21	0	1	1
Law	Environmental law and policies	7.32	1.30	6.94	6.21	11.19	0	0	1
Law	Globalization and global poverty	6.99	1.97	6.88	5.38	11.16	0	1	0
Law	Environment and sustainability	5.64	0.67	5.38	5.06	6.75	0	0	1
STEM	Healthcare	6.88	1.52	6.72	5.01	10.58	2	10	4
STEM	AI and machine learning	6.12	0.83	5.99	5.01	7.81	6	2	5
STEM	Data science and big data	6.47	1.01	6.46	5.02	9.20	3	2	4
STEM	Environment and sustainability	8.36	2.22	9.57	5.05	11.36	0	1	2
STEM	Finance	7.09	1.04	7.68	5.02	7.68	0	3	0
STEM	Gaming	5.51	0.36	5.47	5.14	5.93	1	0	1
STEM	Shopping	9.20	0	9.20	9.20	9.20	1	0	0
STEM	Semiconductor and photonics	7.81	0	7.81	7.81	7.81	0	0	1
STEM	Space and aerospace	7.65	3.23	6.14	5.44	11.36	1	0	0
STEM	Biotechnology	7.27	1.31	7.17	5.23	8.65	0	0	1
STEM	Sports	7.09	0.00	7.09	7.09	7.09	0	0	1
STEM	Performing arts	6.81	0.00	6.81	6.81	6.81	0	1	0
STEM	International law and arbitration	6.37	0.83	6.13	5.36	7.23	1	0	0
STEM	Trade and economic regulations	6.37	0.83	6.13	5.36	7.23	1	0	0
STEM	Miscellaneous technology	6.34	1.52	5.75	5.27	8.59	0	0	1
STEM	Telecommunications and network	5.83	0.60	6.18	5.14	6.18	1	0	0
STEM	Software engineering	5.48	0.55	5.27	5.07	6.10	0	1	0

Table 6.3*Robust ANOVA and Post-hoc Results for Bigram MI Scores Across Pathways*

MI-Score Threshold	<i>F</i>	<i>p-value</i>	HSS vs. STEM	HSS vs. Law	STEM vs. Law
3	0.8026	0.445	NS	NS	NS
4	0.7399	0.479	NS	NS	NS
5	6.4647	0.002	*	***	NS

Note. NS = not significant. Significance levels: * $p < .05$, *** $p < .001$

Table 6.4*Robust Regression Results for MI Scores by Pathway and Proficiency*

Predictor	Estimate	Std. Error	<i>t</i> value	Significance
(Intercept)	7.08	0.73	9.69	***
Pathway (Law)	-4.93	1.42	-3.46	***
Pathway (STEM)	-7.45	1.24	-6.01	***
Presentation score	-0.02	0.01	-1.36	
Pathway (Law): Presentation score	0.08	0.02	3.59	***
Pathway (STEM): Presentation score	0.13	0.02	6.39	***

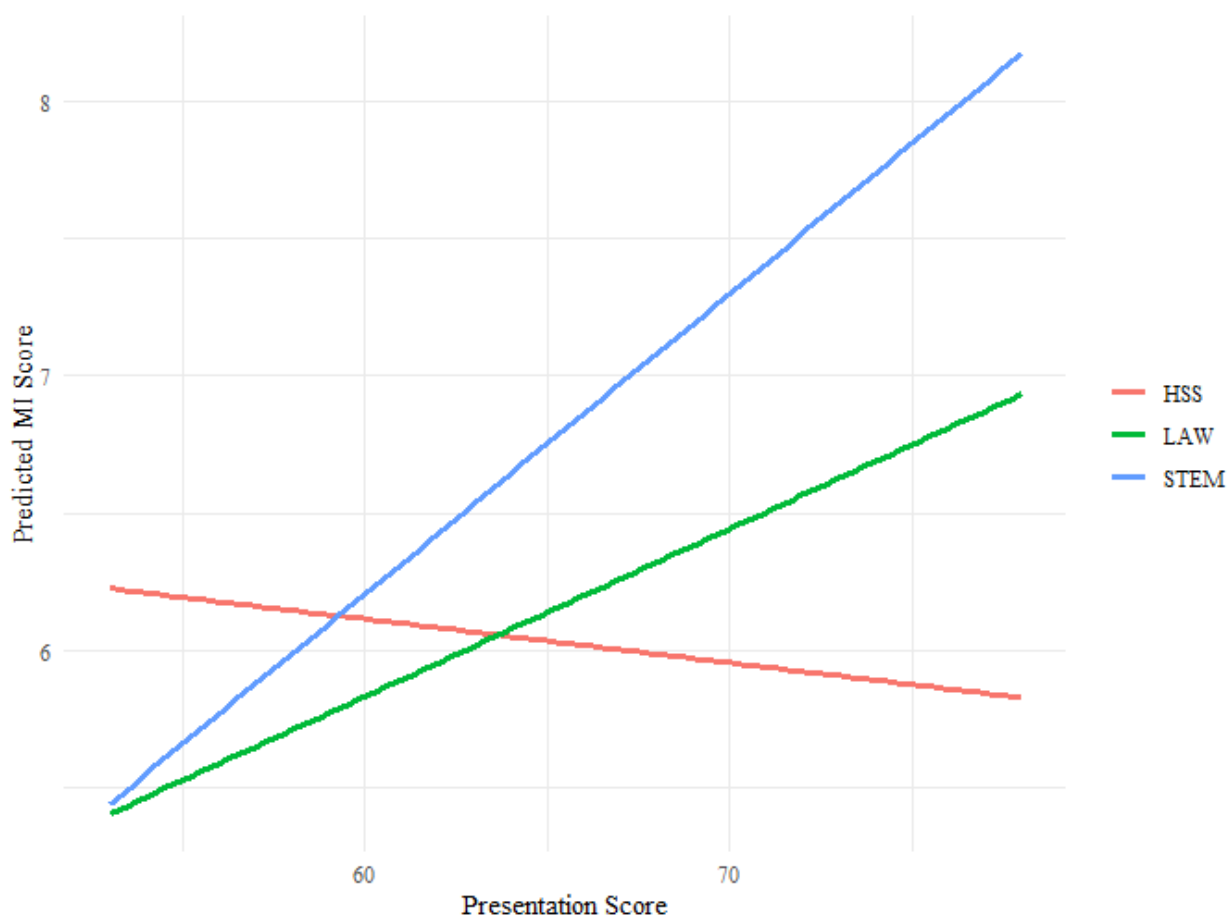
Note. Residual standard error was 0.969 on 679 degrees of freedom. In this model, the HSS pathway was automatically used as the reference category. This means that the coefficients for Law and STEM pathways are interpreted in relation to the HSS pathway. Significance level: *** $p < 0.001$

6.6.3 Qualitative Analysis: Contextual Insights From Selected Participant Transcripts

In this section, I provide a qualitative analysis of the six selected participants' transcripts, focusing on the context in which high MI score bigrams (MI scores above 5) are used (see Section 6.3 for the selection criteria and rationale). The objective of this analysis is to enhance the quantitative results by providing a more comprehensive understanding of the language used in presentations within different pathways and proficiency levels. I examine and detail selected participants' use of bigrams, highlighting specific sentences from their

Figure 6.2

Influence of Proficiency Scores on MI Scores by Pathway (MI Threshold = 5)



transcripts where each bigram appears and analyze these instances to provide contextual insights. I begin with Participant 6, whose low mean MI score provides insights into bigram use within the HSS pathway at IELTS level 6.5.

Participant 6 (HSS pathway, IELTS 6.5, low mean MI score = 5.49)

- Number of high MI-score bigrams produced: 3
- Bigrams and MI scores:
 - *even though* (5.59)

- *large extent* (5.55)
- *links between* (5.32)
- Specific sentences from the transcript where each bigram was used:
 - *Even though there are many problems in implementing the process of this [globalization], however, to a large extent, it promotes the development in aspects of economy, culture, science and technology.*
 - *David Held and Anthony McGrew who are globalization supporters argue that globalization refers to a transformation in the scale of human social organization, which connects remote communities, enhances the range of power in major origins and state of the war. And then Steger, who also a supporter claims that globalization increases the links between countries, strengthens interdependence and communication and is a process that social in its different dimensions.*

Contextual analysis: Participant 6's presentation covers a broad range of topics under the theme of globalization, including economic, cultural, and technological aspects. The language used is relatively general. This general language use is consistent with the participant's IELTS score of 6.5 and the lower mean MI score. The presentation structure is clear and organized, but the vocabulary lacks the specificity seen in higher proficiency levels and in more specialized presentations.

Participant 21 (HSS pathway, IELTS 6.5, low mean MI score = 5.48)

- Number of high MI-score bigrams produced: 2
- Bigrams and MI scores:
 - *global economy* (5.95)
 - *talking about* (5.01)
- Specific sentences from the transcript where each bigram was used:

- *Beginning in the 1990s, globalization was considered in economic terms, it's also in the cultural terms, to reference to increasingly independent world. And in an investment friendly environment, capital flows to the developing world will promote growth and reduce poverty. And therefore, people are quite optimistic about the globali- about the global economy.*
- *And today I will going to talking about does globalization lead to development?*

Contextual analysis: Participant 21's presentation also focuses on globalization and its impacts on development. Basic connectors and transitional phrases like *and therefore* and *and today* are frequently used, making the language simple and straightforward. The participant's IELTS score of 6.5 and the low mean MI score reflect the limited use of complex or specialized vocabulary. The content is well structured but lacks depth in linguistic complexity.

Participant 104 (HSS pathway, IELTS 6.5, low mean MI score = 5.74)

- Number of high MI-score bigrams produced: 4
- Bigrams and MI scores:
 - *each other* (5.73)
 - *talking about* (5.01)
 - *developing countries* (6.46)
 - *get rid* (5.76)
- Specific sentences from the transcript where each bigram was used:
 - *...globalization promotes global economic by deepening linkage and making it freer and easier for countries to interact with each other.*
 - *Secondly, I'll I'll be talking about the impact of globalization on countries.*
 - *Globalization can promote the development of the world economy and help developing countries get rid of poverty.*

Contextual analysis: Participant 104's presentation primarily addresses the economic impacts of globalization. The vocabulary includes some specialized terms related to economics, such as *deepening linkage*, but overall remains within a low complexity range. With the participant's IELTS score of 6.5 and a low mean MI score, the repeated phrases and simple sentence structures are evident. While the content is informative, it lacks a high level of lexical complexity.

Participant 124 (Law pathway, IELTS 7, moderate mean MI score = 6.21)

- Number of high MI-score bigrams produced: 4
- Bigrams and MI scores:
 - *human rights* (7.09)
 - *relationship between* (5.47)
 - *three decades* (5.19)
- Specific sentences from the transcript where each bigram was used:
 - *At the same time, the current legal system of international trade law is not working appropriately. And this system does not consider aspects like the human rights and this system does not consider aspects like human development.*
 - *Well as introduction what is the relationship between globalization and international law.*
 - *And with respect to the human rights and with consideration to the human development so the thesis statement for this presentation is the following. I'm going to argue that there is an increasing global market that needs to be regulated as a system of persons and no more as one of the states. During the last three decades, globalization has raised efficiency and has improved the productivity of many firms all over the world.*

Contextual analysis: Participant 124's presentation discusses the relationship between globalization and international law. *International trade law* and *human rights* are examples of

the more specialized legal terminology used, indicating a higher level of lexical sophistication. The participant's IELTS score of 7 and average MI score demonstrate their skill in employing complex and specific vocabulary. The structure of the presentation is focused on explaining intricate legal concepts and uses specific language.

Participant 150 (STEM pathway, IELTS 7.5, high mean MI score = 7.58)

- Number of high MI-score bigrams produced: 1
- Bigram and MI score: *virtual reality* (7.58)
- Specific sentence from the transcript where the bigram was used:
 - *The first part is about the introduction of the VR system and the robot-assisted minimally invasive surgery. As we know in the present, people have stepped it into a new era. The metaverse age, the technology behind the realization of the metaverse is virtual reality technology.*

Contextual analysis: Participant 150's presentation focuses on advanced technological topics such as VR systems and robot-assisted minimally invasive surgery. Using highly specialized terms like *minimally invasive surgery* and *virtual reality technology* reflects a high level of lexical sophistication. The participant's IELTS score of 7.5 and high mean MI score correspond to the frequent use of technical vocabulary and complex sentence structures. The content displayed in this shows a deep comprehension of cutting-edge technologies, a key hallmark of advanced proficiency in STEM subjects.

Participant 148 (STEM pathway, IELTS 7.5, high mean MI score = 7.27)

- Number of high MI-score bigrams produced: 11
- Bigrams and MI scores:
 - *breast cancer* (8.65)

- *gene therapy* (7.17)
- *side effects* (6.72)
- *cancer cells* (5.70)
- *matters worse* (5.61)
- *spread throughout* (5.23)
- Specific sentences from the transcript where each bigram was used:
 - *In view of the limited, available, limited success of available therapeutic strategies for breast cancer gene therapy, which involves delivering a nucleic acid-based drug to either modify or destroy cancer cells is potentially beneficial.*
 - *Cancer is a life-threatening disease that affects the cells in the body and cause them to grow uncontrollably and spread throughout the body. In breast cancer, the cells in the breast are affected and they grow out of control.*
 - *Current therapeutic strategies include surgery to move the tumor radiotherapy to using high-dose radiation to kill cancer cells. Chemotherapy using toxic compounds to to shrink that chemo and hormonal therapy using hormones to prevent proliferation of the cells. This polypharmacy brings with it side effects that causes emotional distress in patients and not to mention the economic burden it brings. To make matters worse, cancer cells evolve resistance to these drugs at an alarming rate, and in most cases, they develop multi-drug resistance. Gene therapy offers exciting treatment modalities for breast cancer, since changes in genes and gene expression are involved in the development and progression of cancer correction of defective genes.*

Contextual analysis: Participant 148's presentation discusses advanced medical topics such as gene therapy for breast cancer. The language is highly specialized, with use of medical terms like *nucleic acid-based drug* and *therapeutic strategies*. Advanced proficiency and familiarity with complex medical terminology can be inferred from the participant's IELTS score of 7.5 and high mean MI score. The presentation not only demonstrates a thorough understanding of the subject but also showcases a high level of lexical sophistication, which is exactly what is expected in STEM fields.

In summary, the qualitative analysis reveals distinct patterns in bigram use across participants, shaped by their proficiency levels, pathways, and topics. Lower-proficiency participants tended to use more general language, while higher-proficiency participants demonstrated greater lexical sophistication with specialized terminology in STEM and law topics. These observations complement the quantitative findings and set the stage for a discussion of topic influence and proficiency interaction in the next section.

6.7 Discussion

The overarching aim of the current chapter was to explore the influence of presentation topic on the diversity and frequency of high MI-score bigrams used by L2 English speakers and whether this influence varies with proficiency scores. The subsequent discussion is organized to tackle the research question. Section 6.7.1 addresses the influence of presentation topic on lexical sophistication. Section 6.7.2 covers the interaction between topic choice and proficiency scores. Section 6.7.3 discusses the qualitative insights from selected participant transcripts.

6.7.1 *The Influence of Presentation Topic on Lexical Sophistication*

The analysis indicated that the choice of presentation topic (operationalized by pathway) significantly influences the diversity and frequency of high MI-score bigrams. The ANOVAs showed significant differences at the MI score threshold of 5, but not for thresholds of 3 or 4. Post-hoc tests showed that MI scores were significantly higher for STEM compared to HSS and for Law compared to HSS. Presentations in the STEM pathway showed the highest mean MI scores (6.86), followed by Law and HSS pathways (6.47 and 6.24, respectively), when considering MI scores at a cutoff of five. The results of this study support my hypothesis that presentations focusing on specialized and technical topics, like those found in the STEM and Law pathways, show a greater variety and frequency of high

MI-score bigrams in comparison to the broader topics found in the HSS pathway. This supports the idea that specialized and technical topics elicit or require more exclusive and sophisticated language, as measured by MI scores.

It is worth noting that these findings are in line with what has been found in previous research. For instance, Hyland (2008) found significant disciplinary variation in the use of LBs, observing that technical subjects (e.g., electrical engineering) employed more sophisticated and specialized vocabulary compared to non-technical subjects (e.g., business studies). Similarly, Nesi and Basturkmen (2006) conducted a detailed analysis of LBs in academic lectures. They found that academic lectures in technical fields, particularly the Physical Sciences (which generally include disciplines such as physics, chemistry, and earth sciences), utilized more complex and specialized LBs to convey specialized content. The significance of focusing on a specific topic becomes apparent when considering the role it plays in encouraging the utilization of advanced and intricate language in academic discourse.

The findings discussed in this chapter carry potentially weighty implications that can greatly influence the fields of language teaching and curriculum design. They suggest that engaging students with specialized and technical topics can enhance their lexical sophistication in academic and professional settings. There is a need for learners to be exposed to high-level, domain-specific vocabulary to achieve greater linguistic proficiency. Language instructors might consider integrating more specialized and technical content into their curricula to foster advanced lexical development.

6.7.2 The Interaction Between Topic Choice and Proficiency Scores

The second part of my research question asked whether the influence of presentation topic on the diversity and frequency of high MI-score bigrams used by L2 English speakers

varies with proficiency scores. According to the robust regression analysis, presentations in Law and STEM pathways exhibited a faster increase in scores with higher proficiency levels, despite starting with lower MI scores compared to the HSS pathway. These results suggest that higher proficiency speakers are better able to use complex language within specialized topics. This result is also consistent with the second part of my hypothesis, which stated that the effect of presentation topic on the diversity and frequency of high MI-score bigrams would be more pronounced at higher proficiency levels. This finding indicates that as learners' proficiency increases, they develop greater command over the complex and specialized vocabulary necessary for more technical subjects.

This interaction between topic choice and proficiency highlights the importance of tailored language instruction that considers both the proficiency level of learners and the specific demands of the subject matter. Gradually introducing specialized topics to lower proficiency learners can be beneficial as it provides scaffolding to support their linguistic development. As learners become more proficient, they can be exposed to more intricate and technical subjects, which will push them to expand their lexical repertoire and enhance their overall language sophistication.

6.7.3 Qualitative Insights From Selected Participant Transcripts

The qualitative analysis of selected transcripts confirms distinct patterns in the use of high MI-score bigrams across different pathways and proficiency levels. Individuals who are enrolled in the HSS pathway and have lower proficiency levels often exhibit a tendency to use language that is more general in nature and less specialized. In contrast, participants in the Law and STEM pathways use more specialized vocabulary, reflecting their higher proficiency levels and the more technical nature of their presentation topics. These findings align with and complement the quantitative results, which showed significant differences in

mean MI scores across pathways (see Table 6.3). The qualitative insights provide a richer context for understanding these differences, highlighting the importance of both proficiency and topic specialization in influencing lexical sophistication. This integrated approach of combining quantitative metrics with qualitative context offers a detailed understanding of language use in academic presentations, addressing my second research sub-question. By examining specific instances of high MI-score bigram usage, this analysis reveals how linguistic complexity varies with topic choice and speaker proficiency, contributing valuable insights to the field of academic English language learning and teaching.

6.8 Conclusion and Limitation

To conclude, it is evident from the findings in this chapter that the incorporation of high MI-score bigrams is greatly influenced by both the topic being presented and the proficiency level of the speaker. Specialized topics such as those in the STEM and Law pathways tend to require more sophisticated language, particularly among higher proficiency speakers. Language teaching can benefit greatly from these findings as they highlight the importance of encouraging students to explore specialized or technical topics. By doing so, students have the opportunity to enhance their lexical sophistication, ultimately resulting in improved performance on their oral assessments. It is important to acknowledge two limitations that are present in the current chapter. First, the study focused on pre-defined pathways (HSS, Law, STEM) to categorize presentation topics. Although this approach provides a structured analysis, it may oversimplify the complexity and diversity within each pathway. Further research could explore a more granular categorization of topics to address this limitation. Second, the cross-sectional design of the study limits the ability to infer causality. Longitudinal studies tracking the development of lexical sophistication over time would provide deeper insights into how proficiency and topic choice interact to influence language use.

Chapter 7: Discussion

The current chapter synthesizes the key findings from Chapters 3 through 6, focusing on their implications and practical applications. Instead of repeating detailed discussions already covered in each chapter, it highlights the key, overarching themes and offers a comprehensive view of how LB usage and oral fluency interact across proficiency levels. The primary research questions that have guided this thesis are:

- How does LB usage, differentiated by length, vary across different proficiency levels? ([Chapter 3](#))
- To what extent is the use of longer versus shorter LBs associated with three dimensions of oral fluency—speed, breakdown, and repair? ([Chapter 4](#))
- Can LB usage serve as a predictor of oral proficiency scores, and if so, what is the relative contribution of shorter and longer LBs to this prediction? ([Chapter 5](#))
- Does the choice of presentation topic influence the diversity and frequency of high MI-score bigrams used by L2 English speakers, and does this influence vary with proficiency scores? ([Chapter 6](#))

Each of these research questions has been addressed in detail in the preceding chapters, yielding both consistent findings and new insights into the role of LBs in L2 oral fluency. This chapter synthesizes these findings to highlight key themes, such as the interaction between LB length and fluency dimensions, the relationship between proficiency and LB usage, and the impact of presentation topics on LB diversity. The current chapter sets the stage for a broader reflection on L2 fluency and MWS usage, which will be addressed in the final concluding chapter.

7.1 Synthesis of Key Findings

The following sections (7.1.1 to 7.1.4) consolidate the insights gained from experimental Chapters 3 through 6, identifying the most significant patterns in LB usage and its relationship to proficiency and fluency measures. The findings from these chapters highlight key themes concerning how both shorter and longer LBs contribute to oral fluency, with considerations for proficiency development and the role of context.

7.1.1 LB Contribution Across Proficiency Levels: Synthesis From Chapter 3

A key finding from Chapter 3 is that both shorter and longer LBs are used with increasing frequency as proficiency increases, though significant differences lie in specific measures of LB usage. The ANOVAs (Table 3.3) revealed significant differences in bigram MI ($F = 7.466, p = .001$), trigram proportion ($F = 3.428, p = .042$), three-to-five-word usage ($F = 5.906, p = .004$), and three-to-five-word MI ($F = 5.842, p = .005$) across proficiency levels, while other measures, like bigram and trigram frequency, showed no significant differences.

The significant increase in bigram MI as proficiency rose from IELTS 6.5 to 7.5 reflects more sophisticated use of shorter LBs by higher proficiency learners. Advanced speakers are not simply using more frequent bigrams but also deploying bigrams that carry higher mutual information, indicating more nuanced and sophisticated usage. This increase in bigram MI aligns with the broader patterns of proficiency, where more advanced speakers utilize higher quality, lexically rich sequences, leading to smoother and more coherent speech. At the same time, longer sequences, such as three-to-five-word LBs, also showed significant increases in usage and MI scores at higher proficiency levels, contributing to the overall fluency improvements observed as proficiency increases. Consequently, the utilization of both bigram MI and longer LBs plays a crucial role in enhancing the fluency of

more proficient speakers. Such results suggest that as learners progress from IELTS 6.5 to 7.5, they rely on both the strategic use of higher MI bigrams and longer, more complex sequences to manage the cognitive demands of speech production and improve overall fluency.

In summary, Chapter 3 explored how LB usage differs across proficiency levels, demonstrating that both shorter and longer LBs, especially those with higher MI scores, are used more frequently and effectively by higher proficiency learners. The findings from Chapter 3 confirm that proficiency is associated with the quality and complexity of LBs employed in speech, contributing to overall fluency development.

7.1.2 Interplay Between LB Length and Fluency Dimensions: Synthesis From Chapter 4

The results from Chapter 4 highlighted the distinct roles that longer and shorter LBs play in influencing different dimensions of fluency: speed fluency as measured by articulation rate, breakdown fluency as measured by the frequency of mid- and end-clause pauses, and repair fluency as measured by the frequency of total repairs. The robust regression analysis, coupled with dominance and Boruta analyses, sheds light on how LBs of various lengths contribute to the fluency of L2 speakers. These findings confirm existing models of fluency, particularly those proposed by Levelt (1989), Kormos (2006), and Skehan (2014), by providing a more nuanced understanding of the role of multiword sequences in speech production.

7.1.2.1 Longer LBs and Speed Fluency. Longer LBs were found to have a nearly significant influence on articulation rate. The regression analysis found that three-to-five-word usage had a notable positive impact on speed fluency, with a regression coefficient of $b = 2.03$ and a dominance weight of 32.75%. Although the confidence interval [95% CI: $-0.17, 4.23$] narrowly spanned zero, these findings suggest that frequent use of longer LBs allows

speakers to produce speech more fluently by retrieving larger language chunks and reducing the cognitive demands of constructing sentences from shorter segments. The results are consistent with theoretical frameworks such as Levelt's (1989) and Skehan's (2014) theories on speech production. These theories suggest that using pre-constructed language units conserves cognitive resources, allowing speakers to focus on more demanding tasks during communication. However, while three-to-five-word usage positively influenced speed fluency, three-to-five-word MI produced a modest negative effect on articulation rate ($b = -0.08$; dominance weight: 22.30%), suggesting that the complexity of higher MI sequences introduces additional processing demands that may slightly slow speech. This negative effect on articulation rate reflects a cognitive trade-off consistent with Skehan's (1998) trade-off hypothesis, which posits that more complex LBs require greater processing effort but enable more precise communication.

These findings partially align with those of Hougham et al. (2024a), who observed a positive, though statistically non-significant, effect of four-to-five-word MI ($b = 13.07$; 95% CI $[-46.53, 72.67]$) on speed fluency. While Hougham et al.'s study did not report statistical significance for the impact of longer LBs on speed fluency, the findings from Section 4.4 of the current thesis present slightly stronger evidence, showing a nearly significant increase in articulation rate with three-to-five-word usage ($b = 2.03$; 95% CI $[-0.17, 4.23]$, dominance weight: 32.75%). Despite differences in statistical significance, both studies suggest that longer LBs can enhance fluency by promoting more efficient speech production. However, as demonstrated by the current thesis, this benefit can be moderated by the internal complexity of LBs, as seen in the modest negative effect of three-to-five-word MI on articulation rate. This subtle divergence provides new insights into how the cognitive load associated with more advanced LBs may temper fluency gains, expanding our understanding of L2 speech production in line with usage-based theories (Ellis, 2002; Bybee, 2006).

7.1.2.2 Longer LBs and Breakdown Fluency: Mid-ASU Pauses. The findings indicate that three-to-five-word usage was a key factor in reducing mid-ASU pauses, with a negative regression coefficient ($b = -0.58$), significant at the 95% confidence level [95% CI: $-1.15, -0.01$]. Its dominance weight (40.36%) makes it as the most influential predictor of reduced mid-ASU pauses, suggesting that longer LBs help speakers to retrieve pre-assembled language chunks, which reduces the need for real-time formulation and enhances speech flow.

These results align with those of Hougham et al. (2024a), who also reported that four-to-five-word MI had the highest dominance weight (39.59%) in predicting fewer mid-ASU pauses, though their findings narrowly missed statistical significance ($b = -9.99$; 95% CI [$-21.34, 1.37$]). Since the confidence interval in the current thesis does not span zero, these results provide clearer and stronger evidence than those of Hougham et al. regarding the role of longer LBs in reducing mid-clause pauses. The cognitive complexity of high-MI sequences might require more processing, which could offset their fluency benefits in certain contexts, as seen in Hougham et al.'s findings. However, the current results provide stronger support for the idea that frequent use of longer sequences contributes to smoother speech by minimizing mid-ASU pauses. This divergence suggests the need to refine theoretical models to account for variability in LB use across different tasks and contexts, which contributes to a more nuanced understanding of L2 speech fluency development.

7.1.2.3 Shorter LBs and Breakdown Fluency: Mid-ASU Pauses. Similarly, bigram MI also played a significant role in reducing mid-ASU pauses ($b = -8.75$; dominance weight: 16.01%), with its 95% CI [$-16.59, -0.91$] confirming statistical significance. The significant reduction in mid-ASU pauses associated with bigram MI suggests that shorter, high-MI bigrams are quickly accessible and help reduce breakdowns in speech. Despite its lower dominance weight compared to three-to-five-word usage, bigram MI's significant

contribution to reducing pauses highlights its importance in enhancing fluency, particularly in managing mid-clause hesitations. This finding diverges from Tavakoli and Uchihara (2020) and Hougham et al. (2024a), both of whom reported the opposite effect (i.e., bigram MI contributed to increasing mid-clause pauses), though neither of their findings were statistically significant. This divergence is theoretically significant because it challenges the assumption that more frequent or tightly associated bigrams necessarily hinder fluency. Instead, the findings from Section 4.4 suggest that high-MI bigrams might play a more facilitative role in fluid speech production than previously recognized. Furthermore, this difference could be due to the specific task conditions or speaker characteristics in each study, offering valuable insights into the variability of LB use in different contexts. By highlighting the conditions under which shorter LBs promote fluency, these results contribute to refining fluency models and underscore the importance of understanding the task-specific and proficiency-dependent nature of multiword sequence processing in L2 speech.

7.1.2.4 Shorter LBs and Breakdown Fluency: End-ASU Pauses. For end-ASU pauses, bigram frequency emerged as a significant contributor to increased pauses, with a positive regression coefficient ($b = 6.76$), supported by a robust 95% CI [2.20, 11.33] and considerable dominance weight (40.12%). This specific finding reported in Table 4.4 regarding bigram frequency indicates that frequently used bigrams may prompt pauses at the end of ASUs. This could reflect a preparatory pause, where speakers plan the upcoming speech segment, possibly signaling the completion of a thought. Similarly, trigram MI was found to significantly increase end-ASU pauses ($b = 1.87$; 95% CI [0.31, 3.43]; dominance weight: 23.02%). This relationship between trigram MI and end-ASU pauses suggests that strongly associated trigrams may introduce processing demands that delay speech continuation, further contributing to end-ASU pauses. These findings show a contrasting role of shorter LBs, where they may assist with managing mid-sentence fluency but also

contribute to pauses at the ends of ASUs, potentially due to cognitive load as speakers organize subsequent speech segments.

The trigram MI finding (reported in Table 4.4) is consistent with Tavakoli and Uchihara (2020), who similarly reported a positive association between association measure of MI factor and the frequency of end-clause pauses ($r = .302$, $p = .024$), even though their finding missed their corrected alpha level. This convergence with Tavakoli & Uchihara (2020) reinforces the idea that higher MI lexical bundles, particularly trigrams, introduce processing demands that lead to pauses as speakers plan the upcoming segment. This suggests a robust relationship between the complexity of multiword sequences and speech production, aligning with the cognitive models of fluency that emphasize the balance between fluency and cognitive load (Skehan, 1998; Kormos, 2006). In contrast, there is a slight divergence between the trigram MI finding (reported in Table 4.4) and the MI measure findings from Hougham et al. (2024a), who reported weak negative correlations with the frequency of end-clause pauses, though their findings were not statistically significant and not confirmed as important by the Boruta algorithm. This slight divergence from Hougham et al. (2024a) indicates the potential variability in how different types of LBs affect speech across contexts and tasks. These differences may underscore the importance of the specific features of the tasks and speakers in shaping how MI measures influence fluency, thereby contributing to a more nuanced understanding of L2 speech processing and its cognitive underpinnings.

7.1.2.5 Longer LBs and Repair Fluency. In contrast to their impact on speed and breakdown fluency, longer LBs were not found to be strong predictors of repair fluency in the current thesis. The regression and dominance analyses indicated that three-to-five-word usage had a minimal and non-significant effect on repairs. Similarly, the influence of three-to-five-

word MI was negligible, suggesting that longer LBs play a relatively minor role in repair fluency.

The findings here diverge somewhat from Hougham et al. (2024a), who reported a stronger and statistically significant impact of four-to-five-word MI on reducing repairs ($b = -2.41$; 95% CI $[-4.24, -0.59]$; dominance weight: 42.75%). This divergence is theoretically significant because it challenges the hypothesis that longer sequences universally promote fluency by reducing repairs. Hougham et al.'s (2024a) results support a strong link between longer sequences and repair reduction, while the current findings imply that this relationship may not be consistent across all contexts. This inconsistency points to the need for a more context-sensitive model of speech fluency, where task demands and individual speaker differences might moderate the impact of longer sequences on repair fluency. By exploring these divergent results, future research can refine existing models of fluency and deepen our understanding of how linguistic complexity interacts with fluency dimensions across different L2 tasks.

7.1.2.6 Shorter LBs and Repair Fluency. Turning to shorter LBs, the analysis indicated a more complex relationship with repair fluency. Bigram frequency emerged as a significant predictor of increased repair frequency ($b = 3.71$; 95% CI $[0.04, 7.38]$; $t = 1.98$), although its dominance weight was relatively low at 3.48%. Interestingly, despite this statistical significance, the Boruta analysis identified bigram frequency as unimportant, which casts doubt on the actual impact of frequent bigram use on repairs. Bigram MI and trigram MI were both confirmed as important predictors of reduced repair frequency, albeit with confidence intervals that spanned zero, which calls for cautious interpretation. These results suggest that higher MI-score bigrams and trigrams might contribute to fewer speech repairs by facilitating smoother lexical retrieval and reducing the need for corrections during speech. The positive association between bigram frequency and increased repairs observed in Chapter

4 of this thesis was not reported by Hougham et al. (2024a), indicating that frequent use of shorter LBs might, in some cases, disrupt speech flow and lead to a higher number of repairs. Other studies (e.g., Saito et al., 2018; Tavakoli et al., 2020) have also reported inconsistent findings regarding the relationship between repair fluency and oral development, highlighting the still-evolving nature of our understanding in this area. This inconsistency across multiple studies underscores the complexity of repair fluency, suggesting that the influence of MWSs on repairs may not be conclusive and warrants further investigation to account for contextual and individual differences.

Section 7.1.2 has addressed the research question regarding the extent to which longer versus shorter LBs contribute to the three dimensions of fluency—speed, breakdown, and repair. The findings showed that longer LBs positively enhance speed fluency and significantly reduce mid-ASU pauses, supporting their key role in smooth speech production. However, their impact on repair fluency was minimal, suggesting task-specific influences. Conversely, shorter LBs, particularly those with higher MI, played a more complex role, assisting in reducing mid-clause hesitations but also showing a potential for disrupting speech flow in some instances, particularly when frequently used bigrams were associated with increased repairs. These findings highlight the importance of considering both LB length and the cognitive load introduced by high-MI sequences, providing a nuanced understanding of how MWSs influence L2 oral fluency across different tasks and proficiency levels.

7.1.3 The Predictive Power of LBs: Synthesis From Chapter 5

The findings from Chapter 5 emphasize the predictive power of LBs, particularly high-MI bigrams, in determining oral proficiency scores. The robust regression and dominance analyses identified bigram MI (mutual information) as the most significant predictor of raw presentation scores, with a strong positive association ($b = 14.38$; 95% CI

[8.01, 20.76]; $t = 4.42$) and a dominance weight of 58.63%. This suggests that learners who used high-MI bigrams, such as *carbon dioxide* and *global warming*, were more likely to achieve higher presentation outcomes. In addition to bigram MI, the use of three-to-five-word sequences also significantly predicted presentation scores, though with a more modest effect ($b = 0.53$; 95% CI [0.06, 0.99]; $t = 2.22$), contributing 18.80% to the total dominance weight. This highlights the role of both shorter, highly associated bigrams and longer, more complex sequences in proficient speech production.

These results align with usage-based theories of language acquisition and extend prior research by showing that both shorter and longer LBs serve as key predictors of proficiency. For a more comprehensive discussion of how the current findings converge with and diverge from previous studies (Tavakoli and Uchihara, 2020, in particular)—particularly the contrasting trends in bigram MI across proficiency levels—readers are referred to [Section 5.4](#), which explores the potential reasons for the divergence, including methodological differences such as sample size and task type considerations.

7.1.4 Topic-Driven Variability in LB Usage: Synthesis From Chapter 6

The results from Chapter 6 show how the choice of presentation topics significantly influences the use of high MI-score bigrams, highlighting the context-dependent nature of language use and its impact on proficiency scores. The variability introduced by different topics hints at the complex relationship between LB usage and fluency dimensions, showing that more technical or specialized subjects tend to elicit higher lexical sophistication. The STEM and Law pathways consistently showed higher mean MI scores compared to the HSS (Humanities and Social Sciences) pathway, with the STEM pathway showing the highest overall MI scores. For instance, the mean MI score for the STEM pathway was 6.86, while it was 6.47 for Law and 6.24 for HSS at an MI threshold of 5 (Table 6.1). These differences,

statistically significant as shown by the ANOVA and post-hoc tests (Table 6.3), suggest that presentations on specialized topics, such as those found in STEM and Law, promote the use of more sophisticated, high MI-score bigrams, which in turn contributes to higher proficiency scores.

This topic-driven variability in LB usage reflects the context-dependent nature of language performance. Specialized topics often require more precise, technical language, which leads to greater lexical sophistication as measured by MI scores. In contrast, the broader, less technical topics in the HSS pathway tend to evoke more general, lower MI-score bigrams. For example, subcategories within the HSS pathway, such as *globalization and armed conflict*, had relatively low mean MI scores compared to more technical STEM topics like *biotechnology* and *space and aerospace* (Table 6.2). This suggests that the complexity and specificity of the topic itself can drive the diversity and frequency of high MI-score bigrams.

The interaction between topic choice and proficiency also plays a key role. As shown in the robust regression analysis (Table 6.4), although the Law and STEM pathways started with lower MI scores at lower proficiency levels, these scores increased more rapidly as proficiency levels rose, compared to the HSS pathway. This interaction effect implies that higher proficiency learners are better able to navigate the linguistic demands of specialized topics, resulting in a greater frequency and diversity of high MI-score bigrams in their speech. For instance, participants in the STEM pathway who were at IELTS 7.5 produced more complex bigrams like *infectious diseases* and *gas emissions*, while lower proficiency speakers in the HSS pathway used more general, less specialized bigrams like *links between* and *talking about*.

In addition to the clear link between topic choice, proficiency, and lexical sophistication, it is also possible that graders subconsciously awarded higher scores to presenters who used more technical or specialized vocabulary. More technical or specialized topics, such as those in STEM and Law, not only demand the use of more complex language but also offer higher proficiency learners the chance to demonstrate their command of specialized vocabulary. This use of precise, domain-specific vocabulary may have been interpreted by graders as a marker of greater linguistic proficiency and subject expertise, leading to higher presentation scores. Given that technical topics often require speakers to manage more intricate and nuanced language, this could reflect a deeper engagement with both the content and the language itself. Presenters who effectively use MWSs in such contexts are likely perceived as having stronger communication skills, thus influencing the grading process. This interaction between language use, topic complexity, and assessment outcomes highlights how topic-driven variability in LB usage can influence not only fluency but also the scoring of oral presentations.

In summary, the choice of presentation topic exerts a strong influence on both the diversity and frequency of high MI-score bigrams, with specialized topics promoting more advanced language use. This variability highlights the importance of considering topic context when assessing oral proficiency, as it directly impacts LB usage and fluency dimensions. The interaction between proficiency and topic further emphasizes that higher proficiency learners are better equipped to handle the linguistic demands of specialized subjects, resulting in more frequent use of high MI-score bigrams and higher presentation scores.

7.2 Limitations

While the results from experimental chapters 3-6 are positive, it is important to acknowledge several potential limitations. First, the sample size ($N = 150$), although substantial, is limited to participants from 20 L1 backgrounds from a single university, potentially limiting the generalizability to a broader population of L2 English speakers. Second, the participants' proficiency levels were confined to intermediate and higher levels (IELTS bands 6.5 to 7.5), excluding lower proficiency speakers. Third, the cross-sectional design of the experiments limits the ability to track language proficiency development over time or to establish causality between variables.

Fourth, the frequency-based approach using corpus-analysis software has certain limitations, especially when dealing with spoken corpora. The studies reported in the current thesis used a minimum frequency and range of three to identify LBs in the learner corpus, in order to keep the analysis manageable. These minimum frequency cutoffs mean that these studies did not identify all multiword units comprehensively. Some multiword units were used only once or twice or used idiosyncratically or in a nonstandard way by the L2 learners in my corpus. Many MWSs tend to blend into the linguistic context in transcripts, and many are frames or have larger fillable slots, which present real challenges for automatic extraction techniques. Such infrequent and/or semi-fixed units were not detected in the LB measurement and refinement approach taken in the experimental chapters in this thesis. The studies reported in the current thesis focused on bi- and trigrams using TAALES and three-to-five-word phrases using AntConc and Collocate, possibly leaving out other multiword structures that may affect speaking proficiency.

Fifth, the operationalization of LB usage through MI and frequency does not account for the qualitative aspects of contextual appropriateness in conversation. Sixth, since MI

scores were primarily designed for two-word collocations and since they do not consider the order of the words (Biber, 2009; Hyland, 2012), MI scores might not reliably measure longer lexical strings. Seventh, as noted in chapters 5 and 6, the potential influence of different presentation topics on LB usage and presentation scores is an underexplored area that might shed further light on findings, such as the increased use of high MI-score bigrams among higher-scoring learners. Chapter 6 explored this area to some extent by focusing on pre-defined pathways (HSS, Law, STEM) to categorize presentation topics. Although the approach in chapter 6 provided a structured analysis, it may have oversimplified the complexity and diversity within each pathway, and it was limited to specific presentation topics.

The practical implications of the findings from the current thesis, along with broader reflections on the contributions of this research to L2 fluency theory and practice, will be explored in greater detail in the final chapter. Specific suggestions for future research corresponding to each of the above limitations will also be presented in Section 8.4 of the final chapter. The upcoming concluding reflection will help to synthesize the key insights and offer pathways for practical implementation.

Chapter 8: Conclusion

The aim of the current thesis has been to examine the role of lexical bundles (LBs) in L2 oral fluency across different proficiency levels. Throughout this thesis, I have explored various dimensions of LB usage, including the differentiation between shorter and longer bundles and their respective impacts on three aspects of oral fluency (speed, breakdown, and repair). The thesis focused on providing a deeper understanding of how LBs of different lengths contribute to aspects of L2 fluency and how these factors relate to assessed oral proficiency scores.

Building on the discussion in Chapter 7, this concluding chapter aims to provide a broader reflection on L2 fluency and MWS usage, emphasizing the need for ongoing exploration of how LB usage differs across proficiency levels and tasks. The current thesis has not only advanced theoretical understanding but also laid a foundation for more effective instructional and assessment practices in language education. The following sections will provide a brief summary of the key findings, reflect on their implications and importance for language learning and assessment, and outline potential directions for future research. Through these discussions, I aim to underscore the importance of LBs in language acquisition, contributing to a deeper understanding of L2 speech development.

8.1 Summary of Key Findings

The following four sub-sections provide concise summaries of the most significant findings from Chapters 3 to 6, highlighting the overall role of LBs in L2 oral fluency and proficiency.

8.1.1 LB Usage Across Proficiency Levels (Chapter 3)

The findings from Chapter 3 showed that both shorter and longer LBs are used with increasing frequency as proficiency increases. Higher proficiency learners (IELTS 7.5) made

more sophisticated use of shorter LBs (those with higher MI scores), indicating a more nuanced and lexically rich usage. Longer LBs also increased significantly in usage at higher proficiency levels, contributing to smoother speech. These findings suggest that proficiency is associated with the length, quality, and complexity of LBs, which play a key role in enhancing fluency. For a more detailed synthesis, see Section [7.1.1](#).

8.1.2 Influence of LB Length on Fluency Aspects (Chapter 4)

The findings from Chapter 4 highlighted the distinct roles that shorter and longer LBs play in different dimensions of oral fluency. Longer LBs positively influenced speed and breakdown fluency by increasing articulation rates and reducing mid-clause pauses, contributing to smoother and more coherent speech. However, their complexity (as measured by MI scores) also introduced processing demands that led to a modest trade-off in fluency (slightly slower speech). In contrast, shorter LBs showed mixed effects: some of them (e.g., bigram MI) reduced mid-ASU pauses, but others (e.g., bigram frequency and trigram MI) were associated with increased end-ASU pauses, possibly reflecting planning demands. These results underscore the distinct contributions of both shorter and longer LBs in shaping L2 fluency, with variations depending on fluency dimensions and task-specific contexts. For a more detailed synthesis, see Section [7.1.2](#).

8.1.3 Predictive Role of LBs in Oral Proficiency (Chapter 5)

The findings from Chapter 5 emphasized the predictive power of LBs, particularly bigram MI, in determining oral proficiency scores. Bigram MI emerged as the most significant predictor, showing a strong positive association with higher presentation outcomes. Additionally, the use of three-to-five-word sequences contributed to proficiency scores, though with a more modest effect. These results highlighted the role of both shorter,

highly associated bigrams and longer, more complex sequences in proficient speech production. For a more detailed synthesis, see Section [7.1.3](#).

8.1.4 Variability in LB Usage Across Different Topics (Chapter 6)

The findings from Chapter 6 showed that the choice of presentation topic significantly influenced the use of high MI-score bigrams, highlighting the context-dependent nature of language use and its impact on proficiency scores. Specialized topics, such as those in STEM (Science, Technology, Engineering, and Math) and Law pathways, elicited higher mean MI scores compared to broader topics in HSS (Humanities and Social Sciences), suggesting that more technical subjects promote greater lexical sophistication. This variability underscores the importance of considering topic context when assessing oral proficiency, as topic-driven LB usage directly impacts fluency and proficiency outcomes. For a more detailed synthesis, see Section [7.1.4](#).

8.2 Implications for L2 Fluency Theory and Practice

8.2.1 Theoretical Implications

The findings from the current thesis contribute to the theoretical understanding of L2 fluency by shedding light on the roles of both shorter and longer LBs across different fluency aspects (speed, breakdown, and repair). Specifically, the results highlighted the distinct ways in which longer and shorter LBs influence fluency. Longer LBs generally enhanced speed and breakdown fluency by reducing mid-ASU pauses and increasing articulation rates. In contrast, shorter LBs played a more nuanced role, with some (bigram MI) facilitating breakdown fluency by reducing mid-clause pauses, while others (bigram frequency and trigram MI) introduced challenges by increasing end-ASU pauses. These findings align with and expand upon existing models of speech production (Levelt, 1989; Kormos, 2006; Skehan, 2014). Specifically, the results support the idea that using pre-constructed language

units helps conserve cognitive resources for more demanding communication tasks, consistent with Levelt's and Skehan's theories. Longer LBs enhanced speed fluency by facilitating faster articulation and reducing mid-clause pauses, thus freeing up cognitive resources for other aspects of communication. However, three-to-five-word MI had a modest negative effect on articulation rate, reflecting a cognitive trade-off in line with Skehan's (1998) trade-off hypothesis, which suggests that more complex LBs require greater processing effort but enable more precise communication. This trade-off suggests that both the length and complexity of LBs interact to shape various dimensions of L2 fluency. It also highlights the nuanced ways in which different types of LBs contribute to fluency, providing a deeper understanding of L2 speech production.

Moreover, comparisons with previous studies, such as Hougham et al. (2024a) and Tavakoli and Uchihara (2020), underscore the theoretical importance of both convergence and divergence. For example, while both Hougham et al. (2024a) and the current thesis (Chapter 4) identified positive influences of longer LBs on speed fluency, differences and similarities emerged in how shorter LBs influenced breakdown fluency. Specifically, in the current thesis, bigram MI was found to reduce mid-clause pauses, facilitating smoother speech. This finding diverges from those of Hougham et al. (2024a) and Tavakoli and Uchihara (2020), both of whom found that bigram MI contributed to increasing mid-clause pauses, although their findings were not statistically significant. This divergence is theoretically important because it challenges the assumption that more tightly associated bigrams necessarily hinder all aspects of fluency. Instead, the findings from the current thesis (Section 4.4) suggest that high-MI bigrams can facilitate smoother speech production, highlighting the task-specific and proficiency-dependent nature of LB effects in L2 fluency development.

Another example of divergence is found in the current thesis (Chapter 4), where shorter LBs (bigram frequency) were associated with increased end-ASU pauses. This contrasts with the findings of Hougham et al. (2024a) who reported the opposite effect (i.e., bigram frequency was associated with decreased end-ASU pauses, albeit not statistically significant). On the other hand, another example of convergence is that in the current thesis (Chapter 4), shorter LBs (Trigram MI) were associated with increased end-ASU pauses. This aligns with the findings of Tavakoli and Uchihara (2020), who similarly found a positive correlation between the number of end-clause pauses and the association measure of MI (albeit not statistically significant, as it missed reaching their corrected alpha level). These associations of shorter LBs with increased end-ASU pauses found in the current thesis potentially reflect increased planning demands. The differences and similarities with existing research emphasize the need to review theoretical models to better account for variability in LB use across different contexts and speaker characteristics. This suggests that context-specific factors, such as task demands or speaker characteristics, can influence how LBs contribute to fluency, thus necessitating a refinement of theoretical models. The findings of the current thesis thus not only validate but also challenge and extend existing theories, indicating areas that warrant further exploration and explication.

8.2.2 Practical Implications

The findings of the current thesis provide important insights for language teaching, curriculum design, and assessment. By highlighting the roles of both shorter and longer LBs in enhancing oral fluency and proficiency, the studies reported in Chapters 3-6 offer practical and actionable recommendations for educators and assessors.

8.2.2.1 Pedagogical Applications. As discussed in Section 5.6, the results emphasize the need for educators to focus not only on the frequency but also the quality of LBs,

particularly high-MI bigrams and longer sequences. Teachers can apply these findings in the following ways:

- Targeted practice: Designing tasks that raise awareness of high-MI bigrams and longer LBs, encouraging learners to incorporate these sequences into their spoken output.
- Curriculum design: Embedding high-quality LBs into listening, reading, and speaking exercises across different topics.
- Assessment integration: Including criteria related to the quality of LBs in speaking rubrics to monitor and support fluency development.

These practical steps (described in detail in Section 5.6) provide a framework for enhancing learners' speaking proficiency by emphasizing the importance of quality LBs.

8.2.2.2 Language Assessment on a Broader Scale. Taken together, the findings from the four studies in this thesis also suggest expanding the scope of oral fluency assessments on a wider scale. Incorporating LB analysis into standardized tests such as IELTS could offer a more nuanced evaluation of proficiency. Rather than focusing solely on general fluency and coherence, assessing the use of high-MI bigrams and longer LBs could provide insight into a learner's ability to manage complex language chunks and produce fluent, coherent speech. By integrating these elements into assessment criteria, examiners can better gauge the linguistic sophistication of learners' spoken output, as discussed in Chapter 5.

8.2.2.3 Broader Implications for Practice. The findings from Chapter 6 support a more flexible approach to language teaching and assessment, encouraging topic-driven language practice that reflects real-world communication needs while promoting fluency. This aligns with Content and Language Integrated Learning (CLIL) principles, where language development is integrated with subject-specific content, such as STEM or Law,

offering learners opportunities to engage with complex topics relevant to their academic or professional interests.

8.3 Reflections on the Contributions and Strengths of the Current Thesis

As highlighted in the Discussion chapter, the unique contributions of the current thesis lie in its detailed examination of how different types of LBs impact various aspects of fluency across proficiency levels. By providing empirical evidence on both the benefits and limitations of shorter and longer LBs, the studies in the current thesis enhance our understanding of the complexities involved in L2 speech production and fluency development, informing future theoretical frameworks and research directions. The findings emphasize the importance of both shorter, high-MI bigrams and longer sequences in shaping fluency, highlighting their distinct roles in different dimensions of speech performance. Additionally, the results showed both convergence and divergence with those of previous studies (e.g., Hougham et al., 2024a; Tavakoli & Uchihara, 2020), highlighting theoretically significant patterns. These points of convergence and divergence challenge existing assumptions and contribute to refining theoretical models of L2 fluency. Detailed discussions and syntheses of these convergences and divergences can be found in Sections 5.4, 7.1.2, 7.1.3, and 8.2.1.

The methodological strengths of the studies in the current thesis lie in their robust and multifaceted analytical approach, using a range of methods introduced across different chapters. In Chapters 3–5, a two-pronged approach was employed to measure lexical bundles, involving both text-internal and text-external techniques. Text-external analysis utilized an external reference corpus (COCA) with several n-gram measures (proportion, frequency, and association strength), following earlier studies (e.g., Tavakoli & Uchihara, 2020), whereas text-internal analysis focused on identifying and refining longer LBs within the learner

corpus using frequency-based analyses and an innovative LB refinement procedure (adapted from Wood & Appel, 2014).

In Chapters 4, 5, and 6, advanced statistical techniques were used, including robust regression, dominance analysis, and random forests analysis (Boruta algorithm), providing a thorough examination of how shorter and longer LBs contribute to fluency dimensions and oral proficiency scores. These different approaches ensured a thorough and nuanced understanding of the roles that 0LBs play in shaping L2 oral fluency and proficiency. The diverse statistical techniques allowed for a deeper and triangulated understanding of how various LBs contribute to different fluency dimensions, offering a more nuanced picture of L2 speech production. Additionally, the use of a large sample size, with learners from a variety of proficiency levels and L1 backgrounds, enhanced the generalizability of the findings, contributing significantly to the field's understanding of how proficiency and context influence LB usage.

8.4 Suggestions for Future Research

To effectively build upon the findings of the current thesis, here are specific suggestions for future research (corresponding to the list of limitations discussed in Section 7.2):

1. Expand the sample size and diversity to include a broader range of assessed proficiency levels to enhance generalizability.
2. Employ a longitudinal design to track LB usage and proficiency over time, providing insights into developmental trajectories.
3. Expand the scope of frequency-based analysis to include semi-fixed or infrequent multiword units for a more comprehensive understanding of MWS usage.

4. Explore alternative measures for operationalizing LB usage to better account for contextual appropriateness and longer LBs.
5. Investigate the use of longer LBs (e.g., six or more words) and their impact on fluency and proficiency.
6. Conduct a more fine-grained categorization of presentation topics and explore various types of speaking tasks (e.g., spontaneous speech, monologue vs. dialogue) to better understand topic complexity and its effect on LB usage.

8.5 Final Remarks

In conclusion, the studies in the current thesis have shed new light on the complex dynamics of LB usage in L2 oral fluency and provide a foundation for ongoing research and pedagogical applications. Overall, the results from the studies and the related discussions strengthen the hypothesis that LBs are important linguistic building blocks that help to facilitate L2 oral fluency and proficiency:

- LBs are used with increasing frequency and sophistication as proficiency levels rise, indicating a link between LB use and oral proficiency development.
- Both shorter and longer LBs play distinct roles in enhancing L2 oral fluency across different proficiency levels.
- High-MI bigrams contribute significantly to smoother speech production by reducing mid-clause pauses and strongly predict higher oral presentation scores.
- Longer LBs positively influence speed fluency and reduce breakdowns, contributing to more coherent and fluid speech.
- The interaction between LB length and complexity impacts fluency dimensions, with longer sequences introducing both benefits and potential processing challenges.

To conclude, it is worth reiterating the broader significance of these findings for our understanding of L2 oral fluency. By illustrating how both shorter and longer LBs shape different aspects of fluency, this work advances our theoretical grasp of L2 speech production while offering practical insights for language education and assessment. The nuanced roles of LBs highlight their value not just as linguistic tools but as markers of proficiency and fluency development. Moving forward, ongoing research into how LBs can be employed across various learning contexts will continue to refine both our theories of language acquisition and the ways we support learners in achieving greater fluency. These findings represent not just a conclusion but a stepping stone for future exploration and application in language learning and teaching.

Notes

1. A corrected alpha level is an adjusted significance threshold used when multiple comparisons are made to reduce the risk of false positives (Type I errors). In their study, Tavakoli and Uchihara (2020) applied the Bonferroni correction, lowering the threshold to $p < .0083$ ($0.05/6$ for six comparisons per measure).
2. Loewen and Plonsky (2016) have suggested that 5% trimming is a good option because trimming a greater percentage would result in a smaller sample size with the data set. However, for post-hoc comparisons, as of May 2023, there is not an option in the “WRS2” package that allows 5% trimming. The only option built into the “mcppb20” function is 20% trimming. Taking Larson-Hall’s (2015, p. 318) advice into consideration, I therefore chose to implement a 20% trimming approach.
3. Although Hougham et al. (2024a) used Otter.ai for transcription purposes, I opted for Sonix.ai. I considered a few factors when making my decision, and ultimately chose Sonix.ai because of its wider range of functionalities, like text and audio export options, and its user-friendly interface.
4. Foster et al. (2000) define an analysis of speech unit (ASU) as “a single speaker’s utterance consisting of *an independent clause, or subclausal unit*, together with any *subordinate clause(s)* associated with either” (p. 365). While ASUs are designed for spoken language analysis and may occasionally extend beyond traditional grammatical clauses, they generally align with clause boundaries. Thus, for clarity and consistency, the current thesis discusses ASUs and clauses interchangeably.
5. I also tried running a principal component analysis (PCA) to see if I could identify meaningful underlying n-gram factors in a similar way as Tavakoli and Uchihara (2020). However, various attempts at running PCAs found mixed clusters of factor loadings, so I ran multiple regressions instead of PCAs.

6. In addition, I made an attempt to run a second model by incorporating the top subcategories of presentation topics. The objective was to evaluate how these specific subcategories interacted with proficiency scores. This model similarly included interaction terms to explore whether the influence of more detailed subcategories on MI scores varied with proficiency scores. However, after various attempts, I continued to encounter issues such as not enough data points in sub-categories and non-significant results, so discarded the additional regression model.
7. Given that the robust ANOVA showed significant results only at the MI-score threshold of 5, it makes sense to focus on this threshold throughout the results section. This approach keeps the analysis focused and avoids unnecessary comparisons with non-significant thresholds (see Table 6.4 for ANOVA results).

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this thesis, I used ChatGPT4o to improve the text's readability and ensure it conforms with APA 7 style. After using this tool, I reviewed and edited the content as needed and take full responsibility for the content of this work.

References

- Ädel, A., & Erman, B. (2012). Recurrent word combinations in academic writing by native and non-native speakers of English: A lexical bundles approach. *English for Specific Purposes*, 31(2), 81-92. <https://doi.org/10.1016/j.esp.2011.08.004>
- Altenberg, B. (1998). On the Phraseology of Spoken English: The Evidence of Recurrent Word-Combinations. In A. P. Cowie (Ed.), *Phraseology* (pp. 101–123). Oxford University Press. <https://doi.org/10.1093/oso/9780198294252.003.0005>
- Anthony, L. (2022). AntConc (Version 4.2.0) [Computer Software]. Waseda University. Available from <https://www.laurenceanthony.net/software/antconc/>
- Appel, R., & Wood, D. (2016). Recurrent word combinations in EAP test-taker writing: Differences between high- and low-proficiency levels, *Language Assessment Quarterly*, 13(1), 55–71. <https://doi.org/10.1080/15434303.2015.1126718>
- Baayen, R. H. (2007). Storage and computation in the mental lexicon. In G. Jarema & G. Libben (Eds.), *The mental lexicon: Core perspective* (pp. 81–104). Elsevier.
- Ballance, O. J. (2022). Methodological considerations for the use of mutual information: Examining the role of context in collocation research. *Research Methods in Applied Linguistics*, 1(3), 100024. <https://doi.org/10.1016/j.rmal.2022.100024>
- Barlow, M. (2015). Collocate (Version 2.0). [Computer Software]. Athelstan. Available from <https://athel.com/>
- Biber, D. (2006). *University language: A corpus-based study of spoken and written registers*. John Benjamins.
- Biber, D. (2009). A corpus-driven approach to formulaic language in English: Multi-word patterns in speech and writing. *International Journal of Corpus Linguistics*, 14(3), 275–311. <https://doi.org/10.1075/ijcl.14.3.08bib>
- Biber, D., & Barbieri, F. (2007). Lexical bundles in university spoken and written

- registers. *English for Specific Purposes*, 26(3), 263–286.
<https://doi.org/10.1016/j.esp.2006.08.003>
- Biber, D., Conrad, S., & Cortes, V. (2004). If you look at...lexical bundles in academic lectures and textbooks. *Applied Linguistics*, 25, 371–405.
<https://doi.org/10.1093/applin/25.3.371>
- Biber, D., & Gray, B. (2013). Discourse characteristics of writing and speaking task types on the *Toefl iBT*® test: A lexico-grammatical analysis. *ETS Research Report Series*: i–128.
<https://onlinelibrary.wiley.com/doi/pdf/10.1002/j.2333-8504.2013.tb02311.x>
- Biber, D., Johansson, S., Leech, G., Conrad, S., & Finegan, E. (1999). *Longman grammar of spoken and written English*. Longman.
- Bishop, H. (2004). Noticing formulaic sequences: A problem of measuring the subjective. *LSO Working Papers in Linguistics*, 4, 15–19.
<https://langsci.wisc.edu/wp-content/uploads/sites/1012/2019/01/LSOWP4.1-04-Bishop.pdf>
- Bod, R. (2001). Sentence memory: Storage vs. computation of frequent sentences. Paper presented at the CUNY 2001 Conference on Human Sentence Processing.
- Boers, F. (2014). A reappraisal of the 4/3/2 activity. *RELC Journal*, 45(3), 221–235.
<https://doi.org/10.1177/0033688214546964>
- Boers, F., & Lindstromberg, B. (2012). Experimental and intervention studies on formulaic sequences in a second language. *Annual Review of Applied Linguistics* 32, 83–110.
<https://doi.org/10.1017/S0267190512000050>
- Boers, F., Eyckmans, J., Kappel, K., Stengers, H., & Demecheleer, M. (2006). Formulaic sequences and perceived oral proficiency: Putting a lexical approach to the test. *Language Teaching Research*, 10, 245–261.
<https://doi.org/10.1191/1362168806lr195oa>

- Boersma, P., & Weenink, D. (2016) PRAAT: Doing phonetics by computer [Computer program]. Version 6.1.39. Retrieved from <https://www.fon.hum.uva.nl/praat/>
- Bosker, H. R., Pinget, A.-F., Quené, H., Sanders, T., & de Jong, N. H. (2013). What makes speech sound fluent? The contributions of pauses, speed and repairs. *Language Testing*, 30, 159–175. <http://dx.doi.org/10.1177/0265532212455394>
- Bybee, J. (2006). From usage to grammar: The mind's response to repetition. *Language*, 82(4), 711–733.
<https://www.degruyter.com/database/COGBIB/entry/cogbib.1822/html>
- Bybee, J. (2007). *Frequency of use and the organization of language*. Oxford University Press.
- Bygate, M. (2018). *Learning language through task repetition*. John Benjamins.
- Carrol, G., & Conklin, K. (2014). Getting your wires crossed: Evidence for fast processing of L1 idioms in an L2. *Bilingualism: Language and Cognition*, 17, 784–797.
<https://doi.org/10.1017/S1366728913000795>
- Christiansen, M. H., & Arnon, I. (2017). More than words: The role of multiword sequences in language learning and use. *Topics in Cognitive Science*, 9, 542–551.
<https://doi.org/10.1111/tops.12274>
- Church, K. W., & Hanks, P. (1990). Word association norms, mutual information, and lexicography. *Computational Linguistics*, 16, 22–29.
<https://aclanthology.org/J90-1003.pdf>
- Cieślicka, A. (2006). Literal salience in on-line processing of idiomatic expressions by second language learners. *Second Language Research*, 22, 115–144.
<https://doi.org/10.1191/0267658306sr263oa>
- Clenton, J. (2015). Testing the revised hierarchical model: Evidence from word associations. *Bilingualism: Language and Cognition*, 18(1), 118–125.

<https://doi.org/10.1017/S136672891400008X>

- Columbus, G., & Wood, D. (2010). Processing MWUs: Are MWU subtypes psycholinguistically real? In D. Wood (Ed.), *Perspectives on formulaic language: Acquisition and communication* (pp. 194–212). Continuum
- Conklin, K., & Schmitt, N. (2008). Formulaic sequences: Are they processed more quickly than nonformulaic language by native and nonnative speakers? *Applied Linguistics*, 29, 72–89. <https://doi.org/10.1093/applin/amm022>
- Cortes, V. (2004). Lexical bundles in published and student disciplinary writing: Examples from history and biology. *English for Specific Purposes*, 23(4), 397–423. <https://doi.org/10.1016/j.esp.2003.12.001>
- Cowie, A. P. (1981). The treatment of collocations and idioms in learners' dictionaries. *Applied Linguistics* 2(3), 223–235. <https://doi.org/10.1093/applin/II.3.223>
- Cronk, B. C., & Schweigert, W. A. (1992). The comprehension of idioms: The effects of familiarity, literalness, and usage. *Applied Psycholinguistics*, 13, 131–146.
- Dang, T. N. Y., Coxhead, A., & Webb, S. (2017). The academic spoken word list. *Language Learning*, 67, 959–997. <https://doi.org/10.1111/lang.12253>
- Davies, M. (2009). The 385+ million word *Corpus of Contemporary American English* (1990–2008+): Design, architecture, and linguistic insights. *International Journal of Corpus Linguistics*, 14(2), 159–90. <https://doi.org/10.1075/ijcl.14.2.02dav>
- De Cock, S. (1998). A recurrent word combination approach to the study of formulae in the speech of native and non-native speakers of English. *International Journal of Corpus Linguistics*, 3(1), 59–80. <https://doi.org/10.1075/ijcl.3.1.04dec>
- De Cock, S. (2004). Preferred sequences of words in NS and NNS speech. *Belgian Journal of English Language and Literatures, New Series*, 2, 225–246.

- de Jong, N. (2016a). Predicting pauses in L1 and L2 speech: The effects of utterance boundaries and word frequency. *International Review of Applied Linguistics in Language Teaching*, 54, 113–132. <https://doi.org/10.1515/iral-2016-9993>
- de Jong, N. (2016b). Fluency in second language assessment. In D. Tsagari & J. Banerjee (Eds.), *Handbook of second language assessment* (pp. 203–218). De Gruyter.
- de Jong, N., & Bosker, H. R. (2013). Choosing a threshold for silent pauses to measure second language fluency. In R. Eklund (Ed.), *Proceedings of the 6th workshop on disfluency in spontaneous speech* (pp. 17–20). Royal Institute of Technology (KTH).
- de Jong, N., & Perfetti, C. A. (2011). Fluency training in the ESL classroom: An experimental study of fluency development and proceduralization. *Language Learning*, 61(2), 533–568. <https://doi.org/10.1111/j.1467-9922.2010.00620.x>
- de Jong, N., & Tillman, P. C. (2018). Grammatical structures and oral fluency in immediate task repetition: Trigrams across repeated performances. In M. Bygate (Ed.), *Language learning through task repetition* (pp. 43–73). John Benjamins.
- DeKeyser, R. M. (2001). Automaticity and automatization. In P. Robinson (Ed.), *Cognition and Second Language Instruction* (1st ed., pp. 125–151). Cambridge University Press.
- DeKeyser, R. M. (2018). Task repetition for language learning: A perspective from skill acquisition theory. In M. Bygate (Ed.), *Learning language through task repetition* (pp. 27–42). John Benjamins.
- Derwing, T. M., Munro, M. J., Thomson, R. I., & Rossiter, M. J. (2009). The relationship between L1 fluency and L2 fluency development. *Studies in Second Language Acquisition*, 31(4), 533–557. <https://doi.org/10.1017/S0272263109990015>
- Derwing, T. M., Rossiter, M. J., Munro, M. J., & Thomson, R. I. (2004). L2 fluency: Judgments on different tasks. *Language Learning*, 54, 655–679.

<https://doi.org/10.1111/j.1467-9922.2004.00282.x>

- Ding, Y. (2007). Text memorization and imitation: The practices of successful Chinese learners of English. *System* 35, 271–280. <https://doi.org/10.1016/j.system.2006.12.005>
- Dudek, B. (2023). One way ANOVA with R: Completely randomized design - between groups [PDF file]. Retrieved from https://bcdudek.net/anova/oneway_anova_basics.pdf
- Durrant, P. (2008). High frequency collocations and second language learning [Unpublished doctoral dissertation]. University of Nottingham, UK.
- Durrant, P., & Schmitt, N. (2009). To what extent do native and non-native writers make use of collocations? *International Review of Applied Linguistics in Language Teaching*, 47(2), 157–177. <https://doi.org/10.1515/iral.2009.007>
- Ebeling, S., & Hasselgård, H. (2015). Learner corpora and phraseology. In S. Granger, G. Gilquin, & F. Meunier. (Eds.), *The Cambridge handbook of learner corpus research* (pp. 207–229). Cambridge University Press.
- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *Annals of Statistics*, 7, 1–26.
- Eguchi M., & Kyle K. (2020). Continuing to explore the multidimensional nature of lexical sophistication: The case of oral proficiency interviews. *The Modern Language Journal*, 104(2), 381–400. <https://doi.org/10.1111/modl.12637>
- Ellis, N. C. (1996). Sequencing in SLA: Phonological memory, chunking, and points of order. *Studies in Second Language Acquisition*, 18(1), 91–126. <https://doi.org/10.1017/S0272263100014698>
- Ellis, N. C. (2002). Frequency effects in language processing: A review with implications for theories of implicit and explicit language acquisition. *Studies in Second Language Acquisition*, 24(2), 143–188. <https://doi.org/10.1017/S0272263102002024>

- Ellis, N. C. (2012). Formulaic language and second language acquisition: Zipf and the phrasal teddy bear. *Annual Review of Applied Linguistics*, 32, 17–44.
<https://doi.org/10.1017/S0267190512000025>
- Ellis, N. C., Simpson-Vlach, R., & Maynard, C. (2008). Formulaic language in native and second language speakers: Psycholinguistics, corpus linguistics, and TESOL. *TESOL Quarterly*, 42(3), 375–396. <https://doi.org/10.1002/j.1545-7249.2008.tb00137.x>
- Ellis, N. C., Frey, E., & Jalkanen, I. (2009). The psycholinguistic reality of collocation and semantic prosody (1): Lexical access. In U. Römer & R. Schulze (Eds.), *Exploring the lexis-grammar interface* (pp. 89–114). John Benjamins.
- Ellis, N. C., Simpson-Vlach, R., Römer, U., O'Donnell, M. B., & Wulff, S. (2015). Learner corpora and formulaic language in second language acquisition research. In S. Granger, G. Gilquin, & F. Meunier (Eds.), *The Cambridge handbook of learner corpus research* (1st ed., pp. 357–378). Cambridge University Press.
<https://doi.org/10.1017/CBO9781139649414.016>
- Ellis, R. (2005). Measuring implicit and explicit knowledge of a second language: A psychometric study. *Studies in Second Language Acquisition*, 27, 141–172.
<https://doi.org/10.1017/s0272263105050096>
- Fitzpatrick, T. (2006). Habits and rabbits: Word associations and the L2 lexicon. *EUROSLA Yearbook*, 6(1), 121–145. <https://doi.org/10.1075/eurosla.6.09fit>
- Fitzpatrick, T. (2013). Word associations. In C. A. Chapelle (Ed.), *The encyclopedia of applied linguistics* (pp. 1–6). Wiley Blackwell.
- Foster, P., & Skehan, P. (1996). The influence of planning and task type on second language performance. *Studies in Second Language Acquisition*, 18(3), 299–323.
<https://doi.org/10.1017/S0272263100015047>
- Foster, P., Tonkyn, A., & Wigglesworth, G. (2000). Measuring spoken language: A unit for

all reasons. *Applied Linguistics*, 21(3), 354–375.

<https://doi.org/10.1093/applin/21.3.354>

Freed, B. F. (2000). Is fluency, like beauty, in the eyes (and ears) of the beholder? In H. Riggensbach (Ed.), *Perspectives on fluency* (pp. 243–265). University of Michigan Press. <https://doi.org/10.3998/mpub.16109>

Gablasova, D., Brezina, V., & McEnery, T. (2017). Collocations in corpus-based language learning research: Identifying, comparing, and interpreting the evidence. *Language Learning*, 67(S1), 155–179. <https://doi.org/10.1111/lang.12225>

Garner, J., & Crossley, S. (2018). A latent curve model approach to studying L2 N-Gram development. *The Modern Language Journal*, 102(3), 494–511. <https://doi.org/10.1111/modl.12494>

Gatbonton, E., & Segalowitz, N. (1988). Creative automatization: Principles for promoting fluency within a communicative framework. *TESOL Quarterly*, 22, 473–492. <https://doi.org/10.2307/3587290>

Gatbonton, E., & Segalowitz, N. (2005). Rethinking communicative language teaching: A focus on access to fluency. *Canadian Modern Language Review*, 61, 325–353. <https://doi.org/10.3138/cmlr.61.3.325>

Gibbs, R. W., Bogdanovich, J. M., Sykes, J. R., & Barr, D. J. (1997). Metaphor in idiom comprehension. *Journal of Memory and Language*, 37, 141–154. <https://psycnet.apa.org/doi/10.1006/jmla.1996.2506>

Goldberg, A. E. (1995). *Constructions: A construction grammar approach to argument structure*. University of Chicago Press.

Goldberg, A. E. (2006). *Constructions at work: The nature of generalization in language*. Oxford University Press.

Granger, S. (2019). Formulaic sequences in learner corpora: Collocations and lexical bundles.

- In A. Siyanova-Chanturia, & A. Pellicer-Sánchez (Eds.) *Understanding formulaic language: A second language acquisition perspective* (pp. 228–247). Routledge.
- Granger, S., & Bestgen, Y. (2014). The use of collocations by intermediate vs. advanced non-native writers: A bigram-based study. *International Review of Applied Linguistics in Language Teaching*, 52(3), 229–252. <https://doi.org/10.1515/iral-2014-0011>
- Granger, S., & Paquot, M. (2008). Disentangling the phraseological web. In S. Granger & F. Meunier (Eds.), *Phraseology: An interdisciplinary perspective* (pp. 27–49). John Benjamins.
- Gries, S. T. (2013). 50-something years of work on collocations what is or should be next. *International Journal of Corpus Linguistics*, 18, 137–165. <https://doi.org/10.1075/ijcl.18.1.09gri>
- Gries, S. (2022). What do (some of) our association measures measure (most)? Association? *Journal of Second Language Studies* 5(1), 1–33. <https://doi.org/10.1075/jsls.21028.gri>
- Gries, S. T., & Ellis, N. C. (2015). Statistical measures for usage-based linguistics. *Language Learning*, 65, 228–255. <https://doi.org/10.1111/lang.12119>
- Grömping, U. (2006). Relative importance for linear regression in R: The package relaimpo. *Journal of Statistical Software*, 17(1). <https://doi.org/10.18637/jss.v017.i01>
- Groom, N. (2009). Effects of second language immersion on second language collocational development. In A. Barfield & H. Gyllstad (Eds.), *Researching collocations in another language* (pp. 21–33). Palgrave Macmillan.
- Gyllstad, H., & Schmitt N. (2018). Testing formulaic language. In A. Siyanova-Chanturia & A. Pellicer-Sánchez (Eds.) *Understanding formulaic language: A second language acquisition perspective* (pp. 174–191). Routledge.
- Gyllstad, H., & Wolter, B. (2016). Collocational processing in light of the phraseological

- continuum model: Does semantic transparency matter? *Language Learning*, 66, 296–323. <https://doi.org/10.1111/lang.12143>
- Hasselgård, H. (2019). Phraseological teddy bears: Frequent lexical bundles in academic writing by Norwegian learners and native speakers of English. In M. Mahlberg & V. Wiegand (Eds.), *Corpus linguistics, context and culture* (pp. 339–362). De Gruyter.
- Hilton, H. (2008). The link between vocabulary knowledge and spoken L2 fluency. *The Language Learning Journal*, 36(2), 153–166. <https://doi.org/10.1080/09571730802389983>
- Hoang, H., & Boers, F. (2016). Re-telling a story in a second language: How well do adult learners mine an input text for multiword expressions? *Studies in Second Language Learning and Teaching*, 6, 513–535. <https://doi.org/10.14746/ssllt.2016.6.3.7>
- Hougham, D., Clenton, J., & Uchihara, T. (2024a). Disentangling the contributions of shorter vs. longer lexical bundles to L2 oral fluency. *System*, 121, 103243. <https://doi.org/10.1016/j.system.2024.103243>
- Hougham, D., Clenton, J., Uchihara, T., & Higginbotham, G. (2024b). The impact of lexical bundle length on L2 oral proficiency. *Languages*, 9(7), 232. <https://doi.org/10.3390/languages9070232>
- Hyland, K. (2008). As can be seen: Lexical bundles and disciplinary variation. *English for Specific Purposes*, 27(1), 4–21. <https://doi.org/10.1016/j.esp.2007.06.001>
- Hyland, K. (2012). Bundles in academic discourse. *Annual Review of Applied Linguistics*, 32, 150–169. <https://doi.org/10.1017/S0267190512000037>
- JASP Team (2023). JASP (Version 0.17.2) [Computer software]. <https://jasp-stats.org/>
- Jeong, H., & Jiang, N. (2019). Representation and processing of lexical bundles: Evidence from word monitoring. *System*, 80, 188–198.

<https://doi.org/10.1016/j.system.2018.11.009>

Jiang, N., & Nekrasova, T. M. (2007). The processing of formulaic sequences by second language speakers. *The Modern Language Journal*, 91, 433–445.

<https://doi.org/10.1111/j.1540-4781.2007.00589.x>

Kahng, J. (2014). Exploring utterance and cognitive fluency of L1 and L2 English speakers: Temporal measures and stimulated recall. *Language Learning*, 64, 809–854.

<https://doi.org/10.1111/lang.12084>

Kahng, J. (2017). The effect of pause location on perceived fluency. *Applied Psycholinguistics*, 39(3), 569–591.

<https://doi.org/10.1017/S0142716417000534>

Kim, S. H., & Kim, J. H. (2012). Frequency effects in L2 multiword unit processing: Evidence from self-paced reading. *TESOL Quarterly*, 46, 831–841.

<https://doi.org/10.1002/tesq.66>

Koponen, M., & Riggenbach, H. (2000). Overview: Varying perspectives on fluency. In H. Riggenbach (Ed.), *Perspectives on fluency* (pp. 5–24). University of Michigan.

<https://doi.org/10.3998/mpub.16109>

Kormos, J. (2006). *Speech production and second language acquisition*. Routledge.

Kursa, M. B., & Rudnicki, W. R. (2010). Feature selection with the Boruta package. *Journal of Statistical Software*, 36(11). <https://doi.org/10.18637/jss.v036.i11>

Kutas, M., & Federmeier, K. D. (2007). Event-related brain potential (ERP) studies of sentence processing. In M. G. Gaskell (Ed.), *The Oxford handbook of psycholinguistics* (pp. 385–406). Oxford University Press.

Kyle, K., & Crossley, S. A. (2015). Automatically assessing lexical sophistication: Indices, tools, findings, and application. *TESOL Quarterly*, 49, 757–786.

<https://doi.org/10.1002/tesq.194>

- Kyle, K., Crossley, S., & Berger, C. (2018). The tool for the automatic analysis of lexical sophistication (TAALES): Version 2.0. *Behavior Research Methods*, 50, 1030–1046.
<https://doi.org/10.3758/s13428-017-0924-4>
- Langacker, R. W. (2008). *Cognitive grammar: A basic introduction*. Oxford University Press.
- Larson-Hall, J. (2015). *A guide to doing statistics in second language research using SPSS and R*. Routledge.
- Larson-Hall, J. (2017). Moving beyond the bar plot and the line graph to create informative graphics. *The Modern Language Journal*, 101(1), 244–270.
<https://doi.org/10.1111/modl.12386>
- Larson-Hall, J., & Herrington, R. (2010). Improving data analysis in second language acquisition by utilizing modern developments in applied statistics. *Applied Linguistics*, 31, 368–390. <https://doi.org/10.1093/applin/amp038>
- Larson-Hall, J., & Plonsky, L. (2015). Reporting and interpreting quantitative research findings: What gets reported and recommendations for the field. *Language Learning*, 65, 127–159. <https://doi.org/10.1111/lang.12115>
- LeBel, E. P., McCarthy, R. J., Earp, B. D., Elson, M., & Vanpaemel, W. (2018). A unified framework to quantify the credibility of scientific findings. *Advances in Methods and Practices in Psychological Science*, 1(3), 389–402.
<https://doi.org/10.1177/2515245918787489>
- Lennes, M. (2021). *SpeCT* - The speech corpus toolkit for Praat (Version 1.0.0).
 [Computer software script]. <https://lennes.github.io/spect/>
- Lennon, P. (1990). Investigating fluency in EFL: A quantitative approach. *Language Learning*, 40(3), 387–417. <https://doi.org/10.1111/j.1467-1770.1990.tb00669.x>
- Levelt, W. J. M. (1989). *Speaking: From intention to articulation*. The MIT Press.
- Levelt, W. J. M. (1992). Accessing words in speech production: Stages, processes, and

- representations. *Cognition*, 42(1-3), 1–22.
[https://doi.org/10.1016/0010-0277\(92\)90038-J](https://doi.org/10.1016/0010-0277(92)90038-J)
- Lewis, M. (1993). *The lexical approach: The state of ELT and a way forward*. Language Teaching Publications.
- Lewis, M. (ed.). (2000). *Teaching collocation: Further developments in the lexical approach*. Thomson.
- Liakhovitski, D., Bryukhov, Y., & Conklin, M. (2010). Relative importance of predictors: Comparison of Random Forests with Johnson’s Relative Weights. *Model Assisted Statistics and Applications*, 5(4), 235–249. <https://doi.org/10.3233/MAS-2010-0172>
- Liu, D. (2012). The most frequently-used multiword constructions in academic written English: A multi-corpus study. *English for Specific Purposes*, 31(1), 25–35.
<https://doi.org/10.1016/j.esp.2011.07.002>
- Loewen, S., & Plonsky, L. (2016). *An A to Z of applied linguistics research methods*. Palgrave.
- Lynch, T. (2018). Promoting learning from second language speaking tasks. In V. Samuda, K. Van den Branden, & M. Bygate (Eds.), *TBLT as a researched pedagogy* (Vol. 12, pp. 213–234). John Benjamins.
- Mair, P., & Wilcox, R. (2020). Robust statistical methods in R using the WRS2 package. *Behavior Research Methods*, 52, 464–488.
<https://doi.org/10.3758/s13428-019-01246-w>
- Martinez, R., & Schmitt, N. (2012). A phrasal expressions list. *Applied linguistics*, 33(3), 299–320. <https://doi.org/10.1093/applin/ams010>
- McGuire, M., & Larson-Hall, J. (2018). Teaching formulaic sequences in the classroom: Effects on spoken fluency. *TESL Canada Journal*, 34(3), 1–25.
<https://doi.org/10.18806/tesl.v34i3.1271>

- McGuire, M., & Larson-Hall, J. (2021). The contribution of high-frequency multi-word sequences to speech rate and listening perception among EFL learners. *Vocabulary Learning and Instruction*, 10(2), 18–29.
<https://doi.org/10.7820/vli.v10.2.mcguire.larson-hall>
- McNamara, D. S., Grasesser, A. C., McCarthy, P. M., & Cai, Z. (2014). *Automated evaluation of text and discourse with Coh-Metrix*. Cambridge University Press.
<https://doi.org/10.1017/CBO9780511894664>
- Meara, P., & Fitzpatrick T. (2000). Lex30: An improved method of assessing productive Vocabulary in an L2. *System*, 28(1), 19–30.
[https://doi.org/10.1016/S0346-251X\(99\)00058-5](https://doi.org/10.1016/S0346-251X(99)00058-5)
- Mizumoto, A. (2015). Langtest (Version 1.0) [Web application]. <https://langtest.jp/>
- Mizumoto, A. (2023). Calculating the relative importance of multiple regression predictor variables using dominance analysis and random forests. *Language Learning*, 73(1), 161–196. <https://doi.org/10.1111/lang.12518>
- Myles, F., & Cordier, C. (2017). Formulaic sequence (FS) cannot be an umbrella term in SLA: Focusing on psycholinguistic FSs and their identification. *Studies in Second Language Acquisition*, 39, 3–28.
<https://doi.org/10.1017/S027226311600036X>
- Nation, P. (1989). Improving speaking fluency. *System*, 17(3), 377–384.
[https://doi.org/10.1016/0346-251X\(89\)90010-9](https://doi.org/10.1016/0346-251X(89)90010-9)
- Nattinger, J. R., & DeCarrico, J. S. (1992). *Lexical phrases and language teaching*. Oxford University Press.
- Nekrasova, T. M. (2009). English L1 and L2 speakers' knowledge of lexical bundles. *Language Learning*, 59(3), 647–686.
<https://doi.org/10.1111/j.1467-9922.2009.00520.x>

- Nesi, H., & Basturkmen, H. (2006). Lexical bundles and discourse signalling in academic lectures. *International Journal of Corpus Linguistics*, 11(3), 283–304.
<https://doi.org/10.1075/ijcl.11.3.04nes>
- Nesselhauf, N. (2004). What are collocations? In D. Allerton, N. Nesselhauf, & P. Skandera (Eds.), *Phraseological units: Basic concepts and their application* (pp. 1–21). Schwabe.
- O'Brien, I., Segalowitz, N., Freed, B., & Collentine, J. (2007). Phonological memory predicts second language oral fluency gains in adults. *Studies in Second Language Acquisition*, 29(4), 557–582.
<https://doi.org/10.1017/S027226310707043X>
- Öksüz, D., Brezina, V., & Rebuschat, P. (2021). Collocational processing in L1 and L2: The effects of word frequency, collocational frequency, and association. *Language Learning*, 71, 55–98. <https://doi.org/10.1111/lang.12427>
- Paquot, M., & Granger, S. (2012). Formulaic language in learner corpora. *Annual Review of Applied Linguistics*, 32, 130–149.
<https://doi.org/10.1017/S0267190512000098>
- Pawley, A., & Syder, F. H. (1983). Two puzzles for linguistic theory: Nativelike selection and nativelike fluency. In J. C. Richards & R. W. Schmidt (Eds.), *Language and communication* (pp. 191–226). Longman.
- Plonsky, L., & Derrick, D. J. (2016). A meta-analysis of reliability coefficients in second language research. *The Modern Language Journal*, 100, 538–553.
<https://doi.org/10.1111/modl.12335>
- Plonsky, L., & Oswald, F. L. (2014). How big is 'big'? Interpreting effect sizes in L2 research. *Language Learning*, 64(4), 878–912.
- Prodromou, L. (2007). Bumping into creative idiomaticity. *English Today*, 23, 14–25.

<https://doi.org/10.1017/S0266078407001046>

- R Core Team (2019). R: A language and environment for statistical computing [Computer software manual]. Vienna, Austria. Retrieved from <https://www.R-project.org/>
- Rastle, K. (2007). Visual word recognition. In M. G. Gaskell (Ed.), *The Oxford handbook of psycholinguistics* (pp. 71–87). Oxford University Press.
- Renouf, A., & Sinclair, J. (1991). Collocational frameworks in English. In K. Aijmer & B. Altenberg (Eds.), *English corpus linguistics: Studies in honour of Jan Svartvik* (pp. 128–143). Longman.
- Saito, K. (2020). Multi- or single-word units? The role of collocation use in comprehensible and contextually appropriate second language speech. *Language Learning*, 70(2), 548–588. <https://doi.org/10.1111/lang.12387>
- Saito, K., Ilkan, M., Magne, V., Tran, M., & Suzuki, S. (2018). Acoustic characteristics and learner profiles of low, mid and high-level second language fluency. *Applied Psycholinguistics*, 39, 593–617. <https://doi.org/10.1017/S0142716417000571>
- Saito, K., Webb S., Trofimovich P., & Isaacs T. (2016). Lexical profiles of comprehensible second language speech: The role of appropriateness, fluency, variation, sophistication, abstractness, and sense relations. *Studies in Second Language Acquisition*, 38(4), 677–701. <https://doi.org/10.1017/S0272263115000297>
- Schmitt, N. (2010). *Researching vocabulary: A vocabulary research manual*. Palgrave Macmillan.
- Schmitt, N. (2012). Formulaic language and collocation. In C. Chapelle (Ed.), *The encyclopedia of applied linguistics* (1st ed., pp. 1–10). Blackwell. <https://doi.org/10.1002/9781405198431.wbeal0433>
- Schmitt, N., Grandage, S., & Adolphs, S. (2004). Are corpus-derived recurrent clusters psychologically valid? In N. Schmitt (Ed.), *Formulaic sequences* (pp. 127–148).

John Benjamins.

- Schmitt, N., & Underwood, G. (2004). Exploring the processing of formulaic sequences through a self-paced reading task. In N. Schmitt (Ed.), *Formulaic sequences* (pp. 171–189). John Benjamins.
- Segalowitz, N. (2010). *Cognitive bases of second language fluency*. Routledge.
- Segalowitz, N. (2016). Second language fluency and its underlying cognitive and social determinants. *International Review of Applied Linguistics in Language Teaching*, 54(2), 79–95. <https://doi.org/10.1515/iral-2016-9991>
- Segalowitz, N., & Gatbonton, E. (1995). Automaticity and lexical skills in second language fluency: Implications for computer assisted language learning. *Computer Assisted Language Learning*, 8(2-3), 129–149. <https://doi.org/10.1080/0958822940080203>
- Simpson-Vlach, R., & Ellis, N. C. (2010). An academic formulas list. *Applied Linguistics*, 31(4), 487–512. <https://doi.org/10.1093/applin/amp058>
- Sinclair, J. (1991). *Corpus concordance collocation*. Oxford University Press.
- Siyanova-Chanturia, A., Conklin, K., & Schmitt, N. (2011). Adding more fuel to the fire: An eye-tracking study of idiom processing by native and non-native speakers. *Second Language Research*, 27, 251–272. <https://doi.org/10.1177/0267658310382068>
- Siyanova-Chanturia, A., & Spina, S. (2020). Multi-word expressions in second language writing: A large-scale longitudinal learner corpus study. *Language Learning*, 70, 420–463. <https://doi.org/10.1111/lang.12383>
- Siyanova-Chanturia, A., & Van Lancker Sidtis, D. (2018). What on-line processing tells us about formulaic language. In A. Siyanova-Chanturia & A. Pellicer-Sánchez (Eds.), *Understanding formulaic language: A second language acquisition perspective* (pp. 38–61). Routledge.
- Skehan, P. (1998). *A cognitive approach to language learning*. Oxford University Press.

- Skehan, P. (2003). Task-based instruction. *Language Teaching*, 36, 1–14.
<https://doi.org/10.1017/S026144480200188X>
- Skehan, P. (2009). Modelling second language performance: Integrating complexity, accuracy, fluency, and lexis. *Applied Linguistics*, 30(4), 510–532.
<https://doi.org/10.1093/applin/amp047>
- Skehan, P. (Ed.) (2014). *Processing perspectives on task performance*. John Benjamins.
<https://doi.org/10.1075/tblt.5>
- Sonbul, S. (2015). Fatal mistake, awful mistake, or extreme mistake? Frequency effects on off-line/on-line collocational processing. *Bilingualism: Language and Cognition*, 18(3), 419–437. <https://doi.org/10.1017/S1366728914000674>
- Sonbul, S., El-Dakhs, D. A. S., Conklin, K., & Carrol, G. (2023). “Bread and butter” or “butter and bread”? Nonnatives’ processing of novel lexical patterns in context. *Studies in Second Language Acquisition*, 45, 370–392.
<https://doi.org/10.1017/S0272263122000237>
- Staples, S., Egbert, J., Biber, D., & McClair, A. (2013). Formulaic sequences and EAP development: Lexical bundles in the TOEFL iBT writing section. *English for Specific Purposes* 12(3), 214–25. <https://doi.org/10.1016/j.jeap.2013.05.002>
- Stengers, H., Boers, F., Housen, A., & Eyckmans, J. (2010). Does ‘chunking’ foster chunk-uptake? In S. De Knop, F. Boers, & A. De Rycker (Eds.), *Fostering language teaching efficiency through cognitive linguistics* (pp. 99–117). De Gruyter Mouton.
- Stengers, H., Boers, F., Housen, A., & Eyckmans, J. (2011). Formulaic sequences and L2 oral proficiency: Does the type of target language influence the association? *International Review of Applied Linguistics*, 49, 321–343. <http://dx.doi.org/10.1515/iral.2011.017>
- Suzuki, S., Kormos, J., & Uchihara, T. (2021). The relationship between utterance and perceived fluency: A meta-analysis of correlational studies. *The Modern Language*

- Journal*, 105(2), 435–463. <https://doi.org/10.1111/modl.12706>
- Suzuki, Y. (2020). Optimizing fluency training for speaking skills transfer: Comparing the effects of blocked and interleaved task repetition. *Language Learning*, 71, 285–325. <https://doi.org/10.1111/lang.12433>
- Suzuki, Y., & DeKeyser, R. (2015). Comparing elicited imitation and word monitoring as measures of implicit knowledge. *Language Learning*, 65, 860–895. <https://doi.org/10.1111/lang.12138>
- Suzuki, Y., & DeKeyser, R. (2017). The interface of explicit and implicit knowledge in a second language: Insights from individual differences in cognitive aptitudes. *Language Learning*, 67, 747–790. <https://doi.org/10.1111/lang.12241>
- Suzuki, Y., Eguchi, M., & De Jong, N. (2022). Does the reuse of constructions promote fluency development in task repetition? A usage-based perspective. *TESOL Quarterly*, 56(4), 1290–1319. <https://doi.org/10.1002/tesq.3103>
- Tavakoli, P. (2011). Pausing patterns: Differences between L2 learners and native speakers. *ELT Journal*, 65, 71–79. <https://doi.org/10.1093/elt/ccq020>
- Tavakoli, P. (2018). L2 development in an intensive study abroad EAP context. *System*, 72, 62–74. <https://doi.org/10.1016/j.system.2017.10.009>
- Tavakoli, P., & Hunter A. M. (2018). Is fluency being ‘neglected’ in the classroom? Teacher understanding of fluency and related classroom practices. *Language Teaching Research*, 22, 330–349. <https://doi.org/10.1177/1362168817708462>
- Tavakoli, P., Nakatsuhara, F., & Hunter, A. M. (2017). Scoring validity of the Aptis Speaking Test: Investigating fluency across tasks and levels of proficiency. ARAGs Research Reports Online. British Council. <https://www.britishcouncil.org/exam/aptis/research/publications/tavakoli>
- Tavakoli, P., & Skehan, P. (2005). Strategic planning, task structure and performance testing.

- In R. Ellis (Ed.), *Planning and task performance in a second language* (pp. 239–277). John Benjamins.
- Tavakoli, P., & Uchihara, T. (2020). To what extent are multiword sequences associated with oral fluency? *Language Learning*, 70(2), 506–547. <https://doi.org/10.1111/lang.12384>
- Thai, C., & Boers, F. (2016). Repeating a monologue under increasing time pressure: Effects on fluency, complexity, and accuracy. *TESOL Quarterly*, 50(2), 369–393. <https://doi.org/10.1002/tesq.232>
- Thomson, H., Boers, F., & Coxhead, A. (2019). Replication research in pedagogical approaches to spoken fluency and formulaic sequences: A call for replication of Wood (2009) and Boers, Eyckmans, Kappel, Stengers & Demecheleer (2006). *Language Teaching*, 52(3), 406–414. <https://doi.org/10.1017/S0261444817000374>
- Tomasello, M. (2003). *Constructing a language: A usage-based theory of language acquisition*. Harvard University Press.
- Tran, M. N., & Saito, K. (2021). Effects of the 4/3/2 activity revisited: Extending Boers (2014) and Thai & Boers (2016). *Language Teaching Research*, 28(2), 326–345. <https://doi.org/10.1177/1362168821994136>
- Tremblay, A., Derwing, B., Libben, G., & Westbury, C. (2011). Processing advantages of lexical bundles: Evidence from self-paced reading and sentence recall tasks. *Language Learning*, 61(2), 569–613. <https://doi.org/10.1111/j.1467-9922.2010.00622.x>
- Tremblay, A., & Tucker, B. V. (2011). The effects of N-gram probabilistic measures on the recognition and production of four-word sequences. *The Mental Lexicon*, 6(2), 302–324. <https://doi.org/10.1075/ml.6.2.04tre>
- Uchihara, T., & Clenton, J. (2020). Investigating the role of vocabulary size in second

language speaking ability. *Language Teaching Research*, 24(4), 540–556.

<https://doi.org/10.1177/1362168818799371>

Uchihara, T., Eguchi, M., Clenton, J., Kyle, K., & Saito, K. (2022). To what extent is collocation knowledge associated with oral proficiency? A corpus-based approach to word association. *Language and Speech*, 65(2), 311–336.

<https://doi.org/10.1177/00238309211013865>

Uchihara, T., & Harada, T. (2018). Roles of vocabulary knowledge for success in English-medium instruction: Self-perceptions and academic outcomes of Japanese undergraduates. *TESOL Quarterly*, 52, 564–587. <https://doi.org/10.1002/tesq.453>

Uchihara, T., & Saito, K. (2019). Exploring the relationship between productive vocabulary knowledge and second language oral ability. *The Language Learning Journal*, 47(1), 64–75. <https://doi.org/10.1080/09571736.2016.1191527>

Underwood, G., Schmitt, N., & Galpin, A. (2004). The eyes have it: An eye-movement study into the processing of formulaic sequences. In N. Schmitt (Ed.), *Formulaic sequences: Acquisition, processing and use* (pp. 153-172). John Benjamins.

Van Lancker Sidtis, D. (2012). Two-track mind: Formulaic and novel language support a dual-process model. In M. Fraust (Ed.), *The handbook of the neuropsychology of language* (pp. 342–367). Blackwell.

Wajnryb, R. (1990). *Grammar dictation*. Oxford University Press.

Webb, S., & Chang, A. C. S. (2022). How does mode of input affect the incidental learning of collocations? *Studies in Second Language Acquisition*, 44(1), 35–56.

<https://doi.org/10.1017/S0272263120000297>

Wilcox, R. R. (2017a). *Introduction to robust estimation and hypothesis testing* (4th ed.). Academic Press.

Wilcox, R. R. (2017b). *Modern statistics for the social and behavioral sciences: A practical*

- introduction* (2nd ed.). CRC Press.
- Wood, D. (1998). *Making the grade: An interactive course in English for academic purposes*. Prentice Hall.
- Wood, D. (2006). Uses and functions of formulaic sequences in second language speech: An exploration of the foundations of fluency. *Canadian Modern Language Review*, 63(1), 13–33. <https://doi.org/10.3138/cmlr.63.1.13>
- Wood, D. (2007). Mastering the English formula: Fluency development of Japanese learners in a study abroad context. *JALT Journal*, 29, 209–230. <https://doi.org/10.37546/JALTJJ29.2-3>
- Wood, D. (2009). Effects of focused instruction of formulaic sequences on fluent expression in second language narratives: A case study. *Canadian Journal of Applied Linguistics*, 12, 39–57. <https://journals.lib.unb.ca/index.php/CJAL/article/view/19898>
- Wood, D. (2010). *Formulaic language and second language speech fluency: Background, evidence, and classroom applications*. Continuum.
- Wood, D. (2015). *Fundamentals of formulaic language: An introduction*. Bloomsbury.
- Wood, D. (2020). Classifying and identifying formulaic language. In S. Webb (Ed.) *Routledge handbook of vocabulary studies* (pp. 30–45). Routledge.
- Wood, D., & Appel, R. (2014). Multiword constructions in first year business and engineering university textbooks and EAP textbooks, *Journal of English for Academic Purposes*, 15, 1–13. <https://doi.org/10.1016/j.jeap.2014.03.002>
- Wray, A. (2002). *Formulaic language and the lexicon*. Cambridge University Press.
- Wray, A. (2004). ‘Here’s one I prepared earlier’: Formulaic language learning on television. In N. Schmitt (Ed.), *Formulaic sequences: Acquisition, processing and use* (pp. 249–268). John Benjamins.
- Wray, A. (2012). What do we (think we) know about formulaic language? An evaluation of

- the current state of play. *Annual Review of Applied Linguistics*, 32, 231–254.
<https://doi.org/10.1017/S026719051200013x>
- Wray, A., & Fitzpatrick, T. (2008). Why can't you just leave it alone? Deviations from memorized language as a gauge of nativelike competence. In F. Meunier & S. Granger (Eds.), *Phraseology in foreign language learning and teaching* (pp. 123–148). John Benjamins.
- Wray, A., & Namba, K. (2003). Use of formulaic language by a Japanese–English bilingual child: A practical approach to data analysis. *Japan Journal of Multilingualism and Multiculturalism*, 9, 24–51.
- Wray, A., & Pegg, C. (2009). The effect of memorized learning on the writing scores of Chinese IELTS test takers. *IELTS Research Reports* 9, 5, British Council and IELTS Australia. <https://www.ielts.org/for-researchers/research-reports/volume-09-report-5>
- Wray, A., & Perkins, M.R. (2000). The functions of formulaic language: An integrated model. *Language and Communication*, 20(1), 1–28.
- Yi, W. (2018). Statistical sensitivity, cognitive aptitudes, and processing of collocations. *Studies in Second Language Acquisition*, 40, 831–856.
<https://doi.org/10.1017/s0272263118000141>
- Yi, W., Lu, S. Y., & Ma, G. J. (2017). Frequency, contingency and online processing of multiword sequences: An eye-tracking study. *Second Language Research*, 33, 519–549. <https://doi.org/10.1177/0267658317708009>
- Yi, W., & Zhong, Y. (2024). The processing advantage of multiword sequences: A meta-analysis. *Studies in Second Language Acquisition*, 46(2), 427–452.
<https://doi.org/10.1017/S0272263123000542>
- Zareva, A., & Wolter, B. (2012). The 'promise' of three methods of word association analysis to L2 lexical research. *Second Language Research*, 28(1), 41–67.

<https://doi.org/10.1177/0267658311423452>

Zhang, X., Zhao, B., & Li, W. (2021). N-gram use in EFL learners' retelling and monologic tasks. *International Review of Applied Linguistics in Language Teaching*, 61(3), 939–965. <https://doi.org/10.1515/iral-2021-0080>

Appendix A

This appendix provides a side-by-side comparison of the methods used in Tavakoli and Uchiyara (2020) versus those used in Chapters 3 and 4 of the current thesis.

Table A1: *Summary of the Methods Used in Tavakoli and Uchiyara (2020) and in Chapters 3 and 4 of the Current Thesis*

	Tavakoli and Uchiyara (2020)	Chapters 3 and 4 of the Current Thesis
Participants		
Sample size	$N = 56$	$N = 150$
L2	English	English
L2 proficiency	Low B1, High B1, B2, and C1 on the CEFR	6.5, 7, and 7.5 on the IELTS scale.
L1	10 different L1 backgrounds, including Chinese, Arabic, and Japanese	20 different L1 backgrounds including Chinese, Arabic, and Turkish.
Speaking Task		
Task type ¹	A standardized speaking proficiency test called the TEEP. ² The TEEP speaking test includes three tasks but only the first one was used in their study: A monologic task in which the test-taker is expected to speak for 3 mins about a given topic.	An oral presentation consisting of a 7-min presentation (monologic task) with the aid of PowerPoint slides. I took 3-min speech samples starting at the 30-sec timestamp
Spoken sample length	3 mins	3 mins
Topic	The TEEP speaking test includes three tasks, all on the same topic or question. Tavakoli and Uchiyara's study did not explicitly note whether all participants in their study were given the same version of the TEEP test.	While globalization served as the central theme, participants were encouraged to explore this broad topic through various lenses, ranging from economic and legal perspectives (e.g., how globalization affects development and international trade law) to more specialized areas (e.g., acoustic telemetry in fisheries and blockchain technology beyond bitcoin)
Measuring Fluency		
Speed fluency	Articulation rate	Articulation rate
Breakdown fluency	Frequency of mid- and end-clause pauses per minute of speaking time (excluding silent pauses)	Frequency of mid- and end-clause pauses per minute of speaking time (excluding silent pauses)
Repair fluency	Frequency of total repairs	Frequency of total repairs
Lexical Bundle (N-Gram) Analysis		
Text-external n-gram indices to analyze shorter LBs	Proportion, frequency, and association based on two- and three-word contiguous chunks (i.e., bigram and trigram tokens). Computed using TAALES software with the spoken subsection of COCA as the external reference corpus.	Proportion, frequency, and association based on two- and three-word contiguous chunks (i.e., bigram and trigram tokens). Computed using TAALES software with the spoken subsection of COCA as the external reference corpus.

Table A1: Cont.

Lexical Bundle (N-Gram) Analysis: Cont.		
Text-internal quantitative analysis to analyze longer LBs	None	Frequency analyses were conducted using AntConc to generate lists of the most frequently used four-word LBs. An LB refinement procedure was used to identify more structurally complete and useful LBs, primarily comprising core three-word units and four-word structures and several five-word units. An original scoring system was devised to quantify usage of my list of refined LBs. Points were awarded for raw usage counts and MI scores for each refined LB used by each participant.
Text-internal qualitative analysis	Profiles of the most frequently used bigrams and trigrams across proficiency levels were examined. Token counts of the three most frequent discourse markers were conducted.	None
Statistical Analyses		
To examine how n-gram units vary across proficiency levels	Kruskal–Wallis H test (to account for violations of homogeneity of variance)	Robust one-way ANOVA (to account for outliers)
To identify meaningful underlying n-gram factors	Principal component analysis (PCA)	None ³
To isolate the unique contribution of each n-gram factor/measure to each aspect of oral fluency	Pearson correlations between fluency measures and n-gram factors.	Robust multiple regression models between fluency measures and LB measures
To examine the relative importance of each predictor variable	None	Dominance analysis in combination with random forests analysis (Boruta algorithm)

Notes:

1. All participants in both studies did the speaking task as a UK university entrance requirement.
2. The TEEP is a standardized proficiency test used by a number of universities in the UK to obtain information about candidates' proficiency before they start a university degree (See Tavakoli & Uchihara, 2020, pp. 8-10, for a more detailed description of the TEEP).
3. I tried running a PCA in a similar way as Tavakoli and Uchihara. However, various attempts at running PCAs found mixed clusters of factor loadings, so instead of PCA I eventually decided to run robust multiple regression in combination with dominance analysis and random forests.

Appendix B

This appendix shows the written consent form used to request access to the spoken data.

Figure A1: *Written Consent Form for Access to the Spoken Data (Chapters 3-6)*

QMUL Productive Vocabulary Research Project

Dear Students,

We would like to ask for your help with a research project. Specifically, for your permission to analyse the essays and seminar discussions that are part of the pre-sessional course.

This research project will explore the vocabulary that second-language learners (you) use in academic tasks. To do this, we intend to use the written and spoken tasks that you submit at the end of the pre-sessional course to make a corpus. This corpus will be used to analyse lexical diversity and also explore the multi-word-units (phrases) that L2 learners use in academic tasks. The aim is to gain a greater understanding of the vocabulary students such as yourselves can produce.

Note that an early step in the analysis will be to anonymise the data; your name will therefore not appear in any reports or articles concerning this research. Another point to consider is that the analysis will be done after the pre-sessional course has finished, so it will not affect your grade in any way.

As the data will come from tasks that you will do anyway as part of your pre-sessional course, there is nothing extra for you to do (but please give us access to your seminar discussion screencasts).

- If you are happy for us to use your data in this project, please click below to say that you agree.
- If you do not wish to be part of this project, that is fine too; this will not affect your grade on the pre-sessional course (or future academic work) in any way.
- If you decide to withdraw later (you do not need to give a reason), send one of us an email and we will remove your data from the database.

To show our appreciation for your participation in this project, students who agree to take part in this project will have the opportunity to win one of four gift cards worth 25 pounds sterling each. Simply provide your name and email address below, and by clicking on "I agree" you'll be entered into the draw. That's all it takes to participate and be eligible for a chance to win! Once our data collection is complete (around mid-October), we will select four participants's by random draw and will notify the lucky winners by email. The gift cards will be sent to the email address provided in this form.

We wish you the best of luck with the rest of your pre-sessional course!

Kind regards,
 George Higginbotham, Ph.D.
 Dan Hougham, M.A.
 (Co-Investigators)

gmhigginbotham@gmail.com
hougham@hiroshima-u.ac.jp

Figure A1: Cont.

* Required

1. Name * 

Enter your answer

2. Pathway * 

☐ HSS

☐ Law

☐ STEM

3. Reading/Writing Class number 

Enter your answer

4. Do you agree to your essay and seminar discussions being used for research? * 

☐ Yes, I agree.

☐ No, I don't agree.

5. Your email address (if you wish to have a chance to win a gift card, please type your email address here) 

Enter your answer

6. If you have any questions or concerns about the study, please, in the first instance, contact Dr. Higginbotham: gmhigginbotham@gmail.com
Alternatively, you may ask questions to William Tweddle w.tweddle@qmul.ac.uk 

Submit

Appendix C

This appendix shows descriptors used by teachers who graded the pre-sessional presentations at Queen Mary University of London.

Table A2: Presentations Grading Descriptors (Chapter 3)

80 – 100% CEFR: C2		70 - 79%+ CEFR: HIGH C1	60 – 69% CEFR: LOW C1	50 – 59% CEFR: HIGH B2	40 – 49% CEFR: LOW B2	30 – 39% CEFR: B1	1 – 29% CEFR: A1 –A2
Presentation Content [20]	- Purpose of presentation is clear, appropriate and fully achieved. - Presentation is clearly focused and only relevant issues presented. - Excellent research which is clearly demonstrated through illustrations and examples. - All source material cited. - Visual aids are designed to a professional standard in terms of layout, bibliography and contents. - Very good analysis, synthesis and application of research.	- Purpose of presentation is clear, appropriate and fully achieved. - Presentation is focused and only relevant issues presented. - Appropriate research is clearly demonstrated through illustrations and examples. - All source material cited. - Clear, well-designed visual aids; effectively proofread. - Good analysis, synthesis and application of research.	- Purpose of presentation is clear, appropriate and largely achieved. - Presentation focused; issues presented are mainly relevant issues. - Appropriate research is demonstrated through illustrations and examples. - All source material cited, despite minor errors. - Generally clear and well-designed visual aids. Some evidence of proof reading but some errors may persist despite this. - Some evidence of ability to analyse, synthesise and apply research.	- Appropriate and adequately achieved purpose though may lack clarity. - May be occasional loss of focus and irrelevancies in parts. - Presentation shows some evidence of research and an understanding of the topic. - All source material is cited, though with some errors. - Generally satisfactory design of visual aids. Some lack of proof reading may result in careless mistakes. - Presentation may be more descriptive than analytical.	- Purpose of presentation is appropriate but may not be entirely achieved. - Some loss of focus & some irrelevancies may be evident. - Presentation demonstrates evidence of adequate research and some understanding of the topic - Most source material is cited, though with frequent errors. - Adequately designed visual aids. Inadequate proof reading may lead to careless mistakes. - Presentation may be rather descriptive.	- Purpose of presentation may be unclear or inappropriate. - Presentation is generally unfocused with many irrelevancies. - Presentation demonstrates little evidence of research, weak understanding of the topic. - Some citation of source material. - Visual aids may provide inadequate support for the presentation. Inadequate proof reading or lack of proof reading may result in careless errors. - Presentation may be largely descriptive.	- Purpose of presentation unclear or inappropriate. - Presentation is unfocused and contains many irrelevancies. - Presentation demonstrates no evidence of research and limited understanding of the topic. - Little or no citation of source material. - Visual aids non-existent or inadequate. Lack of proof-reading results in incomprehension. - Presentation may be entirely descriptive.

Table A2: *Cont.*

	80 – 100% CEFR: C2	70 – 79%+ CEFR: HIGH C1	60 – 69% CEFR: LOW C1	50 – 59% CEFR: HIGH B2	40 – 49% CEFR: LOW B2	30 – 39% CEFR: B1	1 – 29% CEFR: A1 –A2
Presentation Structure [20]	- Excellent flow; causes no difficulties for listener. - Logical sequencing of ideas very good organisation of presentation. Excellent introduction and Conclusion. Questions invited - Very good organization within sections. - Excellent use of sign-posting expressions to create cohesion and coherence.	- Good flow; causes no difficulties for listener. - Logical sequencing of ideas good organisation of presentation. Very clear introduction and Conclusion. Questions invited - Good organization within sections. - Effective use of sign-posting expressions to create cohesion and coherence.	- Reasonably good flow; causes few difficulties for listener. - Good sequencing of ideas which enables the message to be followed clearly. Good Introduction and Conclusion. - Reasonably good organization within sections although some room for improvement. Questions invited. - Fairly good use of sign-posting language to create cohesion & coherence.	- Reasonable flow causes occasional difficulties for listener. - Some ability to sequence ideas but overall structure may contain flaws. Reasonable Introduction and Conclusion. - Fairly good attempt to organize sections into main and supporting ideas; some use of examples but insufficient. Questions invited. - Attempt at sign-posting language; sometimes inappropriate or inaccurate; parts may lack cohesion.	- Adequate flow but causes some difficulties for listener. - Limited ability to sequence ideas and overall organisation may be flawed but the message can be followed adequately. Introduction and Conclusion may be simplistic, overlong or rushed. Questions not immediately invited. - Sections may lack unity but may show an attempt to use topic and supporting sentences. - Attempt at sign-posting language but it may be inappropriate; there may be some lack of cohesion.	- Lack of flow but causes strain for listener. - Flawed overall sequence of ideas but message can be followed in places. Introduction and Conclusion: simplistic, weak; do not correspond to body. Questions not invited. - Section structure may be weak and disconnected – sections short and disjointed; little use of examples and illustrations. - Limited use of sign-posting language and often inappropriate; some lack of cohesion.	- Absence of flow which often puts strain on listener. - Ideas are poorly sequenced organized, and the message is difficult to follow. Lacks clear organization structure. Little understanding of the purpose of introductions and Conclusions. Questions not invited. - Sections poorly organized and show little understanding of the purpose of structure. - Very limited or inaccurate use of sign-posting language; lack of cohesion.
Seminar Leadership [20]	- There is a totally clear task for seminar participants and the content is all highly focused and relevant. - The student clearly demonstrates a very high level of awareness of his/her audience. - The discussion is excellently controlled throughout. - The student gives a highly lucid summary of the discussion at its conclusion.	- There is a clear task for seminar participants and the content is focused and relevant. - The student demonstrates very good awareness of his/her audience. - The discussion is very well controlled. - The student gives a very good, lucid summary of the discussion at its conclusion.	- There is a fairly clear task for seminar participants and the content is mostly focused and relevant. - The student demonstrates good awareness of his/her audience. - The discussion is well controlled. - The student gives a good summary of the discussion at its conclusion.	- There is a task for seminar participants and the content is mostly relevant, but there may be some lack of clarity. - The student has satisfactory awareness of his/her audience. - An acceptable attempt is made to control the discussion. - The student gives a satisfactory summary of the discussion at its conclusion.	- There is a task for seminar participants, but it may not be presented clearly. Some of the content may lack focus and relevance. - The student may lack awareness of his/her audience. - The discussion may not be well controlled. - The student gives a summary of the discussion at its conclusion, but this may lack clarity.	- There may be some confusion about the task for seminar participants. The content lacks focus and relevance. - The student lacks awareness of his/her audience. - The discussion is only just controlled. - The student gives a summary of the discussion at its conclusion, but this lacks clarity.	- The task for seminar participants may be inappropriate, or unclear and is poorly explained. The content is unfocused and irrelevant. - The student has little or no awareness of his/her audience. - The discussion is not controlled. - The student fails to give a summary of the discussion at its conclusion or does this very poorly.

Table A2: *Cont.*

80 – 100% CEFR: C2		70 - 79%+ CEFR: HIGH C1		60 – 69% CEFR: LOW C1		50 – 59% CEFR: HIGH B2		40 – 49% CEFR: LOW B2		30 – 39% CEFR: B1		1 – 29% CEFR: A1 –A2	
Language Fluency [20]	• Clear pronunciation all the time.	• Clear pronunciation most of the time.	• Generally clear pronunciation.	• Pronunciation is clear but there are some mispronunciations.	• Pronunciation is generally clear enough to be understood despite a noticeable accent.	• Mispronunciation sometimes makes communication difficult.	• Mispronunciation severely impedes communication.						
	• Very good, fluent command of language with almost no hesitations and excellent control of speed.	• Good, fluent command of language with few hesitations and very good control of speed.	• Good, fluent production with some hesitations but good control of speed.	• Speaks with a degree of fluency but with limited control of speed and some hesitations.	• Can speak but with significant hesitation. May require a ‘sympathetic’ interlocutor.	• Hesitations can make communication difficult. Speed may be too fast or too slow. Often requires a ‘sympathetic’ interlocutor.	• Frequent hesitation or lack of control over speed severely impedes communication. Requires a						
	• Excellent use of intonation and stress to convey stance and topic changes	• Very good use of intonation and stress to convey stance and topic changes	• Generally good use of intonation and stress to convey stance and topic changes.	• Reasonable use of intonation and stress to convey topic changes, but stance may not always be evident.	• Intonation and stress may only occasionally be used to convey stance or topic change.	• Stance and topic change not signalled with intonation and stress.	• ‘sympathetic’ and active interlocutor.’						
	• Register always appropriate for this type of interaction.	• Register always appropriate for type of interaction.	• Register generally appropriate for type of interaction.	• Register reasonably appropriate for type of interaction.	• Register is just appropriate; may sometimes be inappropriate for type of interaction.	• Register is often inappropriate for interaction.	• Little control of intonation and stress.						
	• Script independent; very confident and effective use of non-verbal communication (e.g. facial expressions, appropriate appearance);	• Script independent; confident and effective use of non-verbal communication (e.g. facial expressions, appropriate appearance);	• Generally, script independent; effective use of non-verbal communication (e.g. facial expression, appropriate appearance);	• Often script independent; often effective use of non-verbal communication (e.g. facial expressions) and acceptably appropriate appearance;	• Partly script independent; some limited awareness of non-verbal communication (e.g. facial expressions used effectively on occasion, fairly appropriate appearance);	• Script dependent; little awareness of non-verbal communication (e.g. facial expressions sometimes inappropriate, fairly inappropriate appearance);	• Script dependent; poor awareness of non-verbal communication (e.g. inappropriate facial expressions and/or inappropriate appearance);						

Table A2: *Cont.*

	80 – 100% CEFR: C2	70 – 79%+ CEFR: HIGH C1	60 – 69% CEFR: LOW C1	50 – 59% CEFR: HIGH B2	40 – 49% CEFR: LOW B2	30 – 39% CEFR: B1	1 – 29% CEFR: A1 – A2
Language Accuracy [20]	- Student demonstrates mastery of the grammar required for the task; excellent ability to manipulate complex structures, - Excellent use of vocabulary which is appropriate to the task - Excellent academic style with totally appropriate use of register, very good ability to express caution and to avoid overgeneralizing. - Clear evidence of proof-reading (in visuals) and practice in presentation.	- Student demonstrates an authoritative use of the grammar required for the task; good ability to manipulate complex structures, - Good use of vocabulary which is appropriate to the task - Very good academic style with appropriate use of register, good ability to express caution and to avoid overgeneralizing. - Clear evidence of proof-reading (in visuals) and practice in presentation.	- Student shows an above average level of use of grammar required for the task; some use of complex structures but perhaps incorrect use. - Good range of appropriate vocabulary. - Good awareness of academic style (register, expression of caution, few overgeneralizations). - Good evidence of proof-reading (in visuals) and practice in presentation but some errors may persist despite this.	- Student shows a reasonable use of grammar with some ability to manipulate complex structures. There may be a limited number of grammatical errors, but these do not interfere with meaning. - Vocabulary generally appropriate to the task. - Awareness of academic style, but some inappropriate register, expression of caution may be weak, and overgeneralizations may be evident. - Some lack of proof-reading (in visuals) and practice in presentation may result in careless mistakes.	- Student shows a basic grasp of grammar, but limited ability to manipulate complex structures. Errors may interfere with meaning. - Adequate range of appropriate vocabulary: a narrow range of simple language. - Some awareness of academic style but there are likely to be several overgeneralisations and limited ability to express caution. - Inadequate proof-reading (in visuals) and practice in presentation may lead to careless mistakes.	- There may be recurrent grammatical errors and limited ability to manipulate complex structures - Some inappropriate use of vocabulary. - Choice of style and register is often inappropriate. - Inadequate proof reading and practice may result in careless errors.	- Significant, recurrent grammatical errors. Very limited ability to manipulate structures appropriately and frequent errors in basic grammatical structures. - Range of vocabulary is inadequate for the task; errors make the meaning difficult to discern and cause strain for the reader. - Limited or no ability to use academic style - Lack of proof reading and practice results in incomprehension.

Note. Shared with permission from Queen Mary University of London.

Appendix D

This appendix presents the most frequent three-, four-, and five-word recurrent combinations used by all participants, ranked by MI score.

Table A3

The Most Frequent Three-, Four- and Five-Word Recurrent Word Combinations Used by All Speakers (N = 150): Ranked by MI Score (Chapter 3)

Rank	Rank	Bundles	Freq	Range	FREQ as	MI
21	1	if you have any questions	22/18*	22/18	18	33.14
29	2	market forces and free competition (are)	18/18	9/9	18	32.59
73	3	in a socially responsible way	6/5	5/4	5	32.17
98	4	the possibility of armed conflict	4/4	3/3	4	28.58
111	5	Let's start by taking	3	3		25.07
63	6	I shall(will) only take	7(3)**	7(3)		24.89
56	7	the presentation will be brief	10/10	10/10	10	23.62
34	8	(seven) minutes of your time	14	14		23.40
101	9	a wide range of	3	3		22.34
78	10	have a seminar discussion	5	5		21.27
77	11	does globalization lead to	5	5		20.87
37	12	at the same time	12	12		20.78
47	13	as we all know	11	11		20.65
95	14	can be defined as	4	3		20.50
69	15	will last about seven (to eight minutes)	6	6		20.43
86	16	(have/has) a positive impact on	5	4		20.38
97	17	for a long time	4	3		20.32
108	18	it can be seen	3	3		20.10
99	19	a large amount of	3	3		20.06
76	20	the rich and the poor	4/4	4/4	4	19.99
24	21	as you(we) can see (from)	20(7)	17(7)		19.87
80	22	(to) help you understand the	5	5		19.72
102	23	advantage and disadvantage of	3	3		19.71
74	24	on the other hand	6	5		19.55
2	25	I would like to (talk about the/discuss/explain/introduce)	50	39		19.50
61	26	all over the world	8	6		19.31
75	27	we all know that	6	4		19.26
96	28	developed and developing countries	4	3		18.79
100	29	a large number of	3	3		18.72
5	30	(firstly/then) I will talk about	37	24		18.49
118	31	to deal with this	3	3		18.26
85	32	(my) presentation is(will be) divided into	5(4)	5(4)		18.25
50	33	I will focus on	10	10		18.24
117	34	This talk will last	3	3		17.58
13	35	corporate social responsibility (is)	26	12		17.51
105	36	in the United States	3	3		17.08
43	37	we are going to (talk)	12	6		16.76

109	38	it is easy to	3	3		16.32
110	39	my presentation today is	3	3		16.21
9	40	(is) my thesis statement (is/and)	28	22		16.04
67	41	Let's turn to the	7	5		15.88
25	42	at the end of (my/the talk/presentation)	20	19		15.86
88	43	as a result of	4	4		15.58
92	44	I will make a	4	4		15.27
40	45	(I will be) glad(happy) to answer (them)	12(5)	12(5)		15.18
106	46	is a form of	3	3		14.89
58	47	the last one is (the)	8	8		14.79
23	48	and then I(we) will (show)	21(4)	19(4)		14.70
116	49	there will be a	3	3		14.60
107	50	is the most important	3	3		14.39
119	51	will move on to	3	3		14.20
82	52	(I) will start with the	5	5		14.16
87	53	moving on to the	5	4		14.15
10	54	Let's move on (to the)	28	22		14.10
81	55	I will analyze the	5	5		14.00
89	56	in the context of	4	4		13.95
112	57	Let's talk about the	3	3		13.90
1	58	(today/firstly/then) I'm going to (talk about/present/analyze)	52	34		13.87
45	59	Let's start(begin) with (the)	11(5)	11(5)		13.70
114	60	the last part is	3	3		13.24
113	61	So in this presentation	3	3		13.09
30	62	we can see (the/that)	18	11		12.87
14	63	(I have divided) my presentation(talk) into (four/five parts)	25(4)	25(4)		12.78
94	64	So what are the	4	4		12.64
103	65	in the era of	3	3		12.53
104	66	in the field of	3	3		12.40
66	67	a brief introduction (of)	7	7		12.39
18	68	(is/about) the relationship between (a and b)	24	14		12.30
8	69	(first/then/second) I will introduce (the/my/some)	29	18		12.13
4	70	(firstly/and finally) I will give (you/the/a/some)	41	31		11.83
91	71	we move to the	4	4		11.81
93	72	(be) seen as a	4	4		11.66
90	73	that globalization is a	4	4		11.32
65	74	as well as (the)	7	7		11.15
6	75	(Let's) look at the (first part)	34	28		10.97
12	76	I want to (talk about/show)	26	20		10.88
39	77	(this/here) is my outline (and)	12	12		10.82
33	78	(with/about) the background information	15	10		10.78
52	79	(then) I will discuss (the)	10	10		10.75
31	80	We need to (look at)	17	9		10.74
41	81	Let's move to (the)	12	10		10.74
84	82	(the) third one is	5	5		10.51
20	83	first of all (I/the)	23	19		10.38
62	84	in this slide (I)	8	6		10.33
46	85	(and) after that I (will)	11	11		10.24

38	86	(becoming) more and more	12	12		10.12
16	87	(in/and) the second part (is/I)	24	22		10.00
26	88	So what is (globalization/the)	20	19		9.91
42	89	to give you (some/a)	12	10		9.80
51	90	(and) the last one (is the)	10	10		9.69
72	91	to start with (I)	6	6		9.62
115	92	the needs of the	3	3		9.48
3	93	(on to/at) the first part(one) (is/the introduction)	41(12)	31(10)		9.39
59	94	(and) the third part (is)	8	8		9.33
48	95	we will have (a)	11	10		9.30
7	96	(the outline/focus/part) of my presentation (is)	31(4)	29(4)		9.27
79	97	(be) responsible for the	5	5		9.06
70	98	(will) show you the	6	6		8.79
32	99	in this presentation (I will)	15	13		8.43
54	100	in my presentation (I)	10	10		8.39
64	101	(and) the second one (is)	7	7		8.35
35	102	this presentation is (the)	14	13		8.08
60	103	in this part (I will)	8	7		7.99
49	104	(that) I think the	11	9		7.78
53	105	The purpose of (this presentation is)	10	10		7.68
71	106	the effect of (globalization)	6	6		7.38
28	107	(the) impact of globalization (on)	22	12		7.30
17	108	(go/move) to the next (slide/part/point)	24	20		7.04
27	109	(in) the process of (globalization)	19	14		6.97
15	110	the definition of (the/globalization/CSR)	25	19		6.81
44	111	the protection of (women's labor rights)	12	8		6.58
83	112	(presentation) is going to	5	5		6.45
68	113	(and) the third is	7	6		6.39
19	114	(is) one of the (main/most)	24	20		5.98
11	115	(lead to/with) the (rapid) development of	27	19		5.81
55	116	(about) the introduction of (globalization)	10	7		5.28
36	117	the first is (introduction)	13	8		4.98
57	118	of the world (economy)	9	8		4.92
22	119	This is the (outline)	22	19		4.72

Note. Contracted forms (e.g., *I'm*) were counted as a single word; for example, *I'm going to* was considered to be a trigram rather than a quadgram. *Multiple-frequency figures listed in column 4 represent the individual frequencies of the four-word sequences that make up the longer five-word structure. **Frequency of the word combination containing the word in parentheses in the fillable slot.

Appendix E

This appendix provides a summary of the assumption checks for the one-way ANOVA conducted in Chapter 3. It includes the Shapiro-Wilk test results, box plot inspections, and Levene's test outcomes for each measure, indicating whether the assumptions were met.

Table A4: *Assumption Checks for One-Way ANOVA (Chapter 3)*

Measure	Shapiro-Wilk Test (<i>p</i> -value)	Box Plot Inspection	Levene's Test (<i>p</i> -value)	Assumptions Met?
Bigram frequency	0.406	Outliers present	0.45	No
Bigram MI	0.019	Outliers present	0.44	No
Bigram proportion	0.976	Outliers present	0.45	No
Trigram frequency	0.679	Outliers present	0.78	No
Trigram MI	0.921	Outliers present	0.63	No
Trigram proportion	< .001	Outliers present	0.27	No
Three-to-five-word usage	< .001	Outliers present	0.24	No
Three-to-five-word MI	< .001	Outliers present	0.55	No
Articulation rate	< .001	Outliers present	0.02	No
Freq. of mid-ASU pauses	0.002	Outliers present	0.87	No
Freq. of end-ASU pauses	0.065	Outliers present	0.76	No
Freq. of total repair	< .001	Outliers present	0.02	No

Appendix F

This appendix presents the results of post-hoc analyses for N-gram and fluency measures across different assessed speaking levels (IELTS 6.5-7.5) from Chapter 3. It includes effect sizes, confidence intervals, and *p*-values, highlighting significant correlations.

Table A5: Results of Post-Hoc Analysis for N-gram and Fluency Measures Across Different Speaking Levels (Chapter 3)

Measure	Levels	Effect Size	CI	<i>p</i>
Bigram MI	6.5–7	-0.077	-.15, .00	.027
	6.5–7.5	-0.114	-.20, -.05	.000
	7–7.5	-0.037	-.13, .05	.247
Trigram proportion	6.5–7	0.010	-.01, .02	.267
	6.5–7.5	-0.011	-.03, .00	.073
	7–7.5	-0.021	-.04, .00	.023
Three-to-five-word usage	6.5–7	-0.567	-3.0, 1.7	.573
	6.5–7.5	-3.433	-5.4, -.90	.000
	7–7.5	-2.867	-5.2, -.13	.010
Three-to-five-word MI	6.5–7	-4.295	-38.3, 28.6	.761
	6.5–7.5	-41.838	-75.2, -7.0	.000
	7–7.5	-37.543	-70.8, .71	.023
Freq. of End-ASU Pauses	6.5–7	1.836	-1.32, 5.60	0.19
	6.5–7.5	3.200	0.72, 7.77	.000
	7–7.5	2.163	-1.30, 5.84	0.110

Note. All significant correlations are shown in **bold**. MI = mutual information. CI = confidence interval. Post hoc tests were conducted using the “mcppb20” function in the “WRS2” package in R. These post hoc tests use a two-step method based on 20% trimmed means and bootstrapping.

Appendix G

Two-Way ANOVA Method and Full Results (Chapter 3)

This appendix presents the method and rationale for the two-way ANOVA analyses conducted as a complementary approach to the main one-way ANOVA results in Chapter 3. The appendix also includes full tables of results and post hoc comparisons.

The two-way ANOVAs were designed to address specific interaction effects between proficiency level and LB length. The following challenges were identified in the dataset, and the corresponding steps were taken to ensure valid analyses:

1. **Dataset structure:** The dataset included multiple LB measures with differing scales and meanings. To address this, measures were categorized into:
 - Frequency measures: Counting occurrences of bigrams, trigrams, and three-to-five word bundles.
 - Association strength (MI) measures: Quantifying the strength of word associations for these bundles.

Proportion measures (e.g., bigram and trigram proportion) were excluded as they did not cover longer (three-to-five-word) bundles, which are central to the research question.

2. **Restructuring the dataset:** Separate analyses were conducted for frequency and association measures by converting the dataset into a long format, with the following key variables:
 - i. Proficiency level: Categorical (IELTS bands 6.5, 7, 7.5).
 - ii. LB length: Categorical (bigram, trigram, three-to-five-word).
 - iii. Dependent variables: Frequency or association strength (MI scores).
3. **Separate analyses:** Two distinct two-way ANOVAs were performed:
 - Frequency measures: To examine how proficiency level and LB length jointly influence the frequency of LBs.
 - Association strength measures: To explore the interaction between proficiency and LB length in terms of association strength (MI).

These steps ensured that each measure type was analyzed appropriately without conflating variables of differing conceptual or numerical meanings.

Results of Two-Way ANOVAs on Frequency Measures

Table A6 shows the effects of proficiency level (IELTS 6.5, 7, 7.5), LB length (bigram, trigram, three-to-five word), and their interaction on LB frequency. Both main effects are significant ($p = .004$ for proficiency level, $p < .001$ for LB length), and the interaction term is also significant ($p < .001$). These results justify further post hoc comparisons to determine which specific proficiency levels and LB lengths differ, and how those factors combine to influence LB frequency. Table A7 shows pairwise comparisons across the three proficiency levels (6.5, 7, 7.5). The results indicate that proficiency level 7.5 differs significantly from both 6.5 and 7, whereas 6.5 and 7 do not differ significantly. Table A8 presents pairwise comparisons among bigram, trigram, and three-to-five-word bundles. The findings show that three-to-five-word bundles occur significantly more frequently than either bigrams or trigrams, which do not differ significantly from each other. In Table A9, each proficiency

level (6.5, 7, 7.5) is examined separately to compare bigram, trigram, and three-to-five-word bundles. Results show that three-to-five-word bundles are consistently used more often than bigrams and trigrams at all proficiency levels, indicating the strongest differences occur with longer bundles. In Table A10, each bundle type (bigram, trigram, three-to-five word) is examined separately to compare proficiency levels. No significant differences emerge among proficiency levels for bigrams or trigrams, but for three-to-five-word bundles, level 7.5 produces significantly higher frequencies than the other two levels.

Table A6: Two-Way ANOVA Results for LB Frequency

Cases	Sum of Squares	df	Mean Square	<i>F</i>	<i>p</i>
Proficiency Level	92.428	2	46.214	5.636	0.004
LB Length	8668.769	2	4334.384	528.590	< .001
Proficiency Level * LB Length	176.824	4	44.206	5.391	< .001
Residuals	3616.157	441	8.200		

Note. Type III Sum of Squares

Table A7: Post Hoc Comparisons – Proficiency Level

		Mean Difference	SE	df	<i>t</i>	<i>p</i> _{tukey}
6.5	7	–0.096	0.331	441	–0.289	0.955
	7.5	–1.006	0.331	441	–3.041	0.007 **
7	7.5	–0.910	0.331	441	–2.752	0.017 *

* $p < .05$, ** $p < .01$

Note. *P*-value adjusted for comparing a family of 3 estimates. Results are averaged over the levels of LB length.

Table A8: Post Hoc Comparisons – LB Length

		Mean Difference	SE	df	<i>t</i>	<i>p</i> _{tukey}
Bigram	(Three-to-Five Word)	–8.973	0.331	441	–27.137	< .001***
	Trigram	0.642	0.331	441	1.943	0.128
(Three-to-Five Word)	Trigram	9.615	0.331	441	29.079	< .001***

*** $p < .001$

Note. *P*-value adjusted for comparing a family of 3 estimates. Results are averaged over the levels of: Proficiency Level

Table A9: *Post Hoc Comparisons – (Proficiency Level * LB Length) Conditional on Proficiency Level*

Proficiency Level			Mean Difference	SE	df	<i>t</i>	<i>p</i> _{Tukey}
6.5	Bigram	(Three-to-Five Word)	–7.889	0.573	441	–13.774	< .001***
		Trigram	0.632	0.573	441	1.103	0.513
	(Three-to-Five Word)	Trigram	8.520	0.573	441	14.877	< .001***
7	Bigram	(Three-to-Five Word)	–8.186	0.573	441	–14.294	< .001***
		Trigram	0.650	0.573	441	1.135	0.493
	(Three-to-Five Word)	Trigram	8.836	0.573	441	15.429	< .001***
7.5	Bigram	(Three-to-Five Word)	–10.844	0.573	441	–18.934	< .001***
		Trigram	0.645	0.573	441	1.127	0.498
	(Three-to-Five Word)	Trigram	11.489	0.573	441	20.060	< .001***

*** $p < .001$ Note. *P*-value adjusted for comparing a family of 3 estimates.**Table A10:** *Post Hoc Comparisons – (Proficiency Level * LB Length) Conditional on LB Length*

LB Length			Mean Difference	SE	df	<i>t</i>	<i>p</i> _{Tukey}
Bigram	6.5	7	–0.003	0.573	441	–0.005	1.000
		7.5	–0.025	0.573	441	–0.044	0.999
	7	7.5	–0.023	0.573	441	–0.040	0.999
Three-to-Five Word	6.5	7	–0.300	0.573	441	–0.524	0.860
		7.5	–2.980	0.573	441	–5.203	< .001***
	7	7.5	–2.680	0.573	441	–4.680	< .001***
Trigram	6.5	7	0.016	0.573	441	0.028	1.000
		7.5	–0.012	0.573	441	–0.020	1.000
	7	7.5	–0.028	0.573	441	–0.048	0.999

*** $p < .001$ Note. *P*-value adjusted for comparing a family of 3 estimates.

Results of Two-Way ANOVAs on Association Measures

Table A11 shows the effects of proficiency level (IELTS 6.5, 7, 7.5), LB length (bigram, trigram, three-to-five word), and their interaction on association (MI) scores. All three effects are significant ($p = .013$ for proficiency level, $p < .001$ for LB length, and $p = .002$ for the interaction), providing justification for post hoc comparisons. In Table A12, proficiency levels are compared across LB lengths. Tukey-corrected tests indicate that 7.5 differs significantly from both 6.5 and 7, while 6.5 and 7 do not significantly differ from each other. Table A13 compares bigram, trigram, and three-to-five-word bundles across proficiency levels. The results show that three-to-five-word bundles have significantly higher association scores than either bigrams or trigrams, whereas bigrams and trigrams do not significantly differ from each other. Table A14 examines differences among bigram, trigram, and three-to-five-word bundles for each proficiency level (6.5, 7, 7.5). In all three proficiency groups, three-to-five-word bundles emerge as significantly higher in association scores than either bigrams or trigrams, which do not differ from one another. Table A15 compares the three proficiency levels within each bundle type. For bigrams and trigrams, proficiency levels do not differ significantly. However, in the three-to-five-word condition, higher-proficiency learners (particularly at IELTS 7.5) show significantly higher association scores than their lower-proficiency counterparts.

Table A11: Two-Way ANOVA Results for Association Measures

Cases	Sum of Squares	df	Mean Square	<i>F</i>	<i>p</i>
Proficiency Level	15158.450	2	7579.225	4.391	0.013
LB Length	1.699×10 ⁺⁶	2	849647.964	492.226	< .001
Proficiency Level * LB Length	30122.604	4	7530.651	4.363	0.002
Residuals	761224.512	441	1726.133		

Note. Type III Sum of Squares

Table A12: Post Hoc Comparisons – Proficiency Level

	Mean Difference	SE	df	<i>t</i>	<i>p</i> _{Tukey}
6.5 7	0.249	4.797	441	0.052	0.999
7.5	-12.186	4.797	441	-2.540	0.031*
7 7.5	-12.435	4.797	441	-2.592	0.027*

* $p < .05$

Note. *P*-value adjusted for comparing a family of 3 estimates.

Note. Results are averaged over the levels of: LB Length

Table A13: Post Hoc Comparisons – LB Length

	Mean Difference	SE	df	<i>t</i>	<i>p</i> _{Tukey}
Bigram (Three-to-Five Word)	-130.847	4.797	441	-27.275	< .001***
Trigram	-0.986	4.797	441	-0.206	0.977
(Three-to-Five Word) Trigram	129.861	4.797	441	27.069	< .001***

*** $p < .001$

Note. *P*-value adjusted for comparing a family of 3 estimates.

Note. Results are averaged over the levels of: Proficiency Level

Table A14: *Post Hoc Comparisons – (Proficiency Level * LB Length) Conditional on Proficiency Level*

Proficiency Level			Mean Difference	SE	df	<i>t</i>	<i>p</i> _{tukey}
6.5	Bigram	(Three-to-Five Word)	–119.018	8.309	441	–14.323	< .001***
		Trigram	–1.054	8.309	441	–0.127	0.991
7	(Three-to-Five Word)	Trigram	117.964	8.309	441	14.196	< .001***
		Bigram	(Three-to-Five Word)	–118.166	8.309	–14.221	< .001***
7.5	Bigram	Trigram	–0.978	8.309	441	–0.118	0.992
		(Three-to-Five Word)	117.189	8.309	441	14.103	< .001***
	Bigram	(Three-to-Five Word)	–155.358	8.309	441	–18.697	< .001***
		Trigram	–0.927	8.309	441	–0.112	0.993
	(Three-to-Five Word)	Trigram	154.431	8.309	441	18.585	< .001***

*** $p < .001$ Note. *P*-value adjusted for comparing a family of 3 estimates.**Table A15:** *Post Hoc Comparisons – (Proficiency Level * LB Length) Conditional on LB Length*

LB Length			Mean Difference	SE	df	<i>t</i>	<i>p</i> _{tukey}
Bigram	6.5	7	–0.060	8.309	441	–0.007	1.000
		7.5	–0.114	8.309	441	–0.014	1.000
	7	7.5	–0.054	8.309	441	–0.006	1.000
Three-to-Five Word	6.5	7	0.791	8.309	441	0.095	0.995
		7.5	–36.455	8.309	441	–4.387	< .001***
	7	7.5	–37.246	8.309	441	–4.482	< .001***
Trigram	6.5	7	0.016	8.309	441	0.002	1.000
		7.5	0.013	8.309	441	0.002	1.000
	7	7.5	–0.004	8.309	441	-4.501×10^{-4}	1.000

*** $p < .001$ Note. *P*-value adjusted for comparing a family of 3 estimates.

In summary, the results in this appendix confirmed significant interaction effects between proficiency level and LB length for both frequency and association measures. Specifically, the results show that longer LBs (three-to-five-word units) increase substantially at higher proficiency levels (particularly 7.5), whereas bigrams and trigrams show fewer or no significant differences across proficiency bands. A similar pattern is observed for association measures, indicating that as learners move toward more advanced proficiency, they not only use longer bundles more frequently but also use higher MI scores for these longer bundles. Taken together, these findings reinforce the hypothesis that LB length interacts significantly with proficiency level, affecting both the frequency and strength of word associations in learner speech.

Appendix H

This appendix outlines the assumption checks for the regression models used in Chapters 4, 5, and 6. It includes Shapiro-Wilk p -values, visual inspections for linearity and homoscedasticity, and multicollinearity assessments based on the maximum Variance Inflation Factor (VIF) values for each outcome variable, and a correlation matrix for all predictor variables to highlight interrelationships.

Table A16: *Assumption Checks for Regression Models (Chapters 4, 5, and 6)*

Outcome Variable	Shapiro-Wilk p -value	Linearity (Visual inspection)	Homoscedasticity (Visual inspection)	Multicollinearity (Max VIF)
Proficiency level (banded scores)	< .001	Not violated	Not violated	7.38
Presentation raw scores	0.04	Not violated	Not violated	7.38
Articulation rate	< .001	Not violated	Not violated	7.38
Freq. of mid-ASU pauses	< .001	Not violated	Not violated	7.38
Freq. of end-ASU pauses	0.16	Not violated	Not violated	7.38
Freq. of total repair	< .001	Not violated	Not violated	7.38

Note. See sub-table A6.1 below for all VIF values.

Sub-table A16.1: *Variance Inflation Factor (VIF) Values*

Predictor Variable	VIF Value
Bigram frequency	2.63
Bigram MI	1.53
Bigram proportion	4.60
Trigram frequency	2.08
Trigram MI	1.49
Trigram proportion	7.21
Three-to-five-word usage	7.38
Three-to-five-word MI	6.91

Sub-table A16.2: Correlation Matrix for Predictor Variables*Pearson's Correlations*

Variable		BiFreq	BiMI	BiProp	TriFreq	TriMI	TriProp	Three-Five- Word Usage	Three-Five- Word MI
1. BiFreq	Pearson's r	—							
	p-value	—							
2. BiMI	Pearson's r	−0.220**	—						
	p-value	0.007	—						
3. BiProp	Pearson's r	0.572***	−0.089	—					
	p-value	< .001	0.278	—					
4. TriFreq	Pearson's r	0.457***	0.172*	0.162*	—				
	p-value	< .001	0.036	0.047	—				
5. TriMI	Pearson's r	−0.251**	0.498***	−0.154	0.186*	—			
	p-value	0.002	< .001	0.060	0.023	—			
6. TriProp	Pearson's r	0.713***	0.061	0.825***	0.512***	−0.056	—		
	p-value	< .001	0.461	< .001	< .001	0.498	—		
7. Three-Five- Word Usage	Pearson's r	0.326***	−0.035	0.404***	0.132	−0.148	0.413***	—	
	p-value	< .001	0.671	< .001	0.107	0.071	< .001	—	
8. Three-Five- Word MI	Pearson's r	0.244**	0.028	0.353***	0.137	−0.057	0.357***	0.920***	—
	p-value	0.003	0.734	< .001	0.095	0.487	< .001	< .001	—

Note. * $p < .05$, ** $p < .01$, *** $p < .001$

Appendix I

This appendix shows Cook's distance plots from Chapters 4, 5, and 6, used to detect influential observations in the regression models for various fluency measures, including articulation rate, mid-ASU pauses, end-ASU pauses, total repair, and presentation raw scores.

Figure A2: *Cook's Distance Plot for Detecting Influential Observations in the Regression Model of Articulation Rate*

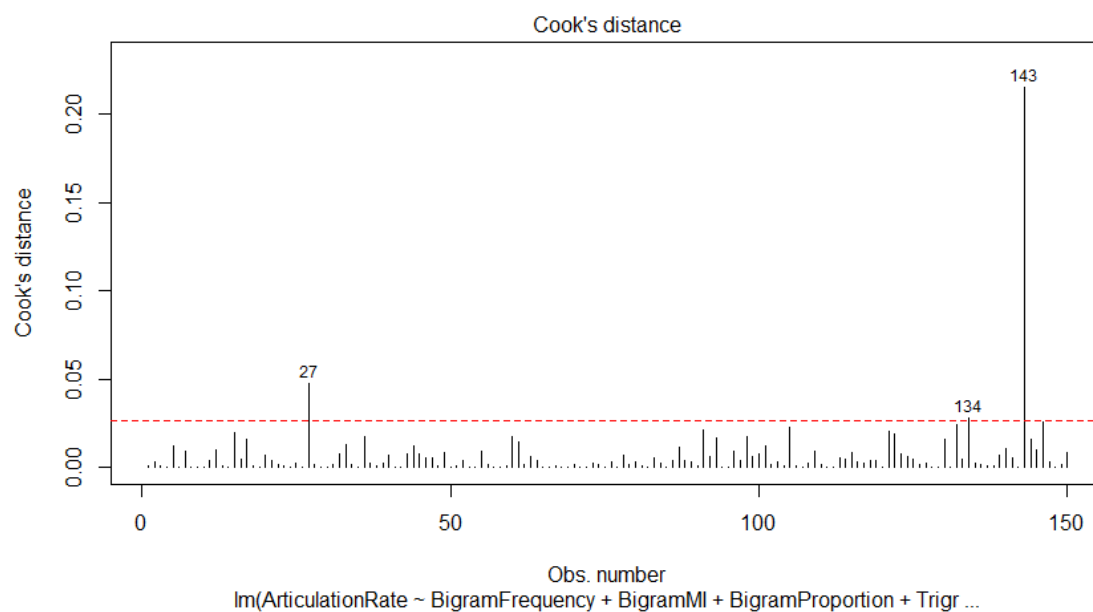


Figure A3: *Cook's Distance Plot for Detecting Influential Observations in the Regression Model of Mid-ASU Pauses*

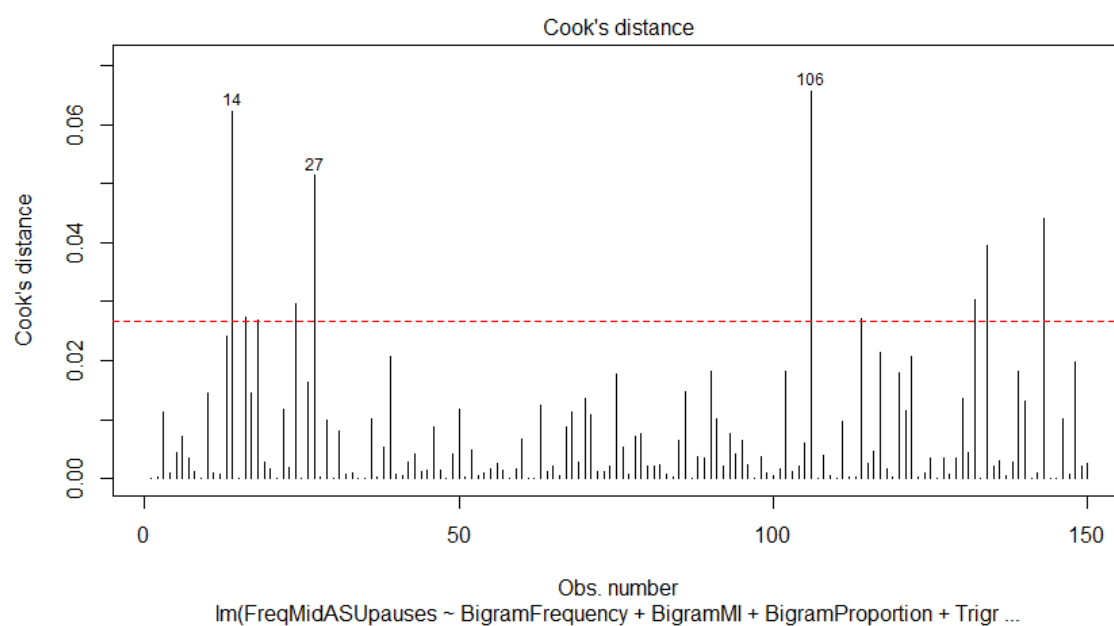


Figure A4: Cook's Distance Plot for Detecting Influential Observations in the Regression Model of End-ASU Pauses

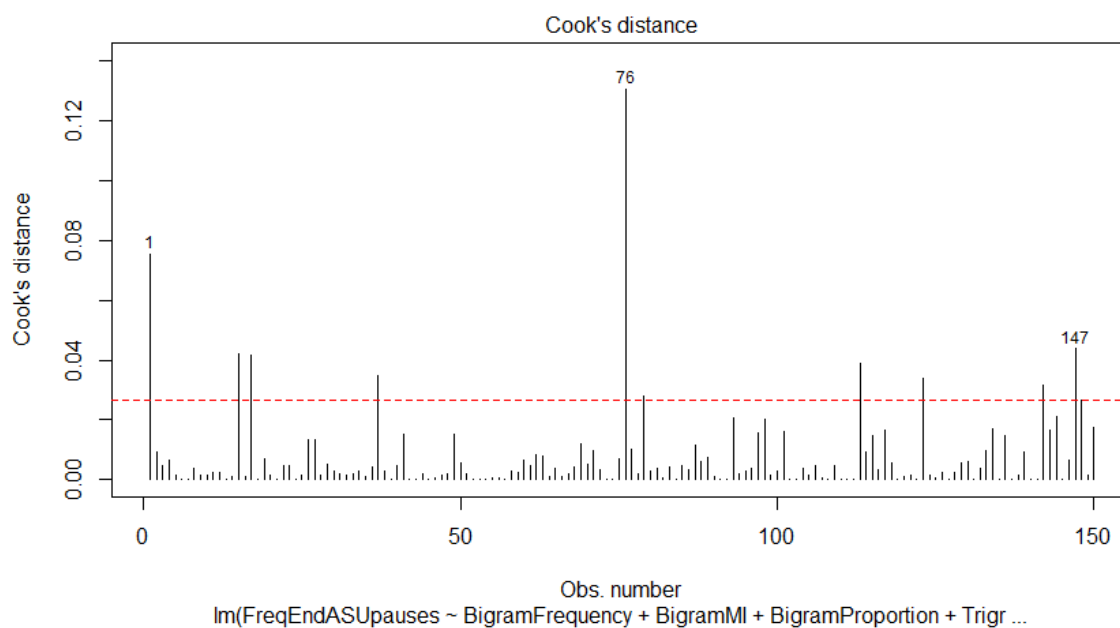


Figure A5: Cook's Distance Plot for Detecting Influential Observations in the Regression Model of Total Repair

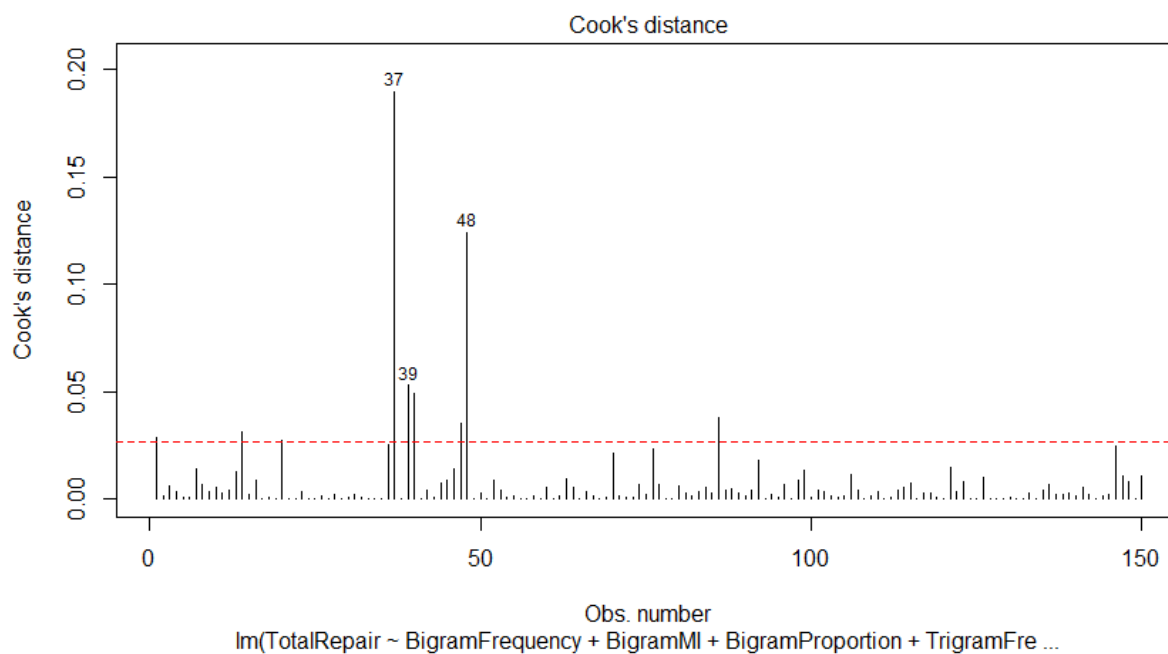
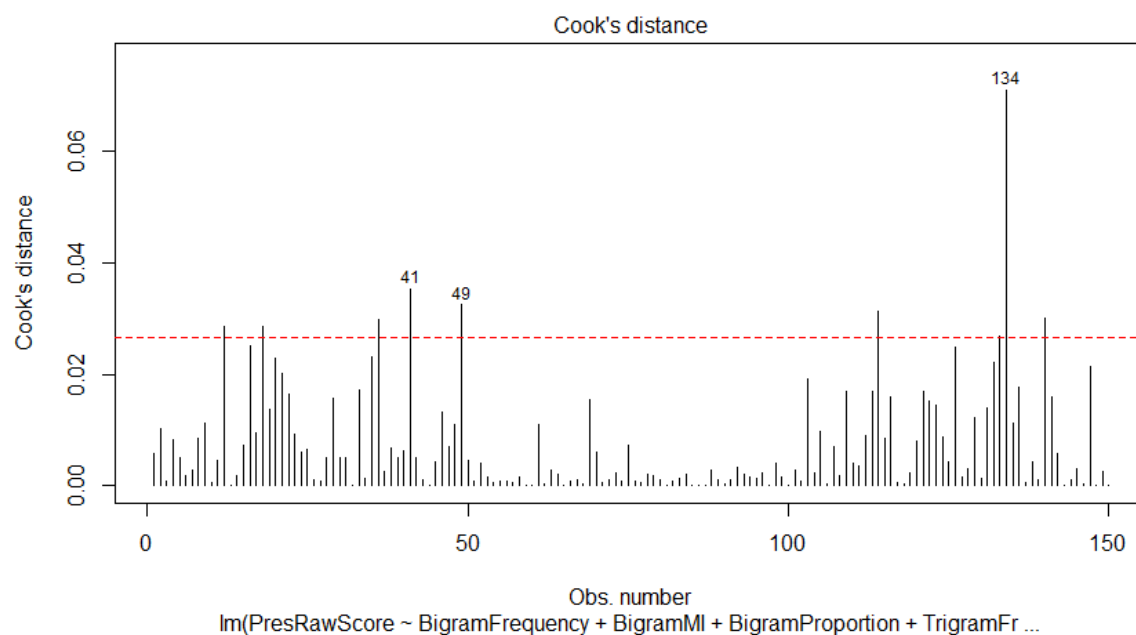


Figure A6: Cook's Distance Plot for Detecting Influential Observations in the Regression Model of Presentation Raw Scores



Appendix J

R Code and Detailed Results of Regression, Dominance, and Random Forests Analyses (Chapters 4 and 5)

This appendix provides the R code and detailed results of the regression, dominance, and random forests analyses conducted in Chapters 4 and 5. It includes relative importance metrics for multiple regression models and random forests models for fluency and n-gram measures. Note that in R code, hashtags (#) are used to write comments, which provide explanations or notes about the code but are not executed as part of the analysis. The hashtags help clarify the purpose of each step, making the analysis easier to follow, especially for readers unfamiliar with coding. R code is colored in blue to make it stand out from the explanations and notes. The printouts of the results have been kept in their original format as they appear in RStudio to maintain authenticity and consistency.

Load Required Packages for Analyses in Appendix I

```
library(MASS)           # For robust linear modeling (rlm)
library(relaimp)         # For dominance analysis
library(randomForest)    # For random forests analysis
library(Boruta)          # For Boruta feature selection
```

Section 1: Regression Analysis for Articulation Rate

Multiple regression analysis using the rlm function (MASS package) with MM estimation
Note: The detailed results for this regression and the dominance analysis are presented in Table 4.2 in Section 4.4.

Fit the robust linear model (rlm) for Articulation Rate

```
model <- rlm(ArticulationRate ~ BigramFrequency + BigramProportion + BigramMI +
  TrigramFrequency + TrigramProportion + TrigramMI +
  Three.Five.WordUsage_RawScores + Three.Five.WordMI_RawScores,
  data = fluencyMWS_subset)
summary(model)           # Print the summary of the rlm model
```

Section 2: Dominance Analysis (= Relative Importance Analysis) for Articulation Rate

```
relimp_results <- calc.relimp(model)      # Calculate relative importance
print(relimp_results)                    # Print relative importance metrics
```

Response variable: ArticulationRate
Total response variance: 819.1017
Analysis based on 150 observations

8 Regressors:

BigramFrequency BigramProportion BigramMI TrigramFrequency TrigramProportion TrigramMI Three.Five.WordUsage_RawScores Three.Five.WordMI_RawScores

Proportion of variance explained by model: 5.73%

Metrics are not normalized (rela=FALSE).

Relative importance metrics:

	lmg
BigramFrequency	0.0102044930

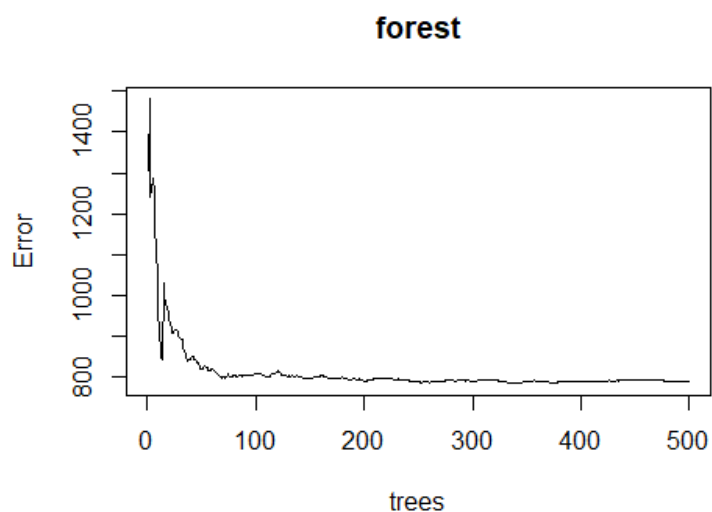
BigramProportion	0.0018063932
BigramMI	0.0005618427
TrigramFrequency	0.0023486413
TrigramProportion	0.0022617589
TrigramMI	0.0085621781
Three.Five.WordUsage_RawScores	0.0187897682
Three.Five.WordMI_RawScores	0.0127709166
Average coefficients for different model sizes:	
	1X 2Xs 3Xs 4Xs 5Xs 6Xs
BigramFrequency	28.79953578 27.58286960 28.77649917 31.39727024 34.52399091 37.40541578
BigramProportion	15.76590897 -4.39979869 -18.05944867 -26.19518412 -29.86985012 -30.16666935
BigramMI	3.36993895 2.75486892 1.84170608 1.01854469 0.44511666 0.09913114
TrigramFrequency	19.42619728 15.31245239 12.43153279 10.10622014 7.83551385 5.55750761
TrigramProportion	57.66112069 36.74060498 16.03741501 -3.91227380 -21.99302739 -37.23714044
TrigramMI	7.87163107 9.20137156 10.29490612 11.27873658 12.23034007 13.13019602
Three.Five.WordUsage_RawScores	1.01955951 1.02387514 1.04929495 1.08861967 1.13309065 1.17792801
Three.Five.WordMI_RawScores	0.06496233 0.05441211 0.04417204 0.03391697 0.02347861 0.01266738
	7Xs 8Xs
BigramFrequency	39.508054181 40.59490174
BigramProportion	-28.036198806 -24.28210246
BigramMI	-0.106163513 -0.26638562
TrigramFrequency	3.480513390 1.83843920
TrigramProportion	-49.075011199 -57.42151341
TrigramMI	13.913948613 14.52363075
Three.Five.WordUsage_RawScores	1.221203674 1.26045638
Three.Five.WordMI_RawScores	0.001370983 -0.01024697

Section 3: Random Forests Analysis for Articulation Rate

```
set.seed(123) # Set the seed for reproducibility
forest <- randomForest(ArticulationRate~., data=fluencyMWS_subset) # Run random forest model
print(forest) # Print the summary of the random forest model
```

Call:
randomForest(formula = ArticulationRate ~ ., data = fluencyMWS_subset)
Type of random forest: regression
Number of trees: 500
No. of variables tried at each split: 2
Mean of squared residuals: 789.01
% Var explained: 3.03

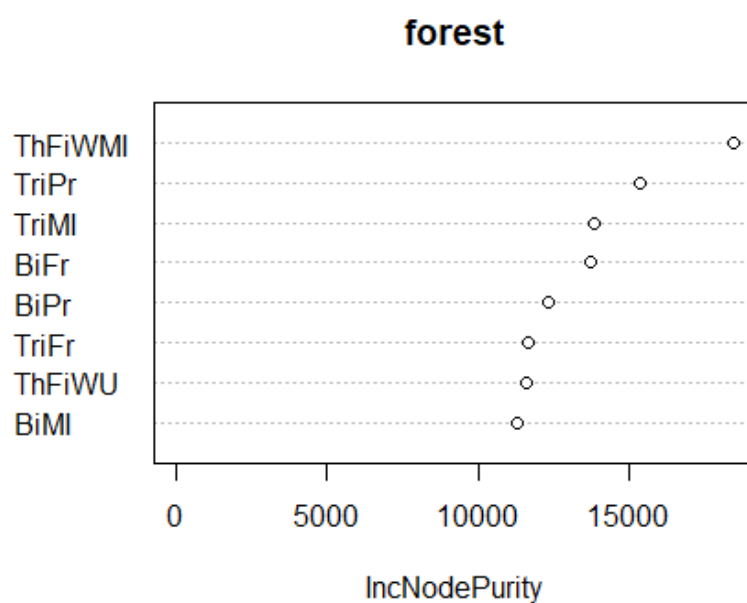
```
plot(forest)      # Plot the error rates for the random forest model
```



```
forest$importance  # Print the importance of each variable in the model
```

	IncNodePurity
BigramFrequency	13718.86
BigramProportion	12345.26
BigramMI	11315.72
TrigramFrequency	11651.23
TrigramProportion	15351.21
TrigramMI	13867.19
Three.Five.WordUsage	11624.52
Three.Five.WordMI	18471.21

```
varImpPlot(forest)  # Create a plot showing the importance of each variable
```



Section 4: Boruta Analysis for Articulation Rate

Conduct Boruta analysis to assess predictor importance

```
set.seed(88)          # Set the seed for reproducibility
boruta <- Boruta(ArticulationRate~., maxRuns = 200, data = fluencyMWS_subset, doTrace = 2)
print(boruta)         # Print the summary of the Boruta results
```

Boruta performed 199 iterations in 4.350658 secs.

2 attributes confirmed important: Three.Five.WordMI, Three.Five.WordUsage

5 attributes confirmed unimportant: BigramFrequency, BigramMI, BigramProportion, TrigramFrequency, TrigramMI;

1 tentative attributes left: TrigramProportion;

```
plot(boruta, cex.axis=0.7) # Plot the Boruta results (See Figure 4.2 in the main text)
```

```
attStats(boruta)           # Print detailed attribute (predictor) statistics
```

	meanImp	medianImp	minImp	maxImp	normHits	decision
BiFr	-0.04332966	-0.05924094	-2.4235997	2.6651545	0.00000000	Rejected
BiPr	1.25470267	1.05334052	-1.0160984	2.9076560	0.05527638	Rejected
BiMI	-1.47645524	-1.62182794	-3.3963249	0.2542719	0.00000000	Rejected
TriFr	1.23714668	1.43431791	-1.3290244	4.4973700	0.06030151	Rejected
TriPr	2.46842431	2.48604684	-1.7158707	6.6207806	0.45728643	Tentative
TriMI	1.89305336	1.92164788	-1.5035154	5.0377616	0.22613065	Rejected
ThFiWU	3.69865970	3.82135910	-2.1195956	7.8794138	0.73869347	Confirmed
ThFiWMI	4.16199613	4.10590882	-0.3682322	8.7989914	0.79396985	Confirmed

Section 5: Regression Analysis for Frequency of Mid-ASU Pauses

Note: The detailed results for this regression and the dominance analysis are presented in Table 4.3 in Section 4.4.

Fit the robust linear model (rlm) for frequency of mid-ASU pauses

```
model2 <- rlm(FreqMidASUpauses ~ BigramFrequency + BigramProportion + BigramMI +
  TrigramFrequency + TrigramProportion + TrigramMI +
  Three.Five.WordUsage_RawScores + Three.Five.WordMI_RawScores,
  data = fluencyMWS_subset)
```

```
summary(model2) # Print the summary of the rlm model2
```

Section 6: Dominance Analysis for Frequency of Mid-ASU Pauses

```
relimp_results2 <- calc.relimp(model2) # Calculate relative importance
```

```
print(relimp_results2) # Print relative importance metrics
```

Response variable: FreqMidASUpauses

Total response variance: 44.96885

Analysis based on 150 observations

8 Regressors:

BigramFrequency BigramProportion BigramMI TrigramFrequency TrigramProportion TrigramMI Three.Five.WordUsage_RawScores Three.Five.WordMI_RawScores

Proportion of variance explained by model: 15.01%

Metrics are not normalized (rela=FALSE).

Relative importance metrics:

	lmg
BigramFrequency	0.0043384207
BigramProportion	0.0048223904
BigramMI	0.0240061217
TrigramFrequency	0.0006353593
TrigramProportion	0.0040758822
TrigramMI	0.0178356763
Three.Five.WordUsage_RawScores	0.0605402430
Three.Five.WordMI_RawScores	0.0338235409

Average coefficients for different model sizes:

	1X	2Xs	3Xs	4Xs	5Xs	6Xs
BigramFrequency	-1.65946130	1.69460120	3.8667498	5.149275090	5.837621858	6.157855304
BigramProportion	-12.21702600	-9.20725811	-7.0267168	-5.582989365	-4.889072222	-5.024547191
BigramMI	-4.77089321	-5.55212057	-6.1789264	-6.711908405	-7.196546368	-7.638030737
TrigramFrequency	-0.55430367	0.87848915	1.3673635	1.253437049	0.836221784	0.225764234
TrigramProportion	-20.47900655	-15.08015473	-10.9445213	-7.620477926	-4.591128444	-1.206606377
TrigramMI	2.96792734	3.11651627	3.3846776	3.703031254	4.017746405	4.304342344
Three.Five.WordUsage_RawScores	-0.41414426	-0.44404882	-0.4736961	-0.502848206	-0.530991612	-0.557713464
Three.Five.WordMI_RawScores	-0.02453181	-0.01909954	-0.0133611	-0.007497687	-0.001689568	0.003958075

	7Xs	8Xs
BigramFrequency	6.246066451	6.16023781
BigramProportion	-6.082867217	-8.15077635
BigramMI	-8.047525123	-8.46405104
TrigramFrequency	-0.572532956	-1.58231316
TrigramProportion	3.190477947	9.21861082
TrigramMI	4.557983818	4.79041008
Three.Five.WordUsage_RawScores	-0.583483919	-0.60989404
Three.Five.WordMI_RawScores	0.009444072	0.01486835

Section 7: Random Forests Analysis for Frequency of Mid-ASU Pauses

`set.seed(123)` # Set the seed for reproducibility

`forest <- randomForest(FreqMidASUpauses ~., data=fluencyMWS_subset)` # Run random forest model

`print(forest)` # Print the summary of the random forest model

Call:

`randomForest(formula = FreqMidASUpauses ~ ., data = fluencyMWS_subset)`

Type of random forest: regression

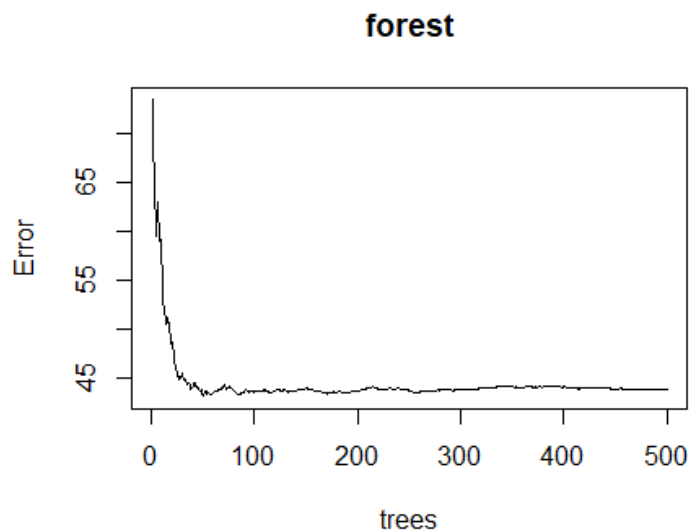
Number of trees: 500

No. of variables tried at each split: 2

Mean of squared residuals: 43.79907

% Var explained: 1.95

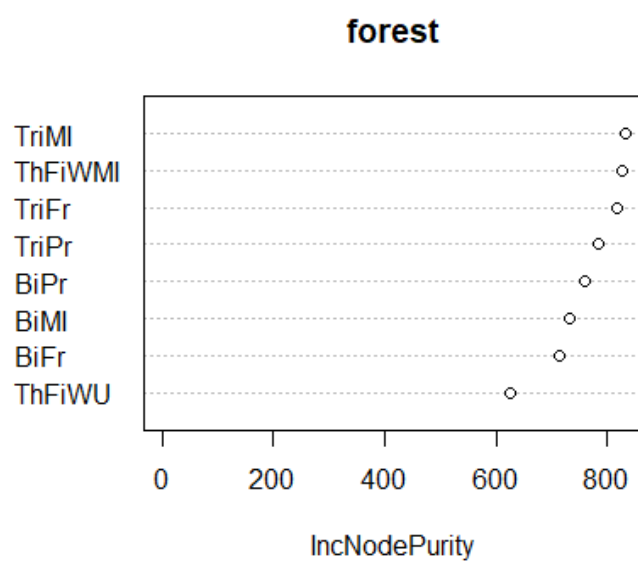
```
plot(forest)      # Plot the error rates for the random forest model
```



```
forest$importance  # Print the importance of each variable in the model
```

	IncNodePurity
BiFr	714.2359
BiPr	759.2030
BiMI	732.8448
TriFr	817.9875
TriPr	782.9590
TriMI	833.7006
ThFiWU	624.7347
ThFiWMI	825.5248

```
varImpPlot(forest)  # Create a plot showing the importance of each variable
```



Section 8: Boruta Analysis for Frequency of Mid-ASU Pauses

Conduct Boruta analysis to assess predictor importance

```
set.seed(88)      # Set the seed for reproducibility
```

```
boruta <- Boruta(FreqMidASUpauses ~., maxRuns = 200, data = fluencyMWS_subset, doTrace = 2)
```

```
print(boruta)     # Print the summary of the Boruta results
```

Boruta performed 199 iterations in 5.25248 secs.
 3 attributes confirmed important: ThFiWMI, ThFiWU, TriMI;
 4 attributes confirmed unimportant: BiFr, BiPr, TriFr, TriPr;
 1 tentative attributes left: BiMI;

```
plot(boruta, cex.axis=0.7)  # Plot the Boruta results (See Figure 4.4 in the main text)
```

```
attStats(boruta)           # Print detailed attribute (predictor) statistics
```

	meanImp	medianImp	minImp	maxImp	normHits	decision
BiFr	1.4712629	1.1617515	-0.7326939	4.628309	0.065326633	Rejected
BiPr	1.9937623	1.9254835	-2.4002027	6.334776	0.341708543	Rejected
BiMI	2.3331165	2.3331414	-1.5822900	5.857585	0.417085427	Tentative
TriFr	0.4233257	0.7054309	-1.8966915	1.539757	0.005025126	Rejected
TriPr	1.1422425	1.1116729	-1.5640482	3.792037	0.015075377	Rejected
TriMI	4.3503650	4.4238240	0.1016909	7.646272	0.718592965	Confirmed
ThFiWU	5.9101142	5.9672373	2.8142504	9.307774	0.854271357	Confirmed
ThFiWMI	6.1610383	6.2052812	2.2038397	10.474472	0.884422111	Confirmed

Section 9: Regression Analysis for Frequency of End-ASU Pauses

Note: The detailed results for this regression and the dominance analysis are presented in Table 4.4 in Section 4.4.

Fit the robust linear model (rlm) for frequency of end-ASU pauses

```
model3 <- rlm(FreqEndASUpauses ~ BigramFrequency + BigramProportion + BigramMI +  
  TrigramFrequency + TrigramProportion + TrigramMI +  
  Three.Five.WordUsage_RawScores + Three.Five.WordMI_RawScores,  
  data = fluencyMWS_subset)
```

```
summary(model3)     # Print the summary of the rlm model3
```

Section 10: Dominance Analysis for Frequency of End-ASU Pauses

```
relimp_results3 <- calc.relimp(model3)  # Calculate relative importance
```

```
print(relimp_results3)  # Print relative importance metrics
```

Response variable: FreqEndASUpauses
 Total response variance: 4.532333
 Analysis based on 150 observations

8 Regressors:
 BigramFrequency BigramProportion BigramMI TrigramFrequency TrigramProportion TrigramMI Three.Fiv
 e.WordUsage_RawScores Three.Five.WordMI_RawScores
 Proportion of variance explained by model: 13.15%
 Metrics are not normalized (rela=FALSE).

Relative importance metrics:

```

                                lmg
BigramFrequency      0.052800494
BigramProportion     0.019782999
BigramMI             0.002051999
TrigramFrequency     0.004283375
TrigramProportion    0.008204861
TrigramMI            0.030320902
Three.Five.WordUsage_RawScores 0.005693843
Three.Five.WordMI_RawScores  0.008380732

Average coefficients for different model sizes:

                                1X      2Xs      3Xs      4Xs      5Xs      6Xs
BigramFrequency      3.05796283 3.902805375 4.803300813 5.694201960 6.506365133 7.176269148
BigramProportion     -2.25066697 -4.355019638 -5.911494409 -6.997944185 -7.718040312 -8.189764909
BigramMI             0.22563631 0.100317286 -0.007731396 -0.078135539 -0.113430222 -0.129743243
TrigramFrequency     1.34817528 1.124344978 0.798412753 0.364878636 -0.162849158 -0.749377378
TrigramProportion    0.78612872 -0.368650692 -1.540329182 -2.629824563 -3.441718634 -3.761198043
TrigramMI            1.19827911 1.313747595 1.433695687 1.560031746 1.691039370 1.818482069
Three.Five.WordUsage_RawScores 0.03922077 0.033274330 0.028885849 0.024963382 0.020432492 0.014703927
Three.Five.WordMI_RawScores  0.00314170 0.003357852 0.003543331 0.003716359 0.003886772 0.004046384

                                7Xs      8Xs
BigramFrequency      7.649616414 7.891954013
BigramProportion     -8.521115383 -8.808608926
BigramMI             -0.149310822 -0.198202094
TrigramFrequency     -1.347048561 -1.912022504
TrigramProportion    -3.405965550 -2.239150732
TrigramMI            1.931692505 2.023604212
Three.Five.WordUsage_RawScores 0.007439104 -0.001850748
Three.Five.WordMI_RawScores  0.004195130 0.004372738

```

Section 11: Random Forests Analysis for Frequency of End-ASU Pauses

```

set.seed(123) # Set the seed for reproducibility
forest <- randomForest(FreqEndASUpauses ~., data=fluencyMWS_subset) # Run random forest model
print(forest) # Print the summary of the random forest model

```

Call:

```
randomForest(formula = FreqEndASUpauses ~ ., data = fluencyMWS_subset)
```

Type of random forest: regression

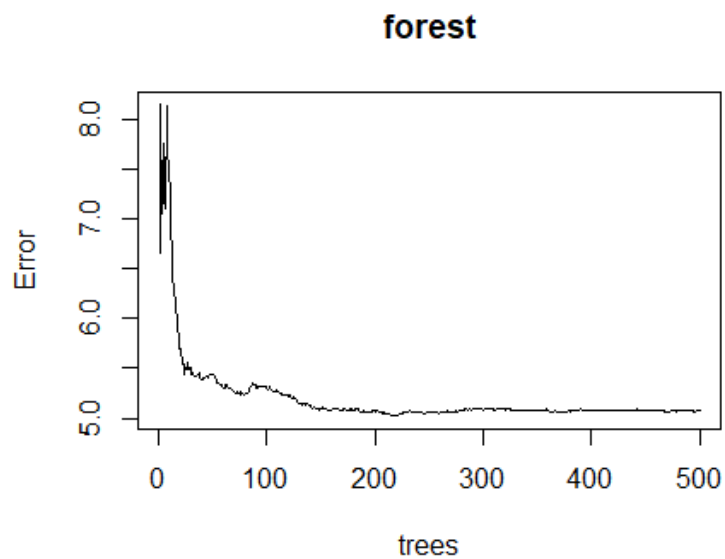
Number of trees: 500

No. of variables tried at each split: 2

Mean of squared residuals: 5.072096

% Var explained: -12.66

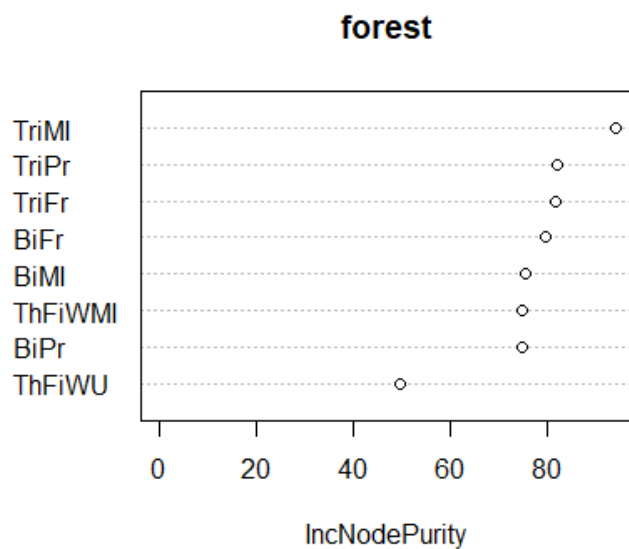
```
plot(forest)      # Plot the error rates for the random forest model
```



```
forest$importance  # Print the importance of each variable in the model
```

IncNodePurity	
BiFr	79.66109
BiPr	74.76083
BiMI	75.57329
TriFr	81.45684
TriPr	81.97912
TriMI	94.06387
ThFiWU	49.71580
ThFiWMI	74.79454

```
varImpPlot(forest)  # Create a plot showing the importance of each variable
```



Section 12: Boruta Analysis for Frequency of End-ASU Pauses

Conduct Boruta analysis to assess predictor importance

`set.seed(88)` # Set the seed for reproducibility

`boruta <- Boruta(FreqEndASUpauses ~., maxRuns = 200, data = fluencyMWS_subset, doTrace = 2)`

`print(boruta)` # Print the summary of the Boruta results

Boruta performed 130 iterations in 4.408205 secs.
 No attributes deemed important.
 8 attributes confirmed unimportant: BiFr, BiMI, BiPr, ThFiWMI, ThFiWU and 3 more;

`plot(boruta, cex.axis=0.7)` # Plot the Boruta results (See Figure 4.6 in the main text)

`attStats(boruta)` # Print detailed attribute (predictor) statistics

	meanImp	medianImp	minImp	maxImp	normHits	decision
BiFr	1.74416189	1.6880253	-1.471296	4.7774319	0.19230769	Rejected
BiPr	-0.50178673	-0.1669662	-2.115745	0.6230742	0.00000000	Rejected
BiMI	-1.46438134	-1.5205647	-3.138131	0.1513090	0.00000000	Rejected
TriFr	1.79739140	1.7754710	-1.638211	6.2423341	0.36153846	Rejected
TriPr	0.55896912	0.4664760	-2.179787	3.2355196	0.04615385	Rejected
TriMI	1.91576226	1.8669697	-2.406670	5.9193868	0.35384615	Rejected
ThFiWU	-0.07150723	0.6275611	-3.086461	1.5313394	0.00000000	Rejected
ThFiWMI	-0.07352297	-0.1638024	-2.997789	3.6081611	0.03076923	Rejected

Section 13: Regression Analysis for Frequency of Total Repairs

Note: The detailed results for this regression and the dominance analysis are presented in Table 4.5 in Section 4.4.

Fit the robust linear model (rlm) for frequency of total repairs

`model4 <- rlm(TotalRepair ~ BigramFrequency + BigramProportion + BigramMI +
 TrigramFrequency + TrigramProportion + TrigramMI +
 Three.Five.WordUsage_RawScores + Three.Five.WordMI_RawScores,
 data = fluencyMWS_subset)`

`summary(model4)` # Print the summary of the rlm model4

Section 14: Dominance Analysis for Frequency of Total Repairs

`relimp_results4 <- calc.relimp(model4)` # Calculate relative importance

`print(relimp_results4)` # Print relative importance metrics

Response variable: TotalRepair
 Total response variance: 4.453663
 Analysis based on 150 observations

8 Regressors:
 BigramFrequency BigramProportion BigramMI TrigramFrequency TrigramProportion TrigramMI Three.Fiv
 e.WordUsage_RawScores Three.Five.WordMI_RawScores

Proportion of variance explained by model: 10.61%
 Metrics are not normalized (rela=FALSE).

Relative importance metrics:

	lmg
BigramFrequency	0.003684052
BigramProportion	0.005256085
BigramMI	0.058753890

```

TrigramFrequency      0.003562159
TrigramProportion     0.008472034
TrigramMI             0.013866432
Three.Five.WordUsage_RawScores 0.005893858
Three.Five.WordMI_RawScores  0.006658045

```

Average coefficients for different model sizes:

	1X	2Xs	3Xs	4Xs	5Xs	6Xs
BigramFrequency	0.647872264	1.155065162	1.37719139	1.410311258	1.33400015	1.205747721
BigramProportion	0.310449691	1.507062236	2.44518120	3.233539468	3.92252661	4.519487362
BigramMI	-3.605391608	-3.566937113	-3.48683231	-3.385073237	-3.27292624	-3.152946073
TrigramFrequency	-1.618851633	-1.435025105	-1.14938417	-0.822934834	-0.47662281	-0.132253036
TrigramProportion	-4.655516827	-6.460532290	-7.86959635	-9.007756831	-9.99104696	-10.876557521
TrigramMI	-1.422804586	-1.290824404	-1.12268872	-0.943312177	-0.77229294	-0.621438543
Three.Five.WordUsage_RawScores	-0.041345394	-0.035954256	-0.03359558	-0.033627062	-0.03447728	-0.034778486
Three.Five.WordMI_RawScores	-0.003347271	-0.003573846	-0.00338889	-0.002941809	-0.00241490	-0.001925917

	7Xs	8Xs
BigramFrequency	1.06467288	0.931711122
BigramProportion	5.02017666	5.420010754
BigramMI	-3.02497069	-2.889963804
TrigramFrequency	0.19133824	0.481652570
TrigramProportion	-11.68753929	-12.419519484
TrigramMI	-0.49626483	-0.395701740
Three.Five.WordUsage_RawScores	-0.03397533	-0.031951230
Three.Five.WordMI_RawScores	-0.00151751	-0.001197109

Section 15: Random Forests Analysis for Frequency of Total Repairs

```

set.seed(123) # Set the seed for reproducibility
forest <- randomForest(TotalRepair ~., data=fluencyMWS_subset) # Run random forest model
print(forest) # Print the summary of the random forest model

```

Call:

```
randomForest(formula = TotalRepair ~ ., data = fluencyMWS_subset)
```

Type of random forest: regression

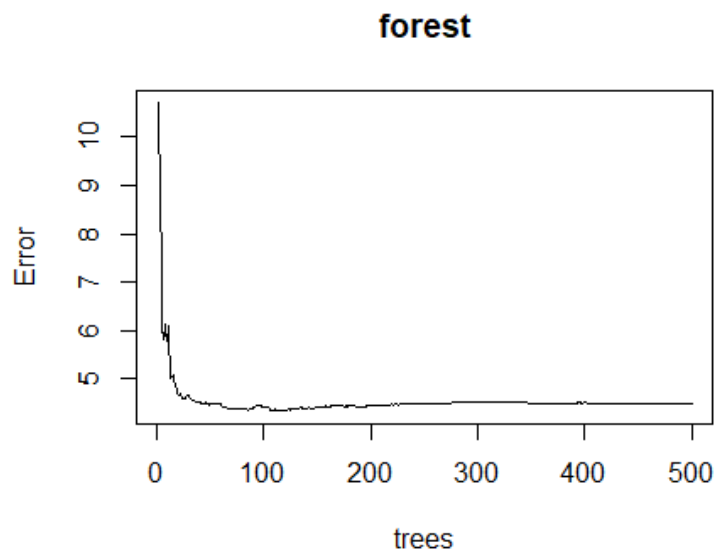
Number of trees: 500

No. of variables tried at each split: 2

Mean of squared residuals: 4.486003

% Var explained: -1.4

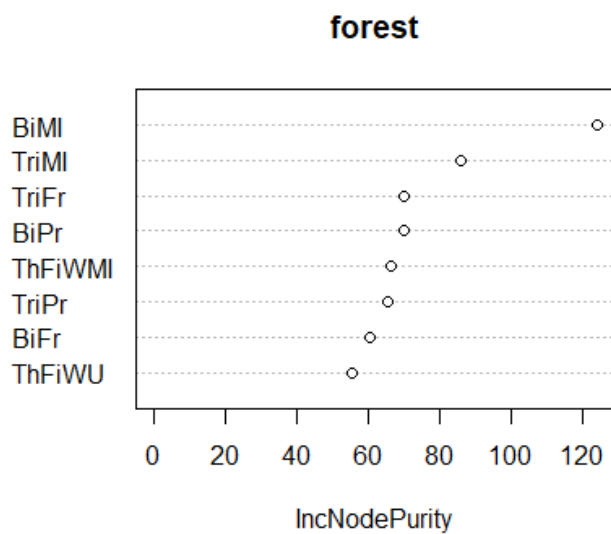
```
plot(forest)      # Plot the error rates for the random forest model
```



```
forest$importance  # Print the importance of each variable in the model
```

	IncNodePurity
BiFr	60.49615
BiPr	70.20959
BiMI	124.39513
TriFr	70.29091
TriPr	65.63111
TriMI	86.14923
ThFiWU	55.66190
ThFiWMI	66.29541

```
varImpPlot(forest)  # Create a plot showing the importance of each variable
```



Section 16: Boruta Analysis for Frequency of Total Repairs

Conduct Boruta analysis to assess predictor importance

`set.seed(88)` # Set the seed for reproducibility

`boruta <- Boruta(TotalRepair ~., maxRuns = 200, data = fluencyMWS_subset, doTrace = 2)`

`print(boruta)` # Print the summary of the Boruta results

Boruta performed 199 iterations in 6.825889 secs.
 6 attributes confirmed important: BiMI, BiPr, ThFiWMI, ThFiWU, TriMI and 1 more;
 1 attributes confirmed unimportant: BiFr;
 1 tentative attributes left: TriFr;

`plot(boruta, cex.axis=0.7)` # Plot the Boruta results (See Figure 4.8 in the main text)

`attStats(boruta)` # Print detailed attribute (predictor) statistics

	meanImp	medianImp	minImp	maxImp	normHits	decision
BiFr	0.3138305	0.428056	-1.4990853	2.167211	0.0201005	Rejected
BiPr	2.5948087	2.641291	-1.1286558	5.233252	0.6231156	Confirmed
BiMI	4.0149314	4.127374	0.5760395	8.104734	0.8241206	Confirmed
TriFr	2.2800190	2.313624	-0.9106139	5.153626	0.4924623	Tentative
TriPr	3.2169534	3.258164	-0.4440952	5.469846	0.7487437	Confirmed
TriMI	3.1737691	3.167879	-0.3482672	6.461633	0.7085427	Confirmed
ThFiWU	2.8193751	2.861073	-0.3162157	4.266796	0.6683417	Confirmed
ThFiWMI	2.7553877	2.762794	0.5287007	4.289852	0.6482412	Confirmed

Section 17 Regression Analysis for Presentation Raw Scores

Note: The detailed results for this regression and the dominance analysis are presented in Table 5.2 in Section 5.3.

Fit the robust linear model (rlm) for presentation raw scores

`model5 <- rlm(PresRawScore ~ BigramFrequency + BigramProportion + BigramMI +
 TrigramFrequency + TrigramProportion + TrigramMI +
 Three.Five.WordUsage_RawScores + Three.Five.WordMI_RawScores,
 data = fluencyMWS_subset)`

`summary(model5)` # Print the summary of the rlm model5

Section 18: Dominance Analysis for Presentation Raw Scores

`relimp_results5 <- calc.relimp(model5)` # Calculate relative importance

`print(relimp_results5)` # Print relative importance metrics

Response variable: PresRawScore
 Total response variance: 30.59512
 Analysis based on 150 observations

8 Regressors:
 BigramFrequency BigramProportion BigramMI TrigramFrequency TrigramProportion TrigramMI Three.Fiv
 e.WordUsage_RawScores Three.Five.WordMI_RawScores
 Proportion of variance explained by model: 21.98%
 Metrics are not normalized (rela=FALSE).

Relative importance metrics:

	lmg
BigramFrequency	0.007450339

```

BigramProportion      0.005816081
BigramMI              0.128826345
TrigramFrequency      0.002702972
TrigramProportion     0.007081414
TrigramMI             0.007240193
Three.Five.WordUsage_RawScores 0.041345049
Three.Five.WordMI_RawScores  0.019292027

```

Average coefficients for different model sizes:

	1X	2Xs	3Xs	4Xs	5Xs
BigramFrequency	3.80901514	2.907595225	2.909519833	3.5245642735	4.470470709
BigramProportion	2.73233279	-2.456032629	-6.573170424	-9.7356206673	-11.904197185
BigramMI	11.29162910	11.871546577	12.428699720	12.9791810099	13.511391696
TrigramFrequency	3.62051751	2.357859560	1.125215711	-0.0278317499	-1.100005716
TrigramProportion	18.28787579	18.440385804	17.957124909	16.8607345718	15.107921272
TrigramMI	0.81550841	0.358790022	-0.168974084	-0.6960828552	-1.170761789
Three.Five.WordUsage_RawScores	0.25189122	0.283930355	0.323958782	0.3688898582	0.413996918
Three.Five.WordMI_RawScores	0.01384419	0.009441345	0.004404227	-0.0009271547	-0.006172841

	6Xs	7Xs	8Xs
BigramFrequency	5.52236836	6.50784542	7.29418638
BigramProportion	-12.97046284	-12.84143381	-11.46732976
BigramMI	13.99119030	14.39475850	14.71759398
TrigramFrequency	-2.00064757	-2.60771017	-2.80877397
TrigramProportion	12.49047075	8.78610415	3.81287391
TrigramMI	-1.57162558	-1.90155682	-2.18573154
Three.Five.WordUsage_RawScores	0.45559038	0.49251677	0.52527146
Three.Five.WordMI_RawScores	-0.01112559	-0.01576029	-0.02012555

Section 19: Random Forests Analysis for Presentation Raw Scores

`set.seed(123)` # Set the seed for reproducibility

`forest <- randomForest(PresRawScore ~., data=fluencyMWS_subset)` # Run random forest model

`print(forest)` # Print the summary of the random forest model

Call:

`randomForest(formula = PresRawScore ~ ., data = fluencyMWS_subset)`

Type of random forest: regression

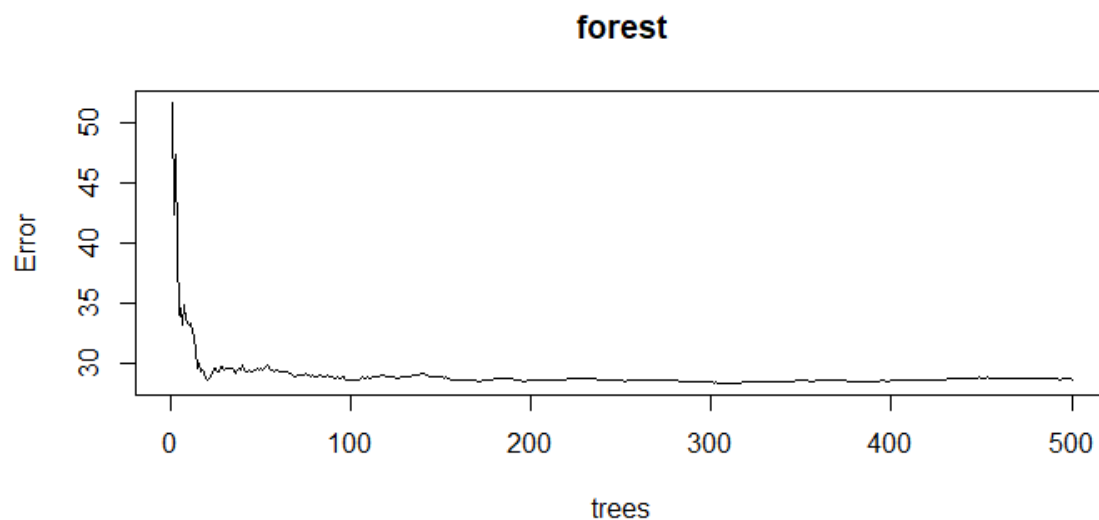
Number of trees: 500

No. of variables tried at each split: 2

Mean of squared residuals: 28.68067

% Var explained: 5.63

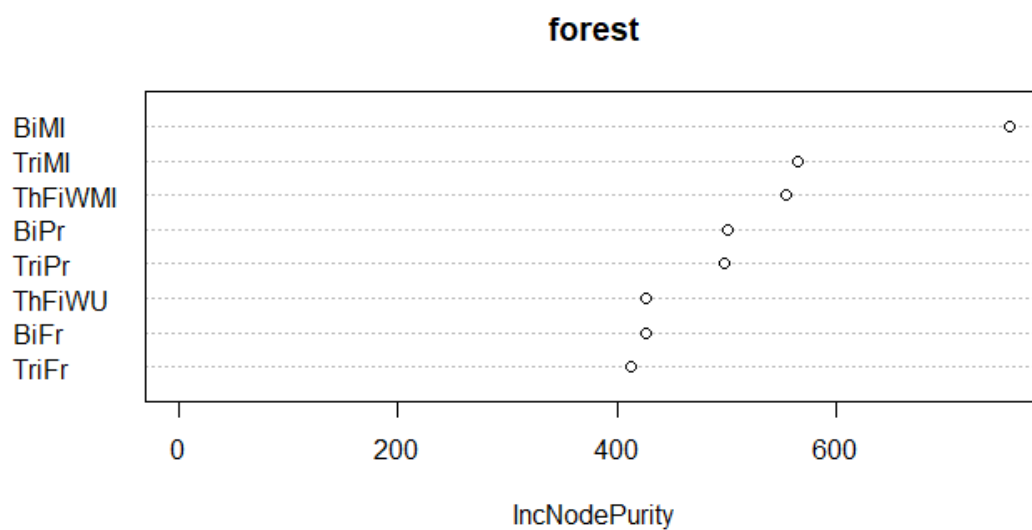
```
plot(forest)      # Plot the error rates for the random forest model
```



```
forest$importance  # Print the importance of each variable in the model
```

IncNodePurity	
BiFr	426.7300
BiPr	501.1696
BiMI	758.0912
TriFr	413.2714
TriPr	497.2588
TriMI	565.0906
ThFiWU	426.9129
ThFiWMI	553.8998

```
varImpPlot(forest)  # Create a plot showing the importance of each variable
```



Section 20: Boruta Analysis for Presentation Raw Scores

Conduct Boruta analysis to assess predictor importance

`set.seed(88)` # Set the seed for reproducibility

`boruta <- Boruta(PresRawScore ~., maxRuns = 200, data = fluencyMWS_subset, doTrace = 2)`

`print(boruta)` # Print the summary of the Boruta results

Boruta performed 199 iterations in 4.023549 secs.
 5 attributes confirmed important: BiMI, BiPr, ThFiWMI, ThFiWU, TriPr;
 2 attributes confirmed unimportant: TriFr, TriMI;
 1 tentative attributes left: BiFr;

`plot(boruta, cex.axis=0.7)` # Plot the Boruta results (See Figure 5.3 in the main text)

`attStats(boruta)` # Print detailed attribute (predictor) statistics

	meanImp	medianImp	minImp	maxImp	normHits	decision
BiFr	2.2010065	2.1907797	-0.8444531	5.2091931	0.407035176	Tentative
BiPr	4.9592707	4.9973312	1.0974161	8.1234120	0.849246231	Confirmed
BiMI	9.2754630	9.2674096	5.2658730	13.1527220	0.989949749	Confirmed
TriFr	-1.0054811	-1.0630377	-1.9453361	0.8583341	0.000000000	Rejected
TriPr	5.5457143	5.5469098	2.7947176	8.8312807	0.884422111	Confirmed
TriMI	0.2555087	0.4776789	-1.1449040	1.5994544	0.005025126	Rejected
ThFiWU	8.3270451	8.3700205	5.3404019	12.1870229	0.979899497	Confirmed
ThFiWMI	5.2638682	5.3396385	2.1541636	8.3487941	0.849246231	Confirmed

Appendix K

This appendix presents a side-by-side comparison of performance metrics for Model A (with bigram MI included) and Model B (with bigram MI excluded). The comparison was conducted using linear regression, as the dependent variable (presentation score) is continuous, making linear regression the most appropriate method for evaluating differences in model performance. Bigram MI was singled out for this comparison because it was the strongest predictor of presentation scores across various analyses in Chapter 5. If Model A performs substantially better than Model B, this suggests that the excluded parameter (bigram MI) is essential for classification. Table A17 highlights differences in Adjusted R^2 , RMSE (root mean squared error), AIC (Akaike information criterion), and BIC (Bayesian information criterion) to demonstrate the impact of excluding bigram MI on model performance. Higher Adjusted R^2 and lower RMSE, AIC, and BIC in Model A indicate that bigram MI is essential for achieving better model performance, as its removal significantly reduces the model's ability to explain variance and fit the data accurately.

Table A17: *Performance Metrics Comparison Between Model A and Model B (Chapter 5)*

Metric	Model A (with bigram MI)	Model B (without bigram MI)	Interpretation
Adjusted R^2	0.175	0.043	Model A explains more variance
RMSE	5.023	5.412	Model A has better prediction accuracy
AIC	920.582	942.053	Model A provides a better model fit
BIC	950.689	969.149	Model A provides a better model fit

Note. MI = mutual information, RMSE = root mean squared error, AIC = Akaike information criterion, and BIC = Bayesian information criterion. Model A includes all predictors, including bigram MI, while Model B excludes bigram MI.

Appendix L

This appendix presents the top 20 MI bigrams produced by the highest-scoring 50 participants in this thesis.

Table A18: *The Top 20 MI Bigrams Produced by the Highest-Scoring 50 Participants (Chapter 5)*

Rank (MI)	Bigram	MI score	Freq	Range
1	carbon dioxide	11.36	2	1
2	Kyoto protocol	11.19	2	2
3	Hong Kong	11.16	1	1
4	Saudi Arabia	9.77	1	1
5	infectious diseases	9.75	1	1
6	global warming	9.57	2	1
7	21st century	9.20	1	1
8	sexual harassment	9.06	7	1
9	19th century	8.84	1	1
10	breast cancer	8.65	8	8
11	human beings	8.59	1	1
12	greenhouse gas	8.16	3	2
13	intellectual property	8.16	16	5
14	environmental protection	7.84	1	1
15	wide range	7.81	3	3
16	pharmaceutical companies	7.67	1	1
17	virtual reality	7.58	2	1
18	substance abuse	7.57	1	1
19	gas emissions	7.48	1	1
20	17th century	7.41	1	1

Appendix M

Assumption Checks for One-Way ANOVA (Chapter 6)

This appendix presents the Shapiro-Wilk test results for normality for each pathway at three different MI score thresholds (Table A8), and Levene's test results for homogeneity of variances for each MI score threshold (Table A9).

Table A19: *Shapiro-Wilk Normality Test Results for Different MI Score Thresholds*

Threshold	Pathway	<i>W</i>	<i>p</i> -value
3	HSS	0.83293	< .001
3	Law	0.80459	< .001
3	STEM	0.75416	< .001
4	HSS	0.84397	< .001
4	Law	0.83084	< .001
4	STEM	0.76720	< .001
5	HSS	0.83774	< .001
5	Law	0.89034	< .001
5	STEM	0.90408	< .001

Table A20: *Levene's Test for Homogeneity of Variances for Different MI Score Thresholds*

Threshold	<i>df</i>	<i>F</i> value	<i>p</i> -value
3	2, 3835	1.4518	.234
4	2, 1576	4.1123	.017
5	2, 682	22.744	< .001

Appendix N

Detailed Results of Robust ANOVA and Post-Hoc Tests (Chapter 6)

Table A21: *Robust ANOVA Results for MI Score Thresholds*

MI Score	<i>F</i>	<i>p</i> -value	Variance Explained	Effect Size
3	0.8026	0.451	0.001	0.027
4	0.7399	0.481	0.002	0.047
5	6.4647	0.002	0.049	0.221

Table A22: *Post-Hoc Comparisons for MI Score Threshold of 5*

Comparison	Mean	95% CI Lower	95% CI Upper	<i>p</i> -value
HSS vs. STEM	−0.22221	−0.45633	0.01260	0.0272
HSS vs. Law	−0.53925	−0.94517	−0.19872	< .001
STEM vs. Law	−0.31704	−0.74031	0.06688	0.0538